

|                                      |                                                            |             |
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| SECRET                               | ENGINEERING STANDARD                                       | SPEC NO.    |
| ESTABLISHED BY                       |                                                            | ES 95486-02 |
| Electronics Network Development Team |                                                            |             |
| REL. DATE                            | ENGINEERING SPECIFICATON                                   | PAGE        |
| 2025-10-16                           |                                                            | 243         |
| TITLE                                | Unified Diagnostic Services (UDS) integrated Specification |             |

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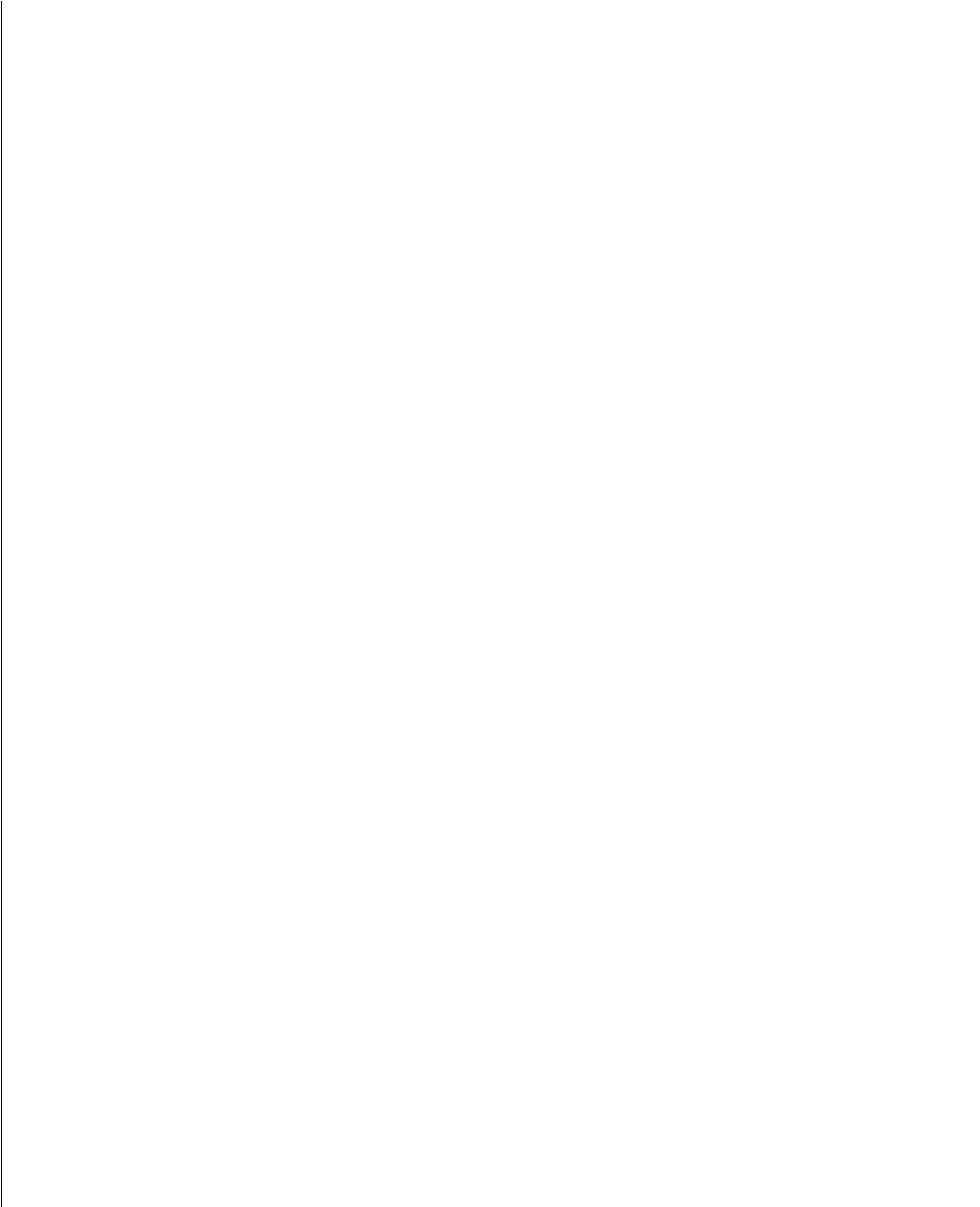
1. Scope
2. Protocol Definition
3. Diagnostic Service Implementation
4. Diagnostic and Communication Management Functional Unit
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- 1) Change Bootloader Optional conditions for OTA-Ready and Update Enable Condition
- 2) Add parameters for CPD intended for App Update
- 3) Add NRC24 condition to OTATLO SwapActiveArea
- 4) Supplement explanation regarding the requirement of Security Access in Routine control

2. In the diagnostic specification document, delete the Quiscent Current (D0, D1) session from Table in section 3.1 and also remove it from the subfunctions of DiagnosticSessionControl.

|                                                           |          |                                                                                                           |  |                       |                       |               |                 |              |
|-----------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------|--|-----------------------|-----------------------|---------------|-----------------|--------------|
| 23                                                        | EESP0612 | Additionally, specify the Safety session only in DiagnosticSessionControl and remove it from section 3.1. |  |                       |                       | 2025-10-16    | INJUN KO        |              |
|                                                           |          | 3. Add NRC24 condition to RequestDownload                                                                 |  |                       |                       |               |                 |              |
|                                                           |          | 4. Add OEUK 13 parameters to Response format and specify that                                             |  |                       |                       |               |                 |              |
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| AUTOONE<br>(STANDARD<br>INFORMATION<br>MAMAGEMENT SYSTEM) |          | REFERENCE                                                                                                 |  | SORT                  | PREPARED              | CHECKED       | CHECKED         | APPROVED     |
|                                                           |          |                                                                                                           |  | APPROVAL              | ses 00), the INJUN KO | JeongHwan Kim | MyeongGyu Jeong | Dongyoul Lee |
|                                                           |          |                                                                                                           |  |                       | 2025-07-24            | 2025-10-14    | 2025-10-16      |              |

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| 23       |  | EESP0612 | <p>1. Detailed Specification of Routine</p> <p>1) Change Bootloader Optional conditions for OTA Ready and Update Enable Condition</p> <p>2) Add parameters for CPD intended for App update</p> <p>3) Add NRC24 condition to OTA1.0 SwapActiveArea</p> <p>4) Supplement explanation regarding the requirement of Security Access in Routine control</p> <p>2. In the diagnostic specification document, delete the Quiscent Current (D0, D1) session from Table in section 3.1 and also remove it from the subfunctions of DiagnosticSessionControl. Additionally, specify the Safety session only in DiagnosticSessionControl and remove it from section 3.1.</p> <p>3. Add NRC24 condition to RequestDownload</p> <p>4. Add OEUK 13 parameters to Response format and specify that OEUK authorization requires ASK + Rollback</p> <p>5. Revise specification so that upon receipt of functional addresses (18DB33F1 / DoIP: E000 / E400), the controller will stop forwarding diagnostic requests to other controllers</p> <p>6. Add data parameters to confirm FIC0 External Bootloader state</p> <p>7. assign DIDs for Fault diagnostic functions, and assign DIDs for Ethernet physical layer diagnostics</p> <p>8. Define engine conditions for SDV architecture / other engine condition changes</p> |             | 2025-10-16 | INJUN KO |
|          |  |          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             |            |          |

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| 22       |  | EESO1769 | 1. 세션 컨트롤 7DF 10 01 대신 7DF 수신시 진단 요청 메시지 전송 중단(2.8절)<br>2. 고전압 전원 사용 제어기의 전원조건 예외/제약 추가 (3.1.1.3절)<br>3. RequestRoutineResult(0x03) subfunction에서 RequestRoutineStatus 허용 (Table 8-1)<br>4. 하나의 제어기에서 진단통신이 논리적으로 구분되어있는 경우, 엔진조건 E2E 기준 일치 (3.1.1.3절)<br>5. IVD의 CRC32 삭제 SHA256 이상으로 변경(Annex C, Annex K)<br>6. Security Access 표 4-81 No9의 Att_cnt +1 조건 삭제, DID F1C0 이름 변경 (Table 4-81, Annex C)                                                                |             | 2024-12-12 | INJUN KO |
| 21       |  | EESO0587 | 1. Security access 사양 상세 명세 (4.3절)<br>2. DID F1C0 추가(Annex. C)<br>3. Routine control RID 0301 추가 (8.1.5절)<br>4. SW Unit 상한 확대 : F1B1~F1BF, F1C1~F1CF (Annex. C)<br>5. CAN-TP DLC 상세 명세 (Table 2-28)<br>6. Authentication NRC54 삭제 (4.5.5절)<br>7. OBD법규 제어기 대상 ClearDiagnosticInformation 서비스의 엔진조건 추가(3.1절)<br>8. OTA Ready Parameter(Optional) 추가(8.1.5.1절)<br>9. SW플랫폼 통신 Log 확인 DID 추가 (Annex C)<br>10. 29bit OBD 법규 진단 ID 영역 추가 ( 2.4.1절)<br>11. 사양서 기타 오기/누락사항 |             | 2024-08-07 | INJUN KO |

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| 20       |  | EESN0821 | <p>1. 미슬립 검출 기능 DID 적용 : AnnexC</p> <p>2. OBD on UDS 법규 반영</p> <p>-1)ReadDTCInformation Service Subfunction 추가 : Table 6-13</p> <p>-2)DTC format Identifier(SAE_J2012-DA_DTCFormat_04) : Table 6-14</p> <p>-3) ClearDiagnosticInformation Group 변경 : Annex D Table D-1</p> <p>-4)Legislative use DID 영역 반영 : Annex C</p> <p>3. 사양서 오기 수정</p> <p>-ReadMemoryByAddress : Default session 삭제 3.1절</p> <p>- RXSWIN figure 수정 : Figure J-13, J-14</p> <p>- Authentication 4.5.6절 lengthOfCertificateClient 오기 수정</p> <p>- Authentication 4.5절 규격 설명 삭제</p> <p>- Authentication NRC list Annex A에 추가</p> <p>- Addressing 범위 오기 수정 2.4.1.2절</p> <p>4. RequestFileTransfer 서비스 추가 9.5절</p> <p>5. Routine control 8.1.4절 Implementation rule의 세션 보안레벨 추가</p> <p>6. EraseMemory parameter 추가 Table 8-96</p> |             | 2023-11-30 | INJUN KO |

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| 19      | EESN0487 | <p>1) 8.1.4 Routine control 수정 :<br/>ES98765-02 리프로그래밍 사양 제정에 따라 기존 리프로그래밍 관련 routine control Old 처리<br/>SwapActiveArea 의 엔진조건 보충설명, NRC F4 추가(다운그레이드 관련)<br/>CheckProgrammingDependencies<br/>NRC F4 추가(다운그레이드관련)</p> <p>2) 8.1.5 FoD관련 implementation rule 삭제 및 리프로그래밍 관련 Implementation rule 신규 :<br/>ES98765-02 리프로그래밍 사양 제정에 따라 리프로그래밍 관련 routine control 신규 생성</p> <p>3) 표8-3. Request message routineIdentifier definition 수정 :<br/>신규 Routine 추가</p> <p>4) Annex C의 DID추가<br/>F1C1~F1CA: Unit이 없을 때 F1C1값의 NRC31 케이스 추가<br/>ED90~ED99 : 미 슬립 DID추가<br/>F1A2 : Application 버전 DID 추가</p> <p>5) 10.1 본 리프로그래밍 사양은 Legacy이며 ES98765-02 참조 문구 추가</p> <p>6) Annex A<br/>NRC F4, NRC F5 추가</p> <p>7) Annex J.4<br/>29bit 진단 ID의 경우 Local RXSWIN 응답 예 추가</p> |             | 2023-07-12 | INJUN KO |
| 18      | EESN0189 | <p>[A] 2.4.1 Addressing 29bit CAN ID (Ethdiag / DoIP) 분리</p> <p>[B] 3.1 ReadMemoryByAddress 보안 레벨 변경 with security access</p> <p>[C] 4.3 Security access OEUK 추가 subfunction(13h/14h)</p> <p>[D] 4.5 Authentication(\$29) 서비스 추가</p> <p>[E] 9.1.1 Exception 추가</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |             | 2023-02-14 | INJUN KO |
| 17      | EESM0695 | <p>1. Annex.C (RXSWINEOLRESULT) DID F1EE 추가</p> <p>2. ReadDTCInformation 의 reportSupportedDTC(0x0A) Subfunction을 Optional에서 Mandatory로 변경</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             | 2022-11-04 | INJUN KO |

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| 16       | EESM0320 | 1) 표 9.1 IMPORTANT 추가<br>(Encrypting method)<br>2) Annex J RXSWIN 사양 상세 업데이트 (Example)<br>3) 2.4.1 절 29bit Addressing 추가<br>4) 3.1.1 엔진 Running/Stop 예시 업데이트<br>5) Annex C 진단기 표출 문구 통일을 위한 Notation 추가                                                                                                                                                                     |             | 2022-07-15 | INJUN KO |
| 15       | EESM0166 | 1. External test equipment 신규 사양 추가 2.9절<br>2. Engine 조건 변경<br>(Communication control의 IGN on 삭제) 3.1절<br>3. Routine control : Copy active area to, EraseTargetArea 신규 Parameter 속성(No specifying) 추가 (8.1.4.4절, 8.1.4.5절)<br>4. Annex C. 다음의 Data들은 ASCII code로 기록(DID F18C~F195)<br>5. 진단기의 DID표시 순서 변경 Table C-4.1. The order of DID in the diagnostic tool and Notation |             | 2022-03-14 | INJUN KO |
| 14       | EESL0710 | a) 3.1절 표3-1 과 표 3-1.1 의 불일치 수정<br>b) Annex C, F1C1~F1CA의 DID 이름 변경 'Hash -> IV D'                                                                                                                                                                                                                                                                                          |             | 2021-11-01 | INJUN KO |
| 13       | EESL0584 | 1. Annex. C 진단 Data identifier 변경<br>: 제어기 Hash DID추가 (F1C1~F1CA)<br><br>DID F1A0 User optional로 변경<br>2. Routine control<br>CheckProgrammingDependancy : Downgrade protection을 위한 NRC 10 설명 추가<br>3. Engine condition 관련 3.1절 사양 추가<br>: 사양 상세화, Communication control 엔진 조건 추가(GN7 부터 적용)                                                                                 |             | 2021-09-07 | INJUN KO |

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| 12       | EESL0492 | <p>1. Annex.J RXSWIN 사양<br/>-RXSWIN format has been modified : 년도 제거<br/>-Central Gateway RXSWIN management 추가<br/>-RXSWIN 수집 조회 분리를 위한 Routine control 추가</p> <p>2. Routine control (CheckProgrammingDependency) 에 누락된 NRC 72 추가</p> <p>3. Annex.C Data Transmission Functional Unit Data Parameter Definitions 의 DID 설명 개선</p> <p>4. Chapter (2.5.2.1) STmin 설명 개선 (내용은 기존과 동일)</p> |             | 2021-07-20 | INJUN KO     |
| 11       | EESL0177 | <p>Chapter 4.3. Security access login lock time<br/>- Requiremet Cvt has been changed from Optional to Mandatory<br/>- Lock time and Attempts has been changed.</p> <p>Chapter 8.1.4.2 EraseTargetArea<br/>- Additional parameter has been added.(Memory block)</p> <p>Chapter Annex. J<br/>- RXSWIN diagnostics requirement</p>                                                      |             | 2021-04-15 | INJUN KO     |
| 10       | EESL0083 | <p>Communication control<br/>-이 서비스는 게이트웨이 라우팅에 영향을 주지 않는다.</p>                                                                                                                                                                                                                                                                                                                       |             | 2021-03-10 | INJUN KO     |
| 9        | EESL0032 | <p>1. FoD RoutineControl 요구사항 추가<br/>2. OTA Swap active area DelayTime parameter 추가<br/>3. Security access 및 RoutineControl 관련 잘못 된 표현 수정</p>                                                                                                                                                                                                                                         |             | 2021-01-20 | INJUN KO     |
| 8        | EESK0507 | <p>버전 관련 DID 변경.<br/>OTA 관련 routine control parameter 추가</p>                                                                                                                                                                                                                                                                                                                          |             | 2020-07-01 | INJUN KO     |
| 7        | EESJ1207 | 2.6.6절 요구사항 C 삭제 (ECU Reset 삭제)                                                                                                                                                                                                                                                                                                                                                       |             | 2019-12-30 | PILYONG PARK |
| 6        | EESJ1055 | <p>1. 게이트웨이 인증 NRC 추가 (F0~F3)<br/>2. OTA DID 수정<br/>3. CANFD TP 사양 문구 오류 수정<br/>4. \$22 NRC13/31 ISO 개정 사양 반영<br/>5. \$2A 응답 메시지 ISO 개정 사양 반영</p>                                                                                                                                                                                                                                     |             | 2019-11-29 | PILYONG PARK |

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| 5        | EESJ0547 | 1. OTA 요구사항 반영<br>- 진단 세션별 서비스 지원 목록 수정<br>- Multiple Concurrent Request 예외 조건 추가<br>- OTA 백그라운드 전송 후 Auto Reset 발생하지 않도록 문구 추가<br>- OTA용 RoutineControl 명세 추가<br>- OTA용 DID 추가<br>- 압축/암호화 방식 적용을 위한 dataFormat 정의<br><br>2. DiagnosticSessionControl 문구 명확화                                                                                                                                                                                           |             | 2019-07-04 | PILYONG PARK |
| 4        | EESI0984 | 1. SecurityAccess(27 hex) service<br>- Unlocked flash level 상태 추가<br>2. RequestDownload(34 hex) service<br>- DataFormatIdentifier 압축방식 및 암호화 사양 추가<br>3. Annex D 5<br>- DTCStatusAvailabilityMask 정의 추가<br>4. InputOutputControlByIdentifier(2F hex) service<br>- controlState definition 표 삭제                                                                                                                                                          |             | 2018-11-22 | PILYONG PARK |
| 3        | EESI0650 | 1. Diagnostic CAN ID 신규 추가 제어기 및 변경사항 일괄 반영<br><br>2. 로트 추적 시스템 구축 강화 위한 Data ID(DID) 명확화 및 조건 변경<br>: F187(차량 제조사 Part Number) / F18B(ECU 제조 일자) / F18C(ECU Serial Number) / F1A1[신규](ECU Supplier 코드)<br><br>3. Extended Session에서 Non-Default Session으로 천이 시 Security Unlock 상태 유지 (ICU, ESU, CCU 대상)<br><br>4. 이더넷 제어기 D-CAN(500kbps) 이용한 일반 진단통신 멀티프레임 전송 시, Flow Control 내 STmin값 10ms 이상 설정<br><br>5. 진단기 P2_Client 값 증대 : [기존] 50ms → [변경] 80ms |             | 2018-09-13 | PILYONG PARK |
| 2        | EESI0109 | Annex. H - 신규 제어기 진단 CAN ID 추가 및 일부 제어기 명칭 변경                                                                                                                                                                                                                                                                                                                                                                                                           |             | 2018-03-08 | HOJIN JUNG   |

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| 1        |  | EESI0061 | 1. 신규 제어기 CAN ID 신규 반영<br>2. 이더넷 관련 내용 추가 : 범위, ISO<br>참고 문서, H/KMC 참고 문서,Protocol<br>정의 등 추가<br>3. 부정응답 메시지 구조 추가<br>4. 진단 서비스 엔진 조건 예외처리 문<br>구 추가 |                | 2018-01-22    | HOJIN JUNG |
| 0        |  | EESH0520 | 신규 UDS 통합사양 제정                                                                                                                                       |                | 2017-05-19    | HOJIN JUNG |

## 1. Scope

This standard specifies requirements of diagnostic services, which allow a diagnostic tester (client) to control diagnostic functions in an on-vehicle Electronic Control Unit (server) in all HMC/KMC vehicles via CAN communication. The CAN-FD application vehicle conforms to this UDS integrated Specification (ES95486-02) regardless of CAN, CAN-FD or Ethernet ECU. The ECU applying this specification shall be evaluated in accordance with the ES95486-12 Diagnostic Communication Test Specification for Electronic Control Unit.

NOTE) This standard specification is described based on ISO 14229-1/2 and ISO 15765-2/3.

This standard does not apply to nondiagnostic message transmission, use of the communication data link between two Electronic Control Units.

This standard does not include CARB-Mode test (Emission-related diagnostic services) separately. For the CARB-Mode test, must comply with the OBD regulation document, such as ISO 15031-5 (Road vehicles – Communication between vehicle and external equipment for emissions-related diagnostics – Part 5: Emission-related diagnostic services) or SAE J1979 (E/E Diagnostic Test Modes - Equivalent to ISO/DIS 15031-5)

| Service Identifier<br>(hex) | Service Type                         |
|-----------------------------|--------------------------------------|
| 00 – 0F                     | Reserved (OBD service request)       |
| 10 – 3E                     | HMC/KMC service request              |
| 3F                          | Reserved                             |
| 40 – 4F                     | Reserved (OBD service response)      |
| 50 – 7E                     | HMC/KMC service positive response    |
| 7F                          | Negative response service identifier |
| 80 – BE                     | HMC/KMC service request              |
| BF                          | Reserved                             |
| C0 - FE                     | HMC/KMC service positive response    |
| FF                          | Reserved                             |

Table 1-1. Service Identifier (SI) Overview

### 1.1 Normative References

This standard and the descriptions of diagnostic communication protocol are based on the following standards. Regarding implementation related specification which is not described in this HMC/KMC standard and other details, refer to the following standards :

#### 1.1.1 ISO Publications

|             |                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| ISO 15765-2 | Road vehicles – Diagnostics Communication over CAN (DoCAN) -<br>Part 2 : Transport protocol and network layer services : 2016 |
| ISO 15765-3 | Road vehicles – Diagnostics on CAN -<br>Part 3 : Implementation of unified diagnostic services<br>(UDS on CAN) : 2004         |
| ISO 14229-5 | Road vehicles – Unified Diagnostic Services (UDS) –                                                                           |

Part 5 : Unified diagnostic services on Internet Protocol implementation  
(UDS on IP) : 2013

ISO 14229-3 Road vehicles – Unified Diagnostic Services (UDS) -  
Part 3 : Unified diagnostic services on CAN implementation  
(UDS on CAN) : 2012

ISO 14229-2 Road vehicles – Unified diagnostic services (UDS) –  
Part 2 : Session layer services: 2013

ISO 14229-1 Road vehicles – Unified diagnostic services (UDS) –  
Part 1 : Specification and requirements : 2013

ISO 15031-6 Road vehicles - Communication between vehicle and external  
equipment for emissions-related diagnostics -  
Part 6: Diagnostic trouble code definitions : 2015

ISO 11898-1 Road vehicles - Controller area network (CAN) -  
Part 1: Data link layer and physical signaling : 2015

ISO 11898-2 Road vehicles - Controller area network (CAN) –  
Part 2: High-speed medium access unit : 2016

ISO 13400-2 Road vehicles – Diagnostic communication over Internet Protocol (DoIP)  
Part 2 : Transport protocol and network layer services

ISO 13400-3 Road vehicles – Diagnostic communication over Internet Protocol (DoIP)  
Part 3 : Wired vehicle interface based on IEEE 802.3

ISO 13400-4 Road vehicles – Diagnostic communication over Internet Protocol (DoIP)  
Part 4 : Ethernet-based high-speed data link connector

### 1.1.2 SAE Publications

SAE J1962 Diagnostic Connector : 2016

SAE J2012 Diagnostic Trouble Code Definitions : 2016

### 1.1.3 HMC/KMC Publications

ES 95480-00 HMC/KMC Engineering Standard –  
High Speed Controller Area Network (CAN) Design Specification

ES 95480-02 HMC/KMC Engineering Standard –  
Controller Area Network – Flexible Data Rate (CAN-FD) Design  
Specification

ES 95489-01 HMC/KMC Engineering Standard –  
ECU Secure Diagnosis Functional Specification

ES 96595-05 HMC/KMC Engineering Standard –



# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)

ES95486-02

페이지

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### IP Middleware Specification

ES 95486-03

HMC/KMC Engineering Standard –  
Ethernet IP Middleware Diagnostic Communication Specification

ES 95486-12

HMC/KMC Engineering Standard –  
Diagnostic Communication Test Specification for Electronic Control Unit

#### 1.1.4 Document Structure

The standard specification is the general specifications establishing the communications concept. For a given controller, it is possible that not everything in the standard specifications will apply.

The device specification (each controller specific) would contains :

- References to the optional parts of standard specifications that are applicable to the controller.
- Specific information (e.g. signals and messages) received and transmitted by that controller.

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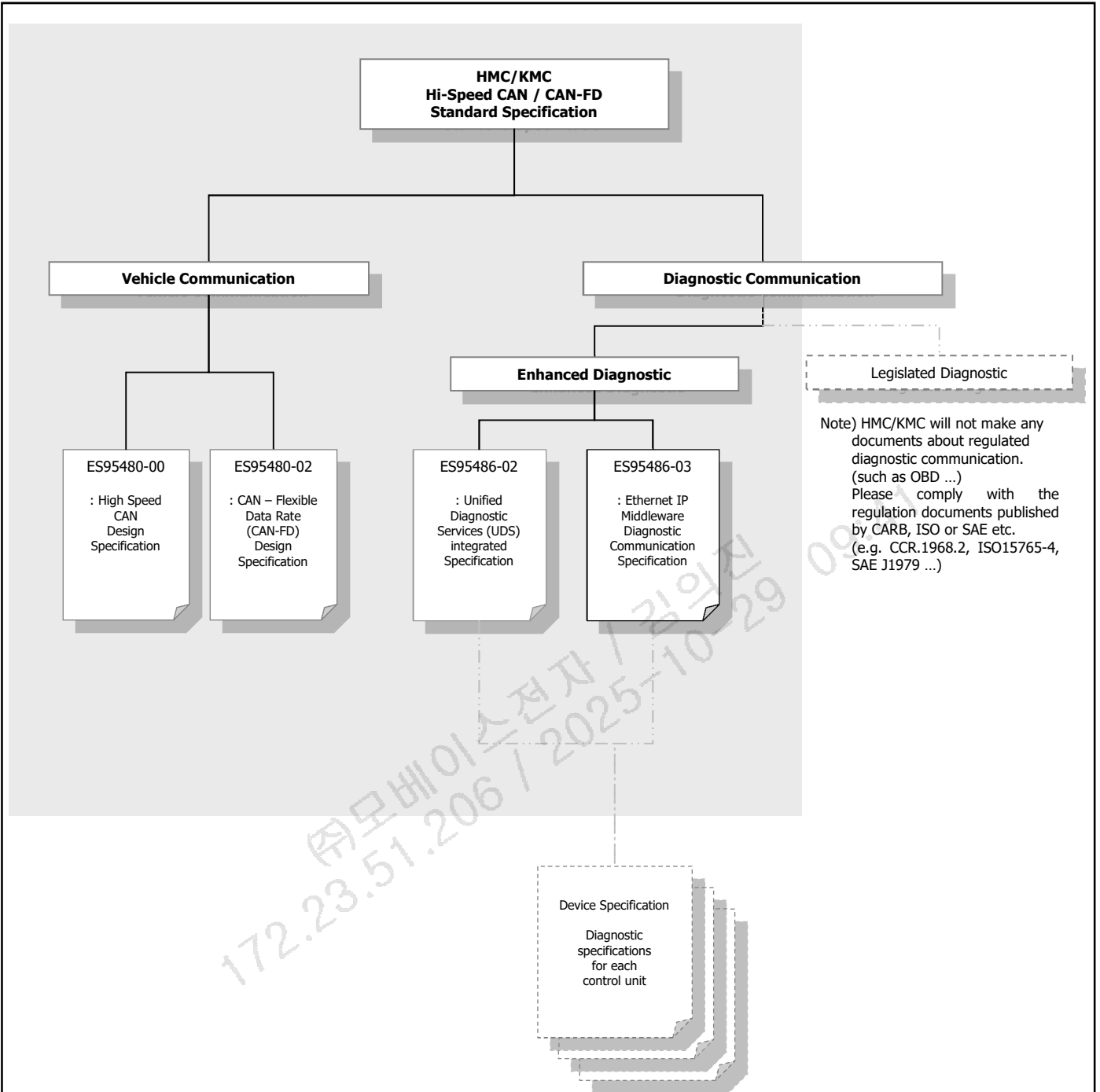


Figure 1-1. HMC/KMC Standard Documents Structure

## 1.2 Conventions

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| M   | : Mandatory [This parameter has to be present.]                                                     |
| U   | : User Optional [This parameter may or may not be present, depending on dynamic usage by the user.] |
| C   | : Conditional [The presence of this parameter depends on other parameters within the service.]      |
| CVT | : Conventions                                                                                       |
| XXh | : Hexadecimal number                                                                                |
| XXd | : Decimal number                                                                                    |
| XXb | : Binary number                                                                                     |

## 1.3 Definitions And Abbreviations

|        |                                                                            |
|--------|----------------------------------------------------------------------------|
| BS     | : Block Size                                                               |
| CAN    | : Controller Area Network                                                  |
| CAN-FD | : Controller Area Network – Flexible Data Rate                             |
| CF     | : Consecutive Frame                                                        |
| DL(C)  | : Data Length (Code)                                                       |
| ECU    | : Electrical Control Unit (a generic term for any electrical control unit) |
| FC     | : Flow Control                                                             |
| FF_DL  | : First Frame Data Length                                                  |
| FS     | : Flow Status                                                              |
| ID     | : Identifier                                                               |
| N_Ar   | : Network layer timing parameter Ar (r = receiver of the message)          |
| N_As   | : Network layer timing parameter As (s = sender of the message)            |
| N_Br   | : Network layer timing parameter Br                                        |
| N_Bs   | : Network layer timing parameter Bs                                        |
| N_Cr   | : Network layer timing parameter Cr                                        |
| N-Cs   | : Network layer timing parameter Cs                                        |
| N_PCI  | : network Protocol Control Information                                     |
| N_PDU  | : Network protocol data unit                                               |



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N\_USData : Network layer Unacknowledged Segmented DATA transfer service name  
SF\_DL : Single Frame Data Length  
SN : Sequence Number  
STmin : Separation Time Minimum

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## 2. Protocol Definition

### 2.1 CAN Physical Layer

The physical layer of diagnostics on CAN shall be compliant to ES95480-00. External Diagnostic Tool should use CAN protocol for diagnostic communication.

### 2.2 CAN-FD Physical Layer

The physical layer of diagnostics on CAN-FD shall be compliant to ES95480-02. Only CAN-FD protocol messages are allowed in the CAN-FD communication line. (Diagnostic Communication of CAN-FD ECUs should be used as CAN-FD communication (Both Tx and Rx))

### 2.3 Ethernet Physical Layer

The physical layer of diagnostics on Ethernet shall be compliant to ES96595-00.

### 2.4 Network Layer

The network layer of the external test equipment and the vehicle ECUs shall be compliant to ISO 15765-2, with the restrictions/additions as defined below.

#### 2.4.1 Addressing

For diagnostic communication on CAN only the normal addressing format as defined in ISO 15765-2 shall be used and 11-bit CAN identifiers and 29-bit CAN identifiers shall be used.

- 11-bit CAN identifiers with normal addressing
- 29-bit CAN identifiers with normal addressing

Two different addressing methods, called physical addressing, and functional addressing are specified for diagnostics.

- Physical addressing (1-to-1 communication) shall be supported for all types of network layer messages.
- Functional addressing (1-to- $n$  communication) shall only be supported for Single Frame transmission.
  - ※ Because the response regarding Functional addressing (07DF) request is applied with Physical addressing (Response ID of ECU), ECUs (Server) should transmit multi-frame.
- 29 bits are divided into addressing of a non-OBD controller and an OBD controller. OBD (SAE J1979 compliant) controllers may use the CAN identifiers 11 bit area or the CAN identifiers 29 bit (non-OBD controllers) area for general diagnostic purposes. CAN identifiers 29 bit - Controllers using the OBD Controller (SAE J1979) shall not respond to an 11-bit OBD (7DF) diagnostic request; see ISO 15765-4 "External test equipment initialization sequence." for details.

##### 2.4.1.1 CAN identifiers 11bit

- Physical addressing ID : 700-7FF, 580-5FF
- Functional addressing ID : 7DF

##### 2.4.1.2 CAN identifiers 29bit

#### Non OBD Controller

- Physical addressing ID : 17FF0001~17FFFE00
- Functional addressing ID : 7DF
- Physical addressing ID format :

17FF[TA 1byte][SA 1byte]

\*TA : Target address

\*SA : Source address

※ SA assign rule

0x00 : External diagnostic tool

0x01~0x10 : Reserved

0x11~0xEF : Unique ECU address

0xF0~0xFF : Reserved

e.g.) ECU Source Address : 0xCA

Diagnostic Request ID : 0x17FFCA00

Diagnostic Response ID : 0x17FF00CA

OBD Controller(SAE J1979)

- Physical addressing ID : 18DA####
- Functional addressing ID : 18DB33F1
- Physical addressing ID format :  
18DA[TA 1byte][SA 1byte]

※ SA assign rule

0x00~0x10 : Reserved

0x11~0x32 : Unique ECU address

0x33 : Reserved for functional group

0x34~0xEF : Unique ECU address

0xF0~0xFF : Reserved (F1: Reserved for external test equipment)

### 2.4.1.3 CAN ID 2byte Conversion

The Ethernet diagnostic logical address is 2 bytes, and the CAN ID is converted to 2 bytes and used as a logical address. In addition to the Ethernet diagnostic logical address, the local RXSWIN response format CAN ID also requires a 2 byte conversion rule.

#### 11bit CAN Identifier

5Bit of MSB is padded to 0 to extend 11bits to 16bits

Request ID is used for logical address 2byte conversion for Ethernet diagnostic

e.g.) hexadecimal : 0x700<sub>(16)</sub> -> 0x0700<sub>(16)</sub>

binary : 0b111 0000 0000<sub>(2)</sub> -> 0b0000 0111 0000 0000<sub>(2)</sub>

#### 29bit CAN Identifier

##### a) Ethdiag

Request ID : 0x17FFCA00

Response ID : 0x17FF00CA

Request Logical address : CA00

Response Logical address : 00CA

Example)

| CAN to                          | Ethdiag      |         |               |               |
|---------------------------------|--------------|---------|---------------|---------------|
| CAN ID                          | Payload type | REQ/RES | TargetAddress | SourceAddress |
| 11bit<br>REQ/RES :<br>700 / 708 | 8001         | REQ     | 0x0700        | -             |
|                                 |              | RES     | 0x0708        | -             |
|                                 | FCBC         | REQ     | 0x0700        | 0x1000        |
|                                 |              | RES     | 0x1000        | 0x0708        |



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| CAN to                                       | Ethdiag |     |        |        |
|----------------------------------------------|---------|-----|--------|--------|
| 29bit<br>REQ/RES :<br>17FFCA00 /<br>17FF00CA | 8001    | REQ | 0xCA00 | -      |
|                                              |         | RES | 0x00CA | -      |
|                                              | FCBC    | REQ | 0xCA00 | 0x1000 |
|                                              |         | RES | 0x1000 | 0x00CA |

| Ethdiag to   |         |               |               | CAN        |
|--------------|---------|---------------|---------------|------------|
| Payload type | REQ/RES | TargetAddress | SourceAddress | CAN ID     |
| 8001         | REQ     | 0x07DF        | -             | 0x7DF      |
|              | REQ     | 0x0700        | -             | 0x700      |
|              | RES     | 0x0708        | -             | 0x708      |
|              | REQ     | 0xCA00        | -             | 0x17FFCA00 |
|              | RES     | 0x00CA        | -             | 0x17FF00CA |
| FCBC         | REQ     | 0xCA00        | 0x1000        | -          |
|              | RES     | 0x1000        | 0x00CA        | -          |

### b) DoIP

For the bits needed to convert 29 bits to 1 byte, refer to the binary numbers below.  
Binary : 1 01DD DDDD DDDD [TA] [SA]

\*D is Don't care

Convert 2 bytes by combining the MSB 3bits and the ECU Address(Source Address)  
0(Fixed 1bit) + 101(MSB 3bit) + 0000(Reserved 4bit) + SA 1byte

e.g.) ECU Source address is 0xCA

Diagnostic Request : 0x17FFCA00

Diagnostic Response : 0x17FF00CA

Logical address for DoIP diagnostic : 0x50CA

### 2.4.2 Standardized Diagnostic CAN Identifiers

See Annex. H

## 2.4.3 Data Stream

### 2.4.3.1 Single frame transmission

Transmission of messages up to six (6) (in case of extended or mixed addressing) or seven (7) (in case of normal addressing) data bytes is performed via transmission of a Single Frame (see Figure 2-1).

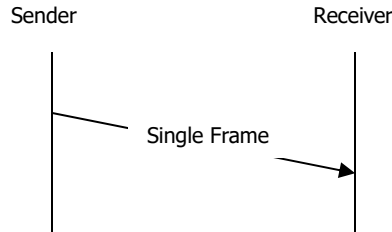


Figure 2-1. Example of unsegmented message

### 2.4.3.2 Multiple frame transmission

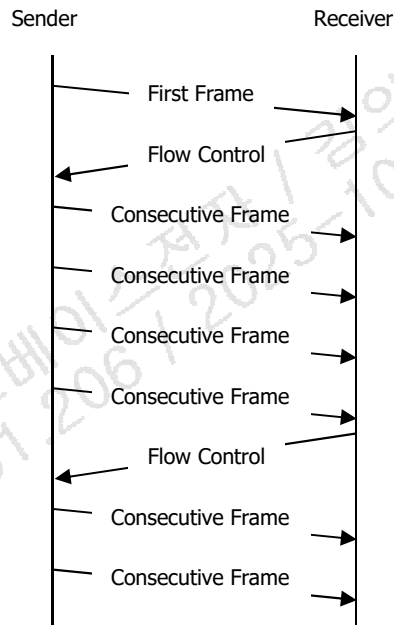


Figure 2-2. Example of segmented message

Transmission of longer messages is performed via segmentation of the message and transmission of multiple messages. Reception of longer messages is performed via reception of multiple messages and reassembly of the received data bytes (concatenation). The multiple messages are called FirstFrame and ConsecutiveFrame. (see Figure 2-2)

The receiver of a multiple message has the possibility of adapting the transmission throughput to its capability by means of the flow control mechanism using the FlowControl protocol data units. Messages longer than six (6) or seven (7) data bytes are segmented into

- a FirstFrame protocol data unit, containing the first five (5) — in the case of extended or mixed addressing — or six (6) — in the case of normal addressing — data bytes, and
- one or more ConsecutiveFrame protocol data units, containing each six (6) or seven (7) data bytes. The CF N\_PDU contains only the remaining data bytes, and may therefore be less than six (6) or seven (7) data bytes long.

### 2.4.4 CAN Message Structure

The DLC (CAN data length code) shall be set to eight (8) and the CAN frame shall include eight (8) bytes . Therefore, all frames must be padded with filler bytes.

- All CAN frames of a request message are padded with a filler byte of 55 hex.
- All CAN frames of a response message are padded with a filler byte of AA hex.

The 8 bytes of the CAN frame defined as follows :

| N_AI                               | N_PCI                           | N_Data                    |
|------------------------------------|---------------------------------|---------------------------|
| Address Information<br>(Not used)  | Protocol Control<br>Information | Databytes for Application |
| 1 – 8 bytes data of CAN Data Field |                                 |                           |

Table 2-3. Data bytes of CAN data field

Because address information(N\_AI) is not needed in the normal addressing mode, the above 8 databytes can be defined as follows (Normal Addressing) :

|                   | Byte 1     |   |   |   | Bit 3-0 | Byte 2 | Byte 3 | Byte 4 ~<br>Byte 7 | Byte 8 |
|-------------------|------------|---|---|---|---------|--------|--------|--------------------|--------|
|                   | N_PCIttype |   |   |   |         |        |        |                    |        |
|                   | 7          | 6 | 5 | 4 |         |        |        |                    |        |
| Single Frame      | 0          | 0 | 0 | 0 | SF_DL   | DATA 1 | DATA 2 | ...                | DATA 7 |
| First Frame       | 0          | 0 | 0 | 1 | FF_DL   |        | DATA 1 | ...                | DATA 6 |
| Consecutive Frame | 0          | 0 | 1 | 0 | SN      | DATA 1 | DATA 2 | ...                | DATA 7 |
| Flow Control      | 0          | 0 | 1 | 1 | FS      | BS     | STmin  | N/A                |        |

Table2-4. Mapping of parameters and data into CAN frame – Normal Addressing

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0   | <b>Single Frame</b><br><br>For unsegmented messages, the network layer protocol provides an optimized implementation of the network protocol with the message length embedded in the PCI byte only. SingleFrame (SF) shall be used to support the transmission of messages that can fit in a single CAN frame.                                                                                                                                |
| 1   | <b>First Frame</b><br><br>A FirstFrame (FF) shall only be used to support the transmission of messages that cannot fit in a single CAN frame, i.e. segmented messages. The first frame of a segmented message is encoded as an FF. On receipt of an FF, the receiving network layer entity shall start assembling the segmented message.                                                                                                      |
| 2   | <b>Consecutive Frame</b><br><br>When sending segmented data, all consecutive frames following the FF are encoded as ConsecutiveFrame (CF). On receipt of a CF, the receiving network layer entity shall assemble the received data bytes until the whole message is received. The receiving entity shall pass the assembled message to the adjacent upper protocol layer after the last frame of the message has been received without error. |
| 3   | <b>Flow Control</b><br><br>The purpose of FlowControl (FC) is to regulate the rate at which CF N_PDUs are sent to the receiver. Three distinct types of FC protocol control information are specified to support this function. The type is indicated by a field of the protocol control information called FlowStatus (FS), as defined hereafter.                                                                                            |

Table 2-5. Definition of N\_PCIttype values

| N_PCI | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SF_DL | Single Frame Data Length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | Value (HEX)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Reserved                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|       | 1 ~ 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | The SF_DL is encoded in the low nibble of N_PCI byte #1 value. It shall be assigned the value of the service parameter <Length>.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|       | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | SF_DL = 7 is only allowed with normal addressing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|       | 8 ~ F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| FF_DL | First Frame Data Length                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | Value (HEX)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|       | 0 – 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|       | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | FF_DL = 7 is only allowed with extended or mixed addressing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|       | 8 ~ FFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | The encoding of the segmented message length results in a twelve (12) bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit “0” of the N_PCI byte #2 and the most significant bit (MSB) is bit three (3) of the N_PCI byte #1. The maximum segmented message length supported is equal to 4095 bytes of user data. It shall be assigned the value of the service parameter <Length>                                                                                                                                                                                               |
| SN    | Sequence Number                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | <p>The SequenceNumber (SN) shall be encoded in the lower nibble bits of N_PCI byte #1. The SN shall be set to a value within the range of zero (0) to fifteen (15).</p> <ul style="list-style-type: none"><li>- The SN shall start with zero (0) for all segmented messages.</li><li>- The FF shall be assigned the value zero (0).</li><li>- The SN of the first CF that immediately follows the FF shall be set to one (1).</li><li>- The SN shall be incremented by one (1) for each new CF that is transmitted during a segmented message transmission.</li><li>- When the SN reaches the value of fifteen (15), it shall wraparound and be set to zero (0) for the next CF.</li></ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| FS    | Flow Status                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | The parameter FlowStatus (FS) indicates whether the sending network entity can proceed with the message transmission.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|       | Value (HEX)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>ContinueToSend (CTS)</b><br>The FlowControl ContinueToSend parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “0”. It shall cause the sender to resume the sending of Consecutive frames. The meaning of this value is that the receiver is ready to receive a maximum of BS number of Consecutive frames.                                                                                                                                                                                                                                                                         |
|       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Wait (WT)</b><br>The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “1”. It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer.                                                                                                                                                                                                                                                                                                                                                                             |
|       | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Overflow (OVFLW)</b><br>The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to “2”. It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. (refer to ISO 15765-3)<br>This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU exceeds the buffer size of the receiving entity. |
| BS    | Block Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|       | The units of BS are the absolute number of CF N_PDUs per block.                                                                             |                                                                                                                                                                                                                                                                                                                             |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | Value (HEX)                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                 |
|       | 00                                                                                                                                          | The BS parameter value zero (0) shall be used to indicate to the sender that no more FC frames shall be sent during the transmission of the segmented message. The sending network layer entity shall send all remaining consecutive frames without any stop for further FC frames from the receiving network layer entity. |
|       | 01 ~ FF                                                                                                                                     | This range of BS parameter values shall be used to indicate to the sender the maximum number of consecutive frames that can be received without an intermediate FC frame from the receiving network entity.                                                                                                                 |
| STmin | <b>Minimum Separation Time</b>                                                                                                              |                                                                                                                                                                                                                                                                                                                             |
|       | The STmin parameter value specifies the minimum time gap allowed between the transmission of consecutive frame network protocol data units. |                                                                                                                                                                                                                                                                                                                             |
|       | Value (HEX)                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                 |
|       | 00 ~ 7F                                                                                                                                     | 0 ms ~ 127 ms, The units of STmin in the range 00 hex ~ 7F hex are absolute milliseconds (ms).                                                                                                                                                                                                                              |
|       | F1 ~ F9                                                                                                                                     | 100 $\mu$ s ~ 900 $\mu$ s, The units of STmin in the range F1 hex ~ F9 hex are even 100 microseconds ( $\mu$ s), where parameter value F1 hex represents 100 $\mu$ s and parameter value F9 hex represents 900 $\mu$ s.                                                                                                     |
|       | FA ~ FF                                                                                                                                     | Reserved                                                                                                                                                                                                                                                                                                                    |

Table 2-6. Definition of N\_PCI

### 2.4.5 CAN-FD Message Structure

The DLC (CAN data length code) shall be set to greater than or equal to eight (8) and the CAN-FD frame can use up to sixty four (64) bytes. The DLC as following table. The DLC must be fixed as eight (8) when ECUs transmit diagnostic response messages.

If diagnostic tester is connected directly to CANFD network, Multiple frames sent by the diagnostic tester are fixed with DLC 64 except for the last consecutive frame.

| Data Length Code (DLC) |   |   |   | Number of data bytes (CAN_DL) |
|------------------------|---|---|---|-------------------------------|
| 1                      | 0 | 0 | 0 | 8                             |
| 1                      | 0 | 0 | 1 | 12                            |
| 1                      | 0 | 1 | 0 | 16                            |
| 1                      | 0 | 1 | 1 | 20                            |
| 1                      | 1 | 0 | 0 | 24                            |
| 1                      | 1 | 0 | 1 | 32                            |
| 1                      | 1 | 1 | 0 | 48                            |
| 1                      | 1 | 1 | 1 | 64                            |

Table 2-7. CAN-FD data length

All frames must be padded with filler bytes according to CAN\_DL in the table above

- All CAN frames of a request message are padded with a filler byte of 55 hex.
- All CAN frames of a response message are padded with a filler byte of AA hex.

The bytes of the CAN-FD frame defined as follows :

| N_AI                                | N_PCI                           | N_Data                    |
|-------------------------------------|---------------------------------|---------------------------|
| Address Information<br>(Not used)   | Protocol Control<br>Information | Databytes for Application |
| 1 – 64 bytes data of CAN Data Field |                                 |                           |

Table 2-8. Data bytes of CAN-FD data field

Because address information(N\_AI) is not needed in the normal addressing mode, CAN-FD frames can be defined as follows (Normal Addressing).

| N_PDU name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | N_PCI bytes       |                   |                        |                   |         |         |         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|------------------------|-------------------|---------|---------|---------|
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Byte #1           |                   | Byte #2                | Byte #3           | Byte #4 | Byte #5 | Byte #6 |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Bits 7-4          | Bits 3-0          |                        |                   |         |         |         |
| SingleFrame (SF)<br>(CAN_DL ≤ 8) (c)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0000 <sub>2</sub> | SF_DL             | DATA 1                 | DATA2             | ...     | ...     | ...     |
| SingleFrame (SF)<br>(CAN_DL > 8) (a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0000 <sub>2</sub> | 0000 <sub>2</sub> | SF_DL                  | DATA 1            | DATA2   | ...     | ...     |
| FirstFrame (FF)<br>(FF_DL ≤ 4095) (d)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0001 <sub>2</sub> | FF_DL             |                        | DATA 1            | DATA2   | ...     | ...     |
| FirstFrame (FF)<br>(FF_DL > 4095) (b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0001 <sub>2</sub> | 0000 <sub>2</sub> | 0000 0000 <sub>2</sub> | FF_DL             |         |         |         |
| Consecutive<br>Frame (CF)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 0010 <sub>2</sub> | SN                | DATA 1                 | DATA2             | ...     | ...     | ...     |
| FlowControl (FC)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0011 <sub>2</sub> | FS                | BS                     | ST <sub>min</sub> | N/A     | N/A     | N/A     |
| NOTE Shaded cells are not utilized for PCI information, but depending on the PDU, they might be utilized for payload data.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |                   |                        |                   |         |         |         |
| <p>(a) Messages with CAN_DL&gt;8 shall use an escape sequence where the lower nibble of Byte #1 is set to 0 (invalid length). This signifies to the network layer that the value of SF_DL is determined based on the next byte in the frame (Byte #2). As CAN_DL is defined to be greater than 8 this definition is only valid for CAN FD type frames.</p> <p>(b) Messages larger than 4095 bytes shall use an escape sequence where the lower nibble of Byte#1 and all bits in Byte #2 are set to 0 (invalid length). This signifies to the network layer that the value of FF_DL is determined based on the next 32 bits in the frame (Byte #3 is the MSB and Byte #6 the LSB)</p> <p>(c) The DLC must be fixed as eight (8) when ECUs transmit diagnostic response messages for compatibility with an existing tester.</p> <p>(d) CAN-FD ECUs do not exceed 4095 bytets, when ECUs transmit diagnostic messages, First Frame must follow format (d)</p> <p>(*) It is allowed to Frame format (a) ~ (d) all when ECUs receive diagnostic messages, however when ECUs transmit diagnostic messages, Only format (c), (d) are allowed like Classical CAN.</p> |                   |                   |                        |                   |         |         |         |

Table 2-9. Mapping of parameters and data into CAN-FD frame – Normal Addressing

| N_PCI Byte #1<br>Bits 7 - 4             | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0000 <sub>2</sub><br>(0 <sub>16</sub> ) | <p><b>Single Frame</b></p> <p>For unsegmented messages with CAN_DL ≤ 8, the message length is embedded in lower nibble of the only PCI byte (Byte #1). For unsegmented messages with CAN_DL &gt; 8, the <i>SingleFrame escape sequence</i> shall be used where the lower nibble of the first PCI byte (Byte #1) is set to 0000<sub>2</sub> and the message length is embedded in the 2<sup>nd</sup> PCI byte (Byte #2). SingleFrame (SF) shall be used to support the transmission of messages that can fit in a single CAN frame.</p> |

|                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0001 <sub>2</sub><br>(1 <sub>16</sub> ) | <b>First Frame</b><br><br>A FirstFrame (FF) shall only be used to support the transmission of messages that cannot fit in a single CAN frame, i.e. segmented messages. The FirstFrame of a segmented message is encoded as a FirstFrame (FF). On receipt of a FirstFrame (FF), the receiving network layer entity shall start assembling the segmented message. For segmented messages with a message length > 4095, the <i>FirstFrame escape sequence</i> shall be used where the lower nibble of the first PCI byte (Byte #1) is set to 0000 <sub>2</sub> and the 2 <sup>nd</sup> PCI byte (Byte #2) is set to zero. The message length is embedded in the following four bytes (Byte #3 .. Byte #6, MSB first). |
| 0010 <sub>2</sub><br>(2 <sub>16</sub> ) | <b>Consecutive Frame</b><br><br>When sending segmented data, all consecutive frames following the FF are encoded as ConsecutiveFrame (CF). On receipt of a ConsecutiveFrame (CF), the receiving network layer entity shall assemble the received data bytes until the whole message is received. The receiving entity shall pass the assembled message to the adjacent upper protocol layer after the last frame of the message has been received without error.                                                                                                                                                                                                                                                   |
| 0011 <sub>2</sub><br>(3 <sub>16</sub> ) | <b>Flow Control</b><br><br>The purpose of FlowControl (FC) is to regulate the rate at which CF N_PDUs are sent to the receiver. Three distinct types of FlowControl (FC) protocol control information are specified to support this function. The type is indicated by a field of the protocol control information called FlowStatus (FS).                                                                                                                                                                                                                                                                                                                                                                         |

Table 2-10. Definition of N\_PCItype values

| N_PCI                   | Description                                    |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------|------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SF_DL<br>(CAN_DL ≤ 8)   | <b>Single Frame Data Length</b>                |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <b>Value (HEX)</b>                             | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | 0                                              | Reserved                                                                                                                                                                                                                                                                                                                                                                                                            |
|                         | 1 ~ 6                                          | The SF_DL is encoded in the low nibble of N_PCI byte #1 value. It shall be assigned the value of the service parameter <Length>.                                                                                                                                                                                                                                                                                    |
|                         | 7                                              | SF_DL = 7 is only allowed with normal addressing.                                                                                                                                                                                                                                                                                                                                                                   |
| SF_DL<br>(CAN_DL > 8)   | 8 ~ F                                          | Invalid                                                                                                                                                                                                                                                                                                                                                                                                             |
|                         | <b>Single Frame Data Length</b>                |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <b>Value (HEX)</b>                             | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | 0 ~ 6                                          | Invalid                                                                                                                                                                                                                                                                                                                                                                                                             |
|                         | 7                                              | <b>SingleFrame DataLength (SF_DL) with extended addressing or mixed addressing</b><br>SF_DL shall be assigned the value of the service parameter <Length>. SF_DL = 7 is only allowed with extended addressing or mixed addressing.                                                                                                                                                                                  |
|                         | 8 .. 61                                        | <b>SingleFrame DataLength (SF_DL)</b><br>SF_DL shall be assigned the value of the service parameter <Length>.                                                                                                                                                                                                                                                                                                       |
| FF_DL<br>(FF_DL ≤ 4095) | 62                                             | <b>SingleFrame DataLength (SF_DL) with normal addressing only</b><br>SF_DL shall be assigned the value of the service parameter <Length>. SF_DL = (CAN_DL - 2) is only allowed with normal addressing.                                                                                                                                                                                                              |
|                         | other values                                   | Invalid                                                                                                                                                                                                                                                                                                                                                                                                             |
|                         | <b>1. First Frame Data Length (If DLC = 8)</b> |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|                         | <b>Value (HEX)</b>                             | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                  |
|                         | 0 - 6                                          | Invalid                                                                                                                                                                                                                                                                                                                                                                                                             |
|                         | 7                                              | FF_DL = 62 is only allowed with extended or mixed addressing                                                                                                                                                                                                                                                                                                                                                        |
|                         | 8 ~ FFF                                        | The encoding of the segmented message length results in a twelve (12) bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit "0" of the N_PCI byte #2 and the most significant bit (MSB) is bit three (3) of the N_PCI byte #1. The maximum segmented message length supported is equal to 4095 bytes of user data. It shall be assigned the value of the service parameter <Length> |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                         | <b>2. First Frame Data Length (If DLC ≠ 8)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <b>Value (HEX)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         | 0 ~ 61                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                         | 62                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | FF_DL = 62 is only allowed with extended or mixed addressing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|                         | 63 ~ FFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | The encoding of the segmented message length results in a twelve (12) bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit "0" of the N_PCI byte #2 and the most significant bit (MSB) is bit three (3) of the N_PCI byte #1. The maximum segmented message length supported is equal to 4095 bytes of user data. It shall be assigned the value of the service parameter <Length>                                                                                                                                                                                               |
| FF_DL<br>(FF_DL > 4095) | <b>First Frame Data Length</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <b>Value (HEX)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         | 0 ~ FFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|                         | FFF+1 ~ FFFF<br>FFFF                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | The encoding of the segmented message length results in a 32 bit length value (FF_DL) where the least significant bit (LSB) is specified to be bit 0 of the sixth N_PCI byte (Byte #6) and the most significant bit (MSB) is bit 7 of the third N_PCI byte (Byte #3). The maximum supported segmented message length is equal to 4 294 967 295 bytes of user data. It shall be assigned the value of the service parameter <Length>                                                                                                                                                                               |
| SN                      | <b>Sequence Number</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <p>The SequenceNumber (SN) shall be encoded in the lower nibble bits of N_PCI byte #1. The SN shall be set to a value within the range of zero (0) to fifteen (15).</p> <ul style="list-style-type: none"> <li>- The SN shall start with zero (0) for all segmented messages.</li> <li>- The FF shall be assigned the value zero (0).</li> <li>- The SN of the first CF that immediately follows the FF shall be set to one (1).</li> <li>- The SN shall be incremented by one (1) for each new CF that is transmitted during a segmented message transmission.</li> <li>- When the SN reaches the value of fifteen (15), it shall wraparound and be set to zero (0) for the next CF.</li> </ul> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| FS                      | <b>Flow Status</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | The parameter FlowStatus (FS) indicates whether the sending network entity can proceed with the message transmission.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | <b>Value (HEX)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>Description</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>ContinueToSend (CTS)</b><br>The FlowControl ContinueToSend parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "0". It shall cause the sender to resume the sending of Consecutive frames. The meaning of this value is that the receiver is ready to receive a maximum of BS number of Consecutive frames.                                                                                                                                                                                                                                                                         |
|                         | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Wait (WT)</b><br>The FlowControl Wait parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "1". It shall cause the sender to continue to wait for a new FlowControl N_PDU and to restart its N_BS timer.                                                                                                                                                                                                                                                                                                                                                                             |
|                         | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Overflow (OVFLW)</b><br>The FlowControl Overflow parameter shall be encoded by setting the lower nibble of the N_PCI byte #1 to "2". It shall cause the sender to abort the transmission of a segmented message and make an N_USData.confirm service call with the parameter <N_Result>=N_BUFFER_OVFLW. (refer to ISO 15765-3)<br>This N_PCI FlowStatus parameter value is only allowed to be transmitted in the FlowControl N_PDU that follows the FirstFrame N_PDU and shall only be used in case the message length FF_DL of the received FirstFrame N_PDU exceeds the buffer size of the receiving entity. |
| BS                      | <b>Block Size</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|                         | The units of BS are the absolute number of CF N_PDUs per block.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

|       | Value (HEX)                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                 |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | 00                                                                                                                                          | The BS parameter value zero (0) shall be used to indicate to the sender that no more FC frames shall be sent during the transmission of the segmented message. The sending network layer entity shall send all remaining consecutive frames without any stop for further FC frames from the receiving network layer entity. |
|       | 01 ~ FF                                                                                                                                     | This range of BS parameter values shall be used to indicate to the sender the maximum number of consecutive frames that can be received without an intermediate FC frame from the receiving network entity.                                                                                                                 |
| STmin | <b>Minimum Separation Time</b>                                                                                                              |                                                                                                                                                                                                                                                                                                                             |
|       | The STmin parameter value specifies the minimum time gap allowed between the transmission of consecutive frame network protocol data units. |                                                                                                                                                                                                                                                                                                                             |
|       | Value (HEX)                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                 |
|       | 00 ~ 7F                                                                                                                                     | 0 ms ~ 127 ms, The units of STmin in the range 00 hex ~ 7F hex are absolute milliseconds (ms).                                                                                                                                                                                                                              |
|       | F1 ~ F9                                                                                                                                     | 100 $\mu$ s ~ 900 $\mu$ s, The units of STmin in the range F1 hex ~ F9 hex are even 100 microseconds ( $\mu$ s), where parameter value F1 hex represents 100 $\mu$ s and parameter value F9 hex represents 900 $\mu$ s.                                                                                                     |
|       | FA ~ FF                                                                                                                                     | Reserved                                                                                                                                                                                                                                                                                                                    |

Table 2-11. Definition of N\_PCI

| Addressing type   | CAN_DL value                   |                                 |                                 |                                 |                                 |                                 |                                 |
|-------------------|--------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|---------------------------------|
|                   | 12                             | 16                              | 20                              | 24                              | 32                              | 48                              | 64                              |
| Normal            | $8 \leq \text{SF\_DL} \leq 10$ | $11 \leq \text{SF\_DL} \leq 14$ | $15 \leq \text{SF\_DL} \leq 18$ | $19 \leq \text{SF\_DL} \leq 22$ | $23 \leq \text{SF\_DL} \leq 30$ | $31 \leq \text{SF\_DL} \leq 46$ | $47 \leq \text{SF\_DL} \leq 62$ |
| Mixed or extended | $7 \leq \text{SF\_DL} \leq 9$  | $10 \leq \text{SF\_DL} \leq 13$ | $14 \leq \text{SF\_DL} \leq 17$ | $18 \leq \text{SF\_DL} \leq 21$ | $22 \leq \text{SF\_DL} \leq 29$ | $30 \leq \text{SF\_DL} \leq 45$ | $46 \leq \text{SF\_DL} \leq 61$ |

Table 2-28. Allowed SF\_DL values for a given CAN\_DL greater than 8

### 2.4.6 Ethernet Message Structure

Ethernet Message Structure shall be compliant to ES95486-03 Ethernet IP Middleware Diagnostic Communication Specification or ES5486-05 DoIP Communication Design Specification according to the applied diagnostic communication protocol.

## 2.5 Initialization and Timing

### 2.5.1 Initialization

No separate initialization is required for diagnostic communication on CAN. The communication can be started by giving the request for any service (either functional request or physical request). After power on, the server (ECU) will be in default session. If no other diagnostic session is started, then the default diagnostic session shall be running as long as the service is powered.

### 2.5.2 Timing

The diagnostic communication on CAN uses two kinds of timing parameters

- Network layer timing parameters and
- Application layer timing parameters

#### 2.5.2.1 Network Layer Timing Parameters

Figure 2-3 shows the CAN message frame flow and timing parameters in the network layer.

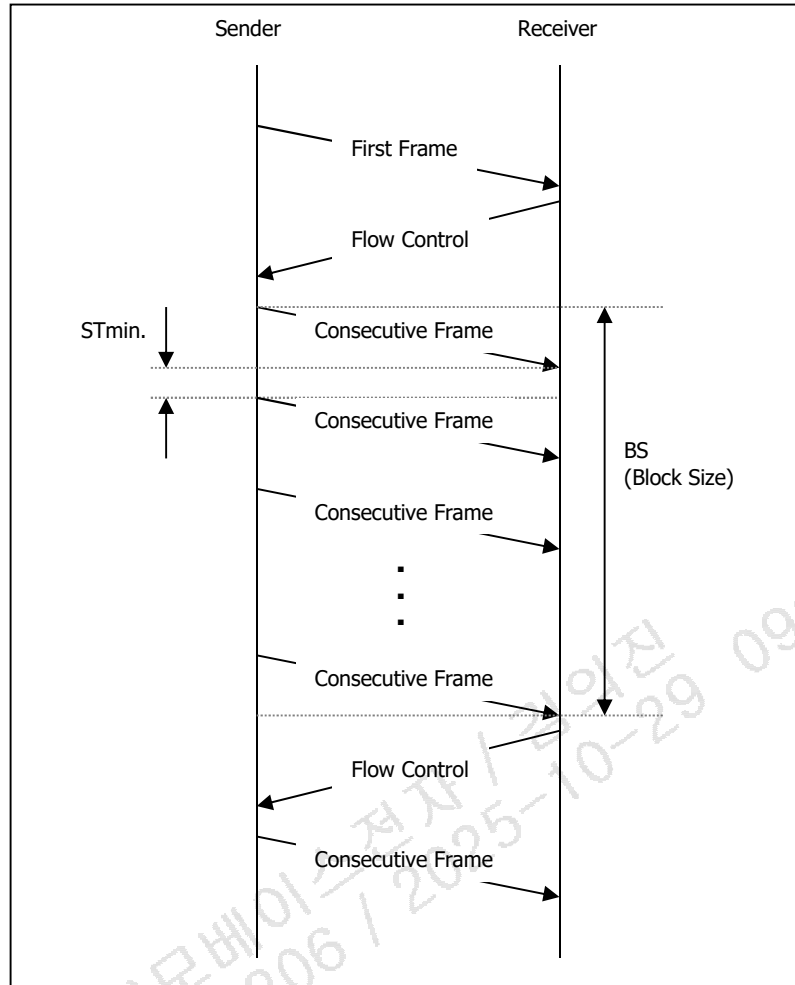


Figure 2-3. Message frame flow diagram

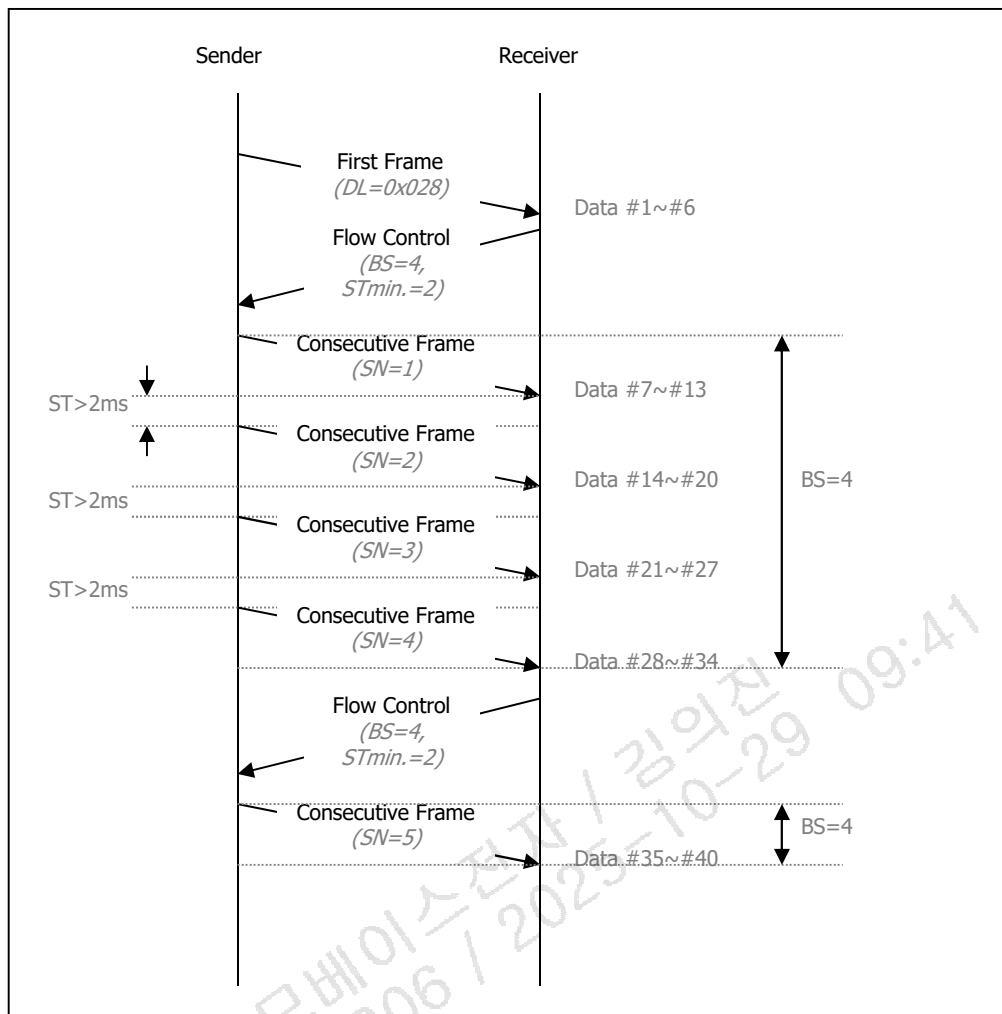


Figure 2-4. Example of message frame flow (STmin = 2, BS = 4)

The timing parameters of Figure 2-3 and each descriptions are defined in the following table respectively.

### [STmin]

| Diagnostic type                                                                                                                                                                                                                                                                                                  | Gateway                                        | Network type   | Description                                                                                                                                | Value        |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| Reprogramming (Boot Area or Programming session)                                                                                                                                                                                                                                                                 | -                                              | -              | ECU shall set the STmin parameter to zero(0). This means that the external test equipment can send consecutive frames as fast as possible. | 0 ms         |
| Non reprogramming                                                                                                                                                                                                                                                                                                | No Gateway / Client – Server direct connection | -              | The ECU (or external test equipment) can set the STmin value from 0 to 8ms.                                                                | 0 ~ 8 ms     |
|                                                                                                                                                                                                                                                                                                                  | Gateway                                        | HS CAN, CAN FD | The ECU (or external test equipment) can set the STmin value from 0.3 to 12ms                                                              | 0.3ms ~ 12ms |
|                                                                                                                                                                                                                                                                                                                  | Gateway                                        | Ethernet       | STmin should be equal or larger than 10ms.                                                                                                 | 10ms ~       |
| ※ If ECU receives STmin Value under 0.3ms, ECU shall transmit Consecutive Frame after waiting STmin Value over 0.3ms. (e.g. OBD ECU STmin 0)<br>※ Background update is not a scope of the Reprogramming because reprogramming is not performed in the BootArea or Programming session with CommunicationControl. |                                                |                |                                                                                                                                            |              |

Table 2-12a. Network layer parameters - STmin

※ In case of transmitting multi-frame, if there were gateway unit(s) between ECU and external test equipment, and Commnuication Control (DisableRxAndTx) is not able to use, external test equipment can extend STmin as 10~12ms regardless of a response of ECU.

### [BS]

| Parameter | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Recommended Value |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| BS        | The ECU (or external test equipment) can set the BS value up to 8. The external test equipment (or ECU) can send less than 8 Consecutive Frames without additional Flow Control Frames.                                                                                                                                                                                                                                                                               | 0 ~ 8             |
|           | In Flash reprogram procedure (when the ECU is in the Boot Area), ECU shall be set the BS parameter to zero(0). This means that the ECU shall be able to receive as many consecutive frames as possible without additional Flow Control Frames.<br><br>※ Actually, the number of Consecutive Frames sending from the external test equipment would be set according to the 'maxNumberOfBlockLength' in the ECU's positive response message of RequestDownload service. | 0<br>(Infinite)   |

Table 2-12b. Network layer parameters - BS

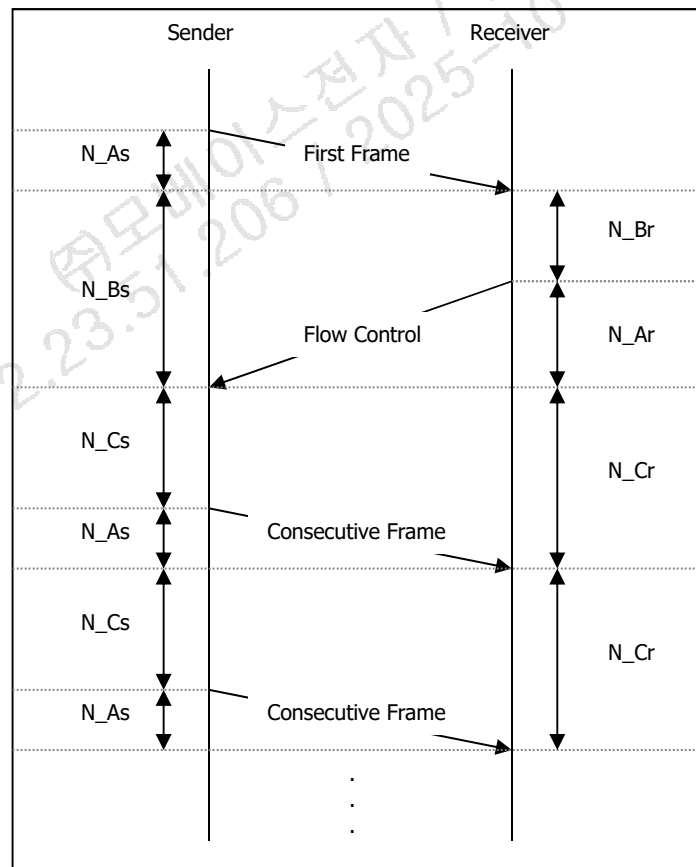


Figure 2-5. Placement of network layer timing parameters

The timing parameters of Figure 2-5 and each descriptions are defined in the following table respectively.

| Timing Parameter | Description | Timeout (ms) |
|------------------|-------------|--------------|
|------------------|-------------|--------------|



# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)  
ES95486-02

페이지  
(SHT/SHTS) 21/244

| N_As | Time for transmission of the CAN frame on the sender side   | 1000 |
|------|-------------------------------------------------------------|------|
| N_Ar | Time for transmission of the CAN frame on the receiver side | 1000 |
| N_Bs | Time until reception of the next Flow Control.              | 1000 |
| N_Br | Time until transmission of the next Flow Control            | N/A  |
| N_Cs | Time until transmission of the next Consecutive Frame.      | N/A  |
| N_Cr | Time until reception of the next Consecutive Frame.         | 1000 |

Table 2-13. Network layer parameters

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주모베이스전자 / 김의진

### 2.5.2.2 Application Layer Timing Parameters

Figure2-6 shows the diagnostic application message flow and timing parameters in the application layer based on ISO 14229-2.

The interval timing parameters (**P2<sub>CAN\_SERVER</sub>** and **S3<sub>SERVER</sub>**) and each descriptions are defined in the following table. The **P2<sub>CAN\_SERVER</sub>** parameter is related to performance of a server and **S3<sub>SERVER</sub>** parameter is related to the timeout of a diagnostic session.

| Parameter                           | Description                                                                                                                                                                                                                                                                                                         | Min (ms)                                                             | Max (ms) |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------|
| <b>P2<sub>CAN_SERVER</sub></b>      | Performance requirement for the server to start with the response message after the reception of a request message.                                                                                                                                                                                                 | 0                                                                    | 50       |
| <b>P2*<sub>CAN_SERVER</sub></b>     | Performance requirement for the server to start with the response message after the transmission of a negative response message with response code 78 hex.                                                                                                                                                          | 0 <sup>1)</sup>                                                      | 5000     |
| <b>P2<sub>CAN_CLIENT</sub></b>      | Timeout for the client to wait after the successful transmission of a request message for the start of incoming response messages (First Frame of a multi-frame message or a SingleFrame message).                                                                                                                  | $P2_{CAN\_SERVER\_max} + \Delta P2_{CAN}$<br>(80ms for Ethernet ECU) | N/A      |
| <b>P2*<sub>CAN_CLIENT</sub></b>     | Enhanced timeout for the client to wait after the reception of a negative response message with response code 78 hex for the start of incoming response messages (First Frame of a multiframe message or a SingleFrame message).                                                                                    | $P2*_{CAN\_SERVER\_max} + \Delta P2_{CAN\_rsp}$                      | N/A      |
| <b>P3<sub>CAN_CLIENT_Phys</sub></b> | Minimum time for the client to wait after the successful transmission of a physically addressed request message with no response required before it can transmit the next physically addressed request message.                                                                                                     | $P2_{CAN\_SERVER\_max} + \Delta P2_{CAN}$                            | N/A      |
| <b>P3<sub>CAN_CLIENT_Func</sub></b> | Minimum time for the client to wait after the successful transmission of a functionally addressed request message before it can transmit the next functionally addressed request message in case no response is required or the requested data is only supported by a subset of the functionally addressed servers. | $P2_{CAN\_SERVER\_max} + \Delta P2_{CAN}$                            | N/A      |

1) During the enhanced response timing, the minimum time between the transmission of consecutive negative messages (each with negative response code 0x78 hex) shall be  $0.3 * P2*_{CAN\_SERVER\_max}$ , in order to avoid flooding the data link with unnecessary negative response code 0x78 hex messages.

Table 2-14. Application layer timing parameters

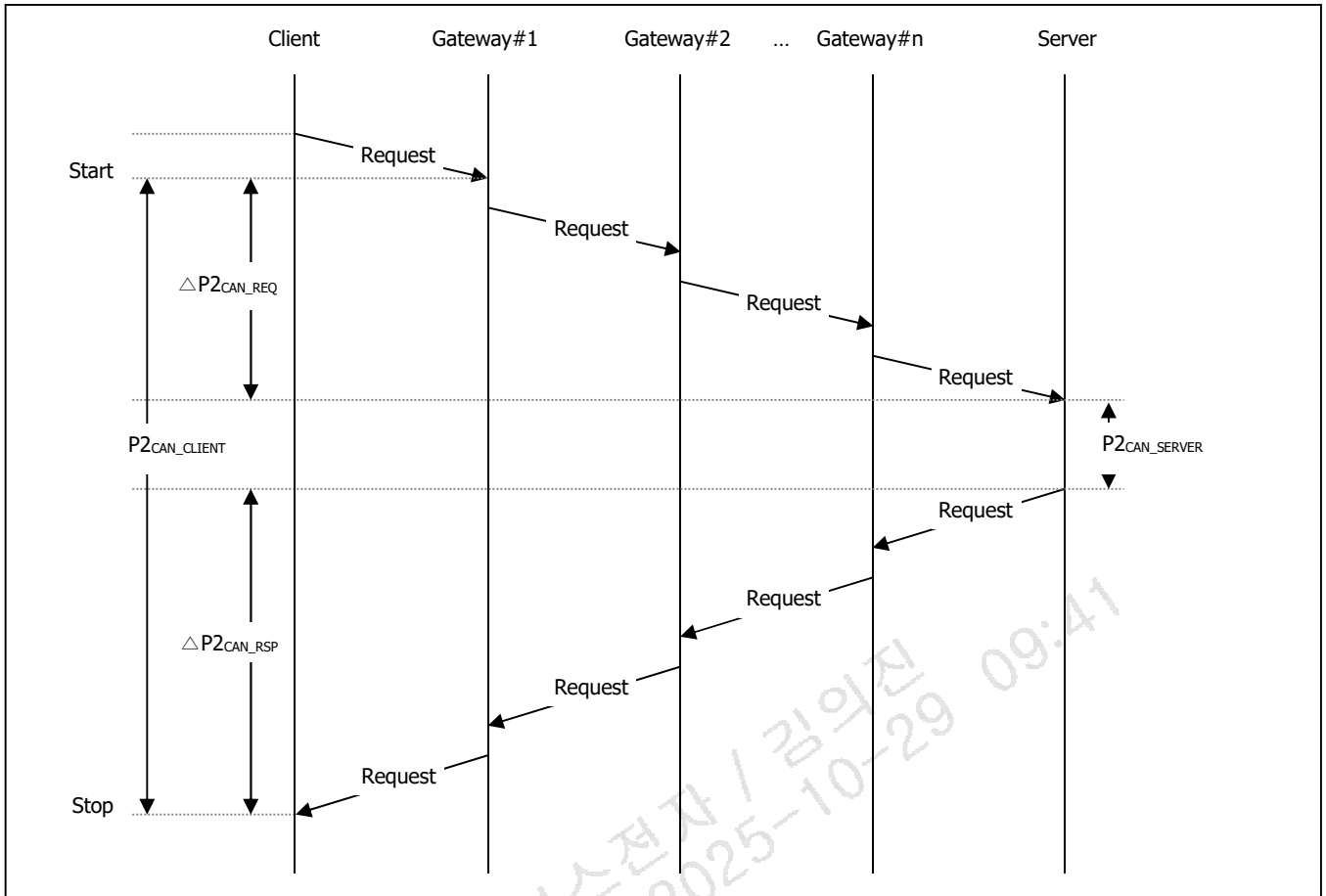


Figure 2-6. Application layer message flow diagram

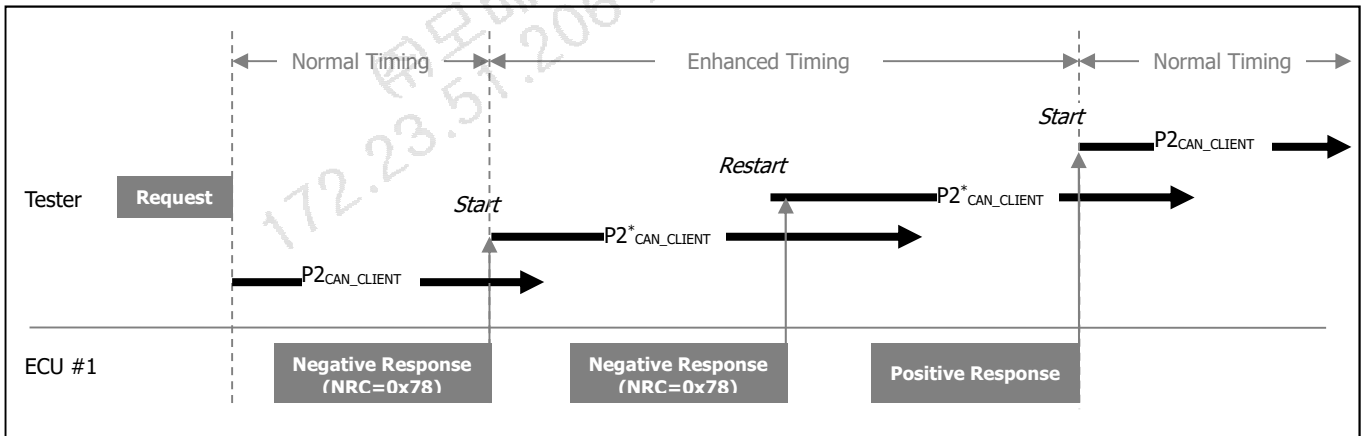


Figure 2-7. P2\*<sub>CAN\_CLIENT</sub> timing parameter Definition

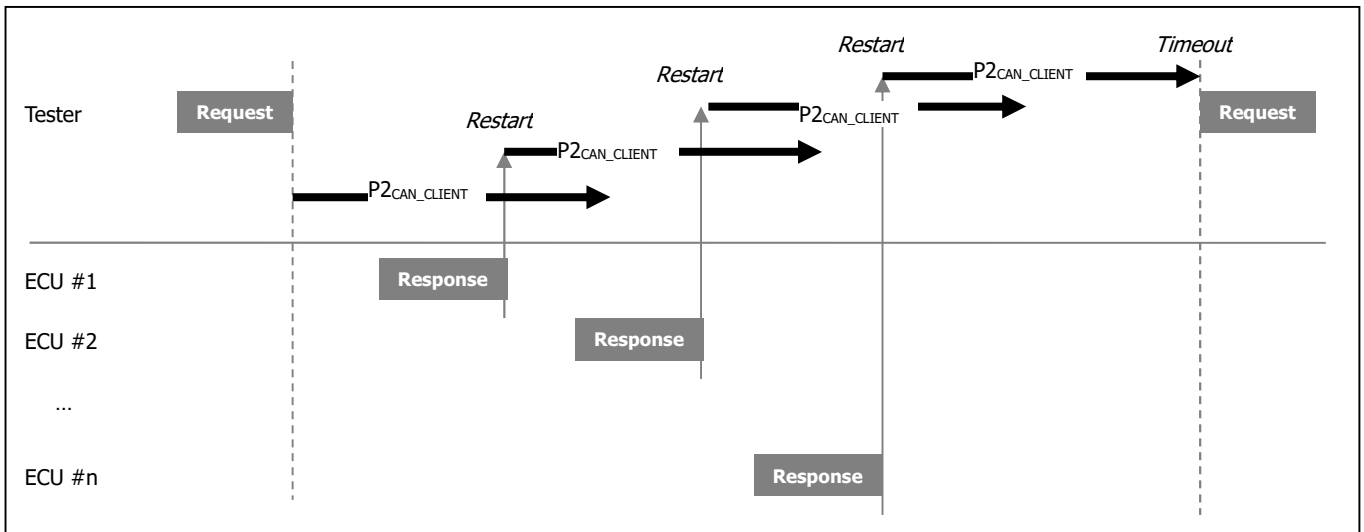


Figure 2-8. P2<sub>CAN\_CLIENT</sub> timing parameter in Functional request

### 2.5.2.3 Session layer timing parameters

When a diagnostic session other than the defaultSession is started, then a session handling is required which is achieved via the session layer timing parameter given in Table 2-13.

Refer to the ISO15765-3 「6.3.5 Detailed timing parameter descriptions」 to get more information about application & session layer timing parameters.

| Timing Parameter     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Type               | Recommended Value | Timeout |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|---------|
| S3 <sub>Client</sub> | Time between functionally addressed TesterPresent (3E hex) request messages transmitted by the client to keep a diagnostic session other than the defaultSession active in multiple servers (functional communication) or maximum time between physically transmitted request messages to a single server (physical communication). The S3 <sub>Client</sub> timeout value includes the travel time travel of the message on the network (gateway delays, etc.) | Timer reload value | 2000ms            | 4000ms  |
| S3 <sub>Server</sub> | Time for the server to keep a diagnostic session other than the defaultSession active while not receiving any diagnostic request message. The tolerance of S3 <sub>Server</sub> is -0ms, +200ms.                                                                                                                                                                                                                                                                | Timer reload value | N/A               | 5000ms  |

Table 2-15. Session layer timing parameter definitions

| Timing Parameter     | Action           | Physical and functional communication, using functionally addressed, periodically transmitted TesterPresent request message                            | Physical communication only, using a physically addressed, sequentially transmitted TesterPresent request message                                                                                                         |
|----------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| S3 <sub>Client</sub> | Initial Start    | The completion of the DiagnosticSession Control (10 hex) request message.<br><br>※ This is only true for if the session type is a non-default session. | The completion of the DiagnosticSession Control (10 hex) request message in case no response is required.<br><br>The reception of the DiagnosticSession Control (10 hex) response message in case a response is required. |
|                      | Subsequent Start | The completion of the functionally addressed TesterPresent (3E hex)                                                                                    | The completion of any request message in case no response is required.                                                                                                                                                    |

|          |                  |                                                                                                                                                                                                                                                                                                                 |                                                                       |
|----------|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| S3Server |                  | request message, which is transmitted each time the S3Client timer times out.                                                                                                                                                                                                                                   | The reception of any response message in case a response is required. |
|          |                  |                                                                                                                                                                                                                                                                                                                 | An error during the reception of a multi-frame response message.      |
|          | Initial Start    | The completion of the transmission of a DiagnosticSessionControl positive response message for a transition from the default session to a non-default session, in case a response message is required.                                                                                                          |                                                                       |
|          |                  | Successful completion of the requested action of the service DiagnosticSessionControl (10 hex) for a transition from the default session to a non-default session, in case no response message is required/allowed.                                                                                             |                                                                       |
|          | Subsequent Stop  | The start of a multi-frame request message or the reception of any SingleFrame request message. If the defaultSession is active, the S3Server timer is disabled.                                                                                                                                                |                                                                       |
|          | Subsequent Start | The completion of any response message that concludes a service execution (final response message) in case a response message is required/allowed to be transmitted (this includes positive and negative response messages). A negative response with response code 78 hex does not restart the S3Server timer. |                                                                       |
|          |                  | The completion of the requested action (service conclusion) in case no response message (positive and negative) is required/allowed.                                                                                                                                                                            |                                                                       |
|          |                  | An error during the reception of a multi-frame request message.                                                                                                                                                                                                                                                 |                                                                       |

Table 2-16. Session layer timing parameter definitions

### 2.5.2.4 Minimum time between client request messages

The minimum time between request messages transmitted by the client is required in order to allow for a polling driven service data interpretation in the server, for example. Based on its normal functionality, a server might process diagnostic request messages with a certain scheduling rate (e.g. 10 ms). The time for the diagnostic service data interpretation scheduler shall be smaller than the performance requirement P2CAN\_Server in order to meet the server requirements.

The timing parameter for the minimum time between request message is divided into the following two timing parameters.

- P3CAN\_Functional : This timing parameter applies to any functionally addressed request message, because it can be the case that a server is not required to respond to a functionally addressed request message if it does not support the requested data.
- P3CAN\_Physical : This timing parameter applies to any physically addressed request message where there is no response required to be transmitted by the server (suppressPosRspMsgIndicationBit = TRUE).

In the case of physical communication where a response is required by the server, the client can transmit the next request immediately after the complete reception of the last response message, because the server has responded completely to the request — which means that the request is completely handled by the server.

The minimum times P3CAN\_Physical and P3CAN\_Functional, between the end of a physically or functionally addressed request message and the start of a new physically or functionally addressed request message, are defined for the client.

a) The value of P3CAN\_Physical will be identical to P2CAN\_Server\_max for the physically addressed server. The timing applies to any physically addressed request message in any diagnostic session (default and nondefault session) and in case no response is required by the server.

The P3CAN\_Physical timer is started in the client each time a physically addressed request message with no response required is successfully transmitted onto the bus, which is indicated via N\_USData.con in the client. When the client wants to transmit a new physically addressed request message following a previous



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request that was completely handled, then this is only allowed in case the  $P3_{CAN\_Physical}$  timer is no longer active at the time the client wants to transmit the physically addressed request message. In case  $P3_{CAN\_Physical}$  would still be active at the point in time the client would like to transmit a new physically addressed request message, then the transmission shall be postponed until  $P3_{CAN\_Physical}$  is timed out.

b) The value of  $P3_{CAN\_Functional}$  will be the maximum (worst-case) value of all functionally addressed server's  $P2_{CAN\_Server\_max}$  for any functionally addressed request message in any diagnostic session (default and non-default session).

The  $P3_{CAN\_Functional}$  timer is started in the client each time a functionally addressed request message with response required or with no response required is successfully transmitted onto the bus, which is indicated via  $N\_USData.con$  in the client. When the client wants to transmit a new functionally addressed request message following a previous request that was completely handled, then this is only allowed in case the  $P3_{CAN\_Functional}$  timer is no longer active at the time the client wants to transmit the functionally addressed request message. In case  $P3_{CAN\_Functional}$  would still be active at the point in time the client would like to transmit a new functionally addressed request message, then the transmission shall be postponed until  $P3_{CAN\_Functional}$  is timed out.

NOTE) "Completely handled" means that either no response is received in case no response is required, all expected responses to a functionally addressed request are received in case the responding servers are known and responses are required or a  $P2_{CAN\_Client}$  timeout occurred in case the responding servers are not known and responses are required.

The requirement for the server is that it shall start with its response message within  $P2_{CAN\_Server}$ . This means that the diagnostic data interpretation rate of the server shall be less than  $P2_{CAN\_Server}$ .

## 2.6 Application Layer

### 2.6.1 Request Message sub-function parameter

The sub-function parameter byte is divided into two part (on-bit-level) as defined in Table below :

| Bit position | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7            | <b>suppressPosRspMsgIndicationBit</b><br>This bit indicates if a positive response message shall be suppressed by the server.<br><br>'0' = FALSE, do not suppress a positive response message (a positive response message is required).<br>'1' = TRUE, suppress response message (a positive response message shall not be sent; the server being addressed shall not send a positive response message).<br>Independent of the suppressPosRspMsgIndicationBit, negative response messages are sent by the server(s) according to the restrictions specified 2.6.3~2.6.5. |
| 6 – 0        | <b>sub-function parameter value</b><br>The bits 0-6 of the sub-function parameter contain the sub-function parameter value of the service (00 - 7F hex). Each service utilizing the sub-function parameter byte, but only supporting the suppressPosRspMsgIndicationBit has to support the zeroSubFunction sub-function parameter value (00 hex).                                                                                                                                                                                                                         |

Table 2-17. Sub-function Parameter Structure

NOTE) When requestCorrectlyReceived-ResponsePending (NRC = 78 hex) response code is used, the server shall always send a final response (positive or negative) independent of the suppressPosRspMsgIndicationBit value.

### 2.6.2 Negative response service

Each diagnostic service has a negative response message according to Table 2-18. The Data Byte #1 is always the specific negative response service identifier. The Data Byte #2 shall be a copy of the service identifier value from the service request message that the negative response message corresponds to.

| Data Byte | Parameter Name               | Cvt | Hex Value | Mnemonic     |
|-----------|------------------------------|-----|-----------|--------------|
| #1d       | <b>Negative Response SID</b> | M   | #7Fh      | <b>SIDNR</b> |
| #2d       | <Service Name> Request SID   | M   | #XXh      | SIDR!        |
| #3d       | Negative Response Code       | M   | #XXh      | NRC_         |

Table 2-18. Negative response message definition

### 2.6.3 General server response behaviour

| Abbreviation                   | Description                                                                                                                             |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| SuppressPosRspMsgIndicationBit | TRUE ('1') = Server shall NOT send a positive response message<br>FALSE ('0') = Server shall send a positive response message           |
| PosRsp                         | Abbreviation for positive response message                                                                                              |
| NegRsp                         | Abbreviation for negative response message                                                                                              |
| NoRsp                          | Abbreviation for NOT sending a positive or negative response message                                                                    |
| NRC                            | Abbreviation for negative response code                                                                                                 |
| ALL                            | All of the requested data parameters (service without sub-function parameter) of the client request message are supported by the server |

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| At Least 1 | At least 1 data parameters (service without sub-function parameter) of the client request message must be supported by the server       |
| NONE       | None of the requested data parameters (service without sub-function parameter) of the client request message is supported by the server |

Table 2-19. Legend for this section

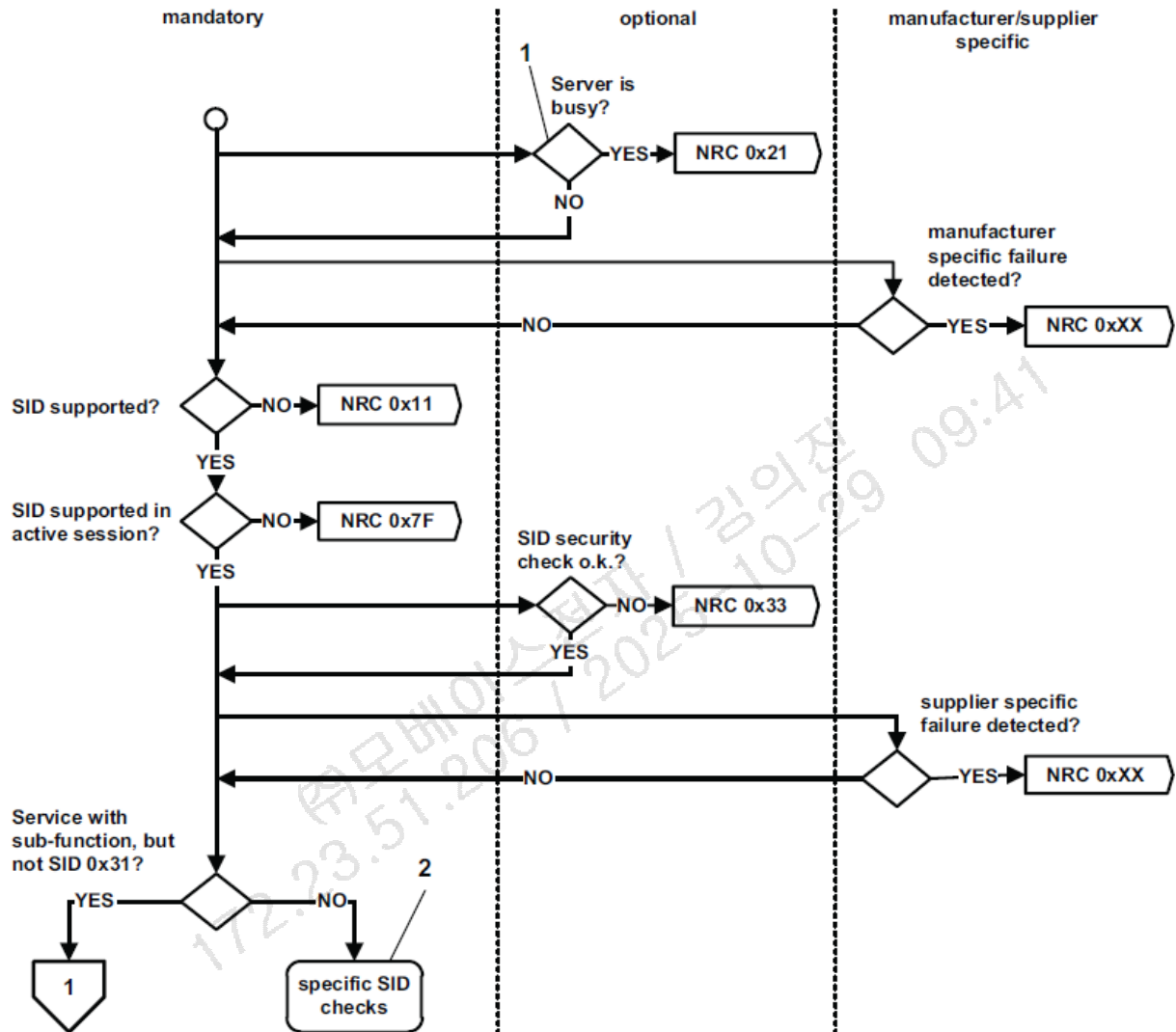


Figure 2-9. General server response behavior

### Key

1 Diagnostic request can not be accepted because another diagnostic task is already and in progress by a different client.

2 refer to the response behavior (supported negative response codes) of each service.

## 2.6.4 Request message with sub-function parameter and server response behaviour

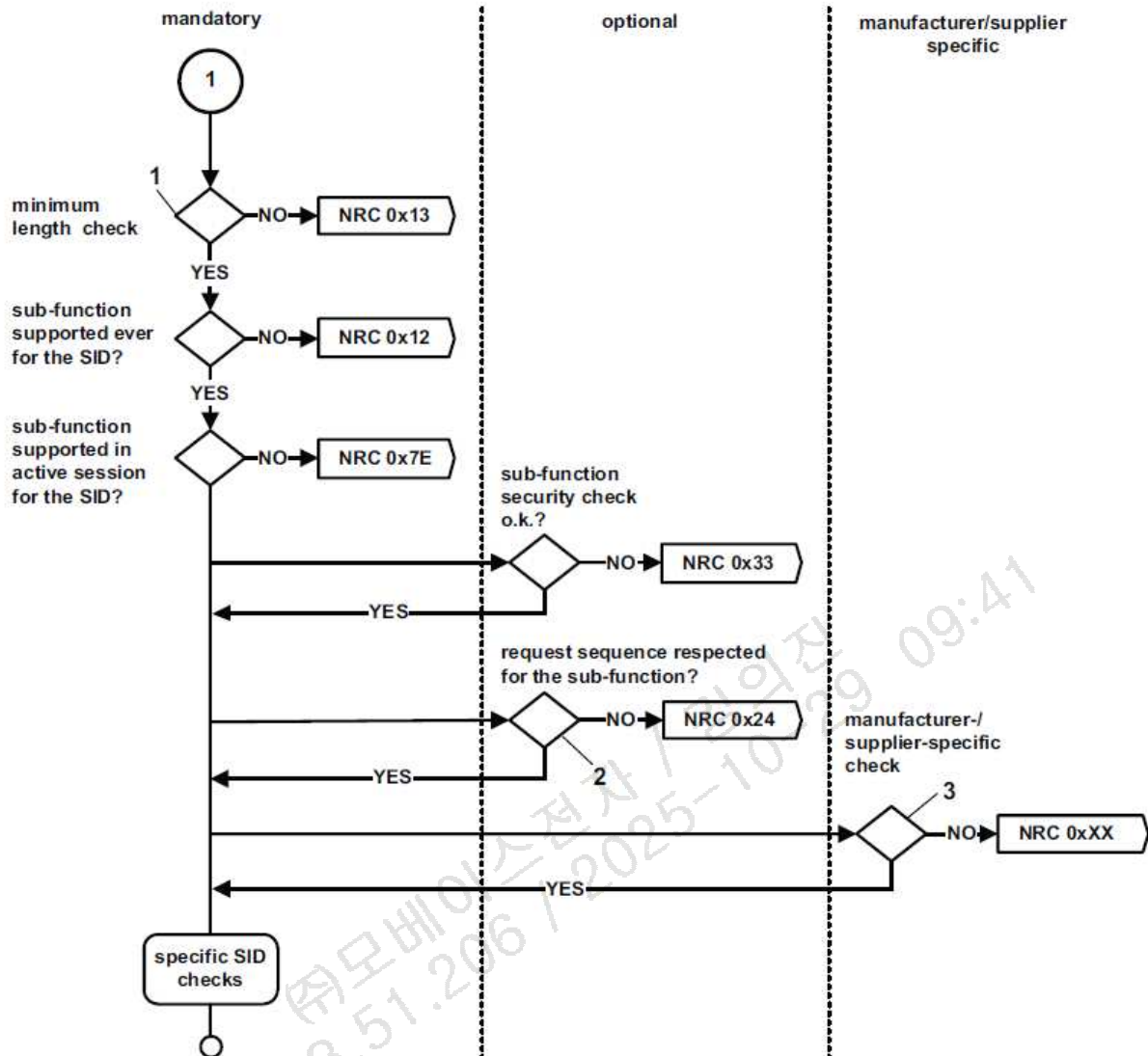


Figure 2-10. General server response behavior for request messages with sub-function parameter

### Key

- 1 at least 2 (SID+SubFunction Parameter).
- 2 if sub-function is subject to sequence check, e.g. LinkControl or SecurityAccess.
- 3 refer to the response behaviour (supported negative response codes) of each service.

## 2.6.4.1 Physically addressed client request message

The server response behaviour specified in this section is referenced in the service description of each service, which supports a sub-function parameter in the physically addressed request message received from the client.

| Server Case # | Client Request Message |                                               | Server Capability   |                        |                                               | Server Response |                     | Comments to Server Response                                                                              |
|---------------|------------------------|-----------------------------------------------|---------------------|------------------------|-----------------------------------------------|-----------------|---------------------|----------------------------------------------------------------------------------------------------------|
|               | Addressing Scheme      | Sub-Function (SuppressPosRspMsgIndicationBit) | ServiceID Supported | Sub-Function Supported | Data Parameter Supported (only if applicable) | Message         | NRC                 |                                                                                                          |
| 1             | Physical               | FALSE (bit=0)                                 | YES                 | YES                    | At Least 1                                    | PosRsp          | -                   | Server sends positive response                                                                           |
| 2             |                        |                                               |                     |                        | At Least 1                                    | NegRsp          | NRC=0xXX            | Server sends negative response because error occurred reading the data parameters of the request message |
| 3             |                        |                                               |                     |                        | None                                          |                 | NRC=ROOR            | Negative response with NRC 0x31                                                                          |
| 4             |                        |                                               | NO                  | -                      | -                                             |                 | NRC=SNS or SNSIAS   | Negative response with NRC 0x11 or NRC 0x7F                                                              |
| 5             |                        |                                               | YES                 | NO                     | -                                             |                 | NRC=SFNS or SFNSIAS | Negative response with NRC 0x12 or NRC 0x7E                                                              |
| 6             |                        | TRUE (bit=1)                                  | YES                 | YES                    | At Least 1                                    | NoRsp           | -                   | Server dose NOT send a response                                                                          |
| 7             |                        |                                               |                     |                        | At Least 1                                    | NegRsp          | NRC=0xXX            | Server send negative response because error occurred reading the data parameters of the request message  |
| 8             |                        |                                               |                     |                        | None                                          |                 | NRC=ROOR            | Negative response with NRC 0x31                                                                          |
| 9             |                        |                                               | NO                  | -                      | -                                             |                 | NRC=SNS             | Negative response with NRC 0x11                                                                          |
| 10            |                        |                                               | YES                 | NO                     | -                                             |                 | NRC=SFNS            | Negative response with NRC 0x12                                                                          |

Table 2-20. Physically addressed request message with sub-function parameter and server response behaviour

Description of server response cases on physically addressed client request messages with subFunction:

- 1) Server sends a positive response message because the service identifier and sub function parameter is supported of the client's request with indication for a response message,
- 2) Server sends a negative response message (e.g. IMLOIF: incorrectMessageLengthOrIncorrectFormat) because the service identifier and sub function parameter is supported of the client's request, but some other error appeared (e.g. wrong PDU length according to service identifier and sub function parameter in the request message) during processing of the sub function.
- 3) Server sends a negative response message with the negative response code ROOR (requestOutOfRange) because the service identifier and sub-function parameter are supported but none of the requested data-parameters are supported by the client's request message.
- 4) Server sends a negative response message with the negative response code SNS (serviceNotSupported) or SNSIAS (serviceNotSupportedInActiveSession) because the service identifier is not supported of the client's request with indication for a response message.
- 5) Server sends a negative response message with the negative response code SFNS (subfunctionNotSupported) or SFNSIAS (sub-FunctionNotSupportedInActiveSession) because the service identifier is supported and the sub-function parameter is not supported for the data-parameters (if applicable) of the client's request with indication for a response message.
- 6) Server sends no response message because the service identifier and sub-function parameter is supported of the client's request with indication for no response message.

NOTE If a negative response code RCRRP (requestCorrectlyReceivedResponsePending) is used, a final response shall be given independent of the suppressPosRspMsgIndicationBit value.

7) Same effect as in 2) (i.e., a negative response message is sent) because the suppressPosRspMsgIndicationBit is ignored for any negative response that needs to be sent upon a physically addressed request messages.

8) Same effect as in 3) (i.e., the negative response message is sent) because the suppressPosRspMsgIndicationBit is ignored for any negative response that needs to be sent upon receipt of a physically addressed request messages.

9) Same effect as in 4) (i.e., the negative response message is sent) because the suppressPosRspMsgIndicationBit is ignored for any negative response that needs to be sent upon a physically addressed request messages.

10) Same effect as in 5) (i.e., the negative response message is sent) because the suppressPosRspMsgIndicationBit is ignored for any negative response that needs to be sent upon a physically addressed request messages.

### 2.6.4.2 Functionally addressed client request message

The server response behaviour specified in this section is referenced in the service description of each service, which supports a sub-function parameter in the functionally addressed request message received from the client.

| Server Case # | Client Request Message |                                               | Server Capability   |                        |                                               | Server Response |          | Comments to Server Response                                                                             |
|---------------|------------------------|-----------------------------------------------|---------------------|------------------------|-----------------------------------------------|-----------------|----------|---------------------------------------------------------------------------------------------------------|
|               | Addressing Scheme      | Sub-Function (SuppressPosRspMsgIndicationBit) | ServiceID Supported | Sub-Function Supported | Data Parameter Supported (only if applicable) | Message         | NRC      |                                                                                                         |
| 1             | Functional             | FALSE (bit=0)                                 | YES                 | YES                    | At Least 1                                    | PosRsp          | -        | Server send positive response                                                                           |
| 2             |                        |                                               |                     |                        | At Least 1                                    | NegRsp          | NRC=0xXX | Server send negative response because error occurred reading the data parameters of the request message |
| 3             |                        |                                               |                     |                        | None                                          | NoRsp           | -        | Server does NOT send a response                                                                         |
| 4             |                        |                                               | NO                  | -                      | -                                             |                 | -        | Server does NOT send a response                                                                         |
| 5             |                        |                                               | YES                 | NO                     | -                                             |                 | -        | Server does NOT send a response                                                                         |
| 6             |                        | TRUE (bit=1)                                  | YES                 | YES                    | At Least 1                                    | NoRsp           | -        | Server does NOT send a response                                                                         |
| 7             |                        |                                               |                     |                        | At Least 1                                    | NegRsp          | NRC=0xXX | Server send negative response because error occurred reading the data parameters of the request message |
| 8             |                        |                                               |                     |                        | None                                          | NoRsp           | -        | Server does NOT send a response                                                                         |
| 9             |                        |                                               | NO                  | -                      | -                                             |                 | -        | Server does NOT send a response                                                                         |
| 10            |                        |                                               | YES                 | NO                     | -                                             |                 | -        | Server does NOT send a response                                                                         |

Table 2-21. Functionally addressed request message with sub-function parameter and server response behaviour

Description of server response cases on functionally addressed client request messages with subFunction:

1) Server sends a positive response message because the service identifier and sub function parameter is supported of the client's request with indication for a response message,

2) Server sends a negative response message (e.g. IMLOIF: incorrectMessageLengthOrIncorrectFormat)

because the service identifier and sub function parameter is supported of the client's request, but some other error appeared (e.g. wrong PDU length according to service identifier and sub function parameter in the request message) during processing of the sub function.

3) Server sends no response message because the negative response code ROOR (requestOutOfRange, which is identified by the server because the service identifier and sub function parameter are supported but a required data parameter is not supported of the client's request) is always suppressed in case of a functionally addressed request message. The suppressPosRspMsgIndicationBit does not matter in such case.

4) Server sends no response message because the negative response codes SNS (serviceNotSupported) and SNSIAS (serviceNotSupportedInActiveSession), which are identified by the server because the service identifier is not supported of the client's request, are always suppressed in case of a functionally addressed request message. The suppressPosRspMsgIndicationBit does not matter in such case.

5) Server sends no response message because the negative response codes SFNS (subfunctionNotSupported) and SFNSIAS (sub-functionNotSupportedInActiveSession), which are identified by the server because the service identifier is supported and the sub-function parameter is not supported for the data-parameters (if applicable) of the client's request, are always suppressed in case of a functionally addressed request. The suppressPosRspMsgIndicationBit does not matter in such case.

6) Server sends no response message because the service identifier and sub function parameter is supported of the client's request with indication for no response message.

NOTE) If a negative response code RCRRP (requestCorrectlyReceivedResponsePending) is used, a final response shall be given independent of the suppressPosRspMsgIndicationBit value.

7) Same effect as in 2) (i.e., a negative response message is sent) because the suppressPosRspMsgIndicationBit is ignored for any negative response. This is also true in case the request message is functionally addressed.

8) Same effect as in 3) (i.e., no response message is sent) because the negative response code ROOR (requestOutOfRange, which is identified by the server because the service identifier and sub-function parameter are supported but a required data-parameter is not supported of the client's request) is always suppressed in case of a functionally addressed request message. The suppressPosRspMsgIndicationBit does not matter in such case.

9) Same effect as in 4) (i.e., no response message is sent) because the negative response codes SNS (serviceNotSupported) and SNSIAS (serviceNotSupportedInActiveSession), which are identified by the server because the service identifier is not supported of the client's request, are always suppressed in case of a functionally addressed request message. The suppressPosRspMsgIndicationBit does not matter in such case.

10) Same effect as in 5) (i.e., no response message is sent) because the negative response codes SFNS (sub-functionNotSupported) and SFNSIAS (sub-functionNotSupportedInActiveSession), which are identified by the server because the service identifier is supported and the sub-function parameter is not supported of the client's request, are always suppressed in case of a functionally addressed request message. The suppressPosRspMsgIndicationBit does not matter in such case.

## 2.6.5 Request message without sub-function parameter and server response behaviour

### 2.6.5.1 Physically addressed client request message

The server response behaviour specified in this section is referenced in the service description of each

service, which does not support a sub-function parameter but a data parameter in the physically addressed request message received from the client.

| Server Case # | Client Request Message | Server Capability   |                                               | Server Response |                   | Comments to Server Response                                                                             |
|---------------|------------------------|---------------------|-----------------------------------------------|-----------------|-------------------|---------------------------------------------------------------------------------------------------------|
|               | Addressing Scheme      | ServiceID Supported | Data Parameter Supported (only if applicable) | Message         | NRC               |                                                                                                         |
| 1             | Physical               | YES                 | ALL                                           | PosRsp          | -                 | Server send positive response                                                                           |
| 2             |                        |                     | At Least 1                                    |                 | -                 | Server send positive response                                                                           |
| 3             |                        |                     | At Least 1                                    | NegRsp          | NRC=0xXX          | Server send negative response because error occurred reading the data parameters of the request message |
| 4             |                        |                     | None                                          |                 | NRC = ROOR        | Negative response with NRC 0x31                                                                         |
| 5             |                        | NO                  | -                                             |                 | NRC=SNS or SNSIAS | Negative response with NRC 0x11 or NRC 0x7F                                                             |

Table 2-22. Physically addressed request message without sub-function parameter and server response behaviour

Description of server response cases on physically addressed client request messages without sub-function (data parameter follows service identifier):

- 1) Server sends a positive response message because the service identifier and all data parameters are supported of the client's request message,
- 2) Server sends a positive response message because the service identifier and a single data parameter is supported of the client's request message,
- 3) Server sends a negative response message (e.g., IMLOIF: incorrectMessageLengthOrIncorrectFormat) because the service identifier is supported and at least one data-parameter is supported of the client's request message, but some other error occurred (e.g. wrong length of the request message) during processing of the service.
- 4) Server sends a negative response message with the negative response code ROOR (requestOutOfRange) because the service identifier is supported and none of the requested data parameters are supported of the client's request message,
- 5) Server sends a negative response message with the negative response code SNS (serviceNotSupported) or SNSIAS (serviceNotSupportedInActiveSession) because the service identifier is not supported of the client's request message.

## 2.6.5.2 Functionally addressed client request message

The server response behaviour specified in this section is referenced in the service description of each service, which does not support a sub-function parameter but a data parameter in the functionally addressed request message received from the client.

| Server Case # | Client Request Message | Server Capability   |                                               | Server Response |     | Comments to Server Response |
|---------------|------------------------|---------------------|-----------------------------------------------|-----------------|-----|-----------------------------|
|               | Addressing Scheme      | ServiceID Supported | Data Parameter Supported (only if applicable) | Message         | NRC |                             |

|   |            |     |            |        |        |                                                                                                         |
|---|------------|-----|------------|--------|--------|---------------------------------------------------------------------------------------------------------|
| 1 | Functional | YES | YES        | PosRsp | -      | Server send positive response                                                                           |
| 2 |            |     | At Least 1 |        | -      | Server send positive response                                                                           |
| 3 |            |     | At Least 1 | NegRsp | NRC=xx | Server send negative response because error occurred reading the data parameters of the request message |
| 4 |            |     | None       | NoRsp  | -      | Server does NOT send a response                                                                         |
| 5 |            | NO  | -          |        | -      | Server does NOT send a response                                                                         |

Table 2-23. Functionally addressed request message without sub-function parameter and server response behaviour

Description of server response cases on functionally addressed client request messages without sub-function (data parameter follows service identifier):

- 1) Server sends a positive response message because the service identifier and single data parameter is supported of the client's request message,
- 2) Server sends a positive response message because the service identifier and at least one data parameter is supported of the client's request message,
- 3) Server sends a negative response message (e.g. IMLOIF: incorrectMessageLengthOrIncorrectFormat) because the service identifier is supported and at least one, more than one, or all data parameters are supported of the client's request message, but some other error occurred (e.g. wrong length of the request message) during processing of the service,
- 4) Server sends no response message because the negative response code ROOR (request out of range; which would occur because the service identifier is supported, but none of the requested data parameters is supported of the client's request) is always suppressed in case of a functionally addressed request,
- 5) Server sends no response message because the negative response codes SNS (serviceNotSupported) and SNSIAS (serviceNotSupportedInActiveSession), which are identified by the server because the service identifier is not supported of the client's request, are always suppressed in case of a functionally addressed request.

## 2.6.6 Multiple Concurrent Request Messages with Physical and Functional Addressing

A common server implementation has only one diagnostic protocol instance available in the server which can only handle one request at a time. The rule is that any received message (regardless of addressing mode physical or functional) occupies this resource until the request message is processed (with final response sent or application call without response).

There are only two (2) exceptions which have to be treated separately:

- a) The keep-alive logic is used by a client to keep a previously enabled session active in one or multiple servers. Keep-Alive-Logic is defined as the functionally addressed valid TesterPresent message with suppressPosRspMsgIndicationBit=true and has to be processed by a bypass logic. It is up to the server to make sure that this specific message can not "block" the server's application layer and that an immediately following addressed message can be processed.
- b) If a server supports one or more legislated diagnostic requests and one of these requests is received while a non legislated service (e.g., enhanced diagnostics) is active, then the active service shall be aborted, the default session shall be started and the legislated diagnostic service shall be processed. This requirement does not apply if the programming session is active.

## 2.7 Error handling

Error handling for the application layer and session management to be fulfilled by the client and the server communicating with the client during physical and functional communication shall be in accordance with Tables 2-14, in respect of which it is assumed that the client and the server implement the application and session layer timing according to this part of ISO 14229-1/2/3.

### 2.7.1 Network / Data Link Layer Error Handling

| Error Type |                              |                                                                                                                                                                                                                                        | Handling                                                                                                     |
|------------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| CAN        | DLC                          | DLC value is smaller than eight (8).                                                                                                                                                                                                   | The reception of a CAN frame shall be ignored by the network layer without any further action.               |
|            | SF_DL                        | SF_DL is equal to zero (0) or SF_DL is greater than seven (7).                                                                                                                                                                         | The network layer shall ignore the received SF.                                                              |
|            | FF_DL                        | FF_DL is greater than the available receiver buffer size.                                                                                                                                                                              | The network layer shall abort the message reception and send FC with the parameter “Flow Status = Overflow”. |
|            |                              | FF_DL is less than eight (8).                                                                                                                                                                                                          | The network layer shall ignore the received FF and not transmit FC.                                          |
| CAN-FD     | DLC                          | DLC other than Table 2-7. CAN_FD data length.                                                                                                                                                                                          | The reception of a CAN-FD frame shall be ignored by the network layer without any further action.            |
|            |                              | DLC is smaller than expected value through SF_DL.                                                                                                                                                                                      |                                                                                                              |
|            | SF_DL                        | Single Frame(SF) where CAN_DL is less or equal to 8 and SF_DL is equal to 0                                                                                                                                                            | The network layer shall ignore the received SF.                                                              |
|            |                              | Single Frame(SF) where CAN_DL is greater than 8 and the low nibble of PCI byte #1 is not 0                                                                                                                                             |                                                                                                              |
|            |                              | Single Frame(SF) where SF_DL is greater than (CAN_DL – 1) in Normal Addressing                                                                                                                                                         |                                                                                                              |
|            |                              | Single Frame(SF) where CAN_DL is greater than 8 and SF_DL is invalid.<br>(Refer to Table 2-11 below to check validity of SF_DL)                                                                                                        |                                                                                                              |
|            | FF_DL                        | First Frame(FF) where CAN_DL is less than 8                                                                                                                                                                                            | The network layer shall ignore the received FF and not transmit FC.                                          |
|            |                              | First Frame(FF) where CAN_DL is larger than 8 and FF_DL is less than the minimum                                                                                                                                                       |                                                                                                              |
|            |                              | First Frame(FF) where all bits of the lower nibble of PCI byte #1 and all bits of PCI byte #2 are set to 0 and FF_DL is equal or less than 4095                                                                                        |                                                                                                              |
|            |                              | FF_DL is greater than the available receiver buffer size.                                                                                                                                                                              | The network layer shall abort the message reception and send FC with the parameter “Flow Status = Overflow”. |
| ST         | Reserved ST parameter value. | The sending network entity shall use the longest ST value specified by this document (7F hex = 127 ms) instead of the value received from the receiving network entity for the duration of the ongoing segmented message transmission. |                                                                                                              |

|    |                                 |                                            |
|----|---------------------------------|--------------------------------------------|
| FS | Invalid (reserved) FS parameter | The message transmission shall be aborted. |
|----|---------------------------------|--------------------------------------------|

Table 2-24. Protocol Control Information Error Handling

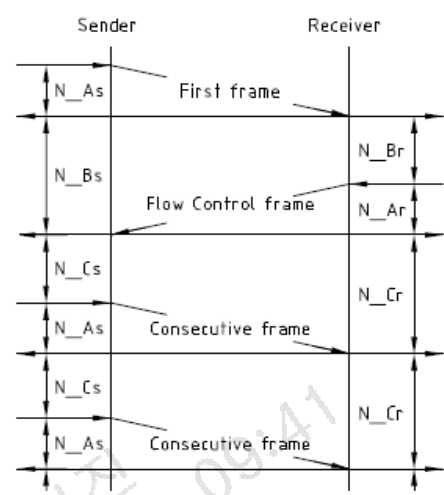
| Error Type |                                                                                                 | Handling                    | Timing Parameter                                                                   |
|------------|-------------------------------------------------------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------|
| N_As       | Any message not transmitted in time on the sender side.                                         | Abort message transmission. |  |
| N_Ar       | Any message not transmitted in time on the receiver side.                                       | Abort message reception.    |                                                                                    |
| N_Bs       | Flow Control not received (lost, overwritten) on the sender side.                               | Abort message transmission. |                                                                                    |
|            | Preceding First Frame or Consecutive Frame not received (lost, overwritten) on the sender side. |                             |                                                                                    |
| N_Cr       | Consecutive Frame not received (lost, overwritten) on the receiver side.                        | Abort message reception.    |                                                                                    |
|            | Preceding Flow Control not received (lost, overwritten) on the receiver side.                   |                             |                                                                                    |

Table 2-25. Timeout Error Handling

### 2.7.2 Application Layer Error Handling

| Error Type                                                  |                                                                   | Client Handling                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                             |                                                                   | Physical Communication                                                                                                                                                                                                                                                                                  | Functional Communication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>Request Transmission</b>                                 | Request message from network layer with a negative result value.  | The client shall repeat the last request, after the time P3 <sub>CAN_Physical</sub> following the error indication.<br>Restart S3Client in the case of a physically addressed and sequentially transmitted TesterPresent (because S3Client has been stopped based on the request message transmission). | The client shall repeat the last request, after the time P3 <sub>CAN_Functional</sub> following the error indication.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>P2<sub>CAN_CLIENT</sub><br/>P2*<sub>CAN_CLIENT</sub></b> | Timeout                                                           | The client shall repeat the last request. Restart S3 <sub>CLIENT</sub> in the case of a physically addressed and sequentially transmitted TesterPresent (because S3 <sub>CLIENT</sub> has been stopped based on the request message transmission).                                                      | Where the client does not know the number of servers responding, then this is the indication for the client that no further response messages are expected. No retry of the request message is required. The client shall completely receive all response messages that are in progress until it can continue with further requests.<br>Where the client knows the number of responding servers, then this is the indication for the client that not all expected servers responded. The client shall repeat the request after it has completely received any response message that is in progress at the point in time the timeout occurs. |
| <b>Response Reception</b>                                   | Response message from network layer with a negative result value. | The client shall repeat the last request. Restart S3 <sub>Client</sub> in the case of a physically addressed and sequentially transmitted Tester Present (because S3 <sub>Client</sub> has been stopped based on the request message transmission).                                                     | The client shall repeat the last request after it has completely received any response message that is in progress at the point in time the error has been indicated.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

The client error handling defined shall be performed for a maximum of two (2) times, which means that the worst-case of service request transmissions is three (3).

Table 2-26. Client Error Handling

| Error Type                                                  |                                                                   | Server Handling                                                                                                                                                                      |
|-------------------------------------------------------------|-------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Request Reception</b>                                    | Request message from network layer with a negative result value.  | Restart S3Server timer (because it has been stopped based on the previously received FirstFrame indication). The server shall ignore the request.                                    |
| <b>P2<sub>CAN_CLIENT</sub><br/>P2*<sub>CAN_CLIENT</sub></b> | Timeout                                                           | N/A                                                                                                                                                                                  |
| <b>Response Transmission</b>                                | Response message from network layer with a negative result value. | Restart S3Server timer (because it has been stopped based on the previously received request message). The server shall <i>not</i> perform a retransmission of the response message. |

Table 2-27. Server Error Handling

## 2.8 External test equipment initialization sequence

The external test equipment shall support the initialization sequence specified in this part.

The purpose of the external test equipment initialization sequence is to prevent the communication collision between external test equipment and a controller which transmits diagnostic request messages to get diagnostic information of other controllers in the vehicle.

If a controller is connected to high-speed CAN network and always transmits diagnostic request message periodically with the same request CAN Identifier of external test equipment, it can cause the communication collision when the external test equipment is connect to the same high-speed CAN network.

To prevent the collision, every time when the external test equipment is connected to the high-speed CAN network and start diagnostic communication, it shall transmit initialization message (functional addressed (7DF hex) DiagnosticSessionControl service (10 hex) with Default Session (81 hex, SPRMIB=True)) at first. Regardless of which service was received, All controllers stop transmit diagnostic request message (if they did) when they received functional addressing (CAN/Ethdiag : 7DF / 18DB33F1 hex, DOIP : E000 / E400 hex) message.

This initialization message has no purpose but the alarm of the external test equipment's connection. Controllers shall always start the standard diagnostic mode when powered up. They still works in the same diagnostic mode (standard diagnostic mode) when the initialization message received.

## 2.9 External test equipment requirement

The purpose of this section is to unify the operation of external test equipment of HMC/KIA (e.g. EOL)  
This requirement doesn't affect the ECU functionality.

### 2.9.1 ECU Identification Data display rules

The content size of the data identifier can be defined as dynamic length. If the data length exceeds the maximum size, external test equipment can display ASCII data with dynamic length.

The display order of Data Identifier is defined in Table C-4.1 The order of DID in the diagnostic tool and Notation. External test equipment shall comply with DID display order. The DID which is not defined in Table C-4.1 is not managed in this specification. It need to be defined in respective ECU specification.

### 2.9.2 Tester present

External Test Equipment shall send Tester present using functional addressing.

Ofcourse Tester present can be sent using physical addressing. But a number of ECUs can be diagnosed at once in the vehicle diagnostic scenario.

And Diagnostic target ECU can be switchable at all time.

If there is no specific problem, Tester present shall be sent using functional addressing.

### 3. Diagnostic service implementation

#### 3.1 Diagnostic Service List

Every ECU should support diagnostic services listed below.

| Diagnostic Service Name                                 | Req. SID (hex) | Cvt. | Diagnostic Modes |                      |                             | Engine / Propulsion system Conditions |
|---------------------------------------------------------|----------------|------|------------------|----------------------|-----------------------------|---------------------------------------|
|                                                         |                |      | Default Session  | Program-ming Session | Extended Diagnostic Session |                                       |
|                                                         |                |      | 01h              | 02h                  | 03h                         |                                       |
| Diagnostic and Communication Management Functional Unit |                |      |                  |                      |                             |                                       |
| DiagnosticSessionControl                                | 10             | M    | ◆                | ◆                    | ◆                           | no restriction, <b>C2</b>             |
| ECUReset                                                | 11             | M    | ◆                | ◆                    | ◆                           | Engine stop, IGN on                   |
| SecurityAccess                                          | 27             | M    | X                | ◆                    | ◆                           | no restriction                        |
| CommunicationControl                                    | 28             | M    | X                | ◆                    | ◆                           | Engine stop                           |
| Authentication                                          | 29             | C3   | ◆                | ◆                    | ◆                           | no restriction                        |
| TesterPresent                                           | 3E             | M    | ◆                | ◆                    | ◆                           | no restriction                        |
| SecuredDataTransmission                                 | 84             | U    | X                |                      | ◆                           | no restriction                        |
| ControlDTCSetting                                       | 85             | M    | X                | ◆                    | ◆                           | no restriction                        |
| ResponseOnEvent                                         | 86             | U    | ◆                |                      | ◆                           | no restriction                        |
| LinkControl                                             | 87             | U    | X                |                      | ◆                           | no restriction                        |
| Data Transmission Functional Unit                       |                |      |                  |                      |                             |                                       |
| ReadDataByIdentifier                                    | 22             | M    | ◆                |                      | ◆                           | no restriction                        |
| ReadMemoryByAddress                                     | 23             | U    | X                |                      | ◇                           | no restriction                        |
| ReadScalingDataByIdentifier                             | 24             | U    | ◆                |                      | ◆                           | no restriction                        |
| ReadDataByPeriodicIdentifier                            | 2A             | U    | X                |                      | ◆                           | no restriction                        |
| DynamicallyDefineDataIdentifier                         | 2C             | U    | ◆                |                      | ◆                           | no restriction                        |
| WriteDataByIdentifier                                   | 2E             | C1   | X                | ◇                    | ◇                           | Engine stop, IGN on                   |
| WriteMemoryByAddress                                    | 3D             | U    | X                | ◇                    | ◇                           | Engine stop, IGN on                   |
| Stored Data Transmission Functional Unit                |                |      |                  |                      |                             |                                       |



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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|----|---|---|---|---------------------------|
| ReadDTCInformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 19 | M  | ◆ |   | ◆ | no restriction            |
| ClearDiagnosticInformation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 14 | M  | ◆ |   | ◆ | no restriction, <b>C4</b> |
| <b>Input/Output Control Functional Unit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |    |    |   |   |   |                           |
| InputOutputControlByIdentifier                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2F | U  | X | X | ◆ | <b>C5</b>                 |
| <b>Remote Activation Of Routine Functional Unit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |    |    |   |   |   |                           |
| RoutineControl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 31 | C1 | ○ | ○ | ○ | Engine stop, IGN on       |
| <b>Upload/Download Functional Unit</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |    |    |   |   |   |                           |
| RequestDownload                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 34 | C1 | X | ◇ | ◇ | Engine stop, IGN on       |
| RequestUpload                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 35 | U  | X | ◇ | ◇ | Engine stop, IGN on       |
| TransferData                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 36 | C1 | X | ◇ | ◇ | Engine stop, IGN on       |
| RequestTransferExit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 37 | C1 | X | ◇ | ◇ | Engine stop, IGN on       |
| RequestFileTransfer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 38 | U  | X | ◇ | ◇ | Engine stop, IGN on       |
| <p>◆ : Function possible in the selected diag. session without security access</p> <p>◇ : Function possible in the selected diag. session with security access</p> <p>○ : Specific sub-function possible in the selected diag.Mode (see each function descriptions)</p> <p>X : Function impossible in selected diag. session.</p> <p>(blank) : No restriction.</p> <p>C1 : This service function shall be supported if the controller is reprogrammable.</p> <p>C2 : When the server transitions from any other session (including default session) to Programming session, Engine / Propulsion system conditions is Engine stop, IGN on.</p> <p>C3 : This service function shall be supported if the controller is the central gateway(CCU2).</p> <p>C4 : Engine / Propulsion system condition is 'Engine Stop' only for OBD controllers. Refer to J1979, J1979-2/3</p> <p>C5 : Engine / Propulsion system condition shall be handled in each controller diagnostic specification. The engine/ Propulsion system conditions could be different depending on the DID. Due to functional safety, restriction of the engine / propulsion system conditions could be exist.</p> <p>※ Secured dataIdentifiers, secured memory areas and secured routines require a SecurityAccess service and therefore a non-default diagnostic session</p> <p>※ ACU controllers can or can not support some diagnostic services at default session regardless of security levels. Details refer to the diagnostic specification of the controllers.</p> <p>※ All ECU need to support all diagnostic services at both physical ID and functional ID.</p> <p>※ "Engine stop" refers to the "Engine stop" state for internal combustion engine (ICE) vehicles, and the "not ready" status for vehicles with electric propulsion systems (applicable to HEV, PHEV, EREV, and EV). For electric propulsion vehicles, "IGN on" means "Power on" and also includes "Service mode." For vehicles with SDV power architecture (e.g., XP2), "IGN on" corresponds to Drive (S4), ON (S3), and Ready (S2) states. In addition, "Engine stop" includes S0, S1, S2, and S3 states. Therefore, both "Engine stop" and "IGN on" correspond to S2 and S3. Note, however, that in S0 and S1 states, diagnostic communication is generally not possible because the ECU power is not supplied.</p> <p>※ In the boot software, 10/11/27/3E/31/34/36/37 hex services must be presented and other services are optionally presented.</p> <p>※ Engine / Propulsion system Condition (Engine stop, IGN on) of some diagnostic services can be changed according to the ECU's specific situation (Exception handling). Refer to the each ECU diagnostic specification in detail.</p> <p>- If OTA target ECUs support memory redundancy and differential data patch, Upload/Download Functional Unit(34/35/36/37) and RoutineControl(31) services can be used in the extended session during engine running or IGN off.</p> <p>- According to OTA specification ECU can change session to programming session under IGN off regardless of C2 restriction.</p> |    |    |   |   |   |                           |

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- RoutineControl(31) services can be used in the extended session during engine running In the production process
- ※ OTA target ECUs need to support ECUReset(11) service and CommunicationControl(28) service during IGN off.

Table 3-1. Diagnostic services overview

### 3.1.1 Summary of Engine / Propulsion system conditions

| Diagnostic Service Name        | Req. SID (hex) | Standard Engine / Propulsion system Condition             | OTA Target ECU                                    |
|--------------------------------|----------------|-----------------------------------------------------------|---------------------------------------------------|
| DiagnosticSessionControl       | 10             | No restriction, Engine stop, IGN on (Programming session) | No restriction, Engine stop (Programming session) |
| ECUReset                       | 11             | Engine stop, IGN on                                       | Engine stop                                       |
| WriteDataByIdentifier          | 2E             | Engine stop, IGN on                                       | Engine stop, IGN on                               |
| WriteMemoryByAddress           | 3D             | Engine stop, IGN on                                       | Engine stop, IGN on                               |
| InputOutputControlByIdentifier | 2F             | System specific                                           | System specific                                   |
| RoutineControl                 | 31             | Engine stop, IGN on                                       | No restriction                                    |
| RequestDownload                | 34             | Engine stop, IGN on                                       | No restriction                                    |
| RequestUpload                  | 35             | Engine stop, IGN on                                       | No restriction                                    |
| TransferData                   | 36             | Engine stop, IGN on                                       | No restriction                                    |
| RequesTransferExit             | 37             | Engine stop, IGN on                                       | No restriction                                    |
| RequestFileTransfer            | 38             | Engine stop, IGN on                                       | Engine stop, IGN on                               |
| CommunicationControl           | 28             | Engine stop                                               | Engine stop                                       |

Table 3-1.1. Diagnostic engine condition overview

#### 3.1.1.1 Engine / Propulsion system signal example

This is a general example, the actual signal may be different from the following. Please refer to Network DB

| Message     | Signal        | Vehicle Type    |
|-------------|---------------|-----------------|
| EMS_02_10ms | ENG_EngSta    | Emission Engine |
| HCU_03_10ms | HCU_HevRdySta | Hybrid          |

|                                                 |                        |     |
|-------------------------------------------------|------------------------|-----|
| VCU_01_10ms                                     | VCU_EvDrvRdySta        | EV  |
| VehiclePowerState_200ms                         | VehiclePowerState      | SDV |
| Vehicle Base API:<br>PowerSystemBase.PowerState | (Refer to Vehicle API) | SDV |

Engine Running/Stop : Except Engine running or Drivable, The others are ENGINE STOP

| EMS 02                  |                                      | HCU 03                       |                                      | VCU 01                      |                                      | VehiclePowerState         |                                      |
|-------------------------|--------------------------------------|------------------------------|--------------------------------------|-----------------------------|--------------------------------------|---------------------------|--------------------------------------|
| Value Description       | Engine / Propulsion system condition | Value Description            | Engine / Propulsion system condition | Value Description           | Engine / Propulsion system condition | Value Description         | Engine / Propulsion system condition |
| 0x00<br>Engine stop     | STOP                                 | 0x00<br>HEV not ready        | STOP                                 | 0x00<br>None                | STOP                                 | 0x00<br>None(Default)     | STOP                                 |
| 0x01<br>Cranking        | STOP                                 | <b>0x01<br/>HEV Drivable</b> | <b>Running</b>                       | <b>0x01<br/>EV Drivable</b> | <b>Running</b>                       | 0x01<br>Standby(S1)       | STOP                                 |
| 0x02<br>Stalled         | STOP                                 | 0x02<br>Not Used             | STOP                                 | 0x02<br>Not used            | STOP                                 | 0x02<br>Ready(S2)         | STOP                                 |
| <b>0x03<br/>Running</b> | <b>Running</b>                       | 0x03<br>Error indicator      | STOP                                 | 0x03<br>Error Indicator     | STOP                                 | 0x03<br>ON(S3)            | STOP                                 |
| 0x07<br>Fault           | STOP                                 |                              |                                      |                             |                                      | <b>0x04<br/>Drive(S4)</b> | <b>Running</b>                       |

Vehicle Base API: PowerSystemBase.PowerState

pub enum PowerState

```
{ off = 0, // STOP   StandBy = 1, // STOP   Ready = 2, // STOP   On = 3, // STOP   Driving = 4, // Running }
```

### 3.1.1.2 Engine / Propulsion system condition by signal status

| Signal status                          | Engine / Propulsion system Condition                   |
|----------------------------------------|--------------------------------------------------------|
| Never received<br>(After start up ECU) | Engine Off                                             |
| Time out                               | The last received engine / propulsion system condition |
| Normal                                 | The last received engine / propulsion system condition |

- Never received (After start up ECU) : ECU has never been received engine / propulsion system signal.
- Time out : Cyclic signal has been time out.

### 3.1.1.3 Limitation and Exception

Bootloader Engine / Propulsion system condition is **Optional**

In the bootloader(FBL), it may be difficult to determine the engine / propulsion system state.  
Because FBL(Flash bootloader) supports minimal functionality for the SW update

Engine / Propulsion system Condition (Engine stop, IGN on) of some diagnostic services can be changed according to the ECU's specific situation (Exception handling). Refer to the each ECU diagnostic specification.

E.g.) If some functions regarding EOL(end of line) calibration are needs Engine ON, Engine / Propulsion system condition would be changed by ECU specific requirement.



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In the power-on state of the electric vehicle power system, the high voltage main relay is turned on. If the controller using the high voltage power enters the programming session or performs communication control, the application message cannot be transmitted for a while. Therefore, even in the main relay on state The controller using high voltage power, which must transmit the application message, can respond with Negative response code 22 when receiving a request for programming session or communication control.

When a single controller is logically divided into two or more diagnostic IDs and UDS servers for each diagnostic ID, unless there is a specific reason, when the validity of the signals used in the engine condition of each UDS server is determined by a CRC or an alive counter, each UDS server shall match the signal determination criteria to be the same.

e.g.) MCU ID : 05D0 AP ID : 05D1

When an invalid engine signal is received, the MCU may respond with a negative response code 22 and the AP may respond with a positive response. As such, the validity of each may be determined differently, resulting in a difference in response results. Therefore, each UDS server must match the criteria for determining signal validity..

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## 4. Diagnostic and Communication Control functional unit

### 4.1 DiagnosticSessionControl (10 hex) service

The DiagnosticSessionControl service is used to enable different diagnostic sessions in the server(s).

There shall always be exactly one diagnostic session active in a server. A server shall always start the default diagnostic session when powered up. If no other diagnostic session is started, then the default diagnostic session shall be running as long as the server is powered.

If the client has requested a diagnostic session, which is already running, then the server shall send a positive response message and behave as shown in Figure 4-1 that describes the server internal behaviour when transitioning between sessions.

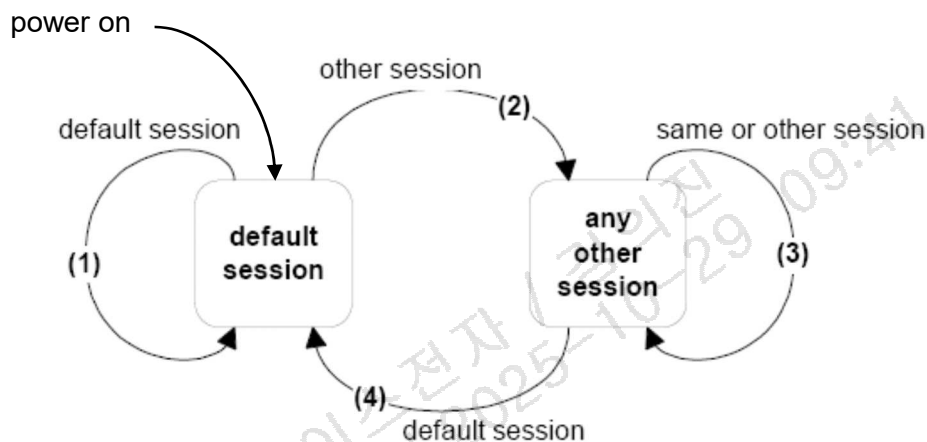


Figure 4-1. Server diagnostic session state diagram

Diagnostic session transition description:

- 1) When the server is in the defaultSession and the client requests to start the defaultSession then the server shall re-initialize the defaultSession completely. The server shall reset all activated/initiated/changed settings/controls during the activated session. This does not include long term changes programmed into non-volatile memory.
- 2) When the server transitions from the defaultSession to any other session than the defaultSession then the server shall only reset the events that have been configured in the server via the ResponseOnEvent (86 hex) service during the defaultSession.
- 3) When the server transitions from any diagnostic session other than the defaultSession to another session other than the defaultSession (including the currently active diagnostic session) then the server shall (re-) initialize the diagnostic session, which means that each event that has been configured in the server via the ResponseOnEvent (86 hex) service shall be reset and Security levels of security access service(\$27) shall be locked. Any configured periodic scheduler shall remain active when transitioning from one non-defaultSession to another or the same non-DefaultSession. The states of the CommunicationControl and ControlDTCSetting services shall not be affected, which e.g. means that normal communication shall remain disabled when it is disabled at the point in time the session is switched.
- 4) When the server transitions from any diagnostic session other than the default session to the defaultSession then the server shall reset each event that has been configured in the server via the ResponseOnEvent (86 hex) service and security shall be enabled. Any configured periodic scheduler shall be disabled. Furthermore the states of the CommunicationControl and ControlDTCSetting

services shall be reset, which e.g. means that normal communication shall be reenabled when it was disabled at the point in time the session is switched to the defaultSession. The server shall reset all activated/initiated/changed settings/controls during the activated session. This does not include long term changes programmed into non-volatile memory.

### 4.1.1 Request message

| Data Byte | Parameter Name                                     | Cvt      | Hex Value   | Mnemonic    |
|-----------|----------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>DiagnosticSessionControl Request Service ID</b> | <b>M</b> | <b>#10h</b> | <b>DSC</b>  |
| #2d       | Sub-function = [<br>diagnosticSessionType ]        | M        | #XXh        | LEV_<br>DS_ |

Table4-1. Request Message

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00  | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M   |          |
| 01  | <b>defaultSession</b><br><br>This diagnostic session enables the default diagnostic session in the server(s) and does not support any diagnostic application timeout handing provisions. (e.g. no Tester Present service is necessary to keep the session active). If any other session than the defaultSession has been active in the server and the defaultSession in once again started, then the following implementation rules shall be followed : <ul style="list-style-type: none"> <li>- The server shall stop the current diagnostic session when it has sent the DiagnosticSessionControl positive response message and shall start the newly requested diagnostic session afterwards.</li> <li>- If the server has sent a DiagnosticSessionControl positive response message it shall have re-locked the server if the client unlocked it during the diagnostic session</li> <li>- If the server sends a negative response message with the DiagnostitcSessionControl request service identifier the active session shall be continued.</li> </ul> | M   | DS       |
| 02  | <b>programmingSession</b><br><br>This diagnosticSession enables all diagnostic services required to support the memory programming of a server. In case the server runs the programmingSession in the boot software, the programmingSession shall only be left via an ECUReset (11 hex) service initiated by the client, a DiagnosticSessionControl (10 hex) service with sessionType equal to defaultSession, or a session layer timeout in the server. In case the server runs in the boot software when it receives the DiagnosticSessionControl (10 hex) service with sessionType equal to defaultSession or a session layer timeout occurs and a valid application software is present for both cases then the server shall restart the application software. This document does not specify the various implementation methods of how to achieve the restart of the valid application software (e.g. a valid application software can be determined directly in the boot software, during the ECU startup phase when performing an ECU reset, etc.).    | C   | ECUPS    |
| 03  | <b>extendedDiagnosticSession</b><br><br>This diagnosticSession can e.g. be used to enable all diagnostic services required to support the adjustment of functions like "Idle Speed, CO Value, etc." in the server's memory. It can also be used to enable diagnostic services, which are not specifically tied to the adjustment of functions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | M   | ECUEDS   |

|       |                                                                                                                                                                               |   |      |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 04    | <b>safetySystemDiagnosticSession</b><br><br>This diagnosticSessin enables all diagnostic services required to support safety system related functions e.g. airbag deployment. | U | SSDS |
| 05~7F | <b>Reserved</b>                                                                                                                                                               | M |      |

C : This session shall be supported if the controller is reprogrammable.

Table4-2. diagnosticSessionType definition

Whenever the service DSC (DiagnosticSessionControl) is called successfully (that includes also calling DSC with the currently active DiagnosticSesstionType), the securityAccess rights and the timing parameters will be initialized again. When the CGW(ICU, ESU, CCU) transitions from the extendedDiagnosticSession to the non-default session, the security right should not be initialized (remain unlock condition). (refer to ES95489-01 in detail.)

## 4.1.2 Positive response message

| Data Byte                                                                                                                                                  | Parameter Name                                                                                                                                                                                                                                 | Cvt | Hex Value                    | Mnemonic     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|------------------------------|--------------|
| #1d                                                                                                                                                        | <b>DiagnosticSessionControl Response Service ID</b>                                                                                                                                                                                            | M   | #50h                         | <b>DSCPR</b> |
| #2d                                                                                                                                                        | diagnosticSessionType                                                                                                                                                                                                                          | M   | #XXh                         | DS_          |
| #3d<br>#4d<br>#5d<br>#6d                                                                                                                                   | SessionParameterRecord =<br>[<br><b>P2</b> <sub>CAN_SERVER_MAX</sub> (high byte)<br><b>P2</b> <sub>CAN_SERVER_MAX</sub> (low byte)<br><b>P2*</b> <sub>CAN_SERVER_MAX</sub> (high byte)<br><b>P2*</b> <sub>CAN_SERVER_MAX</sub> (low byte)<br>] | M   | #XXh<br>#XXh<br>#XXh<br>#XXh | SPREC        |
| ※ <b>P2</b> <sub>CAN_SERVER_MAX</sub> : Range = 0~50ms (resolution = 1ms)<br>※ <b>P2*</b> <sub>CAN_SERVER_MAX</sub> : Range = 0~5000ms (resolution = 10ms) |                                                                                                                                                                                                                                                |     |                              |              |

Table4-3. Positive Response Message definition

## 4.1.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                               |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                        |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the request DiagnosticSessionControl are not met. |

Table4-4. Supported Negative Response Codes

## 4.1.4 Implementaion requirements

This service is used by the diagnostic tool to enable different types of diagnostic sessions in a server. In order to execute a diagnostic service the appropriate session has to be started first.

There shall be only one diagnostic session active at a time.

Default Session (01) shall be enabled automatically by the ECU if no diagnostic session has been requested at power up.

An ECU shall return to /Default Session (01) after timeout of another diagnostic session.

An ECU shall be capable of providing all diagnostic functionality defined for the default diagnostic session under normal operating conditions.

The ECU shall first send a DiagnosticSessionControl Positive Response (50 xx) message before the new session becomes active in the ECU.

A DiagnosticSessionControl Positive Response (50 xx) message shall be returned by an ECU if the diagnostic tool requests a session that is already running. If the client has requested a diagnostic session, which is already running, then the server shall send a positive response message and be have as clause 4.1 that describes the server internal behavior when transitioning between sessions.

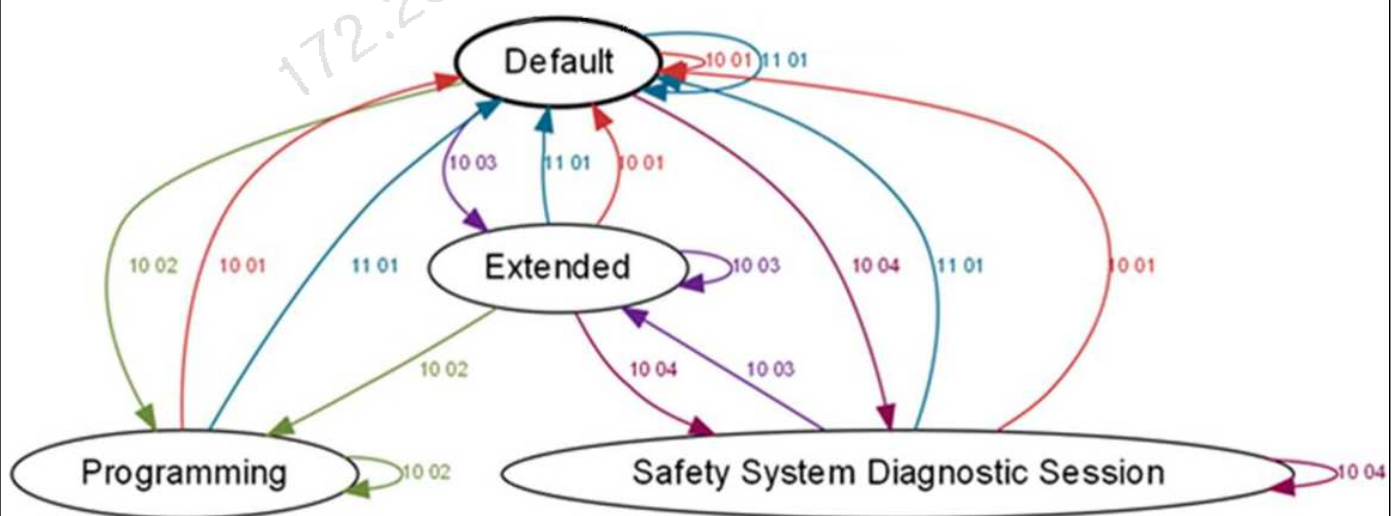
An ECU shall remain in its current diagnostic session if it is not able to switch into the re-requested diagnostic session.

The Tester Present (3E) service shall be used to keep the diagnostic sessions active by re-triggering S3server. Also any other service request keeps the diagnostic session alive.

A functional Tester Present (3E) request without response may be sent at any time, even regardless of any other service in progress.

When receiving or transmitting any diagnostic messages, including \$3E service, the timer will reset. Deactivating a session leads to reinitialisation of all activities, started already.

The following state transitions shall be implemented in the ECU. (In case a session is not available in the ECU, the state transitions from and to the state are omitted):



## 4.2 ECUReset (11hex) service

The ECUReset service is used by the client to request a server reset.

This service requests the server to effectively perform a server reset based on the content of the resetType parameter value embedded in the ECUReset request message. The ECUReset positive response message (if required) shall be sent before the reset is executed in the server(s). After a successful server reset the server shall activate the defaultSession.

The OTA target ECU shall not automatically reset the ECU after receiving OTA background data.

### 4.2.1 Request message

| Data Byte | Parameter Name                     | Cvt | Hex Value | Mnemonic    |
|-----------|------------------------------------|-----|-----------|-------------|
| #1d       | <b>EcuReset Request Service ID</b> | M   | #11h      | ER          |
| #2d       | Sub-function = [ resetType ]       | M   | #XXh      | LEV_<br>RT_ |

Table4-5. Request Message

| Byte pos.<br>In record | Description                                                                                                                                                                                                                                                                                                                                                                                                                    | Cvt | Mnemonic |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00                     | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                | M   |          |
| 01                     | <b>hardReset</b><br>This value identifies a "hard reset" condition which simulates the power-on / start-up sequence typically performed after a server has been previously disconnected from its power supply (i.e. battery). It might result in the re-initialization of both volatile memory and non-volatile memory location to predetermined values.                                                                       | M   | HR       |
| 02                     | <b>keyOffOnReset</b><br>This value identifies a condition similar to the driver turning the ignition key off and back on. This reset condition should simulate a key-off-on sequence (i.e. interrupting the switched power supply). The performed action is implementation specific and not defined by the standard. Typically the values of non-volatile memory locations are preserved; volatile memory will be initialized. | U   | KOFFONR  |
| 03                     | <b>softReset</b><br>This value identifies a "soft reset" condition, which causes the server to immediately restart the application program if applicable. The performed action is implementation specific and not defined by the standard. A typical action is to restart the application without reinitializing of previously learned configuration data, adaptive factors and other long-term adjustments.                   | U   | SR       |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |       |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------|
| 04    | <b>enableRapidPowerShutDown</b><br>This value requests the server to enable and perform a "rapid power shut down" function. The server shall execute the function immediately after "key/ignition" is switched off. While the server executes the power down function, it shall transition either directly or after a defined stand-by time to sleep mode. If the client requires a response message and the server is already prepared to execute the "rapid power shut down" function, the server shall send the positive response message prior to the start of the "rapid power shut down" function. The next occurrence of a "key on" or "ignition on" signal terminates the "rapid power shut down" function. The client shall not send any request messages other than the ECUReset with the sub-function disableRapidPowerShutDown in order to not disturb the rapid power shut down function.<br>NOTE) This sub-function is only applicable to a server supporting a stand by-mode | U | ERPSD |
| 05    | <b>disableRapidPowerShutDown</b><br>This value requests the server to disable the previously enabled "rapid power shut down" function.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | U | DRPSD |
| 06~7F | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | M |       |

Table4-6. Request message sub-function parameter definition

### 4.2.2 Positive response message

| Data Byte | Parameter Name                      | Cvt      | Hex Value   | Mnemonic    |
|-----------|-------------------------------------|----------|-------------|-------------|
| #1d       | <b>EcuReset Response Service ID</b> | <b>M</b> | <b>#51h</b> | <b>ERPR</b> |
| #2d       | resetType                           | M        | #XXh        | RT_         |
| #3d       | powerDownTime                       | C        | #XXh        | PDT         |

C : This parameter is present if the sub-function parameter is set to the enableRapidPowerShutDown value (04hex);

Table4-7. diagnosticSessionControl Positive Response Message definition

| Definition                                                                                                                                                  | Value       | Description                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------|
| <b>powerDownTime</b><br>This parameter indicates to the client the minimum time of the stand-by-sequence the server will remain in the power down sequence. | 00 – FE hex | 1 second per count<br>(0-254 seconds powerDowntime) |
|                                                                                                                                                             | FF hex      | Failure or time not available                       |

Table4-8. powerDownTime parameter definition

### 4.2.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                              |
|-----|---------------------------------------|-----|----------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported       |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                               |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the ECUReset request is not met. |

Table4-9. Supported Negative Response Codes

### 4.3 SecurityAccess (27 hex) service

The purpose of this service is to provide a means to access data and/or diagnostic services, which have restricted access for security, emissions, or safety reasons. The security concept uses a seed and key relationship.

A typical example of the use of this service is as follows ;

- client requests the "Seed"
- server sends the "Seed"
- client sends the "Key" (appropriate for the Seed received)
- server responds that the "Key" was valid and that it will unlock itself

"Seed" should be a random (not fixed) non-zero(0000 hex) and non-FFFF hex number, and its size should be more than 2-byte.

The security key shall be derived from the security seed using an encryption algorithm (security algorithm). The Security Algorithm applies the ASK and the OEUK. OEUK has all the permissions that ASK possesses, along with additional permissions to perform rollback.

After 3 (Three) false attempts of security access 'sendKey' request message by the tester, the ECU is locked for 180 seconds. It is **Mandatory** that the ECU is locked for 180 seconds. If the tester request seed to the ECU during this time delay, the ECU will send negative response code , "requiredTimeDelayNotExpired".

See Table 4-80 for a description of the relevant parameters.

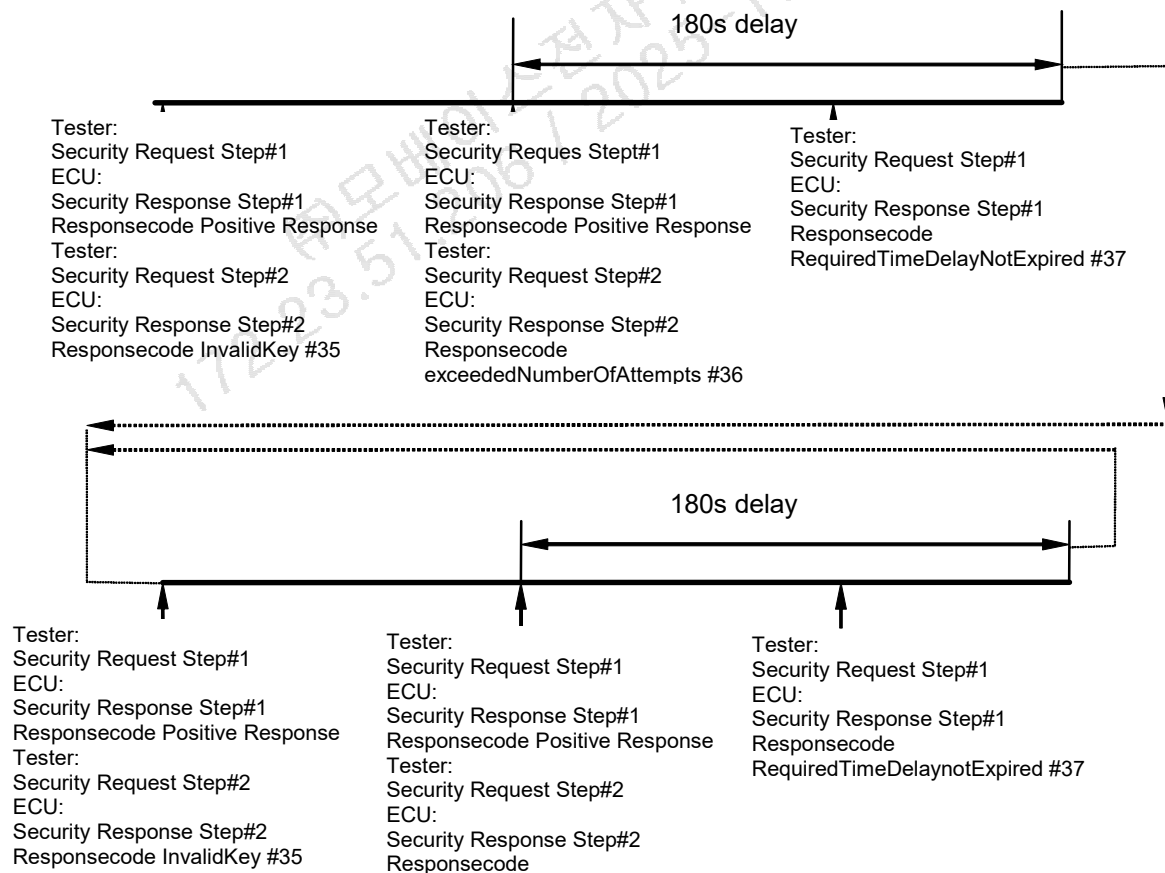


Figure 4-2. Example of SecurityAccess service procedure (false attempt)

| Name        | Description                                                                                                                             | Value             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Delay_Timer | This value represents the required minimum time between security access attempts. This timer shall started when Att_Cnt reaches Att_Cnt | (volatile) 180sec |

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|
|               | <p>Limit(Att_Cnt == Att_Cnt_Limit).</p> <p><b>A server shall implement a single timer for all security levels(e.g. ASK, OEUK), which means that all security levels share a single timer.</b></p> <p>When memory initialization such as ECU reset occurs when the delay timer has not yet expired, the controller starts the timer again from 180 seconds. Therefore, the parameter may be managed as volatile.</p> <p><b>IMPORTANT : this value shall not be initialized by Diagnostic Session Control Service(0x10)</b></p>                                                                                                                                             |                                                                     |
| Att_Cnt_Limit | This value represents the number of false security access attempts before a delay time (Start_Delay) is inserted.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | (constant) 3 times                                                  |
| Att_Cnt       | <p>this variable tracks the current value of the number of false access attempts. A separate counter is required for each individual security level. A server have same limits for each security level. Att_Cnt shall be implemented for a given SubFunction. If a valid key is received, the count is initialized to zero. And Delay_Timer expired then Att_cnt of security level which have reached Att_Cnt_Limit shall be initialized to zero.</p> <p><b>IMPORTANT : this value shall not be initialized by ECU reset or Diagnostic Session Control Service(0x10)</b></p> <p><b>NOTE : Counters can be managed separately for the boot loader and application.</b></p> | <p>(non-volatile)0~3</p> <p>NOTE : This parameters can be used.</p> |

Table 4-80 SecurityAccess parameters

The state transition diagram of SecurityAccess is shown in Figure 4-11, and the detailed operation is defined as shown in Table 4-81.

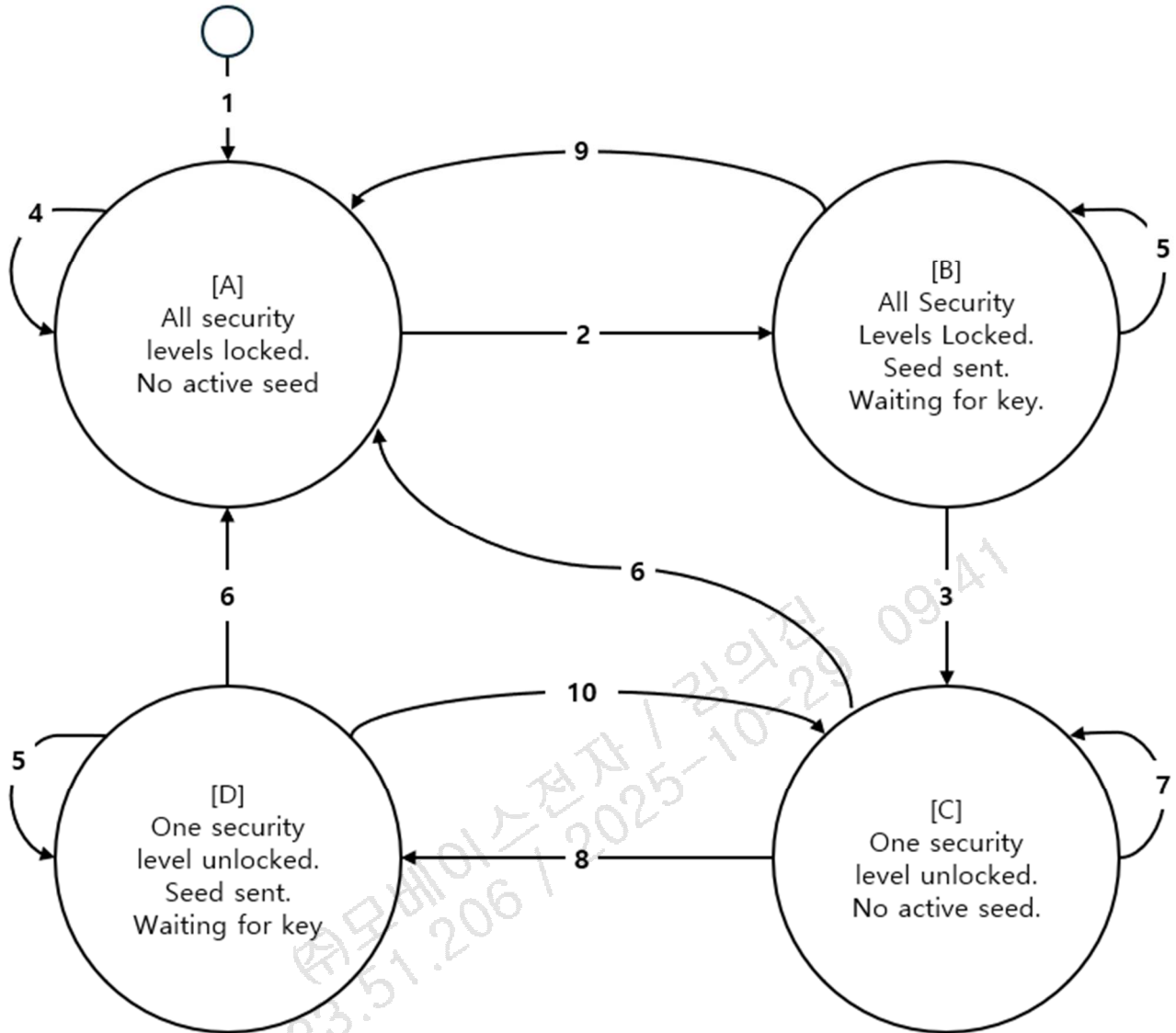


Figure 4-11 SecurityAccess state chart

| No | Operation | Condition                                                                                                                                                                                                            | Action                                                                                                                                                                                                      |
|----|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  |           | Start/Restart of ECU Application(e.g. ECU reset, Power cycle, Key cycle, Sleep - > wake transition, etc)                                                                                                             | Initialize Att_Cnt (Load Att_Cnt from NVM)<br>If at least one of the Att_Cnt reaches Att_Cnt_Limit then Delay_Timer shall be started.                                                                       |
| 2  |           | AND<br>SecurityAccess requestSeed received<br>Message length OK<br>Delay expired                                                                                                                                     | Generate and store Seed for the requested securityAccessType(e.g. ASK, OEUK).<br>Transmit SecurityAccess positive response on requestSeed request with Seed as active for the requested securityAccessType. |
| 3  |           | AND<br>SecurityAccess sendKey received<br>SubFunction: yy == xx+1<br>Sendkey Subfunction must be of the expected securityAccessType (the active stored seed is for the corresponding requestSeed securityAccessType) | Att_Cnt = 0 for SubFunction xx<br>Store Att_Cnt in non-volatile memory<br>Unlock security level for SubFunction xx.<br>Transmit SecurityAccess positive response on sendKey request.                        |



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|   |    |     |                                                                                                                                                                                    |                                                                                                                                                                       |
|---|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                       |
|   |    |     | Key OK                                                                                                                                                                             |                                                                                                                                                                       |
| 4 | OR | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC13                                                                                                                                      |
|   |    |     | Message Length NOK                                                                                                                                                                 |                                                                                                                                                                       |
|   |    |     | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC 24                                                                                                                                     |
|   |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC 37                                                                                                                                     |
|   |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                       |
|   |    |     | Delay NOT expired                                                                                                                                                                  |                                                                                                                                                                       |
|   |    |     | SecurityAccess request results in a general negative response code (e.g. minimum length, SubFunction supported) according to the general negative response handling (see Annex A). | Transmit negative response code as defined in Annex A                                                                                                                 |
|   |    |     | Delay Timer Expiration Occurs for SubFunction xx.                                                                                                                                  | Att_Cnt = 0 for SubFunction xx.<br>Store Att_Cnt in non-volatile memory                                                                                               |
| 5 |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Generate new seed and transmit SecurityAccess positive response with the new seed for the requested securityAccessType<br>Save SubFunction: xx = securityAccess Type. |
|   |    |     | Static_Seed == False<br>NOTE : Only randomseed (Static_Seed == False) is allowed for HMC. For Static seed is described in ISO14229-1                                               |                                                                                                                                                                       |
| 6 |    |     | DiagnosticSessionControl accepted or session timeout occurs.                                                                                                                       | Start appropriate diagnostic session.<br>Lock ECU.                                                                                                                    |
| 7 |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC13                                                                                                                                      |
|   |    |     | Message Length NOK                                                                                                                                                                 |                                                                                                                                                                       |
|   |    |     | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC24                                                                                                                                      |
|   |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC37                                                                                                                                      |
|   |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                       |
|   |    |     | Delay NOT expired                                                                                                                                                                  |                                                                                                                                                                       |
|   |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit SecurityAccess positive response with zero seed.                                                                                                             |
|   |    |     | Requested level is unlocked                                                                                                                                                        |                                                                                                                                                                       |
|   |    |     | SecurityAccess request results in a general negative response code (e.g. minimum length, SubFunction supported) according to the general negative response handling (see Annex A). | Transmit negative response code as defined in Annex A                                                                                                                 |
|   |    |     | Delay Timer Expiration Occurs for SubFunction xx.                                                                                                                                  | Att_Cnt = 0 for SubFunction xx<br>Store Att_Cnt in non-volatile memory                                                                                                |
| 8 |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Generate and store Seed for the requested securityAccessType                                                                                                          |
|   |    |     | Request level is NOT unlocked                                                                                                                                                      | Save SubFunction: xx = securityAccess Type.                                                                                                                           |
|   |    |     | Message length OK                                                                                                                                                                  | Transmit SecurityAccess positive response on requestSeed request with Seed as active for the requested securityAccessType.                                            |
|   |    |     | Delay expired                                                                                                                                                                      |                                                                                                                                                                       |

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|    |    |     |                                                                                                                                                                                    |                                                                                                                                                                     |
|----|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9  | OR | AND | SecurityAccess sendKey received                                                                                                                                                    | Att_Cnt++ for SubFunction xx<br>Store Att_Cnt in non-volatile memory<br>Transmit negative response NRC 35.                                                          |
|    |    |     | SubFunction: yy == xx+1                                                                                                                                                            |                                                                                                                                                                     |
|    |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                     |
|    |    |     | Key NOK                                                                                                                                                                            |                                                                                                                                                                     |
|    |    |     | (Att_Cnt+1) < Att_Cnt_Limit                                                                                                                                                        |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Att_Cnt = Att_Cnt_Limit for SubFunction xx.<br>Start Delay_Timer.<br>Store Att_Cnt in non-volatile memory.<br>Transmit negative response NRC 36.                    |
|    |    |     | SubFunction:yy==xx+1                                                                                                                                                               |                                                                                                                                                                     |
|    |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                     |
|    |    |     | Key NOK                                                                                                                                                                            |                                                                                                                                                                     |
|    |    |     | (Att_Cnt+1) >= Att_Cnt_Limit                                                                                                                                                       |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC 24                                                                                                                                   |
|    |    |     | SubFunction:yy<>xx+1                                                                                                                                                               |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC 13                                                                                                                                   |
|    |    |     | SubFunction:yy==x+1                                                                                                                                                                |                                                                                                                                                                     |
|    |    |     | Message length NOK                                                                                                                                                                 |                                                                                                                                                                     |
|    |    |     | DiagnosticSessionControl accepted or session timeout occurs.                                                                                                                       | Start appropriate diagnostic session.                                                                                                                               |
|    |    |     | SecurityAccess request results in a general negative response code (e.g. minimum length, SubFunction supported) according to the general negative response handling (see Annex A). | Transmit negative response code as defined in Annex A.                                                                                                              |
|    |    | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC13                                                                                                                                    |
|    |    |     | Message Length NOK                                                                                                                                                                 |                                                                                                                                                                     |
| 10 | OR | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit SecurityAccess positive response with zero seed                                                                                                            |
|    |    |     | Requested level is unlocked                                                                                                                                                        |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Att_Cnt = 0 forSubFunction xx.<br>Store Att_Cnt in non-volatile memory.<br>Lock currently unlocked security level.<br>Unlock security level forSubFunction xx.      |
|    |    |     | SubFunction: yy==xx+1                                                                                                                                                              |                                                                                                                                                                     |
|    |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                     |
|    |    |     | Key OK                                                                                                                                                                             |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Att_Cnt++ for SubFunction xx.<br>Store Att_Cnt in non-volatile memory.<br>Transmit negative response NRC 35.                                                        |
|    |    |     | SubFunction: yy==xx+1                                                                                                                                                              |                                                                                                                                                                     |
|    |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                     |
|    |    |     | Key NOK                                                                                                                                                                            |                                                                                                                                                                     |
|    |    |     | (Att_Cnt+1) < Att_Cnt_Limit                                                                                                                                                        |                                                                                                                                                                     |
|    |    |     | SecurityAccess sendKey received                                                                                                                                                    |                                                                                                                                                                     |
|    |    | AND | SecurityAccess sendKey received                                                                                                                                                    | Att_Cnt = Att_Cnt_Limit for SubFunction xx.<br>Start Delay_Timer for SubFunction xx.<br>Store Att_Cnt in non-volatile memory.<br>Transmit negative response NRC 36. |
|    |    |     | SubFunction:yy=xx+1                                                                                                                                                                |                                                                                                                                                                     |
|    |    |     | Message length OK                                                                                                                                                                  |                                                                                                                                                                     |
|    |    |     | Key NOK                                                                                                                                                                            |                                                                                                                                                                     |
|    |    |     | (Att_Cnt+1) >= Att_Cnt_Limit                                                                                                                                                       |                                                                                                                                                                     |

|  |     |                                                                                                                                                                                    |                                                        |
|--|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
|  | AND | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC 24                      |
|  |     | SubFunction: yy <> xx+1                                                                                                                                                            |                                                        |
|  | AND | SecurityAccess sendKey received                                                                                                                                                    | Transmit negative response NRC 13                      |
|  |     | SubFunction: yy == xx+1                                                                                                                                                            |                                                        |
|  |     | Message length NOK                                                                                                                                                                 |                                                        |
|  |     | SecurityAccess request results in a general negative response code (e.g. minimum length, SubFunction supported) according to the general negative response handling (see Annex A). | Transmit negative response code as defined in Annex A. |
|  | AND | SecurityAccess requestSeed received                                                                                                                                                | Transmit negative response NRC 13                      |
|  |     | Message Length NOK                                                                                                                                                                 |                                                        |

Table 4-81 SecurityAccess State transition

When the Delay Timer expires, the value must be initialized to 0 only for Att\_Cnt(X) that started the Delay Timer.

Example)

| Sequence | Att_Cnt(ASK) | Att_Cnt(OEUK) | DelayTimer               | Action       |
|----------|--------------|---------------|--------------------------|--------------|
| 1        | 0            | 0             | -                        | -            |
| 2        | 1            | 0             | -                        | ASK key NOK  |
| 3        | 2            | 0             | -                        | ASK key NOK  |
| 4        | 2            | 1             | -                        | OEUK key NOK |
| 5        | 2            | 2             | -                        | OEUK key NOK |
| 6        | 3            | 2             | Start 180sec Delay_Timer | ASK key NOK  |
| 7        | 0            | 2             | Timer Expired            | -            |
| 8        | 0            | 3             | Start 180sec Delay_Timer | OEUK key NOK |
| 9        | 0            | 0             | Timer Expired            |              |

Table 4-82 Att\_Cnt(X) Example

DelayTimer uses Single Time and shares DelayTimer at all levels.

That is, in the case of the controller applying the ASK and the OEUK, the timer of the ASK and the timer of the OEUK are the same, and if the DelayTimer is operated by a specific security level, it is judged that the timer is also operating at another level. Therefore, if there is a request for seed when the DelayTimer does not expire, Respond with NRC 37 regardless of the security level unlock status.

NOTE: Since Att\_Cnt is managed separately, only Att\_Cnt of the security level that started DelayTimer is initialized to 0 when DelayTimer expires.

Boot SW (Programming session) and Application SW can be allowed to manage Att\_Cnt(X) values separately from Boot SW and Application if they are separated by separate SW. However, it is strongly recommended that Att\_Cnt(X) values be shared between Boot SW and Application.

If a server supported security, but is already unlocked when a SecurityAccess 'requestSeed' message is received, that server shall respond with a SecurityAccess 'requestSeed' positive response message service with a seed value equal to zero (0). The client shall use this method to determine if a server is locked by checking for a non-zero seed.

The 'requestSeed' subfunction parameter value shall always be an odd number and the corresponding

'sendKey' subfunction parameter value for the same security level shall equal the 'requestSeed' sub-function parameter value plus one.

### 4.3.1 Request message

| Data Byte                                                                                                                                                                                                                                                                                                                                                              | Parameter Name                                                                          | Cvt      | Hex Value                                | Mnemonic                     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|----------|------------------------------------------|------------------------------|
| #1d                                                                                                                                                                                                                                                                                                                                                                    | <b>SecurityAccess Request Service ID</b>                                                | <b>M</b> | <b>#27h</b>                              | <b>SA</b>                    |
| #2d                                                                                                                                                                                                                                                                                                                                                                    | Sub-function = [<br>securityAccessType =<br>requestSeedType#1<br>requestSeedType#2<br>] | M        | #XXh=<br>[<br>#01h/0Bh<br>#11h/#13h<br>] | LEV_<br>SAT_RSD1<br>SAT_RSD2 |
| ※ Type#1 : Normal Seed/Key Algorithm (size ≤ 5-byte)<br>If a ECU supports a specific state for flash(Unlock flash state), 0B hex can be used.<br>※ Type#2 : Advanced Seed/Key (ASK) Algorithm (size > 5-byte)<br>※ ASK is an improved version of Normal Seed/Key Algorithm. Type#1 can be replaced with Type#2 to improve Security.<br>※ Type#3 : OEUK Algorithm - 13h |                                                                                         |          |                                          |                              |

Table4-10. Request Message definition – sub-function = #01h / #0Bh / #11h / #13h

| Data Byte                                                                                                           | Parameter Name                                                              | Cvt      | Hex Value          | Mnemonic              |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------|--------------------|-----------------------|
| #1d                                                                                                                 | <b>SecurityAccess Request Service ID</b>                                    | <b>M</b> | <b>#27h</b>        | <b>SA</b>             |
| #2d                                                                                                                 | Sub-function = [<br>securityAccessType =<br>sendCertificateandrequestSeed ] | M        | #41h               | LEV_<br>SAT_SCR<br>SD |
| #3d<br>..<br>#nd                                                                                                    | securityAccessDataRecord[] = [<br>Certificate<br>]                          | M        | #XXh<br>..<br>#XXh | SECACCD<br>R          |
| ※ In case of application of Secure Access function in CGW, sub-function #41h must be used.<br>(refer to ES95489-01) |                                                                             |          |                    |                       |

Table4-11. Request Message definition – sub-function = #41h

| Data Byte                                                                                                                                                                                                                                                                                                               | Parameter Name                                             | Cvt          | Hex Value                                 | Mnemonic                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|--------------|-------------------------------------------|--------------------------------|
| #1d                                                                                                                                                                                                                                                                                                                     | <b>SecurityAccess Request Service ID</b>                   | <b>M</b>     | <b>#27h</b>                               | <b>SA</b>                      |
| #2d                                                                                                                                                                                                                                                                                                                     | Sub-function = [<br>securityAccessType = sendKey<br>]      | M            | #XXh=[<br>#02h/0Ch<br>#12h/#14h<br>#42h ] | LEV_<br>SAT_SK                 |
| #3d<br>..<br>#nd                                                                                                                                                                                                                                                                                                        | sendKey[] = [<br>key (high byte)<br>..<br>key (low byte) ] | M<br>..<br>U | #XXh<br>..<br>#XXh                        | SEKEY_<br>KEYHB<br>..<br>KEYLB |
| The data size of key from the client if the securityAccessType parameter(=02/0C hex) shall grater than 2-byte / securityAccessType parameter(=12 hex) shall be 8-byte / securityAccessType parameter(= 14/42 hex) shall be 256-byte.<br>※ 256-byte Key is the value of "PKCS#1 Signature" by "the tester's private key" |                                                            |              |                                           |                                |

Table4-12. Request Message definition – sub-function = #02h / #0Ch / #12h / #14h / #42h

### 4.3.2 Positive response message

| Data Byte | Parameter Name | Cvt | Hex Value | Mnemonic |
|-----------|----------------|-----|-----------|----------|
|-----------|----------------|-----|-----------|----------|

| #1d | SecurityAccess Response Service ID | M  | #67h | SAPR     |
|-----|------------------------------------|----|------|----------|
| #2d | securityAccessType                 | M  | #XXh | SAT_     |
| #3d | securitySeed[] = [                 | C  | #XXh | SECSEED_ |
| ..  | seed (high byte)                   | .. | ..   | SEEDHB   |
| #6d | seed (low byte) ]                  | C  | #XXh | SEEDLB   |

C : The presence of this parameter depends on the securityAccessType parameter. It is mandatory to be present if the securityAccessType parameter(=01/0B/11/13/41 hex) indicates that the client wants to retrieve the seed from the server. The data size of seed from the server if the securityAccessType parameter (=01/0B hex) shall greater than 2-byte. The data size of seed from the server if the securityAccessType parameter(=11/13/41 hex) shall be 8-byte.

Table4-13. Positive Response Message definition

### 4.3.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                 |
|-----|---------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                                                                          |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                  |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the request Security Access are not met.                                                            |
| 24  | requestSequenceError                  | M   | Send if the 'sendKey' sub-function is received without first receiving a 'requestSeed' request message.                                             |
| 31  | requestOutOfRange                     | M   | This code shall be sent if the user optional securityAccessDataRecord contains invalid data.                                                        |
| 35  | invalidKey                            | M   | Send if an expected 'sendKey' sub-function value is received and the value of the key does not match the server's internally stored/calculated key. |
| 36  | exceededNumberOfAttempts              | M   | Send if the delay timer is active due to exceeding the maximum number of allowed false access attempts.                                             |
| 37  | requiredTimeDelayNotExpired           | M   | Send if the delay timer is active and request is transmitted.                                                                                       |

Table4-14. Supported Negative Response Codes

### 4.3.4 Message Flow Example SecurityAccess – server is in a “locked” state

In the following example, it is assumed:

- the node has a physical request CANId of \$7E0
- the node has a physical response CANId of \$7E8
- AABB hex is a valid seed value for the target node
- CCDD hex is a valid key value for the target node

#### 4.3.4.1 SecurityAccess – requestSeed

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 02 27 01 55 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 04 67 01 AA BB AA AA AA | Single Frame |

Table 4-15 SecurityAccess(requestSeed) Message Example

#### 4.3.4.2 SecurityAccess – sendKey



# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)  
ES95486-02

페이지  
(SHT/SHTS) 58/244

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 04 27 02 CC DD 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 02 67 02 AA AA AA AA AA | Single Frame |

Table 4-16 SecurityAccess(sendKey) Message Example

### 4.3.5 Message Flow Example SecurityAccess – server is in a “unlocked” state

#### 4.3.5.1 SecurityAccess – requestSeed

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 02 27 01 55 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 04 67 01 00 00 AA AA AA | Single Frame |

Table 4-17 SecurityAccess(requestSeed) Message Example

172.23.51.206 / 2025-10-29 09:41  
주모베이스전자 / 김의진

### 4.4 CommunicationControl (28 hex) service

The purpose of this service is to switch on/off the transmission and/or the reception of certain messages of (a) server(s) (e.g. application communication messages).

Note) This service doesn't affect transmission and/or the reception of message routing for the gateway ECU.

This service doesn't affect transmission and/or the reception of the diagnostic message.

After Communication Control (28hex) with disableRxAndTx (03hex) request, if the server does not receive any request for S3<sub>server</sub>, it returns to Default Session and resumes Normal message transmission and "Communication Time Out" monitoring within 500ms.

CAN Fault Diagnosis corresponding server(s) is conducted after communication resumption, because communication resumption time is different depending on the server(s). (Communication is resumed within 500ms, and CAN Fault Diagnosis is conducted after 500ms.)

According to the ES95480-00 (Initialization Time Parameter), after 'Communication Control (28hex) with enableRxAndTx (00hex)' service, the server(s) shall restart its communication and "Communication Time Out" function within 500msec.

#### 4.4.1 Request message

| Data Byte | Parameter Name                                 | Cvt | Hex Value | Mnemonic   |
|-----------|------------------------------------------------|-----|-----------|------------|
| #1d       | <b>CommunicationControl Request Service ID</b> | M   | #28h      | CC         |
| #2d       | Sub-function = [ controlType ]                 | M   | #XXh      | LEV_CTRLTP |
| #3d       | communicationType                              | M   | #XXh      | CTP        |

Communication Type : See annex B.1 for the coding of the communicationType data parameter in detail.

Table4-18. Request Message definition

| Hex   | Description                                                                                                                                                                  | Cvt | Mnemonic |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00    | <b>enableRxAndTx</b><br>This value indicates that the reception and transmission of messages shall be enabled for the specified communicationType.                           | M   | ERXTX    |
| 01    | <b>enableRxAndDisableTx</b><br>This value indicates that the reception of messages shall be enabled and transmission shall be disabled for the specified communication Type. | U   | ERXDTX   |
| 02    | <b>diableRxAndEnableTx</b><br>This value indicates that the reception of messages shall be disabled and transmission shall be enabled for the specified communication Type.  | U   | DRXETX   |
| 03    | <b>disableRxAndTx</b><br>This value indicates that the reception and transmission of messages shall be disabled for the specified communicationType.                         | M   | DRXTX    |
| 04~7F | <b>Reserved</b>                                                                                                                                                              | M   |          |

Table4-19. Request message sub-function (controlType) parameter definition

| HEX | Parameter Name                                                                                                                                                            | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 01  | <b>normalCommunicationMessages</b><br>This value references all application-related communication (interapplication signal exchange between multiple in-vehicle servers). | M   | NCM      |
| 02  | <b>networkManagementCommunicationMessages</b><br>This value references all network management related communication.                                                      | M   | NMCM     |

|    |                                                                                                                                                                      |   |         |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
| 03 | <b>networkManagementCommunicationMessages and normalCommunicationMessages</b><br>This value references all network management and application-related communication. | M | NMCMNCM |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|

Table4-20. Request message communication Type definition

### 4.4.2 Positive response message

| Data Byte | Parameter Name                                  | Cvt | Hex Value | Mnemonic    |
|-----------|-------------------------------------------------|-----|-----------|-------------|
| #1d       | <b>CommunicationControl Response Service ID</b> | M   | #68h      | <b>CCPR</b> |
| #2d       | controlType                                     | M   | #XXh      | CTRLTP      |

Table4-21. Positive Response Message definition

### 4.4.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                  |
|-----|---------------------------------------|-----|------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                           |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                   |
| 22  | conditionNotCorrect                   | M   | Used when the server is condition that cannot disable/enable the requested communication type.       |
| 31  | requestOutOfRange                     | M   | The server shall use this response code, if it detects and error in the communicationType parameter. |

Table4-22. Supported Negative Response Codes

### 4.4.4 Message Flow Example

#### 4.4.4.1 Communication Control – EnableRxAndTx

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7DF    | Functional Request               | 8   | 02 28 00 03 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 02 68 00 AA AA AA AA AA | Single Frame |
| 7E9    | Response                         | 8   | 02 68 00 AA AA AA AA AA | Single Frame |

Table 4-23 CommunicationControl Message Example

#### 4.4.4.2 Communication Control – DisableRxAndTx

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7DF    | Functional Request               | 8   | 03 28 03 03 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 02 68 03 AA AA AA AA AA | Single Frame |
| 7E9    | Response                         | 8   | 02 68 03 AA AA AA AA AA | Single Frame |

Table 4-24 CommunicationControl Message Example

## 4.5 Authentication (29 hex) service

HMC have conventional security methods with SecurityAccess(\$27), Nomal seed key(shall not be used), Advanced Seed key(ASK), and C-SAC(Certificate based security access control). Those security methods utilize securityAccessType of Security access(\$27) service. Conventional method might be replaced by Authentication(\$29) and it give some advantage of removing exception of SecurityAccess(\$27) for HMC specific. The reason why CCU / ICU / ESU SecurityAccess login shall not be initialized whenever DSC(Diatnositic Session Control) is called is that C-SAC shall not be initilize by functionally addressed DSC causes initialization of the SecurityAccess login. If login is initialized whenever DSC is called, it is hard to route diagnostic services to the target ECU because functionally addressed DSC can cause initialization. So this exception is defined for the Central Gateway. Authentication would be an option to overcome this exception.

To send diagnostic services to the target ECU, Client shall optain right through SecurityAccess(C-SAC). This is conventional method and it shall be changed to APCE method of the Authentication(\$29) service.

The purpose of this service is to provide a means for the client to prove its identity, allowing it to access data and/or diagnostic services, which have restricted access for, for example security, emissions, or safety reasons. Diagnostic services for downloading/uploading routines or data into a server and reading specific memory locations from a server are situations where authentication may be required. Improper routines or data downloaded into a server could potentially damage the electronics or other vehicle components or risk the vehicle's compliance to emission, safety, or security standards. On the other hand, data security might be violated when retrieving data from a server.

NOTE 1 The generation, distribution and storage of cryptographic material is out of scope for this document.

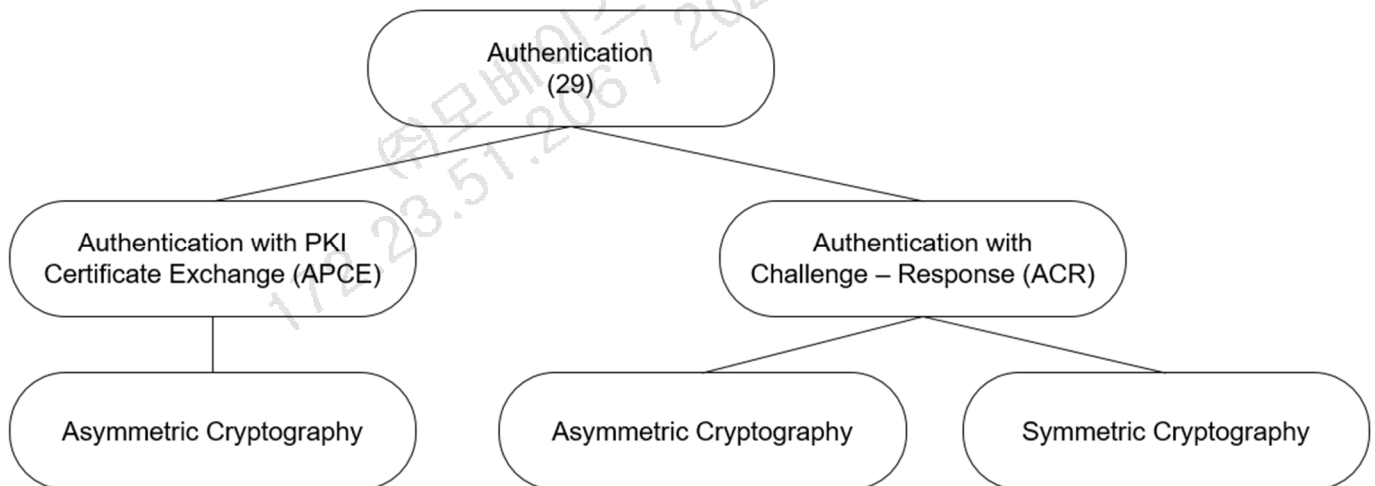


Figure 4-8 Overview over the Authentication service security concepts

NOTE 2 Figure 4-8 shows only the major options of authentication.

To identify the supported concept in the server, the client can send this service with the SubFunction parameter 'authenticationConfiguration'.

NOTE 3 Bidirectional authentication is not a scope for the HMC specification.

NOTE 4 Authentication with Challenge - Response (ACR) is not a scope for the HMC specification

### 4.5.1 Authentication with PKI Certificate Exchange (APCE)

The Authentication service is used for authentication, deAuthentication and explicit certificate transmission. The 'authenticationTask' SubFunction parameter identifies the corresponding task to process.

“deAuthenticate”, this SubFunction parameter actively ends the authenticated state.

“verifyCertificateUnidirectional”, this SubFunction parameter initiates the unidirectional Authentication procedure to only authenticate the client against the server.

“verifyCertificateBidirectional”, this SubFunction parameter initiates the bidirectional Authentication procedure to authenticate the client against the server and the server against the client.

“proofOfOwnership”, this SubFunction parameter is used to transmit the proof of ownership data to the client.

“transmitCertificate”, this SubFunction parameter transfers a certificate independently or subsequent to a previous authentication.

**Prerequisites:**

Different sets of certificates (and corresponding private keys) shall be available at both sides (client and server):

In case of unidirectional authentication, the client needs a certificate client with its private key, which allows the client to identify itself as a legitimate client. Depending on the trust model of the public key infrastructure (PKI), the server may need a certificate of the certificate authority (CA) which issued and signed the certificate client.

In case of bidirectional authentication, the client needs a certificate client with its private key, which allows the client to identify itself as a legitimate client. Additionally, the server also needs a certificate server with its private key, which allows the server to identify itself as a legitimate server. Depending on the trust model of the public key infrastructure (PKI), the client and the server may need a certificate of the certificate authority (CA) which issued and signed the certificate client and certificate server.

The authentication process triggered by “verifyCertificateUnidirectional” or “verifyCertificateBidirectional” takes place in two steps as depicted in Figure 4-9.

**4.5.1.1 Unidirectional authentication:**

- Client creates challenge client if used in proof of ownership client (1).

It is recommended to create the challenge based on ISO/IEC 9798-3 (unilateral, two pass authentication) or as a security-related equivalent challenge to the one as per ISO/IEC 9798-3.

- Client sends its certificate client and, if generated in (1), its challenge client via SubFunction verifyCertificateUnidirectional (2).

- Server verifies the certificate client (3).

- Server creates a challenge server (4).

- If session key establishment based on the Ephemeral Diffie-Hellman key agreement is indicated by the client in (2), then the server generates an ephemeral private/public key pair to later derive a session key for further secure communication (5).

NOTE 1 Diffie-Hellman key agreement is necessary if algorithms are used in the certificates which are only usable for signature calculation, but are not usable for a key agreement protocol.

- Server sends the challenge server and, if generated in (5), its ephemeral public key (7).

- If session key establishment based on the Ephemeral Diffie-Hellman key agreement is indicated by the

client in (2), then the client also generates an ephemeral private/public key pair to later derive a session key for further secure communication (9).

NOTE 2 Diffie-Hellman key agreement is necessary if algorithms are used in the certificates which are only usable for signature calculation, but are not usable for a key agreement protocol.

– Client calculates proof of ownership client by building an appropriate authentication token whose content to be signed include at least (parts of) the challenge server and, if generated in (9), its ephemeral public key (10).

It is recommended to build the authentication token based on ISO/IEC 9798-3 (unilateral, two pass authentication) or as a security-related equivalent authentication token to the one as per ISO/IEC 9798-3.

– Client sends the proof of ownership client and, if generated in (9), its ephemeral public key via SubFunction proofOfOwnership (11).

– Server verifies the proof of ownership client with the public key from the received certificate client (12).

– If session key establishment is indicated by the client in (2), then the server creates or derives and enables the session key(s) for further secure communication and sets up the session key info (13).

– The server grants access to diagnostic objects according to access rights (14).

– The server responds that the authentication was successful and, if present from (13), sends the session key info (15).

– If session key establishment is indicated by the client in (2), then the client extracts the session key(s) from the session key Info or derives the session key(s) for further secure communication (16).

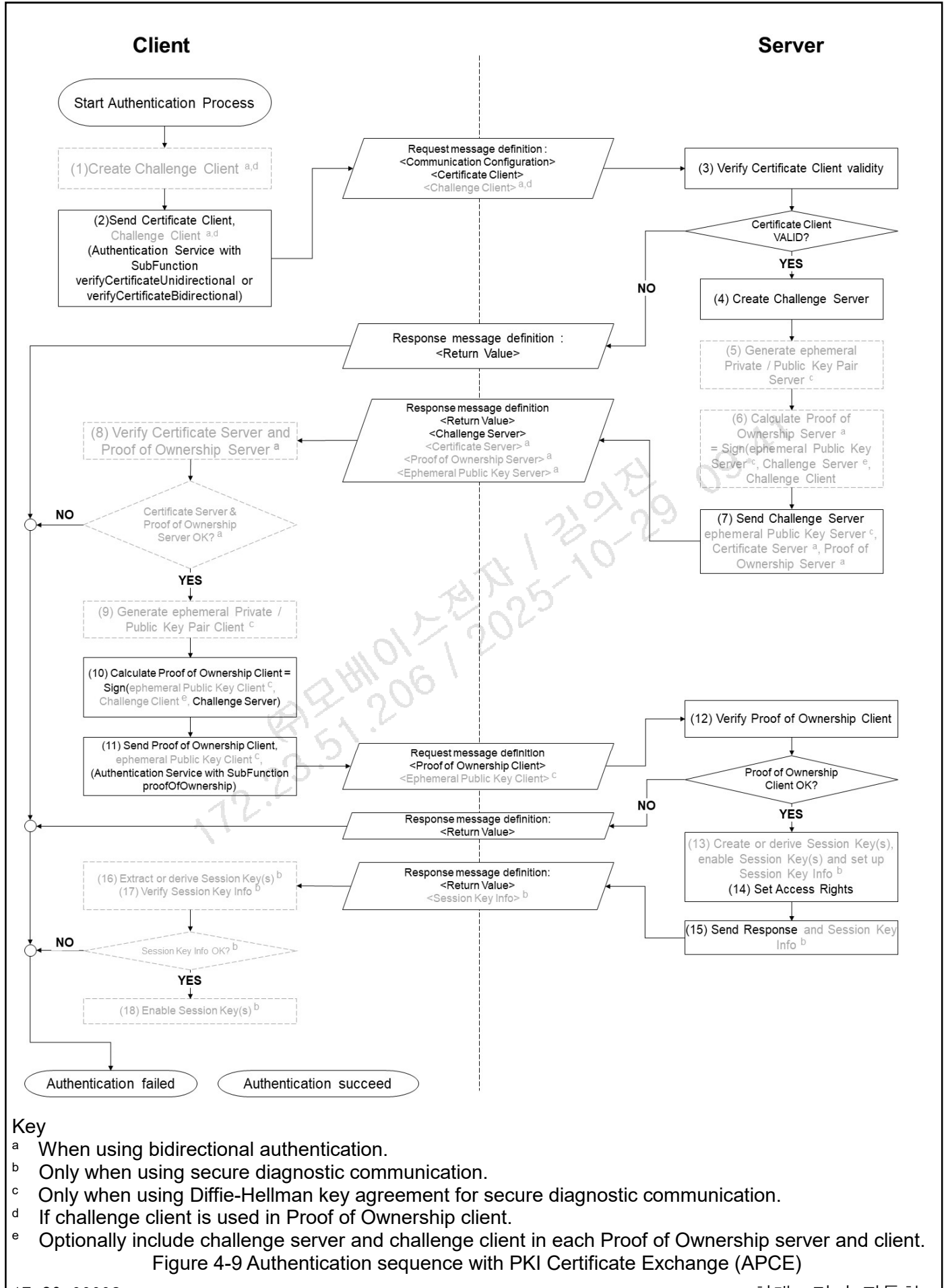
– If session key establishment is indicated by the client in (2), then the client verifies the session key info using the session key(s) (17).

NOTE 3 Step (17) ensures that the session key establishment is complete and valid.

– If session key establishment is indicated by the client in (2), then the client enables the session key(s) for further secure diagnostic communication (18).

NOTE 4 When using unidirectional authentication, the server does not authenticate against the client. Thus, the client cannot be sure that it is communicating with the correct server.

.



#### 4.5.2 Common requirements

The authentication is designed to secure applicable diagnostic sessions/functions/services. Therefore the following sequence of services shall be required:

- Authentication service,
- Any diagnostic service that is secured or restricted by authentication.

For authentication there is no direct relationship to diagnostic session or security level. Once authenticated, the server shall be in an authenticated state which it only shall leave if a security timeout occurs, a mileage offset limit is reached or the authenticated state is intentionally left by the “deAuthenticate” request. Applicable diagnostic services assigned to the corresponding authentication settings shall be accessible as long as the client is authenticated.

There are several possibilities to leave the authenticated state:

- Explicitly: the authentication request shall be called with the SubFunction parameter “deAuthenticate”.
- Implicitly using a timeout as fallback: a timer shall be installed. The timer shall start when transitioning to the authenticated state. If the timer times out, then the authenticated state shall be ended. The timeout can be set individually or can be bound to an existing timing parameter (e.g. session layer timing parameter definitions). At least every request message of the same diagnostic protocol received by the transport protocol layer shall keep the authenticated state active and reset the timeout period.

NOTE 1 The timer value is vehicle manufacturer specific.

It is recommended that the usage of the timeout is based on a reliable internal time.

- Implicitly using a mileage offset limit as fallback: a mileage monitoring shall be installed. The mileage monitoring shall start when transitioning to the authenticated state. If the mileage limit is reached, then the authenticated state shall be ended.

NOTE 2 The mileage offset limit is vehicle manufacturer specific.

It is recommended that the usage of the mileage monitoring is based on reliable mileage information.

The explicit exit condition shall be mandatory, the implementation of at least one of the implicit exit conditions shall be mandatory, both are optionally possible.

If a server supports authentication and is already in an authenticated state when an authentication request message is received from the same client, that server shall stay in the authenticated state until the authentication is again successfully completed. The server shall change its state to the new received authentication information.

Unsuccessful attempts to access an authenticated state shall not prevent other diagnostic communication. An authenticated state shall be linked to a certain diagnostic channel. Multiple clients can be handled on multiple channels with different authentication settings. The basis for this would be a multi-user server system.

Servers which provide security shall support the NRC 34 “authenticationRequired”, if a secured service is requested while the server is in a non-authenticated state.

As an option, session keys can be established and used to further secure communication between client and server.

This can be achieved, for example, using one of the following approaches:

- by creation of the session key on the server side and an encrypted transmission to the client, or
- by using an asymmetric key agreement protocol, or
- by derivation of existing pre-shared keys on both sides (in case of symmetric cryptography).

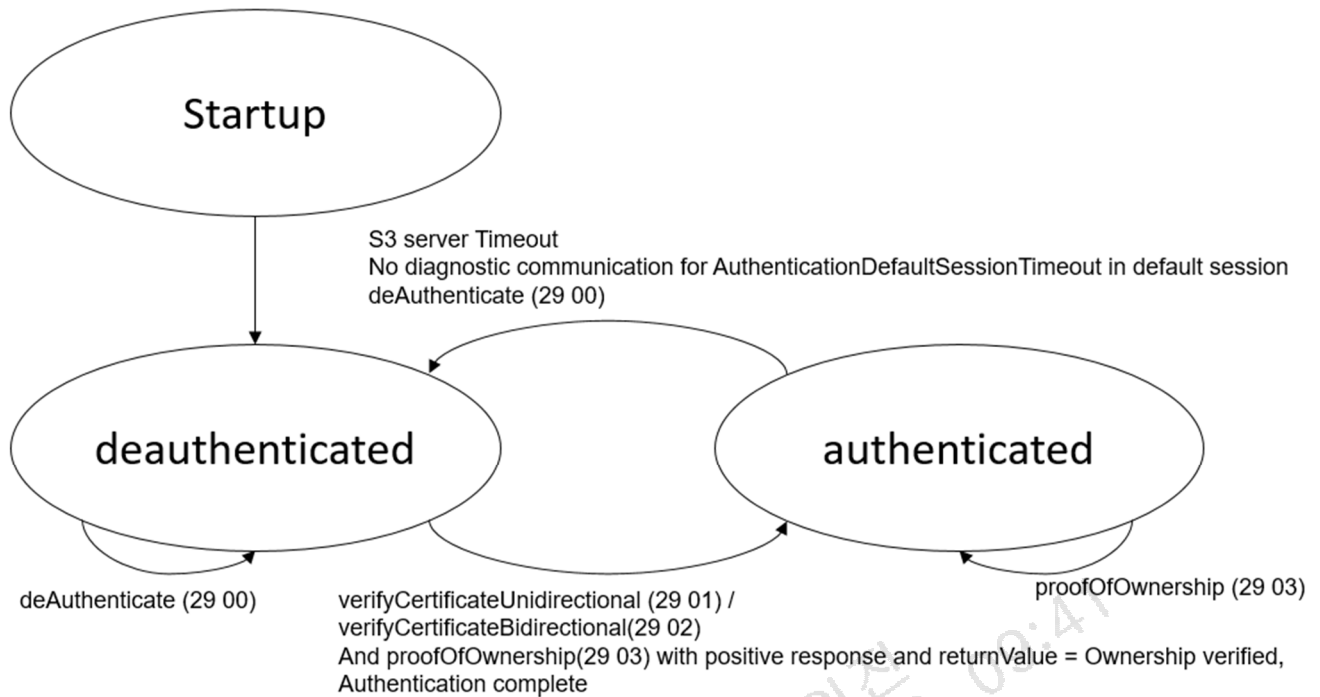


Figure 4-10 Authentication state

| Timing Parameter                    | Description                                                                                                                                                         | Type               | Recommended Value | Timeout |
|-------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|-------------------|---------|
| AuthenticationDefaultSessionTimeout | Time for the server to keep a authenticated state session in defaultSession while not receiving valid diagnostic request message.<br>The tolerance is -0ms, +200ms. | Timer reload value | N/A               | 10000ms |

Table4-60. Timing parameter definitions - AuthenticationDefaultSessionTimeout

| Timer parameter              | Action | Description                                                                                                                                                |
|------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AuthenticationDefaultSession | Start  | ECU is in default session with authenticated state.                                                                                                        |
|                              | Reset  | valid diagnostic request was received within 10000ms(-0ms/+200ms) after the last Server_T.data confirmed.<br>Persisted authentication mode is deactivated. |
|                              | Stop   | ECU is in non-default session.<br>ECU is in deauthenticated state.                                                                                         |

Table4-61. Timer of the AuthenticationDefaultSession

### 4.5.3 Request message

| Data Byte | Parameter Name                                        | Cvt | Hex Value | Mnemonic  |
|-----------|-------------------------------------------------------|-----|-----------|-----------|
| #1d       | Authentication Request SID                            | M   | #29h      | ARS       |
| #2d       | Sub-function = [ authenticationTask = deAuthenticate] | M   | #00h      | LEV_AT_DA |

Table4-62. Request message - sub-function = #00h

| Data Byte | Parameter Name             | Cvt | Hex Value | Mnemonic |
|-----------|----------------------------|-----|-----------|----------|
| #1d       | Authentication Request SID | M   | #29h      | ARS      |

|                       |                                                                           |   |                     |                |
|-----------------------|---------------------------------------------------------------------------|---|---------------------|----------------|
| #2d                   | Sub-function = [ authenticationTask =<br>verifyCertificateUnidirectional] | M | #01h                | LEV_AT_V<br>CU |
| #3d                   | communicationConfiguration[] =[<br>noSecuredCommunication]                | M | #00h                | COCO           |
| #4d<br>#5d            | lengthOfCertificateClient[] = [<br>byte#1(MSB)<br>byte#2(LSB)]            | M | #XXh<br>#XXh        | LOCECL         |
| #6<br>...<br>#m+5     | certificateClient[] = [<br>byte#1<br>...<br>byte#m]                       | M | #XXh<br>...<br>#XXh | CECL           |
| #m+6<br>#m+7          | lengthOfChallengeClient[] = [<br>byte#1(MSB)<br>byte#2(LSB)]              | M | #00h<br>#00h        | LOCHCL         |
| #m+8<br>...<br>#n+m+7 | challengeClient[] = [<br>byte#1<br>...<br>byte#n]                         | C | #XXh<br>...<br>#XXh | CHCL           |

C : The presence of this parameter depends on the length information parameter of the lengthOfChallengeClient. If lengthOfChallengeClient is equal to 0x0000 then the parameter challengeClient will not be transmitted.

Table4-63. Request message - sub-function = #01h

| Data Byte             | Parameter Name                                                        | Cvt      | Hex Value           | Mnemonic        |
|-----------------------|-----------------------------------------------------------------------|----------|---------------------|-----------------|
| #1d                   | <b>Authentication Request SID</b>                                     | <b>M</b> | <b>#29h</b>         | <b>ARS</b>      |
| #2d                   | Sub-function = [ authenticationTask =<br>proofOfOwnership]            | M        | #03h                | LEV_AT_P<br>OWN |
| #3d<br>#4d            | lengthOfProofOfOwnershipClient[] = [<br>byte#1(MSB)<br>byte#2(LSB)]   | M        | #XXh<br>#XXh        | LPOWNCL         |
| #5<br>...<br>#m+4     | certificateClient[] = [<br>byte#1<br>...<br>byte#m]                   | M        | #XXh<br>...<br>#XXh | POWNCL          |
| #m+5<br>#m+6          | lengthOfEphemeralPublicKeyClient[] = [<br>byte#1(MSB)<br>byte#2(LSB)] | M        | #00h<br>#00h        | LOEPKCL         |
| #m+7<br>...<br>#n+m+6 | ephemeralPublicKeyClient[] = [<br>byte#1<br>...<br>byte#n]            | C        | #XXh<br>...<br>#XXh | EPKCL           |

C : The presence of this parameter depends on the length information parameter of the lengthOfEphemeralPublicKeyClient. If lengthOfEphemeralPublicKeyClient is equal to 0x0000 then the parameter ephemeralPublicKeyClient will not be transmitted.

Table4-64. Request message - sub-function = #03h

| Data Byte | Parameter Name                                                        | Cvt      | Hex Value   | Mnemonic      |
|-----------|-----------------------------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>Authentication Request SID</b>                                     | <b>M</b> | <b>#29h</b> | <b>ARS</b>    |
| #2d       | Sub-function = [ authenticationTask =<br>authenticationConfiguration] | M        | #08h        | LEV_AT_A<br>C |

Table4-65. Request message - sub-function = #08h

| Hex   | Description                                                                                                                                                | Cvt | Mnemonic |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00    | <b>deAuthenticate</b><br>Request to leave the authenticated state.                                                                                         | M   | DA       |
| 01    | <b>verifyCertificateUnidirectional</b><br>Initiate Authentication by verifying the Certificate                                                             | M   | VCU      |
| 02    | <b>verifyCertificateBidirectional</b><br>Initiate Authentication by verifying the Certificate and generating a Proof of Ownership from the server.         | U   | VCB      |
| 03    | <b>proofOfOwnership</b><br>Verify the Proof of Ownership from the client.                                                                                  | M   | POWN     |
| 04    | <b>transmitCertificate</b><br>Verify Certificate and extract information from Certificate to handle it according to its contents.                          | U   | TC       |
| 05    | <b>requestChallengeForAuthentication</b><br>Initiate the Authentication process by requesting server to output a challenge.                                | U   | RCFA     |
| 06    | <b>verifyProofOfOwnershipUnidirectional</b><br>Request server to verify the POWN for unidirectional authentication.                                        | U   | VPOWNU   |
| 07    | <b>verifyProofOfOwnershipBidirectional</b><br>Request server to verify the client side POWN and provide server-side POWN for bidirectional authentication. | U   | VPOWNB   |
| 08    | <b>authenticationConfiguration</b><br>Indicates the provided authentication configuration of the server.                                                   | M   | AC       |
| 09~7F | <b>Reserved</b>                                                                                                                                            | M   | RESRVD   |

Table4-66. Request message sub-function parameter definition

| Hex   | Description                                                                        | Cvt | Mnemonic |
|-------|------------------------------------------------------------------------------------|-----|----------|
| 00    | <b>noSecuredCommunication</b><br>No Secure communication for further communication | M   | NSC      |
| 01~FF | <b>Reserved</b>                                                                    | M   | RESRVD   |

Table4-67. Request message communicationConfiguration parameter definition

#### 4.5.4 Positive response message

| Data Byte | Parameter Name                                        | Cvt | Hex Value | Mnemonic      |
|-----------|-------------------------------------------------------|-----|-----------|---------------|
| #1d       | <b>Authentication Response SID</b>                    | M   | #69h      | <b>ARS</b>    |
| #2d       | Sub-function = [ authenticationTask = deAuthenticate] | M   | #00h      | LEV_AT_D<br>A |
| #3d       | returnValue [] = [authenticationReturnParameter]      | M   | #XXh      | RV            |

Table4-68. Positive Response Message definition - sub-function = #00h

| Data Byte | Parameter Name                                                            | Cvt | Hex Value | Mnemonic       |
|-----------|---------------------------------------------------------------------------|-----|-----------|----------------|
| #1d       | <b>Authentication Response SID</b>                                        | M   | #69h      | <b>ARS</b>     |
| #2d       | Sub-function = [ authenticationTask =<br>verifyCertificateUnidirectional] | M   | #01h      | LEV_AT_V<br>CU |
| #3d       | returnValue[] =[<br>authenticationReturnParameter]                        | M   | #XXh      | RV             |

|                       |                                                                       |   |                     |         |
|-----------------------|-----------------------------------------------------------------------|---|---------------------|---------|
| #4d<br>#5d            | lengthOfChallengeServer[] = [<br>byte#1(MSB)<br>byte#2(LSB)]          | M | #XXh<br>#XXh        | LOCHSE  |
| #6<br>...<br>#m+5     | challengeServer[] = [<br>byte#1<br>...<br>byte#m]                     | M | #XXh<br>...<br>#XXh | CHSE    |
| #m+6<br>#m+7          | lengthOfEphemeralPublicKeyServer[] = [<br>byte#1(MSB)<br>byte#2(LSB)] | M | #00h<br>#00h        | LOEPKSE |
| #m+8<br>...<br>#n+m+7 | ephemeralPublicKeyServer[] = [<br>byte#1<br>...<br>byte#n]            | C | #XXh<br>...<br>#XXh | EPKSE   |

C : The presence of this parameter depends on the length information parameter of the lengthOfEphemeralPublicKeyServer. If lengthOfEphemeralPublicKeyServer is equal to 0x0000 then the parameter ephemeralPublicKeyServer shall not be transmitted.

Table4-69. Positive Response Message definition - sub-function = #01h

| Data Byte         | Parameter Name                                               | Cvt      | Hex Value           | Mnemonic        |
|-------------------|--------------------------------------------------------------|----------|---------------------|-----------------|
| #1d               | <b>Authentication Response SID</b>                           | <b>M</b> | <b>#69h</b>         | <b>ARS</b>      |
| #2d               | Sub-function = [ authenticationTask =<br>proofOfOwnership]   | M        | #03h                | LEV_AT_P<br>OWN |
| #3d               | returnValue [] =[<br>authenticationReturnParameter]          | M        | #XXh                | RV              |
| #4d<br>#5d        | lengthOfSessionKeyInfo [] = [<br>byte#1(MSB)<br>byte#2(LSB)] | M        | #00h<br>#00h        | LOSKI           |
| #6<br>...<br>#m+5 | sessionKeyInfo [] = [<br>byte#1<br>...<br>byte#m]            | C        | #XXh<br>...<br>#XXh | SKI             |

C : The presence of this parameter depends on the length information parameter of the lengthOfSessionKeyInfo. If lengthOfSessionKeyInfo is equal to 0x0000 then the parameter sessionKeyInfo shall not be transmitted.

Table4-70. Positive Response Message definition - sub-function = #03h

| Data Byte | Parameter Name                                                        | Cvt      | Hex Value   | Mnemonic      |
|-----------|-----------------------------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>Authentication Response SID</b>                                    | <b>M</b> | <b>#69h</b> | <b>ARS</b>    |
| #2d       | Sub-function = [ authenticationTask =<br>authenticationConfiguration] | M        | #08h        | LEV_AT_A<br>C |
| #3d       | returnValue [] =[<br>authenticationReturnParameter]                   | M        | #XXh        | RV            |

Table4-71. Positive Response Message definition - sub-function = #08h

| Hex | Description                                                                                                                                               | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00  | <b>RequestAccepted</b><br>Request was successful.                                                                                                         | U   | RA       |
| 01  | <b>GeneralReject</b><br>Request was not successful                                                                                                        | U   | GR       |
| 02  | <b>AuthenticationConfiguration APCE</b><br>Indicates the provided authentication configuration as Authentication with<br>PKI Certificate Exchange (APCE). | M   | ACAPCE   |

|         |                                                                                                                                                                                                         |   |         |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
| 03      | <b>AuthenticationConfiguration ACR with asymmetric cryptography</b><br>Indicates the provided authentication configuration as Authentication with Challenge-Response (ACR) and asymmetric cryptography. | U | ACACRAC |
| 04      | <b>AuthenticationConfiguration ACR with symmetric cryptography</b><br>Indicates the provided authentication configuration as Authentication with Challenge-Response(ACR) and symmetric cryptography.    | U | ACACRSC |
| 05 ~ 0F | <b>Reserved</b>                                                                                                                                                                                         | M | RESRVD  |
| 10      | <b>DeAuthentication successful</b><br>DeAuthentication was successful, server is protected again.                                                                                                       | M | DAS     |
| 11      | <b>CertificateVerified, OwnershipVerificationNecessary</b><br>Certificate could be verified in first step, second step is pending                                                                       | M | CVOVN   |
| 12      | <b>OwnershipVerified, AuthenticationComplete</b><br>Proof of Ownership could be verified, Authentication is complete.                                                                                   | M | OVAC    |
| 13      | <b>CertificateVerified</b><br>Certificate could be verified                                                                                                                                             | U | CV      |
| 14 ~ 9F | <b>Reserved</b>                                                                                                                                                                                         | M | RESRVD  |
| A0 ~ CF | <b>HMCSpecific</b><br>This range of values is reserved for vehicle manufacturer specific use.                                                                                                           | U | HMCS    |
| D0 ~ FE | <b>SystemSupplierSpecific</b><br>This range can be used while development time and shall not be used after production.                                                                                  | U | SSS     |
| 09 ~ 7F | <b>Reserved</b>                                                                                                                                                                                         | M | RESRVD  |

Table4-72. Positive Response Message - authenticationReturnParameter parameter definition

## 4.5.5 Supported negative response codes

| Hex | NegativeResponseCode                                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----|-------------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported                               | M   | Send if the sub-function parameter in the request message is not supported                                                                                                                                                                                                                                                                                                                                                                                                             |
| 13  | incorrectMessageLengthOrInvalidFormat                 | M   | The length of the message is wrong                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 22  | conditionsNotCorrect                                  | M   | This NRC shall be returned if the criteria for the request Authentication are not met.                                                                                                                                                                                                                                                                                                                                                                                                 |
| 24  | requestSequenceError                                  | M   | This NRC shall be returned in case of the following scenarios:<br>- If the 'proofOfOwnership' SubFunction is received without first successfully processing either a 'verifyCertificateUnidirectional' or 'verifyCertificateBidirectional' request message, or<br>- If either the 'verifyProofOfOwnershipUnidirectional' or 'verifyProofOfOwnershipBidirectional' SubFunction is received without first successfully processing a 'requestChallengeForAuthentication' request message. |
| 50  | Certificate verification failed – Invalid Time Period | M   | Date and time of the server does not match the validity period of the Certificate.                                                                                                                                                                                                                                                                                                                                                                                                     |

|    |                                                                 |   |                                                                                                                                                                                               |
|----|-----------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 51 | Certificate verification failed – invalid Signature             | M | Signature of the Certificate could not be verified.                                                                                                                                           |
| 56 | Certificate verification failed – Invalid Scope                 | M | The scope of the Certificate does not match the contents of the server.                                                                                                                       |
| 57 | Certificate verification failed – Invalid Certificate (revoked) | M | Certificate received from client is invalid, because the server has revoked access for some reason.                                                                                           |
| 58 | Ownership verification failed                                   | M | Delivered Ownership does not match the provided challenge or could not verified with the own private key                                                                                      |
| 59 | Challenge calculation failed                                    | M | The challenge could not be calculated on the server side.                                                                                                                                     |
| 5A | Setting Access Rights failed                                    | M | The server could not set the access rights.                                                                                                                                                   |
| 5C | Configuration data usage failed                                 | M | Configuration data usage failed.                                                                                                                                                              |
| 5D | DeAuthentication failed                                         | M | DeAuthentication was not successful, server could still be unprotected.                                                                                                                       |
| F0 | CRLintegrityFailed                                              | M | CRL integrity check (Verification of CRL Signature) is failed. The hash value of SHA_256 of Certificate authority's public key and the CRL Issuer Public Key Identifier value are not matched |
| F1 | CRLvalidityPeriodFailed                                         | M | CRL is expired                                                                                                                                                                                |
| F2 | RoleandRightofCertificateDenied                                 | M | Role and Right of Certificate verification failed. The request service or the target from the external diagnostic tool are not matched with Certificate Holder<br>Reference : Authorization   |

Table4-73. Supported Negative Response Codes

### 4.5.6 Message Flow Example – Unidirectional Authentication with PKI Certificate Exchange without session key establishment

In the following example, it is assumed:

NOTE : Assumption might be different from ES95489-01. Please refer to ES95489-01

- Step #1 : authentication configuration of the server is optional step
- Step #2 : No secure communication for further communication – communicationConfiguration : 00h
- Step #2 : Length of Certificate client : 600dec(0x258) [Certificate] + 501dec(0x1F5) [CRL]: 1101dec (0x44D)
- Step #2 : Parameter returnValue in the positive response indicating Certificate verified, Ownership verification necessary
- Length of the challenge Client : 0dec (0x0000)
- Length of Proof of Ownership : 256dec (0x100)

#### #Step #1 : RequestAuthentication Configuration

| Data byte | Description                                      | Byte Value | Mnemonic  |
|-----------|--------------------------------------------------|------------|-----------|
| #1        | Authentication Request SID                       | 29h        | ARS       |
| #2        | authenticationTask = authenticationConfiguration | 08h        | LEV_AT_AC |

Table4-74. Authentication Configuration request message flow example – step #1

| Data byte | Description                                      | Byte Value | Mnemonic  |
|-----------|--------------------------------------------------|------------|-----------|
| #1        | Authentication Request SID                       | 69h        | ARS       |
| #2        | authenticationTask = authenticationConfiguration | 08h        | LEV_AT_AC |
| #3        | returnValue = AuthenticationConfiguration APCE   | 02h        | RV_ACAPCE |

Table4-75. Authentication Configuration positive response message flow example – step #1

### #Step #2 : Send Certificate Client

| Data byte | Description                                          | Byte Value | Mnemonic   |
|-----------|------------------------------------------------------|------------|------------|
| #1        | Authentication Request SID                           | 29h        | ARS        |
| #2        | authenticationTask = verifyCertificateUnidirectional | 01h        | LEV_AT_VCU |
| #3        | communicationConfiguration = no secure communication | 00h        | COCO       |
| #4        | lengthOfCertificateClient[Byte #1]                   | 04h        | LOCECL     |
| #5        | lengthOfCertificateClient[Byte #2]                   | 4Dh        |            |
| #6        | certificateClient[byte#1]                            | XXh        | CECL       |
| ...       | ...                                                  | ...        |            |
| #1106     | certificateClient[byte#1101]                         | XXh        |            |
| #1107     | lengthOfChallengeClient [ byte#1 ]                   | 00h        | LOCHCL     |
| #1108     | lengthOfChallengeClient [ byte#2 ]                   | 00h        |            |

Table4-76. Authentication request message flow example – step #2

| Data byte | Description                                                | Byte Value | Mnemonic   |
|-----------|------------------------------------------------------------|------------|------------|
| #1        | Authentication Request SID                                 | 69h        | ARS        |
| #2        | authenticationTask = verifyCertificateUnidirectional       | 01h        | LEV_AT_VCU |
| #3        | returnValue = Certificate verified, Ownership verification | 11h        | RV_CVOVN   |
| #4        | lengthOfChallengeServer[Byte #1]                           | 00h        | LOCHSE     |
| #5        | lengthOfChallengeServer [Byte #2]                          | 08h        |            |
| #6        | challengeServer[byte#1]                                    | XXh        | CHSE       |
| ...       | ...                                                        | ...        |            |
| #13       | challengeServer [byte#8]                                   | XXh        |            |
| #14       | lengthOfEphemeralPublicKeyServer [ byte#1 ]                | 00h        | LOEPKSE    |
| #15       | lengthOfEphemeralPublicKeyServer [ byte#2 ]                | 00h        |            |

Table4-77. Authentication positive response message flow example – step #2

### #Step #3 : Validate the Proof of Ownership

| Data byte | Description                                 | Byte Value | Mnemonic    |
|-----------|---------------------------------------------|------------|-------------|
| #1        | Authentication Request SID                  | 29h        | ARS         |
| #2        | authenticationTask = proofOfOwnership       | 03h        | LEV_AT_POWN |
| #3        | lengthOfProofOfOwnershipClient[Byte #1]     | 01h        | LPOWNCL     |
| #4        | lengthOfProofOfOwnershipClient [Byte #2]    | 00h        |             |
| #5        | proofOfOwnershipClient[byte#1]              | XXh        | POWNCL      |
| ...       | ...                                         | ...        |             |
| #260      | proofOfOwnershipClient [byte#256]           | XXh        |             |
| #261      | lengthOfEphemeralPublicKeyClient [ byte#1 ] | 00h        | LOEPKCL     |
| #262      | lengthOfEphemeralPublicKeyClient [ byte#2 ] | 00h        |             |

Table4-78. Authentication request message flow example – step #3

| Data byte | Description                                                 | Byte Value | Mnemonic    |
|-----------|-------------------------------------------------------------|------------|-------------|
| #1        | Authentication Request SID                                  | 69h        | ARS         |
| #2        | authenticationTask = proofOfOwnership                       | 03h        | LEV_AT_POWN |
| #3        | returnValue = Certificate verified, Authentication complete | 12h        | RV_CVOVN    |
| #4        | lengthOfSessionKeyInfo[Byte #1]                             | 00h        | LOSKI       |
| #5        | lengthOfSessionKeyInfo [Byte #2]                            | 00h        |             |

Table4-79. Authentication positive response message flow example – step #3

## 4.6 TesterPresent (3E hex) service

This service is used to indicate to a server (or servers) that a client is still connected to the vehicle and that certain diagnostic services and/or communication that have been previously activated are remain active.

This service is used to keep one or multiple servers in a diagnostic session other than defaultSession.

This can either be done by transmitting the TesterPresent request message periodically or in case of the absence of other diagnostic services to prevent the server(s) from automatically returning to the defaultSession.

### 4.6.1 Request message

| Data Byte | Parameter Name                          | Cvt      | Hex Value    | Mnemonic  |
|-----------|-----------------------------------------|----------|--------------|-----------|
| #1d       | <b>TesterPresent Request Service ID</b> | <b>M</b> | <b>#3Eh</b>  | <b>TP</b> |
| #2d       | Sub-function = [ zeroSubFunction ]      | M        | #00h or #80h | LEV_ZSUBF |

Table4-28. Request Message

| Hex | Description                                                                                                                                                           | Cvt | Mnemonic |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00  | <b>zeroSubFunction</b><br>This parameter value is used to indicate that no sub-function value beside the suppressPosRspMsgIndicationBit is supported by this service. | M   | ZSUBF    |

Table4-29. Request message sub-function parameter definition

### 4.6.2 Positive response message

| Data Byte | Parameter Name                           | Cvt      | Hex Value   | Mnemonic    |
|-----------|------------------------------------------|----------|-------------|-------------|
| #1d       | <b>TesterPresent Response Service ID</b> | <b>M</b> | <b>#7Eh</b> | <b>TPPR</b> |
| #2d       | zeroSubFunction                          | M        | #00h        | ZSUBF       |

Table4-30. Positive Response Message definition

### 4.6.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                         |
|-----|---------------------------------------|-----|-----------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported. |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                          |

Table4-31. Supported Negative Response Codes

## 4.7 SecuredDataTransmission (84 hex) service

### 4.7.1 Security Sub-layer

The purpose of this service is to transmit data that is protected against attacks from third parties – which could endanger data security – according to ISO 15764.

The SecuredDataTransmission service is applicable if a client intends to use diagnostic services defined in this document in a secured mode. It may also be used to transmit external data, which conform to some other application protocol, in a secured mode between a client and a server. A secured mode in this context means that the data transmitted is protected by cryptographic methods.

This section briefly describes the security sub-layer as defined in ISO 15764 (see ISO 15764 for details).

Figure 4-3 illustrates the security sub-Layer as defined in ISO 15764. The security sub-layer has to be added in the server and client application for the purpose of performing diagnostic services in a secured mode.

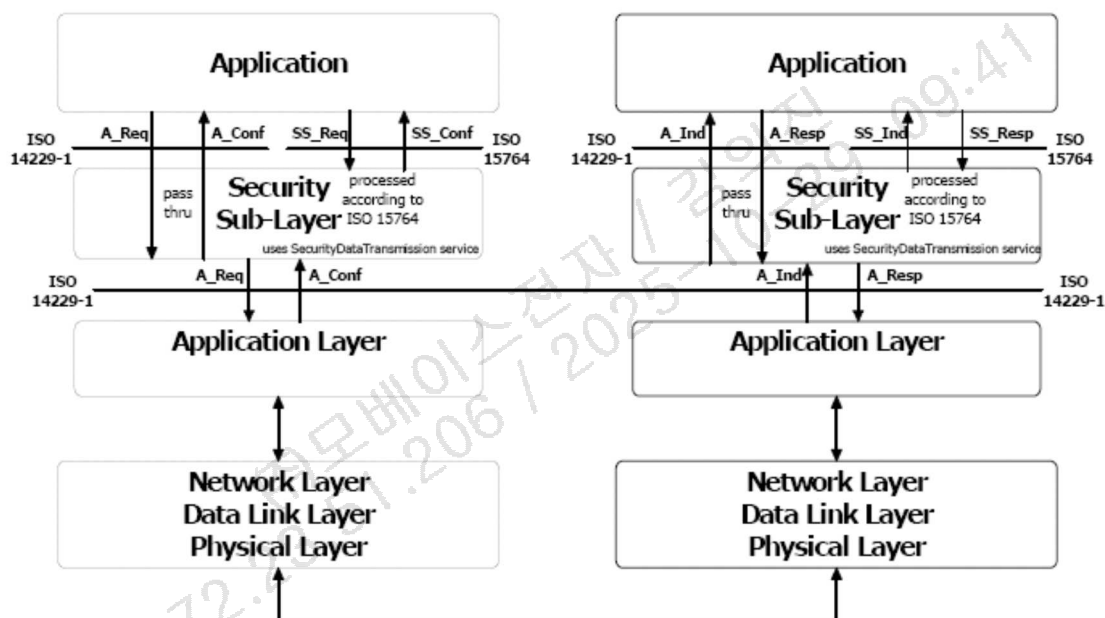


Figure 4-3. Security Sub-Layer implementation

There are two (2) methods to perform diagnostic service data transfer between the client and server(s):

#### - Unsecured data transmission mode

The application uses the diagnostic Services and Application Layer Service Primitives defined in this document to exchange data between a client and a server. The Security Sub-Layer performs a "Pass-Thru" of data between "Application" and "Application Layer" in the client and the server.

#### - Secured data transmission mode

The application uses the diagnostic Services or external services and the Security Sub-Layer Service Primitives defined in ISO 15764 to exchange data between a client and a server. The security sub-layer uses the SecuredDataTransmission service for the transmission/reception of the secured data. Secured links must be point-to-point communication. Therefore only physical addressing is allowed, which means that only one server is involved.

The interface of the security sub-layer to the application is according to the ISO/OSI model conventions and therefore provides the following four (4) security sub-layer (SS\_) service primitives

- SS\_SecuredMode.req : Security Sub-Layer Request
- SS\_SecuredMode.ind : Security Sub-Layer Indication
- SS\_SecuredMode.resp : Security Sub-Layer Response
- SS\_SecuredMode.conf : Security Sub-Layer Confirmation

ISO 14229-1 defines both, confirmed and unconfirmed services. In a secured mode only confirmed services are allowed (suppressPosRspMsgIndicationBit = FALSE). Based on this requirement the following services are not allowed to be executed in a secured mode:

- ResponseOnEvent (86 hex)
- ReadDataByPeriodicIdentifier (2A hex)
- TesterPresent (3E hex)

The confirmed services (suppressPosRspMsgIndicationBit = FALSE) use the four (4) application layer service primitives request, indication, response and confirmation. Those are mapped onto the four (4) security sublayer service primitives and vice versa when executing a confirmed diagnostic service in a secured mode.

The task of the Security Sub-Layer when performing a diagnostic service in a secured mode is to encrypt data provided by the "Application", to decrypt data provided by the "Application Layer" and to add, check, and remove security specific data elements. The Security Sub-Layer uses the SecuredDataTransmission (84 hex) service of the application layer to transmit and receive the entire diagnostic message or message according to an external protocol (request and response), which shall be exchanged in a secured mode.

### 4.7.2 Security Sub-layer Access

The concept of accessing the security sub-layer for a secured service execution is similar to the application layer interface as described in this document. The security sub-layer makes use of the application layer service primitives.

The following describes the execution of confirmed diagnostic service in a secured mode:

- The client application uses the security sub-layer SecuredServiceExecution service request to perform a diagnostic service in a secured mode. The security sub-layer performs the required action to establish a link with the server(s), adds the specific security related parameters, encrypts the service data of the diagnostic service to be executed in a secured mode if needed and uses the application layer SecuredDataTransmission service request to transmit the secured data to the server.
- The server receives an application layer SecuredDataTransmission service indication, which is handled by the security sub-layer of the server. The security sub-layer of the server checks the security specific parameters, decrypts encrypted data and presents the data of the service to be executed in a secured mode to the application via the security sub-layer SecuredServiceExecution service indication. The application executes the service and uses the security sub-layer SecuredServiceExecution service response to respond to the service in a secured mode. The security sub-layer of the server adds the specific security related parameters, encrypts the response message data if needed and uses the application layer SecuredDataTransmission service response to transmit the response data to the client.
- The client receives an application layer SecuredDataTransmission service confirmation primitive, which is handled by the security sub-layer of the client. The security sub-layer of the client checks the security specific parameters, decrypts encrypted response data and presents the data via the security sub-layer SecuredServiceExecution confirmation to the application.

Figure 11 graphically shows the interaction of the security sub-layer, the application layer, and the application when executing a confirmed diagnostic service in a secured mode.

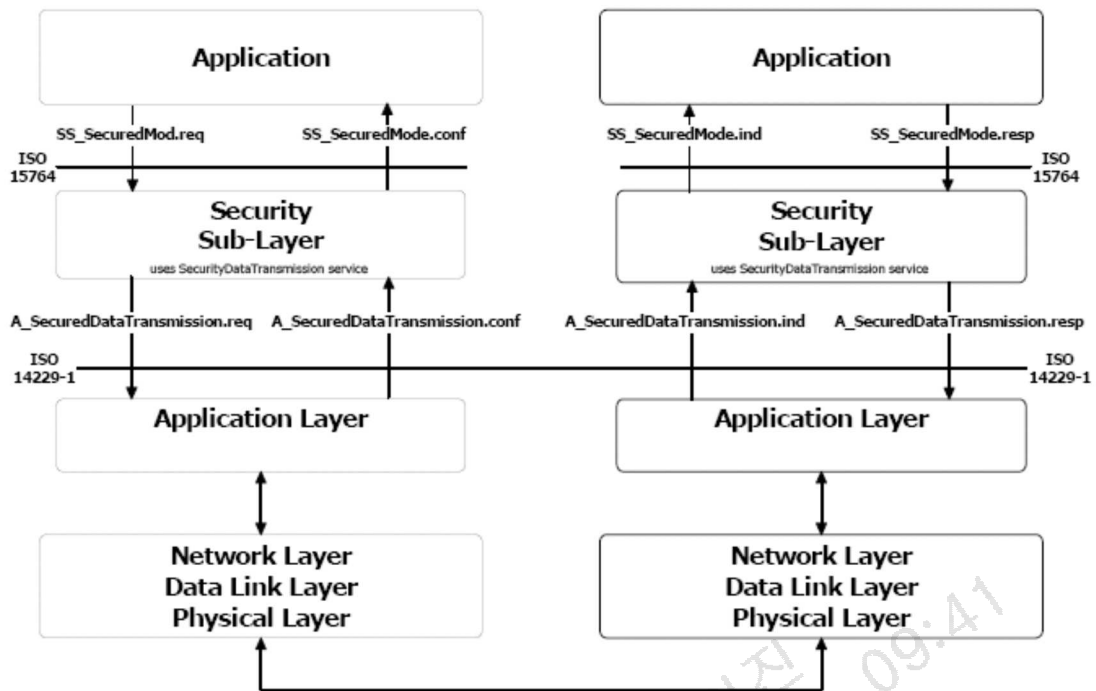


Figure 4-4 Security sub-layer, application layer, and application interaction

#### 4.7.3 Request message

| Data Byte | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic     |
|-----------|-------------------------------------------------------|----------|-------------|--------------|
| #1d       | <b>SecuredDataTransmission Request Service ID</b>     | <b>M</b> | <b>#84h</b> | <b>SDT</b>   |
| #2d       | securityDataRequestRecord = [ securityDataParameter#1 | M        | #XXh        | SECDRQR_SDP_ |
| ..        | ..                                                    | ..       | ..          | ..           |
| #nd       | securityDataParameter#m ]                             | M        | #XXh        | SDP_         |

Table4-32. Request Message

#### 4.7.4 Positive response message

| Data Byte | Parameter Name                                          | Cvt      | Hex Value   | Mnemonic     |
|-----------|---------------------------------------------------------|----------|-------------|--------------|
| #1d       | <b>SecuredDataTransmission Response Service ID</b>      | <b>M</b> | <b>#C4h</b> | <b>SDTPR</b> |
| #2d       | securedDataResponseRecord[] = [ securityDataParameter#1 | M        | #XXh        | SECDRQR_SDP_ |
| ..        | ..                                                      | ..       | ..          | ..           |
| #nd       | securityDataParameter#m ]                               | M        | #XXh        | SDP_         |

Table4-33. Positive Response Message definition

#### 4.7.5 Supported negative response codes

| Hex   | NegativeResponseCode                           | Cvt | Cause of Occurrence                                             |
|-------|------------------------------------------------|-----|-----------------------------------------------------------------|
| 13    | incorrectMessageLengthOrInvalidFormat          | M   | The length of the message is wrong                              |
| 38-4F | reservedByExtendedDataLinkSecurityDocu<br>ment | M   | This range of values is reserved by extended data link security |

Table4-34. Supported Negative Response Codes

## 4.8 ControlDTCSetting (85 hex) service

The ControlDTCSetting service shall be used by a client to stop or resume the setting of diagnostic trouble codes (DTC's) in the server(s).

The ControlDTCSetting request message can be used to stop the setting of diagnostic trouble codes in an individual server or a group of servers. If the server being addressed is not able to stop the setting of diagnostic trouble codes, it shall respond with a ControlDTCSetting negative response message indicating the reason for the reject.

The update of the DTC status bit information shall continue once a ControlDTCSetting request is performed with sub-function set to "on" or a session layer timeout occurs (server transitions to defaultSession).

If a clearDiagnosticInformation (14 hex) service is sent by the client the ControlDTCSetting shall not prohibit resetting the server's DTC memory. In case a successful ECUReset is performed then this re-enables the setting of DTC's.

### 4.8.1 Request message

| Data Byte | Parameter Name                       | Cvt      | Hex Value   | Mnemonic        |
|-----------|--------------------------------------|----------|-------------|-----------------|
| #1d       | <b>ControlDTCSetting Service ID</b>  | <b>M</b> | <b>#85h</b> | <b>CDTCS</b>    |
| #2d       | Sub-function = [<br>DTCSettingType ] | M        | #XXh        | LEV_<br>DTCSTP_ |

Table4-35. Request Message

| Hex   | Description | Cvt | Mnemonic |
|-------|-------------|-----|----------|
| 00    | Reserved    | M   |          |
| 01    | On          | M   | ON       |
| 02    | Off         | M   | OFF      |
| 03~7F | Reserved    | M   |          |

Table4-36. Request message sub-function parameter definition

### 4.8.2 Positive response message

| Data Byte | Parameter Name                               | Cvt      | Hex Value   | Mnemonic       |
|-----------|----------------------------------------------|----------|-------------|----------------|
| #1d       | <b>ControlDTCSetting Response Service ID</b> | <b>M</b> | <b>#C5h</b> | <b>CDTCSRP</b> |
| #2d       | DTCSettingType                               | M        | #XXh        | DTCSTP         |

Table4-37. Positive Response Message definition

### 4.8.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                           |
|-----|---------------------------------------|-----|-----------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                    |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                            |
| 22  | conditionNotCorrect                   | M   | Used when the server is condition that cannot perform the requested DTC control functionality |

Table4-38. Supported Negative Response Codes

## 4.9 ResponseOnEvent (86 hex) service

The ResponseOnEvent service requests a server to start or stop transmission of response on a specified event. This service provides the possibility to automatically execute a diagnostic service in case a specified event occurs in the server. The client specifies the event (including optional event parameters) and the service (including service parameters) to be executed in case the event occurs.

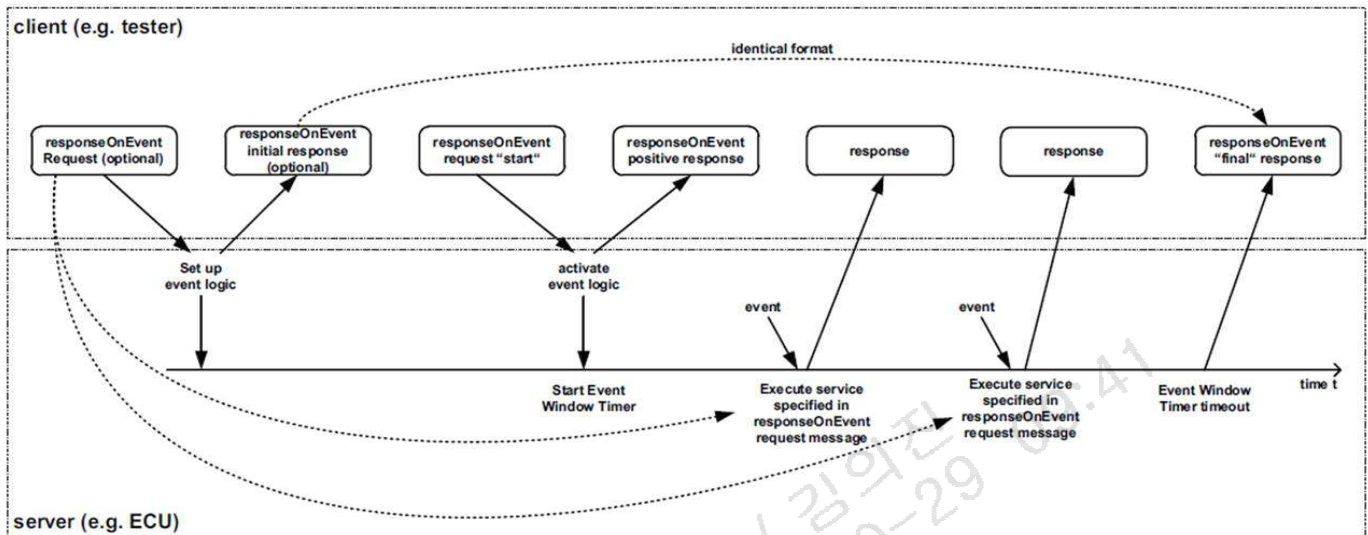


Figure 4-5 ResponseOnEvent service - client and server behaviour

The server shall evaluate the sub-function and data content of the ResponseOnEvent request message at the time of the reception. This includes the following sub-function and parameters:

- eventType,
- eventWindowTime,
- eventTypeRecord (eventTypeParameter #1-#m).

In case of invalid data in the ResponseOnEvent request message a negative response with the negative response code 31 hex shall be sent. The serviceToRespondToRecord is not part of this evaluation. The serviceToRespondToRecord parameter will be evaluated when the specified event occurs, which triggers the execution of the service contained in the serviceToRespondToRecord. At the time the event occurs the serviceToRespondToRecord (diagnostic service request message) shall be executed. In case conditions are not correct a negative response message with the appropriate negative response code shall be sent. Multiple events shall be signalled in the order of their occurrence.

The following requirements shall apply for this service when implemented on CAN.

- a) Multiple ResponseOnEvent services may run concurrently with different requirements (different EventTypes, serviceToRespondTo-Records, ...) to start and stop diagnostic services.
- b) While the ResponseOnEvent service is active, the server shall be able to process concurrent diagnostic request and response messages accordingly. This should be accomplished with a (different) pair of serviceToRespondTo-request/response CAN identifiers. See Figure 4-2. If the same diagnostic request/response CAN identifiers are used for diagnostic communication and the serviceToRespondToresponses, the following restrictions shall apply.
  - 1) The server shall ignore an incoming diagnostic request after an event has occurred and the serviceToRespondTo-response is in progress, until the serviceToRespondTo-response is completed.

- 2) After the client receives any response after sending a diagnostic request, the response shall be classified according to the possible serviceToRespondTo-responses and the expected diagnostic responses that have been sent.
  - 3) If the response is a serviceToRespondTo-response (one of the possible responses set up with ResponseOnEvent-service), the client shall repeat the request after the serviceToRespondToresponse has been received completely.
  - 4) If the response is ambiguous (i.e. the response could originate from the serviceToRespondTo initiated by an event or from the response to a diagnostic request), the client shall present the response both as a serviceToRespondTo-response and as the response to the diagnostic request. The client shall not repeat the request with the exception of NegativeResponseCode busyRepeatRequest (21 hex). (See the negative response code definitions in ISO 14229-1.)
- c) The ResponseOnEvent service shall only be allowed to use those diagnostic services available in the active diagnostic session.
- d) While the ResponseOnEvent service is active, any change in a diagnostic session shall terminate the current ResponseOnEvent service(s). For instance, if a ResponseOnEvent service has been set up during extendedDiagnosticSession, it shall terminate when the server switches to the defaultSession.
- e) If a ResponseOnEvent (86 hex) service has been set up during defaultSession, then the following shall apply :
- 1) If Bit 6 of the eventType subfunction parameter is set to 0 (do not store event), then the event shall terminate when the server powers down. The server shall not continue a ResponseOnEvent diagnostic service after a reset or power on (i.e. the ResponseOnEvent service is terminated).
  - 2) If Bit 6 of the eventType subfunction parameter is set to 1 (store event), it shall resume sending serviceToRespondTo-responses according to the ResponseOnEvent-set up after a power cycle of the server.

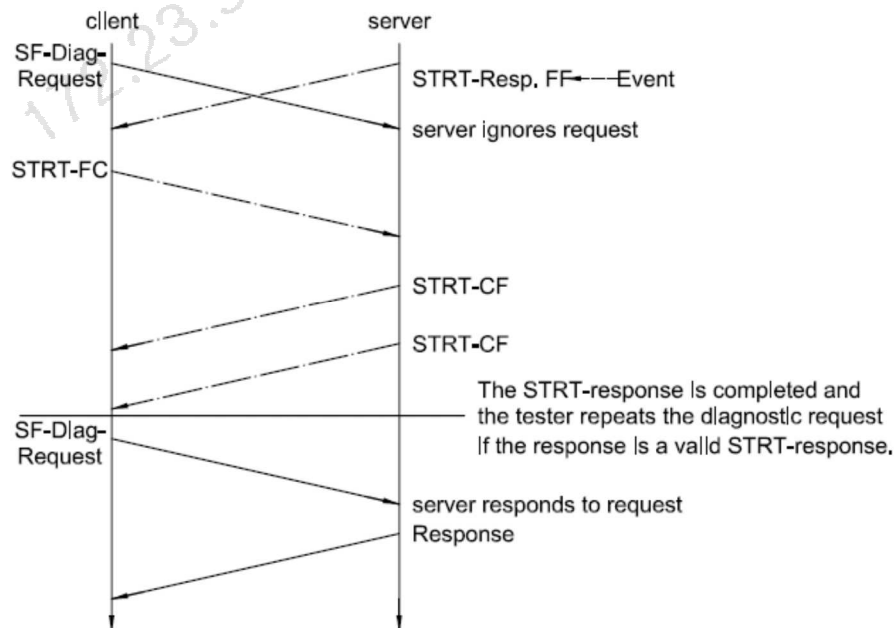


Figure 4-6 Concurrent request when the event occurs

- f) The subfunction parameter value responseRequired = "no" should only be used for the eventType = stopResponseOnEvent, startResponseOnEvent or clearResponseOnEvent. The server shall always

return a response to the event-triggered response when the specified event is detected.

- g) The server shall return a final positive response to indicate the ResponseOnEvent (86 hex) service has reached the end of the finite event window, unless one of the following conditions apply:
- 1) if eventTypes do not setup ResponseOnEvent, such as stopResponseOnEvent, startResponseOnEvent, clearResponseOnEvent or reportActivatedEvents;
  - 2) if the infinite event window was established
    - if the Service has been deactivated before the event window was closed,
    - Bit 6 of the eventType subfunction parameter is set to 0 (do not store) and the server powers down or resets.
- h) When the specified event is detected, the server shall respond immediately with the appropriate serviceToRespondTo-response message. The immediate serviceToRespondTo-response message shall not disrupt any other diagnostic request or response transmission already in progress (i.e. the serviceToRespondTo-response shall be delayed until the current message transmission has been completed — see Figure 4-7).

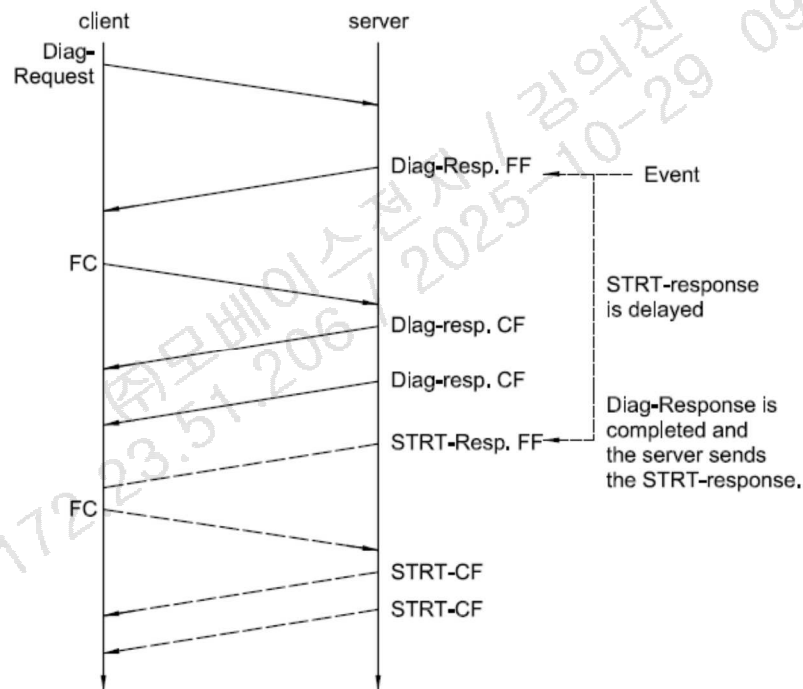


Figure 4-7 Event occurrence during a message in progress

- i) The ResponseOnEvent service shall only apply to transient events and conditions. The server shall return a response once per event occurrence. For a condition that is continuously sustained over a period of time, the response service shall be executed only one time at the initial occurrence. In case the eventType is defined so that serviceToRespondTo-responses could occur at a high frequency, then appropriate measures have to be taken in order to prevent back to back serviceToRespondTo-responses. A minimum separation time between serviceToRespondTo-responses could be part of the eventTypeRecord (vehicle-manufacturer-specific).

The following table defines the data parameters applicable for the implementation of this service on CAN.

| Recommended services (ServiceToRespondTo) | RequestService Identifier (SId) |
|-------------------------------------------|---------------------------------|
| ReadDataByIdentifier                      | 22 hex                          |

|                                |        |
|--------------------------------|--------|
| ReadDTCInformation             | 19 hex |
| RoutineControl                 | 31 hex |
| InputOutputControlByIdentifier | 2F hex |

Table4-39. Recommended services to be used with the ResponseOnEvent service

### 4.9.1 Request message

| Data Byte   | Parameter Name                             | Cvt      | Hex Value   | Mnemonic   |
|-------------|--------------------------------------------|----------|-------------|------------|
| #1d         | <b>ResponseOnEvent Request Service ID</b>  | <b>M</b> | <b>#86h</b> | <b>ROE</b> |
| #2d         | Sub-function = [ eventType ]               | M        | #XXh        | LEV_ETP    |
| #3d         | eventWindowTime                            | M        | #XXh        | EWT        |
| #4d         | eventTypeRecord[] = [ eventTypeParameter 1 | C1       | #XXh        | ETR_ETP1   |
| ..          | ..                                         | ..       | ..          | ..         |
| #(m-1)+4d   | eventTypeParameter m ]                     | C1       | #XXh        | ETPm       |
| #n-(r-1)-1d | servieToRespondToRecord[] = [ serviceId    | C2       | #XXh        | STRTR_SI   |
| #n-(r-1)d   | serviceParameter 1                         | C3       | #XXh        | SP1        |
| ..          | ..                                         | ..       | ..          | ..         |
| #nd         | serviceParameter r ]                       | C3       | #XXh        | SPr        |

C1 : present if the eventType requires additional parameters to be specified for the event to respond to.  
C2 : mandatory to be present if the sub-function parameter is not equal to reportActivateEvents, stopResponseOnEvent, startResponseOnEvent, ClearResponseOnEvent  
C3 : present if the service request of the service to respond to requires additional service parameters  
eventWindowTime : See annex B-2 for this parameter  
eventTypeRecord : This parameter record contains additional parameters for the specified eventType.  
See definition of each eventType(table 4-33).

Table4-40. Request Message definition

| Bit 6 value | Description                                                                                                                                                                                                                                                  | Cvt | Mnemonic |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 0           | <b>doNotStoreEvent</b><br>This value indicates that the event shall terminate when the server powers down and the server shall not continue a ResponseOnEvent diagnostic service after a reset or power on (i.e. the ResponseOnEvent service is terminated). | M   | DNSE     |
| 1           | <b>storeEvent</b><br>This value indicates that the event shall resume sending serviceToRespondTo-responses according to the ResponseOnEvent-set up after a power cycle of the server.                                                                        | U   | SE       |

Table4-41. sub-function (eventType) bit 6 definition - StorageState

| Hex (bit 5-0) | Description                                                                                                                                                                                                                                                         | Cvt | Mnemonic |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00            | <b>stopResponseOnEvent</b><br>This value is used to stop the server sending responses on event. The event logic that has been set up is not cleared but can be restarted with the startResponseOnEvent sub-function parameter.<br>Length of eventTypeRecord: 0 byte | U   | STPROE   |

|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |   |         |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------|
| 01    | <b>onDTCStatusChange</b><br>This value identifies the event as a new DTC detected matching the DTCStatusMask specified for this event. This eventType requires the specification of the DTCStatusMask in the request message (eventTypeParameter#1).<br>Length of eventTypeRecord: 1 byte                                                                                                                                                                                                                                                                                                                                              | U | ONDTCS  |
| 02    | <b>onTimerInterrupt</b><br>This value identifies the event as a timer interrupt, but the timer and its values are not part of the ResponseOnEvent service.<br>This eventType requires the specification of more details in the request message (eventTypeRecord).<br>Length of eventTypeRecord: 1 byte                                                                                                                                                                                                                                                                                                                                 | U | OTI     |
| 03    | <b>onChangeOfDataIdentifier</b><br>This value identifies the event as a new internal data record identified by dataIdentifier. The data values are vehicle manufacturer specific. This eventType requires the specification of more details in the request message (eventTypeRecord).<br>Length of eventTypeRecord: 3 bytes                                                                                                                                                                                                                                                                                                            | U | OCOCID  |
| 04    | <b>reportActivatedEvents</b><br>This value is used to indicate that in the positive response all events are reported that have been activated in the server with the ResponseOnEvent service (and are currently active).<br>Length of eventTypeRecord: 0 bytes                                                                                                                                                                                                                                                                                                                                                                         | U | RAE     |
| 05    | <b>startResponseOnEvent</b><br>This value is used to indicate to the server to activate the event logic (including event window timer) that has been set up and start sending responses on event.<br>Length of eventTypeRecord: 0 byte.                                                                                                                                                                                                                                                                                                                                                                                                | M | STRTROE |
| 06    | <b>clearResponseOnEvent</b><br>This value is used to clear the event logic that has been set up in the server (This also stops the server sending responses on event.)<br>Length of eventTypeRecord: 0 byte.                                                                                                                                                                                                                                                                                                                                                                                                                           | U | CLRROE  |
| 07    | <b>onComparisonOfValues</b><br>A defined alteration of a data value out of a specific record identified by a dataIdentifier which identifies a data value event. With this sub-function the user shall have the possibility to define an event at the occurrence of a specific result gathered from a defined measurement value comparison. A specific measurement value included in a data record assigned to a defined dataIdentifier is compared with a given comparison value. The specified operator defines the kind of comparison. The event occurs if the comparison result is positive.<br>Length of eventTypeRecord: 10 Byte | U | OCOV    |
| 08~3F | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M |         |

Table4-42. sub-function (eventType) bit5-0 definition

### 4.9.2 Positive response message

| Data Byte   | Parameter Name                             | Cvt      | Hex Value   | Mnemonic     |
|-------------|--------------------------------------------|----------|-------------|--------------|
| #1d         | <b>ResponseOnEvent Response Service ID</b> | <b>M</b> | <b>#C6h</b> | <b>ROEPR</b> |
| #2d         | eventType                                  | M        | #XXh        | ETP          |
| #3d         | numberOfIdentifiedEvents                   | M        | #XXh        | NOIE         |
| #4d         | eventWindowTime                            | M        | #XXh        | EWT          |
| #5d         | eventTypeRecord[] = [                      | C        | #XXh        | ETR_         |
| ..          | eventTypeParameter 1                       | ..       | ..          | ETP1         |
| ..          | ..                                         | ..       | ..          | ..           |
| ..          | eventTypeParameter m ]                     | C        | #XXh        | ETPm         |
| #n-(r-1)-1d | servieToRespondToRecord[] = [              | C1       | #XXh        | STRTR_       |
| #n-(r-1)d   | serviceld                                  | C        | #XXh        | SI           |
| ..          | serviceParameter 1                         | ..       | ..          | SP1          |
| ..          | ..                                         | ..       | ..          | ..           |
| #nd         | serviceParameter r ]                       | C        | #XXh        | SPr          |

Table4-43. Positive Response Message definition for all sub-functions but reportActivatedEvents

| Data Byte         | Parameter Name                             | Cvt      | Hex Value   | Mnemonic     |
|-------------------|--------------------------------------------|----------|-------------|--------------|
| #1d               | <b>ResponseOnEvent Response Service ID</b> | <b>M</b> | <b>#C6h</b> | <b>ROEPR</b> |
| #2d               | eventType = reportActivatedEvents          | M        | #04h        | ETP_RAE      |
| #3d               | numberOfIdentifiedEvents                   | M        | #XXh        | NOIE         |
| #4d               | eventTypeOfActivateEvent #1                | C1       | #XXh        | EVOAE        |
| #5d               | eventWindowTime #1                         | C1       | #XXh        | EWT          |
| #6d               | eventTypeRecord#1[] = [                    | C2       | #XXh        | ETR_         |
| ..                | eventTypeParameter 1                       | ..       | ..          | ETP1         |
| ..                | ..                                         | ..       | ..          | ..           |
| ..                | eventTypeParameter m ]                     | C2       | #XXh        | ETPm         |
| #p-(o-1)-1d       | servieToRespondToRecord#1[] = [            | C3       | #XXh        | STRTR_       |
| #p-(o-1)d         | serviceld                                  | C4       | #XXh        | SI           |
| ..                | serviceParameter 1                         | ..       | ..          | SP1          |
| ..                | ..                                         | ..       | ..          | ..           |
| #pd               | serviceParameter o ]                       | C4       | #XXh        | SPo          |
| ..                | ..                                         | ..       | ..          | ..           |
| #n-(r-1)-4-(q-1)d | eventTypeOfActivateEvent #k                | C1       | #XXh        | EVOAE        |
| #n-(r-1)-3-(q-1)d | eventWindowTime #k                         | C1       | #XXh        | EWT          |
| #n-(r-1)-2-(q-1)d | eventTypeRecord#k[] = [                    | C2       | #XXh        | ETR_         |
| ..                | eventTypeParameter 1                       | ..       | ..          | ETP1         |
| ..                | ..                                         | ..       | ..          | ..           |
| ..                | eventTypeParameter q ]                     | C2       | #XXh        | ETPq         |
| #n-(r-1)-2d       | servieToRespondToRecord#k[] = [            | C3       | #XXh        | STRTR_       |
| #n-(r-1)-1d       | serviceld                                  | C4       | #XXh        | SI           |
| #n-(r-1)d         | serviceParameter 1                         | ..       | ..          | SP1          |
| ..                | ..                                         | ..       | ..          | ..           |
| #nd               | serviceParameter r ]                       | C4       | #XXh        | SPr          |

C1: present if an active event is reported

C2: present if the reported eventType of the active event (eventTypeOfActiveEvent) requires additional parameters to be specified for the event to respond to.

C3: mandatory to be present when reporting an active event.

C4: present if the report service request of the service to respond to requires additional service parameters.

Table4-44. Positive Response Message definition, sub-function = reportActivatedEvents

## 4.9.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                        |
|-----|---------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                                                                                 |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                         |
| 22  | conditionNotCorrect                   | M   | Used when the server is condition that cannot perform the requested functionality                                                                          |
| 31  | requestOutOfRange                     | M   | The server shall use this response code :<br>1. if it detects an error in the eventTypeRecord parameter<br>2. if the specified eventWindowTime is invalid. |

Table4-45. Supported Negative Response Codes

## 4.9.4 Message flow example(s) ResponseOnEvent (Finite Event Window)

In the following example, it is assumed:

- the node has a physical request CANId of \$7E0
- the node has a physical response CANId of \$7E8
- storageStatus = doNotStoreEvent (0)
- suppressPosRspMsgIndicationBit = FALSE (0)

### 4.9.4.1 Step #1 : Set Up Event Logic

- eventWinowTime equal to 08 hex defines an event window of 80 seconds (eventWindowTime \* 10 s)
- eventType = onDTCStatusChange (01 hex)
- eventTypeParameter = testFailed status (01 hex)
- serviceToRespondToRecord :  
serviceld = ReadDTCInformation (19 hex)  
sub-function = reportNumberOfDTCByStatusMask (01 hex)  
DTCStatusMask = testFailed status (01 hex)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments              |
|--------|----------------------------------|-----|-------------------------|-----------------------|
| 7E0    | Physical Request                 | 8   | 07 86 01 08 01 19 01 01 | Single Frame          |
| 7E8    | Response                         | 8   | 10 08 C6 01 00 08 01 19 | First Frame           |
| 7E0    | Physical Request                 | 8   | 30 00 00 55 55 55 55 55 | Flow Control          |
| 7E8    | Response                         | 8   | 21 01 01 AA AA AA AA AA | 1st Consecutive Frame |

Table 4-46 ResponseOnEvent (FiniteEventWindow) Message Example – Set Up Event Logic

### 4.9.4.2 Step #2 : Activate Event Logic

- eventType = startResponseOnEvent (05 hex)
- DTCStatusAvailibilityMask = FF hex
- DTCCount = 4 (00 04 hex)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 02 86 04 55 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 04 C6 01 00 08 AA AA AA | Single Frame |
| 7E8    | Response                         | 8   | 04 59 FF 00 04 AA AA AA | Single Frame |



# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)  
ES95486-02

페이지  
(SHT/SHTS) 85/244

|     |          |   |                         |              |
|-----|----------|---|-------------------------|--------------|
| 7E8 | Response | 8 | 04 59 FF 00 04 AA AA AA | Single Frame |
|-----|----------|---|-------------------------|--------------|

Table 4-47 ResponseOnEvent (FiniteEventWindow) Message Example – Activate Event Logic

### 4.9.4.3 Step #3-1 (during the active event window) : Report Currently Active Events

- eventType = reportActivatedEvents (04 hex)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments              |
|--------|----------------------------------|-----|-------------------------|-----------------------|
| 7E0    | Physical Request                 | 8   | 03 86 05 08 55 55 55 55 | Single Frame          |
| 7E8    | Response                         | 8   | 10 09 C6 04 02 01 08 01 | First Frame           |
| 7E0    | Physical Request                 | 8   | 30 00 00 55 55 55 55 55 | Flow Control          |
| 7E8    | Response                         | 8   | 21 19 01 01 AA AA AA AA | 1st Consecutive Frame |

Table 4-48 ResponseOnEvent (FiniteEventWindow) Message Example –Report Currently Active Events

### 4.9.4.4 Step #3-2 (Event Window Time has expired) : Event Window Timer Timeout

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments              |
|--------|----------------------------------|-----|-------------------------|-----------------------|
| 7E8    | Response                         | 8   | 10 08 C6 01 02 08 01 19 | First Frame           |
| 7E0    | Physical Request                 | 8   | 30 00 00 55 55 55 55 55 | Flow Control          |
| 7E8    | Response                         | 8   | 21 01 01 AA AA AA AA AA | 1st Consecutive Frame |

Table 4-49 ResponseOnEvent (FiniteEventWindow) Message Example –Event Window Timer Timeout

## 4.10 LinkControl (87 hex) service

The LinkControl service is used to control the communication link baudrate between the client and the server(s) for the exchange of diagnostic data. This service optionally applies to those data link layers, which allow for a baudrate transition during an active diagnostic session.

- Step #1 : The client verifies if the transition can be performed and informs the server(s) about the baudrate to be used. Each server has to respond positively (suppressPosRspMsgIndicationBit = FALSE) before the client performs step#2. This step actually does not perform the baudrate transition.
- Step #2: The client actually requests the transition of the baudrate. This step shall only be performed if it is verified that the baudrate transition can be performed (step #1 performed). In case of functional communication it is recommended that there shall not be any response from a server when the baudrate is transitioned (suppressPosRspMsgIndicationBit = TRUE), because one server might already have been transitioned to the new baudrate while others still need to transmit their response message(s) (baudrate mismatch avoidance).

Any baudrate transition shall occur as follows:

- suppressPosRspMsgIndicationBit = TRUE: after the successful transmission/reception of the client request message, which requests the baudrate transition.
- suppressPosRspMsgIndicationBit = FALSE: after the successful transmission/reception of the server positive response message, which confirms the successful reception of the request, which requests the baudrate transition.

This service is tied to a non-defaultSession. A session layer timer timeout will transition the server(s) back to its(their) normal speed of operation. The same applies in case an ECUReset service (11 hex) is performed. The transition into another non-defaultSession shall not influence the baudrate.

### 4.10.1 Request message

| Data Byte | Parameter Name                        | Cvt      | Hex Value   | Mnemonic      |
|-----------|---------------------------------------|----------|-------------|---------------|
| #1d       | <b>LinkControl Request Service ID</b> | <b>M</b> | <b>#87h</b> | <b>LC</b>     |
| #2d       | Sub-function = [ linkControlType ]    | M        | #XXh        | LEV_<br>LCTP_ |
| #3d       | baudrateIdentifier                    | C1       | #XXh        | BI_           |

Table4-50. Request Message definition  
(linkControlType = verifyBaudrateTransitionWithFixedBaudrate)

| Data Byte | Parameter Name                                                                          | Cvt      | Hex Value   | Mnemonic      |
|-----------|-----------------------------------------------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>LinkControl Request Service ID</b>                                                   | <b>M</b> | <b>#87h</b> | <b>LC</b>     |
| #2d       | Sub-function = [ linkControlType ]                                                      | M        | #XXh        | LEV_<br>LCTP_ |
| #3d       | linkBaudrateRecord[] = [<br>baudrateHighByte<br>baudrateMiddleByte<br>baudrateLowByte ] | C        | #XXh        | LBR_<br>BRHB  |
| #4d       |                                                                                         | C        | #XXh        | BRMB          |
| #5d       |                                                                                         | C        | #XXh        | BRLB          |

Table4-51. Request Message definition  
(linkControlType = verifyBaudrateTransitionWithSpecificBaudrate)

| Hex | Description | Cvt | Mnemonic |
|-----|-------------|-----|----------|
|-----|-------------|-----|----------|

|         |                                                                                                                                                                                                                        |   |          |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------|
| 00      | <b>Reserved</b>                                                                                                                                                                                                        | M |          |
| 01      | <b>verifyBaudrateTransitionWithFixedBaudrate</b><br>This parameter is used to verify if a transition to a pre-defined baudrate, which is specified by the baudrateIdentifier data parameter can be performed.          | C | VBTFWBR  |
| 02      | <b>verifyBaudrateTransitionWithSpecificBaudrate</b><br>This parameter is used to verify if a transition to a specially defined baudrate, which is specified by the linkBaudrateRecord data parameter can be performed. | C | VBTFWSBR |
| 03      | <b>transitionBaudrate</b><br>This sub-function parameter requests the server(s) to transition the baudrate to the one that was specified in the preceding verification message.                                        | U | TB       |
| 04 ~ 7F | <b>Reserved</b>                                                                                                                                                                                                        | M |          |

C : More than one identifier shall be supported.

Table4-52. Request message sub-function (LinkControlType) parameter definition

| Hex   | Description                                                                  | Cvt | Mnemonic   |
|-------|------------------------------------------------------------------------------|-----|------------|
| 00    | Reserved                                                                     | M   |            |
| 01    | PC9600Baud<br>This value specifies the standard PC baudrate of 9.6KBaud.     | U   | PC9600     |
| 02    | PC19200Baud<br>This value specifies the standard PC baudrate of 19.2KBaud.   | U   | PC19200    |
| 03    | PC38400Baud<br>This value specifies the standard PC baudrate of 38.4KBaud.   | U   | PC38400    |
| 04    | PC57600Baud<br>This value specifies the standard PC baudrate of 57.6KBaud.   | U   | PC57600    |
| 05    | PC115200Baud<br>This value specifies the standard PC baudrate of 115.2KBaud. | U   | PC115200   |
| 10    | CAN125000Baud<br>This value specifies the standard CAN baudrate of 123KBaud. | U   | CAN125000  |
| 11    | CAN250000Baud<br>This value specifies the standard CAN baudrate of 250KBaud. | U   | CAN250000  |
| 12    | CAN500000Baud<br>This value specifies the standard CAN baudrate of 500KBaud. | U   | CAN500000  |
| 13    | CAN1000000Baud<br>This value specifies the standard CAN baudrate of 1MBaud.  | U   | CAN1000000 |
| 14~FF | Reserved                                                                     | M   |            |

Table4-53. Definition of baudrateIdentifier value

## 4.10.2 Positive response message

| Data Byte | Parameter Name | Cvt | Hex Value | Mnemonic |
|-----------|----------------|-----|-----------|----------|
|-----------|----------------|-----|-----------|----------|

|     |                                 |   |      |      |
|-----|---------------------------------|---|------|------|
| #1d | LinkControl Response Service ID | M | #C7h | LCPR |
| #2d | linkControlType                 | M | #XXh | LCTP |

Table4-54. Positive Response Message definition

### 4.10.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                     |
|-----|---------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | Send if the sub-function parameter in the request message is not supported                                                                                              |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                      |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the requested LinkControl are not met.                                                                                  |
| 24  | requestSequenceError                  | M   | This code shall be returned if the client requests the transition of the baudrate without a preceding verification step, which specifies the baudrate to transition to. |
| 31  | requestOutOfRange                     | M   | This code shall be returned if,<br>1. the requested fixed baudrate (baudrateIdenfitier) is invalid<br>2. the specific baudrate (linkBaudrateRecord) is invalid.         |

Table4-55. Supported Negative Response Codes

### 4.10.4 Message Flow Examples LinkControl

In the following example, it is assumed:

- the node has a physical request CANId of \$7E0
- the node has a physical response CANId of \$7E8

#### 4.10.4.1 Example #1 – Transition baudrate to fixed baudrate

- PC baudrate = 115200 kBit/s (05 hex)

##### 4.10.4.1.1 Step #1 : Verify if all criteria are met for a baudrate switch

- linkControlType = verifyBaudrateTransitionWithFixedBaudrate (01 hex)
- suppressPosRspMsgIndicationBit = FALSE (0)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 03 87 01 05 55 55 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 02 C7 01 AA AA AA AA AA | Single Frame |

Table 4-56 LinkControl Message Example – VerifyBaudrateTransitionWithFixedBaudrate

##### 4.10.4.1.2 Step #2 : Transition the baudrate

- linkControlType = transitionBaudrate (03 hex)
- suppressPosRspMsgIndicationBit = True (1)

| CAN ID | Client Request / Server Response | DLC | Data Bytes | Comments |
|--------|----------------------------------|-----|------------|----------|
|--------|----------------------------------|-----|------------|----------|



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|     |                  |   |                         |              |
|-----|------------------|---|-------------------------|--------------|
| 7E0 | Physical Request | 8 | 02 87 83 55 55 55 55 55 | Single Frame |
|-----|------------------|---|-------------------------|--------------|

Table 4-57 LinkControl Message Example – TransitionBaudrateWithFixedBaudrate

### 4.10.4.2 Example #2 – Transition baudrate to specific baudrate

- PC baudrate = 150 kBit/s (02 49 F0 hex)

#### 4.10.4.2.1 Step #1 : Verify if all criteria are met for a baudrate switch

- linkControlType = verifyBaudrateTransitionWithSpecificBaudrate (02 hex)
- suppressPosRspMsgIndicationBit = FALSE (0)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 05 87 02 02 49 F0 55 55 | Single Frame |
| 7E8    | Response                         | 8   | 02 C7 02 AA AA AA AA AA | Single Frame |

Table 4-58 LinkControl Message Example – VerifyBaudrateTransitionWithSpecificBaudrate

#### 4.10.4.2.2 Step #2 : Transition the baudrate

- linkControlType = transitionBaudrate (03 hex)
- suppressPosRspMsgIndicationBit = True (1)

| CAN ID | Client Request / Server Response | DLC | Data Bytes              | Comments     |
|--------|----------------------------------|-----|-------------------------|--------------|
| 7E0    | Physical Request                 | 8   | 02 87 83 55 55 55 55 55 | Single Frame |

Table 4-59 LinkControl Message Example – TransitionBaudrateWithSpecificBaudrate

## 5. Data Transition functional unit

### 5.1 ReadDataByIdentifier (22 hex) service

The ReadDataByIdentifier service allows the client to request data record values from the server identified by one or more dataIdentifiers. Upon receiving a ReadDataByIdentifier request, the server shall access the data elements of the records specified by the dataIdentifier parameter(s) and transmit their value in one single ReadDataByIdentifier positive response containing the associated dataRecord parameter(s). The request message may contain the same dataIdentifier multiple times. The server shall treat each dataIdentifier as a separate parameter and respond with data for each dataIdentifier as often as requested.

#### 5.1.1 Request message

| Data Byte                      | Parameter Name                                      | Cvt | Hex Value | Mnemonic   |
|--------------------------------|-----------------------------------------------------|-----|-----------|------------|
| #1d                            | ReadDataByIdentifier Request Service ID             | M   | #22h      | RDBI       |
| #2d                            | dataIdentifier[] #1 = [<br>byte#1 (MSB)<br>byte#2 ] | M   | #XXh      | DID_<br>HB |
| #3d                            |                                                     | M   | #XXh      | LB         |
| ..                             | ..                                                  | ..  | ..        | ..         |
| #(n-1)d                        | dataIdentifier[] #m = [<br>byte#1 (MSB)<br>byte#2 ] | U   | #XXh      | DID_<br>HB |
| #nd                            |                                                     | U   | #XXh      | LB         |
| dataIdentifier : see Annex C.1 |                                                     |     |           |            |

Table5-1. Request Message

#### 5.1.2 Positive response message

| Data Byte   | Parameter Name                                      | Cvt      | Hex Value   | Mnemonic        |
|-------------|-----------------------------------------------------|----------|-------------|-----------------|
| #1d         | <b>ReadDataByIdentifier Response Service ID</b>     | <b>M</b> | <b>#62h</b> | <b>RDBIPR</b>   |
| #2d         | dataIdentifier[] #1 = [<br>byte#1 (MSB)<br>byte#2 ] | M        | #XXh        | DID_<br>HB      |
| #3d         |                                                     | M        | #XXh        | LB              |
| #4d         | dataRecord[] #1 = [<br>data#1<br>..<br>data#k ]     | M        | #XXh        | DREC_<br>DATA_1 |
| ..          |                                                     | ..       | ..          | ..              |
| #(k-1)+4d   |                                                     | U        | #XXh        | DATA_k          |
| ..          | ..                                                  | ..       | ..          | ..              |
| #n-(o-1)-2d | dataIdentifier[] #m = [<br>byte#1 (MSB)<br>byte#2 ] | U        | #XXh        | DID_<br>HB      |
| #n-(o-1)-1d |                                                     | U        | #XXh        | LB              |
| #n-(o-1)d   | dataRecord[] #m = [<br>data#1<br>..<br>data#k ]     | U        | #XXh        | DREC_<br>DATA_1 |
| ..          |                                                     | ..       | ..          | ..              |
| #nd         |                                                     | U        | #XXh        | DATA_o          |

Table5-2. Positive Response Message definition

#### 5.1.3 Supported negative response codes

| Hex | NegativeResponseCode | Cvt | Cause of Occurrence |
|-----|----------------------|-----|---------------------|
|-----|----------------------|-----|---------------------|



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|    |                                       |   |                                                                                                                                                                                                                                                       |
|----|---------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | incorrectMessageLengthOrInvalidFormat | M | This response code shall be sent if the length of the request message is invalid or the client exceeded the maximum number of dataIdentifiers allowed to be requested at a time.                                                                      |
| 22 | conditionNotCorrect                   | U | This response code shall be sent if the operating condition of the server are not met to perform the required action.                                                                                                                                 |
| 31 | requestOutOfRange                     | M | This code shall be sent if none of the requested dataIdentifier values are supported by the device; none of the requested dataIdentifiers are supported in the current session; the requested dynamicDefinedDataIdentifier has not been assigned yet; |
| 33 | securityAccessDenied                  | M | This code shall be sent if at least one of the dataIdentifier is secured and the server is not in and unlocked state.                                                                                                                                 |
| 14 | responseTooLong                       | M | This NRC shall be if the total length of the response exceeds the limit of the underlying transport protocol (e.g., when multiple DIDs are requested in a single request)                                                                             |

Table5-3. Supported Negative Response Codes

## 5.2 ReadMemoryByAddress (23 hex) service

The ReadMemoryByAddress service allows the client to request memory data from the server via provided starting address and size of memory to be read.

### 5.2.1 Request message

| Data Byte | Parameter Name                                | Cvt      | Hex Value   | Mnemonic    |
|-----------|-----------------------------------------------|----------|-------------|-------------|
| #1d       | <b>ReadMemoryByAddress Request Service ID</b> | <b>M</b> | <b>#23h</b> | <b>RMBA</b> |
| #2d       | addressAndLengthFormatIdentifier              | M        | #XXh        | ALFID       |
| #3d       | memoryAddress[] = [<br>byte#1 (MSB)           | M        | #XXh        | MA_<br>B1   |
| ..        | ..                                            | ..       | ..          | ..          |
| #(m-1)+3  | byte#m ]                                      | C1       | #XXh        | Bm          |
| #n-(k-1)d | memorySize[] = [<br>byte#1 (MSB)              | M        | #XXh        | MS_<br>B1   |
| ..        | ..                                            | ..       | ..          | ..          |
| #nd       | byte#k ]                                      | C2       | #XXh        | Bk          |

addressAndLengthFormatIdentifier : This parameter is a one byte value with each nibble encoded separately.  
- bit 7-4 : Length(number of byte) of the memorySize parameter  
- bit 3-0 : Length(number of byte) of the memoryAddress parameter  
C1: The presence of this parameter depends on address length information parameter of the addressAndLengthFormatIdentifier.  
C2: The presence of this parameter depends on the memory size length information of the addressAndLengthFormatIdentifier.

Table5-4. Request Message

### 5.2.2 Positive response message

| Data Byte | Parameter Name                                 | Cvt      | Hex Value   | Mnemonic        |
|-----------|------------------------------------------------|----------|-------------|-----------------|
| #1d       | <b>ReadMemoryByAddress Response Service ID</b> | <b>M</b> | <b>#63h</b> | <b>RMBAPR</b>   |
| #2d       | dataRecord[] = [<br>data#1                     | M        | #XXh        | DREC_<br>DATA_1 |
| ..        | ..                                             | ..       | ..          | ..              |
| #nd       | data#m ]                                       | U        | #XXh        | DATA_m          |

Table5-5. Positive Response Message definition

### 5.2.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                   |
|-----|---------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                    |
| 22  | conditionNotCorrect                   | M   | This response code shall be sent if the operating conditions of the server are not met to perform the required action |



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|----|----------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | requestOutOfRange    | M | This response code shall be sent if ;<br>1. any memory address within the interval [\$MA, (\$MA+\$MS-\$1)] is invalid.<br>2. any memory address within the interval [\$MA, (\$MA+\$MS-\$1)] is restricted.<br>3. the memorySize parameter value in the request message is greater than the maximum value supported by the server.<br>4. the specified addressAndLengthFormatIdentifier is not valid. |
| 33 | SecurityAccessDenied | M | This code shall be sent if any memory address within the interval [\$MA, (\$MA+\$MS-\$1)] is secure and the server is locked.                                                                                                                                                                                                                                                                        |

Table5-6. Supported Negative Response Codes

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주모베이스전자 / 김의진

### 5.3 ReadScalingDataByIdentifier (24 hex) service

The ReadScalingDataByIdentifier service allows the client to request scaling data record information from the server identified by a dataIdentifier.

Upon receiving a ReadScalingDataByIdentifier request, the server shall access the scaling information associated with the specified dataIdentifier parameter and transmit the scaling information values in one ReadScalingDataByIdentifier positive response.

### 5.3.1 Request message

| Data Byte                      | Parameter Name                                   | Cvt | Hex Value | Mnemonic   |
|--------------------------------|--------------------------------------------------|-----|-----------|------------|
| #1d                            | ReadScalingDataByIdentifier Request Service ID   | M   | #24h      | RSDBI      |
| #2d                            | dataIdentifier[] = [<br>byte#1 (MSB)<br>byte#2 ] | M   | #XXh      | DID_<br>HB |
| #3d                            |                                                  | M   | #XXh      | LB         |
| dataIdentifier : see Annex C.1 |                                                  |     |           |            |

Table5-7. Request Message

### 5.3.2 Positive response message

| Data Byte              | Parameter Name                                                                                             | Cvt            | Hex Value          | Mnemonic                   |
|------------------------|------------------------------------------------------------------------------------------------------------|----------------|--------------------|----------------------------|
| #1d                    | <b>ReadScalingDataByIdentifier Response Service ID</b>                                                     | <b>M</b>       | <b>#64h</b>        | <b>RSDBI PR</b>            |
| #2d<br>#3d             | dataIdentifier[] = [<br>byte#1 (MSB)<br>byte#2 ]                                                           | M<br>M         | #XXh<br>#XXh       | DID_<br>HB<br>LB           |
| #4d                    | scalingByte#1                                                                                              | M              | #XXh               | SB_1                       |
| #3d<br>..<br>#(p-1)+5d | scalingByteExtension[]#1 = [<br>scalingByteExtensionParameter#1<br>..<br>scalingByteExtensionParameter#p ] | C1<br>..<br>C1 | #XXh<br>..<br>#XXh | SBE_<br>PAR1<br>..<br>PARp |
| #(n-r)d                | scalingByte#k                                                                                              | C2             | #XXh               | SB_k                       |
| #n-(r-1)d<br>..<br>#nd | scalingByteExtension[]#1 = [<br>scalingByteExtensionParameter#1<br>..<br>scalingByteExtensionParameter#r ] | C1<br>..<br>C1 | #XXh<br>..<br>#XXh | SBE_<br>PAR1<br>..<br>PARr |

C1 : The presence of this parameter depends on the scalingByte high nibble. It is mandatory to be present if the scalingByte high nibble is encoded as formula, unit/format, or bitMappedReported/WithOutMask.

C2 : The presence of this parameter depends on whether the encoding of the scaling information requires more than one byte

scalingByte : The parameter scalingByte consists of one byte (high and low nibble). The scalingByte high nibble defines the data type, which is used to represent the dataIdentifier (DID). The scalingByte low nibble defines the number of bytes used to represent the parameter in a datastream. (see Annex C.2)

scalingByteExtension : This parameter is used to provide additional information for scalingBytes with a high nibble encoded as formula, unit/format, or bitmappedReportedWithOutMask. (see Annex C.3)

Table5-8. Positive Response Message definition

### 5.3.3 Supported negative response codes

| Hex | NegativeResponseCode | Cvt | Cause of Occurrence |
|-----|----------------------|-----|---------------------|
|-----|----------------------|-----|---------------------|



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|----|---------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13 | incorrectMessageLengthOrInvalidFormat | M | This response code shall be sent if the length of the request message is invalid.                                                                                                                                                                                                     |
| 22 | conditionNotCorrect                   | M | This code shall be sent if the operating conditions of the server are not met to perform the required action.                                                                                                                                                                         |
| 31 | requestOutOfRange                     | M | This return code shall be sent if ;<br>1. the requested dataIdentifier value is not supported by the device (physical addressing only)<br>2. the requested dataIdentifier value is supported by the device, but no scaling information is available for the specified dataIdentifier. |
| 33 | securityAccessDenied                  | M | This code shall be sent if the dataIdentifier is secured and the server is not in an unlocked state.                                                                                                                                                                                  |

Table5-9. Supported Negative Response Codes

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## 5.4 ReadDataByPeriodicIdentifier (2A hex) service

The ReadDataByPeriodicIdentifier service allows the client to request the periodic transmission of data record values from the server identified by one or more periodicDataIdentifiers.

The client request message contains one or more 1-byte periodicDataIdentifier values that identify data record(s) maintained by the server. The periodicDataIdentifier represents the low byte of a dataIdentifier out of the dataIdentifier range reserved for this service (0xF2XX, see table C-4 for allowed periodicDataIdentifier values), e.g. the periodicDataIdentifier 0xE3 used in this service is the identifier 0xF2E3.

The format and definition of the dataRecord shall be vehicle manufacturer specific, and may include analog input and output signals, digital input and output signals, internal data, and system status information if supported by the server.

Upon receiving a ReadDataByPeriodicIdentifier request other than stopSending the server shall check whether the conditions are correct to execute the service.

A periodicDataIdentifier shall only be supported with a single transmissionMode at a given time. A change to the schedule of a periodicDataIdentifier shall be performed on reception of a request message with the transmissionMode parameter set to a new schedule for the same periodicDataIdentifier. Multiple schedules for different periodicDataIdentifiers shall be supported upon vehicle manufacturer's request.

**IMPORTANT** — If the conditions are correct then the server shall transmit a positive response message, including only the service identifier. The server shall never transmit a negative response message once it has accepted the initial request message by responding positively.

Following the initial positive response message the server shall access the data elements of the records specified by the periodicDataIdentifier parameter(s) and transmit their value in separate periodic data response messages for each periodicDataIdentifier containing the associated dataRecord parameters.

The separate periodic data response messages defined to transmit the periodicDataIdentifier data to the client following the initial positive response message shall include the periodicDataIdentifier and the data of the periodicDataIdentifier, but not the positive response service identifier. The mapping of the periodic response message onto certain data link layers is described in the appropriate implementation specifications of ISO 14229.

The documented periodic rate for a specific transmissionMode is defined as the time between any two consecutive response messages with the same periodicDataIdentifier, when only a single periodicDataIdentifier is scheduled. If multiple periodicDataIdentifiers are scheduled concurrently, the effective period between the same periodicDataIdentifier will vary based upon the following design parameters:

- the call rate of the periodic scheduler,
- the number of available protocol specific periodic data response message address information IDs allocated per scheduler call (e.g., CAN identifier on CAN)
- the number of periodicDataIdentifiers that can be defined in parallel to be transmitted concurrently.

These parameter values will impact the extent of how much the effective period between the same periodicDataIdentifier will increase if multiple periodicDataIdentifiers are transmitted concurrently. Therefore, all of the previously mentioned design parameters shall be specified by the vehicle manufacturer. Each time the periodic scheduler is called it shall determine if any periodicDataIdentifiers are ready to transmit.

Example, two distinct ECU implementations may both support a fast transmissionMode with a periodic rate of 10 ms and a single unique periodic data response message address information ID. If the first

implementation calls the periodic scheduler every 10ms, the time between the same periodicDataIdentifier would increase to 20 ms when two periodicDataIdentifiers are scheduled and would increase to 40 ms when four periodicDataIdentifiers are scheduled. If the second implementation calls the periodic scheduler every 5 ms, the time between the same periodicDataIdentifier would remain at 10 ms when two periodicDataIdentifiers are scheduled and would increase to 20 ms when four periodicDataIdentifiers are scheduled.

Upon receiving a ReadDataByPeriodicIdentifier request including the transmissionMode stopSending the server shall either stop the periodic transmission of the periodicDataIdentifier(s) contained in the request message or stop the transmission of all periodicDataIdentifier if no specific one is specified in the request message. The response message to this transmissionMode only contains the service identifier.

The server may limit the number of periodicDataIdentifiers that can be simultaneously supported as agreed upon by the vehicle manufacturer and system supplier. Exceeding the maximum number of periodicDataIdentifier that can be simultaneously supported shall result in a single negative response and none of the periodicDataIdentifiers in that request shall be scheduled. Repetition of the same periodicDataIdentifier in a single request message is not allowed and the server shall ignore them all except one periodicDataIdentifier if the client breaks this rule.

### 5.4.1 Request message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic     |
|-----------|--------------------------------------------------------|----------|-------------|--------------|
| #1d       | <b>ReadDataByPeriodicIdentifier Request Service ID</b> | <b>M</b> | <b>#2Ah</b> | <b>RDBPI</b> |
| #2d       | transmissionMode                                       | M        | #XXh        | TM           |
| #3d       | periodicDataIdentifier[] #1                            | C        | #XXh        | PDID1        |
| ..        | ..                                                     | ..       | ..          | ..           |
| #m+2d     | periodicDataIdentifier[] #m                            | U        | #XXh        | PDIDm        |

C: The first periodicDataIdentifier is mandatory to be present in the request message if the transmissionMode is equal to sendAtSlowRate, sendAtMediumRate, or sendAtFastRate. In case the transmissionMode is equal to stopSending there can either be no periodicDataIdentifier present in order to stop all scheduled periodicDataIdentifier or the client can explicitly specify one or more periodicDataIdentifier(s) to be stopped.

Table5-10. Request Message

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00  | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                             | M   |          |
| 01  | <b>sendAtSlowRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a slow rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend).<br>The repetition rate specified by the transmissionMode parameter slow is vehicle manufacturer specific, and predefined in the server.     | C   | SASR     |
| 02  | <b>sendAtMediumRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a medium rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend). The repetition rate specified by the transmissionMode parameter medium is vehicle manufacturer specific, and pre-defined in the server. | C   | SAMR     |

|       |                                                                                                                                                                                                                                                                                                                                                                                      |   |      |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 03    | <b>sendAtFastRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a fast rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend). The repetition rate specified by the transmissionMode parameter fast is vehicle manufacturer specific, and predefined in the server. | C | SAFR |
| 04    | <b>stopSending</b><br>The server stops transmitting positive response messages send periodically/repeatedly. Note that maximumNumberOfResponsesToSend parameter should be set to 0x01 if transmissionMode = stopSending (otherwise, server operation could be undefined).                                                                                                            | M | SS   |
| 05~FF | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                      | M |      |

C : More than one identifier shall be supported.

Table5-11. Data parameter definition - TransmissionMode

#### 5.4.2 Positive response message

It has to be distinguished between the initial positive response, which indicates that the server accepts the service and subsequent positive response message, which include periodicDataIdentifier data.

Table5-12/13 defines the initial positive response message to be transmitted by the server when it accepts the request.

| Data Byte | Parameter Name                                          | Cvt      | Hex Value   | Mnemonic       |
|-----------|---------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>ReadDataByPeriodicIdentifier Response Service ID</b> | <b>M</b> | <b>#6Ah</b> | <b>RDBPIPR</b> |

Table5-12. Positive Response Message definition

The data of a periodicDataIdentifier is transmitted periodically (with updated data) at a rate determined by the transmissionMode parameter of the request.

After the initial positive response, for each supported periodicDataIdentifier in the request the server shall start sending a single periodic data response message as defined below.

Table 5-13 defines the periodic data response message data definition.

| Data Byte | Parameter Name          | Cvt | Hex Value | Mnemonic    |
|-----------|-------------------------|-----|-----------|-------------|
| #1d       | periodicDataIdentifier  | M   | #XXh      | PDID        |
| #2d       | dataRecord[] = [ data#1 | M   | #XXh      | DREC_DATA_1 |
| ..        | ..                      | ..  | ..        | ..          |
| #k+2d     | data#k ]                | U   | #XXh      | DATA_k      |

Table5-13. Periodic data response message data definition

#### 5.4.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                            |
|-----|---------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | This NRC shall be sent if the length of the request message is invalid or the client exceeded the maximum number of periodicDataIdentifiers allowed to be requested at a time. |



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|    |                      |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|----------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | conditionNotCorrect  | M | This NRC shall be sent if the operating conditions of the server are not met to perform the required action. E.g. this could occur if the client requests periodicDataIdentifiers with different transmissionModes and the server does not support multiple transmissionModes simultaneously.                                                                                                                                                          |
| 31 | requestOutOfRange    | M | This NRC shall be sent if none of the requested periodicDataIdentifier values are supported by the device;<br>none of the requested periodicDataIdentifiers are supported in the current session;<br>The specified transmissionMode is not supported by the device;<br>the requested dynamicDefinedDataIdentifier has not been assigned yet;<br>the client exceeded the maximum number of periodicDataIdentifiers allowed to be scheduled concurrently |
| 33 | securityAccessDenied | M | This NRC shall be sent if at least one of the periodicDataIdentifier is secured and the server is not in an unlocked state.                                                                                                                                                                                                                                                                                                                            |

Table5-14. Supported Negative Response Codes

## 5.5 DynamicallyDefinedDataIdentifier (2C hex) service

The DynamicallyDefinedDataIdentifier service allows the client to dynamically define in a server a data identifier that can be read via the ReadDataByIdentifier service at a later time. The intention of this service is to provide the client with the ability to group one or more data elements into a data superset that can be requested en masse via ReadDataByIdentifier or ReadDataByPeriodicIdentifier service. The data elements to be grouped together can either be referenced by ;

- a source data identifier, a position and size or,
- a memory address and a memory length, or,
- a combination of the two methods listed above using multiple requests to define the single data element. The dynamically defined dataIdentifier will then contain a concatenation of the data parameter definitions.

This service allows greater flexibility in handling ad hoc data needs of the diagnostic application that extend beyond the information that can be read via statically defined data identifiers, and can also be used to reduce bandwidth utilization by avoiding overhead penalty associated with frequent request/response transactions. The definition of the dynamically defined data identifier can either be done via a single request message or via multiple request messages. This allows for the definition of a single data element referencing source identifier(s) and memory addresses. The server has to concatenate the definitions for the single data element. A redefinition of a dynamically defined data identifier can be achieved by clearing the current definition and start over with the new definition.

Although this service does not prohibit such functionality, it is not recommended for the client to reference one dynamically defined data record from another, because deletion of the referenced record could create data consistency problems within the referencing record. This service also provides the ability to clear an existing dynamically defined data record. Requests to clear a data record shall be positively responded to if the specified data record identifier is within the range of valid dynamic data identifiers supported by the server (see ISO 14229-1.2 annex C.1 for more details).

The server shall maintain the dynamically defined data record until it is cleared or as specified by the vehicle manufacturer (e.g., deletion of dynamically defined data records upon session transition or upon power down of the server).

The server can implement data records in two different ways:

- 1) Composite data records containing multiple elemental data records which are not individually referenced.
- 2) Unique 2-byte identification "tag" or parameter identifier (PID) value for individual, elemental data records supported within the server (an example elemental data record, or PID, is engine speed or intake air temperature). This implementation of data records is a subset of a composite data record implementation, because it only references a single elemental data record instead of a data record including multiple elemental data records.

Both types of implementing data records are supported by the DynamicallyDefineDataIdentifier service to define a dynamic data identifier.

- 1) Composite block of data: The position parameter has to reference the starting point in the composite block of data and the size parameter has to reflect the length of data to be placed in the dynamically defined data identifier. The tester is responsible to not include only a portion of an elemental data record of the composite block of data in the dynamic data record.
- 2) 2-byte PID: The position parameter has to be set to one (1) and the size parameter has to reflect the length of the PID (length of the elemental data record). The tester is responsible to not include only a portion of the 2-byte PID value in the dynamic data record.

The ordering of the data within the dynamically defined data record shall be of the same order as it was specified in the client request message(s). Also, first position of the data specified in the client's request

shall be oriented such that it occurs closest to the beginning of the dynamic data record, in accordance with the ordering requirement mentioned in the preceding sentence.

In addition to the definition of a dynamic data identifier via a logical reference (a record data identifier) this service provides the capability to define a dynamically defined data identifier via an absolute memory address and a memory length information. This mechanism of defining a dynamic data identifier is recommended to be used only during the development phase of a server.

## 5.5.1 Request message

| Data Byte      | Parameter Name                                                        | Cvt      | Hex Value    | Mnemonic           |
|----------------|-----------------------------------------------------------------------|----------|--------------|--------------------|
| #1d            | <b>DynamicallyDefinedDataIdentifier Request Service ID</b>            | <b>M</b> | <b>#2Ch</b>  | <b>DDDI</b>        |
| #2d            | sub-function = [ defineByIdentifier ]                                 | M        | #01h         | LEV_<br>DBID       |
| #3d<br>#4d     | dynamicallyDefinedDataIdentifier[] = [ byte#1 (MSB)<br>byte#2 (LSB) ] | M<br>M   | #XXh<br>#XXh | DDDDI_<br>HB<br>LB |
| #5d<br>#6d     | sourceDataIdentifier[] #1 = [ byte#1 (MSB)<br>byte#2 (LSB) ]          | M<br>M   | #XXh<br>#XXh | SDI_<br>HB<br>LB   |
| #7d            | positionInSourceDataRecord #1                                         | M        | #XXh         | PISDR#1            |
| #8d            | memorySize #1                                                         | M        | #XXh         | MS#1               |
| ..             | ..                                                                    | ..       | ..           | ..                 |
| #n-3d<br>#n-2d | sourceDataIdentifier[] #m = [ byte#1 (MSB)<br>byte#2 (LSB) ]          | U<br>U   | #XXh<br>#XXh | SDI_<br>HB<br>LB   |
| #n-1d          | positionInSourceDataRecord #m                                         | U        | #XXh         | PISDR#m            |
| #nd            | memorySize #m                                                         | U        | #XXh         | MS#m               |

positionInSourceDataRecord : This 1-byte parameter is used to specify the starting byte position of the except of the source data record to be included in the dynamic data record. A position of one(1) shall reference the first byte of the data record referenced by the sourceDataIdentifier

Table5-15. Request message definition – sub-function = defineByIdentifier

| Data Byte                        | Parameter Name                                                        | Cvt                | Hex Value                | Mnemonic                    |
|----------------------------------|-----------------------------------------------------------------------|--------------------|--------------------------|-----------------------------|
| #1d                              | <b>DynamicallyDefinedDataIdentifier Request Service ID</b>            | <b>M</b>           | <b>#2Ch</b>              | <b>DDDI</b>                 |
| #2d                              | sub-function = [ defineByMemoryAddress ]                              | M                  | #02h                     | LEV_<br>DBMA                |
| #3d<br>#4d                       | dynamicallyDefinedDataIdentifier[] = [ byte#1 (MSB)<br>byte#2 (LSB) ] | M<br>M             | #XXh<br>#XXh             | DDDDI_<br>HB<br>LB          |
| #5d                              | addressAndLengthFormatIdentifier                                      | M1                 | #XXh                     | ALFID                       |
| #6d<br>..<br>#(m-1)+6d           | memoryAddress[] = [ byte#1 (MSB)<br>..<br>byte#m ]                    | M<br>..<br>C       | #XXh<br>..<br>#XXh       | MA_<br>B1<br>..<br>Bm       |
| #m+6d<br>..<br>#m+6+(k-1)d<br>.. | memorySize[] = [ byte#1 (MSB)<br>..<br>byte#k ]                       | M<br>..<br>C<br>.. | #XXh<br>..<br>#XXh<br>.. | MS_<br>B1<br>..<br>Bk<br>.. |

|                            |                                                       |                |                    |                       |
|----------------------------|-------------------------------------------------------|----------------|--------------------|-----------------------|
| #n-k-(m-1)d<br>..<br>#n-kd | memoryAddress[] = [<br>byte#1 (MSB)<br>..<br>byte#m ] | U<br>..<br>U/C | #XXh<br>..<br>#XXh | MA_<br>B1<br>..<br>Bm |
| #n-(k-1)d<br>..<br>#nd     | memorySize[] = [<br>byte#1 (MSB)<br>..<br>byte#k ]    | U<br>..<br>U/C | #XXh<br>..<br>#XXh | MS_<br>B1<br>..<br>Bk |

M1 : The addressAndLengthFormatIdetifier parameter is only present once at the very beginning of the request message and defines the length of the address and length information for each memory location reference throughout the whole request message.

Table5-16. Request message definition – sub-function = defineByMemoryAddress

| Data Byte  | Parameter Name                                                           | Cvt      | Hex Value    | Mnemonic          |
|------------|--------------------------------------------------------------------------|----------|--------------|-------------------|
| #1d        | <b>DynamicallyDefinedDataIdentifier Request Service ID</b>               | <b>M</b> | <b>#2Ch</b>  | <b>DDDI</b>       |
| #2d        | sub-function = [<br>clearDynamicallyDefinedDataIdentifier ]              | M        | #03h         | LEV_<br>CDDID     |
| #3d<br>#4d | dynamicallyDefinedDataIdentifier[] = [<br>byte#1 (MSB)<br>byte#2 (LSB) ] | C<br>C   | #XXh<br>#XXh | DDDI_<br>HB<br>LB |

C : The present of this parameter requires the server to clear the dynamicallydefinedDataIdentifier included in byte#1 and byte#2. If the parameter is not present all dynamicallyDefinedDataIdentifier in the server shall be cleared.

Table5-17. Request message definition – sub-function = clearDynamicallyDefinedDataIdentifier

| Hex   | Description                          | Cvt | Mnemonic |
|-------|--------------------------------------|-----|----------|
| 00    | Reserved                             | M   |          |
| 01    | defineByIdentifier                   | U   | DBID     |
| 02    | defineByMemoryAddress                | U   | DBMA     |
| 03    | clearDynamicallyDefineDataIdentifier | U   | CDDDI    |
| 04~7F | Reserved                             | M   |          |

Table5-18. request message sub-function parameter definition

| Parameter                | Description                                                                                              | # of bytes | Resolution | Min value | Max value |
|--------------------------|----------------------------------------------------------------------------------------------------------|------------|------------|-----------|-----------|
| <b>P2CAN_SERVER_MAX</b>  | Default <b>P2CAN_SERVER_MAX</b> timing supported by the server for the activated diagnostic session      | 2          | 1ms        | 0ms       | 65535ms   |
| <b>P2*CAN_SERVER_MAX</b> | Enhanced (NRC 78h) <b>P2CAN_SERVER_MAX</b> supported by the server for the activated diagnostic session. | 2          | 10ms       | 0ms       | 655350ms  |

Table5-19. sessionParameterRecord definition

### 5.5.2 Positive response message

| Data Byte  | Parameter Name                                                            | Cvt      | Hex Value    | Mnemonic          |
|------------|---------------------------------------------------------------------------|----------|--------------|-------------------|
| #1d        | <b>DynamicallyDefineDataIdentifier Response Service ID</b>                | <b>M</b> | <b>#6Ch</b>  | <b>DDDI PR</b>    |
| #2d        | definitionType                                                            | M        | #XXh         | DM                |
| #3d<br>#4d | dynamicallyDefinedDataIdentifier [] = [<br>byte#1 (MSB)<br>byte#2 (LSB) ] | C<br>C   | #XXh<br>#XXh | DDDI_<br>HB<br>LB |

Table5-20. Positive Response Message definition

### 5.5.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | This response code shall sent if the sub-function parameter is not supported.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 22  | conditionNotCorrect                   | M   | This response code shall be sent if the operating conditions of the server are not met to perform the required action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 31  | requestOutOfRange                     | M   | This response code shall be sent if ;<br>1. Any data identifier (dynamicallyDefinedData Identifier or any sourceDataIdentifier) in the request message is not supported/invalid.<br>2. The positionSourceDataRecord was incorrect (less than 1, or greater than maximum allowed by server)<br>3. Any memory address in the request message is not supported in the server<br>4. The specified memorySize was invalid.<br>5. The amount of data to be packed into the dynamic data exceeds the maximum allowed by server.<br>6. The specified addressAndLengthFormat Identifier is not valid. |
| 33  | securityAccessDenied                  | M   | This code shall be sent if;<br>1. any data identifier (dynamicallyDefinedData Identifier) in the request message is secured and the server is not in an unlocked state.<br>2. any memory address in the request message is secured and the server is not in unlocked state.                                                                                                                                                                                                                                                                                                                  |

Table5-21. Supported Negative Response Codes

## 5.6 WriteDataByIdentifier (2E hex) service

The WriteDataByIdentifier service allows the client to write information into the server at an internal location specified by the provided data identifier.

Dynamically defined dataIdentifier(s) shall not be used with this service.

### 5.6.1 Request message

| Data Byte | Parameter Name                                  | Cvt      | Hex Value   | Mnemonic        |
|-----------|-------------------------------------------------|----------|-------------|-----------------|
| #1d       | <b>WriteDataByIdentifier Request Service ID</b> | <b>M</b> | <b>#2Eh</b> | <b>WDBI</b>     |
| #2d       | dataIdentifier = [<br>byte#1 (MSB)<br>byte#2 ]  | M        | #XXh        | DID_<br>HB      |
| #3d       |                                                 | M        | #XXh        | LB              |
| #4d       | dataRecord = [<br>data#1<br>..<br>data#m ]      | M        | #XXh        | DREC_<br>DATA_1 |
| ..        |                                                 | ..       | ..          | ..              |
| #m+3d     |                                                 | U        | #XXh        | DATA_m          |

dataIdentifier : see Annex C.1

Table5-23. Request Message

### 5.6.2 Positive response message

| Data Byte | Parameter Name                                   | Cvt      | Hex Value   | Mnemonic      |
|-----------|--------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>WriteDataByIdentifier Response Service ID</b> | <b>M</b> | <b>#6Eh</b> | <b>WDBIPR</b> |
| #2d       | dataIdentifier = [<br>byte#1 (MSB)<br>byte#2 ]   | M        | #XXh        | DID_<br>HB    |
| #3d       |                                                  | M        | #XXh        | LB            |

Table5-23. Positive Response Message definition

### 5.6.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                |
|-----|---------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                                                                                                                                 |
| 22  | conditionNotCorrect                   | M   | This response code shall be sent if the operating condition of the server are not met to perform the required action.                                                                                                                                                                                                              |
| 31  | requestOutOfRange                     | M   | This response code shall be sent if<br>1. The dataIdentifier in the request message is not supported in the server or the dataIdentifier is supported for read only purpose (via ReadDataByIdentifier service).<br>2. Any data transmitted in the request message after the dataIdentifier is invalid (if applicable to the node). |
| 33  | securityAccessDenied                  | M   | This code shall be sent if the dataIdentifier, which reference a specific address, is secured and the server is not in an unlocked state.                                                                                                                                                                                          |
| 72  | generalProgrammingFailure             | M   | This return code shall be sent if the server detects an error when writing to a memory location.                                                                                                                                                                                                                                   |

Table5-24. Supported Negative Response Codes

## 5.7 WriteMemoryByAddress (3D hex) service

The WriteMemoryByAddress service allows the client to write information into the server at one or more contiguous memory locations.

The WriteMemoryByAddress request message writes information specified by the parameter dataRecord[] into the server at memory locations specified by parameters memoryAddress and memorySize. It is also possible to use a fixed addressAndLengthFormatIdentifier and unused bytes within the memoryAddress or memorySize parameter are padded with the value 00 hex in the higher range address locations.

### 5.7.1 Request message

| Data Byte     | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic        |
|---------------|-------------------------------------------------------|----------|-------------|-----------------|
| #1d           | <b>WriteMemoryByAddress Request Service ID</b>        | <b>M</b> | <b>#3Dh</b> | <b>WMBA</b>     |
| #2d           | addressAndLengthFormatIdentifier                      | M        | #XXh        | ALFID           |
| #3d           | memoryAddress[] = [<br>byte#1 (MSB)<br>..<br>byte#m ] | M        | #XXh        | MA_<br>B1       |
| ..<br>#m+2d   |                                                       | ..<br>C1 | ..<br>#XXh  | ..<br>Bm        |
| #n-r-2-(k-1)d | memorySize[] = [<br>byte#1 (MSB)<br>..<br>byte#k ]    | M        | #XXh        | MS_<br>B1       |
| ..<br>#n-r-2d |                                                       | ..<br>C2 | ..<br>#XXh  | ..<br>Bk        |
| #n-(r-1)d     | dataRecord[] = [<br>data#1<br>..<br>data#r ]          | M        | #XXh        | DREC_<br>DATA_1 |
| ..<br>#nd     |                                                       | ..<br>U  | ..<br>#XXh  | ..<br>DATA_r    |

C1: The presence of this parameter depends on address length information parameter of the addressAndLengthFormatIdentifier.  
C2: The presence of this parameter depends on the memory size length information of the addressAndLengthFormatIdentifier.

Table5-25. Request Message

### 5.7.2 Positive response message

| Data Byte      | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic      |
|----------------|-------------------------------------------------------|----------|-------------|---------------|
| #1d            | <b>WriteMemoryByAddress Response Service ID</b>       | <b>M</b> | <b>#7Dh</b> | <b>WMBAPR</b> |
| #2d            | addressAndLengthFormatIdentifier                      | M        | #XXh        | ALFID         |
| #3d            | memoryAddress[] = [<br>byte#1 (MSB)<br>..<br>byte#m ] | M        | #XXh        | MA_<br>B1     |
| ..<br>#(m-1)+3 |                                                       | ..<br>C1 | ..<br>#XXh  | ..<br>Bm      |
| #n-(k-1)d      | memorySize[] = [<br>byte#1 (MSB)<br>..<br>byte#k ]    | M        | #XXh        | MS_<br>B1     |
| ..<br>#nd      |                                                       | ..<br>C2 | ..<br>#XXh  | ..<br>Bk      |

C1: The presence of this parameter depends on address length information parameter of the addressAndLengthFormatIdentifier.  
C2: The presence of this parameter depends on the memory size length information of the addressAndLengthFormatIdentifier.

Table5-26. Positive Response Message definition

### 5.7.3 Supported negative response codes



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| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                                                |
|-----|---------------------------------------|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                                                                                                                                                                 |
| 22  | conditionNotCorrect                   | M   | This response code shall be sent if the operating condition of the server are not met to perform the required action.                                                                                                                                                                                                                                              |
| 31  | requestOutOfRange                     | M   | 1. Any memory address within the interval [\$MA, (\$MA + \$MS -\$1)] is invalid.<br>2. Any memory address within the interval [\$MA, (\$MA + \$MS -\$1)] is restricted.<br>3. The memorySize parameter value in the request message is greater than the maximum value supported by the server.<br>4. The specified addressAndLengthFormat Identifier is not valid. |
| 33  | securityAccessDenied                  | M   | This code shall be sent if any memory address within the interval [\$MA, (\$MA + \$MS -\$1)] is secure and the server is locked.                                                                                                                                                                                                                                   |
| 72  | generalProgrammingFailure             | M   | This return code shall be sent if the server detects an error when writing to a memory location.                                                                                                                                                                                                                                                                   |

Table5-27. Supported Negative Response Codes

## 6. Stored Data Transmission Functional Unit

| DTC High Byte |   |                 |   |                                   |   |   |   | DTC Middle Byte                   |   |   |   |                                   |   |   |   | DTC Low Byte         |   |   |   |                      |   |   |   |
|---------------|---|-----------------|---|-----------------------------------|---|---|---|-----------------------------------|---|---|---|-----------------------------------|---|---|---|----------------------|---|---|---|----------------------|---|---|---|
| 7             | 6 | 5               | 4 | 3                                 | 2 | 1 | 0 | 7                                 | 6 | 5 | 4 | 3                                 | 2 | 1 | 0 | 7                    | 6 | 5 | 4 | 3                    | 2 | 1 | 0 |
| 00 : P        |   | 1 <sup>st</sup> |   | 2 <sup>nd</sup> Character of Code |   |   |   | 3 <sup>rd</sup> Character of Code |   |   |   | 4 <sup>th</sup> Character of Code |   |   |   | DTC Failure Category |   |   |   | DTC Failure Sub Type |   |   |   |
| 01 : C        |   | Char.           |   |                                   |   |   |   |                                   |   |   |   |                                   |   |   |   |                      |   |   |   |                      |   |   |   |
| 10 : B        |   | of              |   |                                   |   |   |   |                                   |   |   |   |                                   |   |   |   |                      |   |   |   |                      |   |   |   |
| 11 : U        |   | Code            |   |                                   |   |   |   |                                   |   |   |   |                                   |   |   |   |                      |   |   |   |                      |   |   |   |

### Table6-1. 3 Byte Diagnostic Trouble Code Structure

Existing ISO documents for defining DTCs such as ISO 15031-6 currently define DTCs for existing systems. If a standard DTC is already defined for a component / system and that DTC description already comprehends the DTC Failure Type information, then the standard DTC number can be used and the DTCFailureTypeByte shall be set to a value of 00 hex. A DTCFailureTypeByte value of 00 hex indicates that no additional information is contained in the DTCFailureTypeByte.

The following example shows the three (3) principle combinations of DTC and DTCFailureTypeByte.

- DTC which does not require any additional description included in the DTCFailureTypeByte (no DTC Failure Category name and no DTC Failure Sub Type)  
 <e.g.> emissions-related DTC (012700 hex): P0127 Intake Air Temperature Too High,
- DTC which requires additional description included in the DTCFailureTypeByte (DTC Failure Category name and no DTC Failure Sub Type)  
 <e.g.> DTC (803901): B0039-01 Second Row Right Frontal Stage 1 Deployment Control – General Electrical Failure
- DTC, which requires additional description included in the DTCFailureTypeByte (DTC Failure Category name and DTC Failure Sub Type)  
 <e.g.> DTC (403123): C0031-23 Left Front Wheel Speed Sensor -General Signal Failure - Signal Stuck Low DTC Failure Category Definition

Refer to the ISO 15031-6 Annex. D “DTC Failure Category and Sub Type definition”, if you need more information.

### 6.1 ClearDiagnosticInformation (14 hex) service

The ClearDiagnosticInformation service is used by the client to clear diagnostic information in one or multiple servers' memory. The server shall send a positive response even if no DTC's are stored. If a server supports multiple copies of DTC status information in memory (e.g. one copy in RAM and one copy in EEPROM), the server shall clear the copy used by the ReadDTCInformation status reporting service. Additional copies, e.g. backup copy in long-term memory, are updated according to the appropriate backup strategy (e.g. in the power-latch phase).

The request message of the client contains 1 parameter. The parameter groupOfDTC allows the client to clear a group of DTC's (e.g. Powertrain, Body, Chassis, etc.), or a specific DTC. Unless otherwise stated, the server shall clear both emission-related and non emission-related DTC information from memory for the requested group.

DTC information reset / cleared via this service includes but is not limited to the following :

- DTC status byte
- Captured DTC snapshot data
- Captured DTC extended data
- Other DTC related data such as first/most recent DTC, flags, counters, timers, etc. specific to DTC's

Any DTC information stored in an optionally available DTC mirror memory in the server is not affected by this service.

Note) Mirror Memory : The DTC mirror memory is an additional optional error memory in the server that cannot be erased by the ClearDiagnosticInformation (14 hex) service. The DTC mirror memory mirrors the normal DTC memory and can be used for example if the normal error memory is erased. To clear Mirror Memory an according Routine Identifier shall be used.

#### 6.1.1 Request message

| Data Byte                  | Parameter Name                                                                        | Cvt | Hex Value | Mnemonic     |
|----------------------------|---------------------------------------------------------------------------------------|-----|-----------|--------------|
| #1d                        | ClearDiagnosticInformation Request Service ID                                         | M   | #14h      | CDTCI        |
| #2d                        | groupOfDTC[] = [<br>groupOfDTCHighByte<br>groupOfDTCMiddleByte<br>groupOfDTCLowByte ] | M   | #XXh      | GODTC_<br>HB |
| #3d                        |                                                                                       | M   | #XXh      | MB           |
| #4d                        |                                                                                       | M   | #XXh      | LB           |
| groupOfDTC : see Annex D.1 |                                                                                       |     |           |              |

Table6-2. Request Message

#### 6.1.2 Positive response message

| Data Byte | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic       |
|-----------|-------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>ClearDiagnosticInformation Response Service ID</b> | <b>M</b> | <b>#54h</b> | <b>CDTCIPR</b> |

Table6-3. Positive Response Message definition

#### 6.1.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                             |
|-----|---------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                              |
| 22  | conditionNotCorrect                   | M   | This response code shall be used if internal conditions within the server prevent the clearing of DTC related information stored in the server. |



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|    |                           |   |                                                                                              |
|----|---------------------------|---|----------------------------------------------------------------------------------------------|
| 31 | requestOutOfRange         | M | The return code shall be sent if the specified groupOfDTC parameter is not supported.        |
| 72 | generalProgrammingFailure | U | This NRC shall be returned if the server detects an error when writing to a memory location. |

Table6-4. Supported Negative Response Codes

### 6.1.4 Implementaion rule

The ECU behavior shall be consistent: The DTCs shall be cleared before the positive response is sent. This does not require that the non volatile memory is cleared, but any memory cache has to be.

With the parameter groupOfDTC it is also possible to clear only a specific group of DTCs (e.g. Powertrain) or a specific DTC.

A subsequent call to read out the fault memory shall not retrieve any DTC which has been set before the last execution of ClearDiagnosticInformation.

Even if no DTC was stored, the ECU shall return with a positive response.

There shall be no sequence dependency to any other service. Even if the fault memory was not read, it may be cleared.

A DTC can just be cleared by tester with service #14h.

## 6.2 ReadDTCInformation (19 hex) service

This service allows a client to read the status of server resident Diagnostic Trouble Code (DTC) information from any server, or group of servers within a vehicle. Unless otherwise stated, the server shall return both emission-related and non emission-related DTC information. This service allows the client to do the following :

- Retrieve the number of DTC's matching a client defined DTC status mask (at the point of the request),
- Retrieve the list of all DTC's matching a client defined DTC status mask,
- Retrieve DTCSnapshot data associated with a client defined DTC and status mask combination: DTC Snapshots are specific data records associated with a DTC, that are stored in the server's memory. The content of the DTC Snapshots is not defined by this standard, but typical usage of DTC Snapshots is to store data upon detection of a system malfunction. The DTC Snapshots will act as a snapshot of data values from the time of the system malfunction occurrence.
- Retrieve DTCExtendedData associated with a client defined DTC and status mask combination out of the DTC memory or the DTC mirror memory. DTC Extended Data consist of extended status information associated with a DTC. DTC Extended Data contains DTC parameter values, which have been identified at the time of the request. A typical use of DTC Extended Data is to store dynamic data associated with the DTC, e.g.
  - DTC occurrence counter,
  - current threshold values,
  - time of last occurrence (etc.).
  - fault validation counters (e.g. counts number of reported "test failed" and possible other counters if the validation is performed in several steps),
  - uncompleted test counters (e.g. counts numbers of driving cycles since the test was latest completed i.e. since the test reported "test passed" or "test failed"),
  - fault occurrence counters (e.g. counts number of driving cycles in which "testFailed" have been reported),
  - DTC aging counter (e.g. counts number of driving cycles since the fault was latest failed excluding the driving cycles in which the test haven't report "test pass" or the test report "test fail"),
  - specific counters for OBD (e.g. number of remaining driving cycles until the "check engine" lamp is switched off).
- Retrieve the number of DTC's matching a client defined severity mask (at the point of the request),
- Retrieve the list of DTC's matching a client defined severity mask record,
- Retrieve severity information for a client defined DTC,
- Retrieve the status of all DTC's supported by the server,
- Retrieve the first DTC failed by the server,
- Retrieve the most recently failed DTC within the server,
- Retrieve the first DTC confirmed by the server,
- Retrieve the most recently confirmed DTC within the server,
- Retrieve the list of DTC's out of the DTC mirror memory matching a client defined DTC status mask,
- Retrieve mirror memory DTCExtendedData record data for a client defined DTC mask and a client defined DTCExtendedData record number out of the DTC mirror memory,
- Retrieve the number of DTC's out of the DTC mirror memory matching a client defined DTC status mask,
- Retrieve the number of "only" emissions-related OBD DTC's matching a client defined DTC status mask. Emissions-related OBD DTC's cause the malfunction indicator to be turned on/display a message in case such DTC is detected,
- Retrieve the status of "only" emissions-related OBD DTC's matching a client defined DTC status mask. Emissions-related OBD DTC's cause the malfunction indicator to be turned on/display a message in case such DTC is detected,
- Retrieve all current "prefailed" DTCs which have not yet been detected as "pending" or "confirmed",
- Retrieve all DTCs with "permanentDTC" status. These DTCs have been previously cleared by the clearDiagnosticInformation service but remain in the non-volatile memory of the server until the appropriate monitors for each DTC have successfully passed.

This service uses a sub-function to determine which type of diagnostic information the client is requesting. Further details regarding each sub-function parameter are provided in the following sections.

This service makes use of the following terms:

- Enable Criteria: Server/vehicle manufacturer/system supplier specific criteria used to control when the server actually performs a particular internal diagnostic.
- Test Pass Criteria: Server/vehicle manufacturer/system supplier specific conditions, that define, whether a system being diagnosed is functioning properly within normal, acceptable operating ranges (e.g. no failures exist and the diagnosed system is classified as “OK”).
- Test Failure Criteria: Server/vehicle manufacturer/system supplier specific failure conditions, that define, whether a system being diagnosed has failed the test.
- Confirmed Failure Criteria: Server/vehicle manufacturer/system supplier specific failure conditions that define whether the system being diagnosed is definitively problematic (confirmed), warranting storage of the DTC record in long term memory.
- Occurrence Counter: A counter maintained by certain servers that records the number of instances in which a given DTC test reported a unique occurrence of a test failure.
- Aging: A process whereby certain servers evaluate past results of each internal diagnostic to determine if a confirmed DTC can be cleared from long-term memory, e.g. in the event of a calibrated number of failure free cycles. (see ISO 14229-1.2 “10.3 ReadDTCInformation Service” for more details)

### 6.2.1 Request message

| Data Byte                     | Parameter Name                                                             | Cvt      | Hex Value    | Mnemonic                   |
|-------------------------------|----------------------------------------------------------------------------|----------|--------------|----------------------------|
| #1d                           | <b>ReadDTCInformation Request Service ID</b>                               | <b>M</b> | <b>#19h</b>  | <b>RDTCI</b>               |
| #2d                           | Sub-function = [<br>reportNumberOfDTCByStatusMask<br>reportDTCByStatusMask | M        | #01h<br>#02h | LEV_<br>RNOTCBSM<br>RDTCSM |
| #3d                           | DTCStatusMask                                                              | M        | #XXh         | DTCSM                      |
| DTCStatusMask : see Annex D.2 |                                                                            |          |              |                            |

Table6-5. Request Message definition – sub-function = #01h / 02h

| Data Byte                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Parameter Name                                                                              | Cvt      | Hex Value    | Mnemonic                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|----------|--------------|------------------------------|
| #1d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>ReadDTCInformation Request Service ID</b>                                                | <b>M</b> | <b>#19h</b>  | <b>RDTCI</b>                 |
| #2d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Sub-function = [<br>reportDTCSnapshotIdentification<br>reportDTCSnapshotRecordByDTCNumber ] | M        | #03h<br>#04h | LEV_<br>RDTCSSI<br>RDTCSBDBC |
| #3d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | DTCMaskRecord[] = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte ]                         | C        | #XXh         | DTCMREC_<br>DTCHB            |
| #4d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             | C        | #XXh         | DTCMB                        |
| #5d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                             | C        | #XXh         | DTCLB                        |
| #6d                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | DTCSnapshotRecordNumber                                                                     | C        | #XXh         | DTCSSRN                      |
| DTCSnapshotRecordNumber : DTCSnapshotRecordNumber is a 1-byte value indicating the number of the specific DTCSnapshot data record requested for a client defined DTCMaskRecord via the reportDTCSnapshotByDTCNumber / reportDTCSnapshotByRecordNumber sub-functions. For emissions-related servers (OBD compliant ECUs) the DTCSnapshot data record number 00 hex shall be the equivalent data record as specified in ISO 15031-5 service 02 hex frame number 00 hex. If the server supports multiple DTCSnapshot data records the range of 01 hex through FE hex shall be used. A value of FF hex requests the server to report all stored DTCSnapshot data records at once. |                                                                                             |          |              |                              |

Table6-6. Request Message definition – sub-function = #03/ 04h

| Data Byte | Parameter Name                               | Cvt      | Hex Value   | Mnemonic     |
|-----------|----------------------------------------------|----------|-------------|--------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b> | <b>M</b> | <b>#19h</b> | <b>RDTCI</b> |

|     |                                                             |   |      |                  |
|-----|-------------------------------------------------------------|---|------|------------------|
| #2d | Sub-function = [<br>reportDTCSnapshotRecordByRecordNumber ] | M | #05h | LEV_<br>RDTCSBRN |
| #3d | DTCSnapshotRecordNumber                                     | M | #XXh | DTCSSRN          |

Table6-7. Request Message definition – sub-function = #05h

| Data Byte | Parameter Name                                                      | Cvt      | Hex Value   | Mnemonic           |
|-----------|---------------------------------------------------------------------|----------|-------------|--------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>                        | <b>M</b> | <b>#19h</b> | <b>RDTCI</b>       |
| #2d       | Sub-function = [<br>reportDTCEntendedDataRecordByDTCNumber]         | M        | #06h        | LEV_<br>RDTCEDRBDN |
| #3d       | DTCMaskRecord[] = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte ] | M        | #XXh        | DTCMREC_<br>DTCHB  |
| #4d       |                                                                     | M        | #XXh        | DTCMB              |
| #5d       |                                                                     | M        | #XXh        | DTCLB              |
| #6d       | DTCEntendedDataRecordNumber                                         | M        | #XXh        | DTCEDRN            |

DTCEntendedDataRecordNumber : DTCEntendedDataRecordNumber is a 1-byte value indicating the number of the specific DTCEntendedData record requested for a client defined DTCMaskRecord via the reportDTCEntendedDataRecordByDTCNumber sub-function. For emissions-related servers (OBD compliant ECUs) the DTCEntendedDataRecordNumber 00 hex shall be reserved for future OBD use.

Table6-8. Request Message definition – sub-function = #06

| Hex   | Description                                                                                   | Cvt             |
|-------|-----------------------------------------------------------------------------------------------|-----------------|
| 00    | Reserved (for future OBD use)                                                                 | M               |
| 01~8F | Vehicle Manufacturer Specific Stored DTCEntendedData Records                                  | (To be defined) |
| 90~EF | Reserved (for Legislated OBD Stored DTCEntendedData Records)                                  | M               |
| F0~FD | Reserved (for future)                                                                         | M               |
| FE    | Reserved (for All Legislated OBD Stored DTCEntendedData Records in a Single Response Message) | M               |
| FF    | All Stored DTCEntendedData Records in a Single Response Message                               | M               |

Table6-9. DTCEntendedDataRecordNumber definition

| Data Byte | Parameter Name                                                                                                                                                                                                                    | Cvt      | Hex Value                                            | Mnemonic                                                                            |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>                                                                                                                                                                                      | <b>M</b> | <b>#19h</b>                                          | <b>RDTCI</b>                                                                        |
| #2d       | Sub-Function = [<br>reportSupportedDTC<br>reportFirstTetFailedDTC<br>reportFirstConfirmedDTC<br>reportMostRecentTestFailedDTC<br>reportMostRecentConfirmedDTC<br>reportDTCFaultDetectionCounter<br>reportDTCWithPermanentStatus ] | M        | #0Ah<br>#0Bh<br>#0Ch<br>#0Dh<br>#0Eh<br>#14h<br>#15h | LEV_<br>RSUPDTC<br>RFTFDTC<br>RFCDDTC<br>RMRTFDTC<br>RMRCDDTC<br>RDTCFDC<br>RDTCWPS |

Table6-10. Request Message definition – sub-function = #0A~15

| Data Byte | Parameter Name                                                                               | Cvt      | Hex Value    | Mnemonic                       |
|-----------|----------------------------------------------------------------------------------------------|----------|--------------|--------------------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>                                                 | <b>M</b> | <b>#19h</b>  | <b>RDTCI</b>                   |
| #2d       | Sub-function = [<br>reportNumberOfDTCBySeverityMaskRecord<br>reportDTCBySeverityMaskRecord ] | M<br>M   | #07h<br>#08h | LEV_<br>RNODTCBSMR<br>RDTCBSMR |
| #3d       | DTCSeverityMaskRecord[] = [<br>DTCSeverityMask<br>DTCStatusMask ]                            | M        | #XXh         | DTCSVMREC_<br>DTCSVM           |
| #4d       |                                                                                              | M        | #XXh         | DTCSM                          |

DTCSeverityMask, DTCSeverity : See Annex D.3

Table6-11. Request Message definition – sub-function = #07 / 08h

| Data Byte | Parameter Name                                                      | Cvt      | Hex Value   | Mnemonic          |
|-----------|---------------------------------------------------------------------|----------|-------------|-------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>                        | <b>M</b> | <b>#19h</b> | <b>RDTCI</b>      |
| #2d       | Sub-function = [<br>reportSeverityInformationOfDTC ]                | M        | #09h        | LEV_<br>RSIODTC   |
| #3d       | DTCMaskRecord[] = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte ] | M        | #XXh        | DTCMREC_<br>DTCHB |
| #4d       |                                                                     | M        | #XXh        | DTCMB             |
| #5d       |                                                                     | M        | #XXh        | DTCCLB            |

Table6-12. Request Message definition – sub-function = #09h

| Data Byte | Parameter Name                                       | Cvt      | Hex Value   | Mnemonic        |
|-----------|------------------------------------------------------|----------|-------------|-----------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>         | <b>M</b> | <b>#19h</b> | <b>RDTCI</b>    |
| #2d       | Sub-function = [<br>reportSupportedDTCExtDataRecord] | M        | #1Ah        | LEV_<br>RDTCEDI |
| #3d       | DTCExtDataRecordNumber                               | M        | #01~FDh     | DTCEDRN         |

Table6-23. Request Message definition – sub-function = #1Ah

| Data Byte | Parameter Name                                                    | Cvt      | Hex Value   | Mnemonic            |
|-----------|-------------------------------------------------------------------|----------|-------------|---------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>                      | <b>M</b> | <b>#19h</b> | <b>RDTCI</b>        |
| #2d       | Sub-function = [<br>reportWWHOBDDTCByMaskRecord]                  | M        | #42h        | LEV_<br>ROBDDTCBMR  |
| #3d       | FunctionalGroupIdentifier                                         | M        | #XX         | FGID                |
| #4d       | DTCSeverityMaskRecord[] = [<br>DTCStatusMask<br>DTCSeverityMask ] | M        | #XX         | DTCSVMREC_<br>DTCSM |
| #5d       |                                                                   |          | #XX         | DTCSVM              |

Table6-24. Request Message definition – sub-function = #42h

| Hex   | Description              | Cvt | Mnemonic     |
|-------|--------------------------|-----|--------------|
| 00~32 | ISO/SAE reserved         | M   | ISOSAERESRVD |
| 33    | Emissions-system group   | M   | EMSYSGRP     |
| 34~CF | ISO/SAE reserved         | M   | ISOSAERESRVD |
| D0    | Safety-system group      | M   | SAFESYSGRP   |
| D1~DF | Legislative system group | M   | LEGSYSGRP    |
| E0~FD | ISO/SAE reserved         | M   | ISOSAERESRVD |
| FE    | VOBD system              | M   | VOBDSYSGRP   |
| FF    | ISO/SAE reserved         | M   | ISOSAERESRVD |

Table6-25. FunctionalGroupIdentifiers definition

| Data Byte | Parameter Name                                          | Cvt      | Hex Value   | Mnemonic                  |
|-----------|---------------------------------------------------------|----------|-------------|---------------------------|
| #1d       | <b>ReadDTCInformation Request Service ID</b>            | <b>M</b> | <b>#19h</b> | <b>RDTCI</b>              |
| #2d       | Sub-function = [<br>reportWWHOBDDTCWithPermanentStatus] | M        | #55h        | LEV_<br>RWWHOBDDT<br>CWPS |
| #3d       | FunctionalGroupIdentifier                               | M        | #XXh        | FGID                      |

Table6-26. Request Message definition – sub-function = #55h

| Data Byte | Parameter Name | Cvt | Hex Value | Mnemonic |
|-----------|----------------|-----|-----------|----------|
|-----------|----------------|-----|-----------|----------|

| #1d | ReadDTCInformation Request Service ID                                  | M | #19h | RDTCl            |
|-----|------------------------------------------------------------------------|---|------|------------------|
| #2d | Sub-function = [<br>reportDTCInformationByDTCReadinessGroupIdentifier] | M | #56h | LEV_<br>RDTCBrgI |
| #3d | FunctionalGroupIdentifier                                              | M | #XX  | FGID             |
| #4d | DTCReadinessGroupIdentifier                                            | M | #XX  | DTCRGI           |

Table6-27. Request Message definition – sub-function = #56h

| Hex | Description                                                                                                                                                                                                                                                                                                                                                                                                                          | Cvt | Mnemonic    |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-------------|
| 00  | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                      | M   |             |
| 01  | <b>reportNumberOfDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client the number of DTC's matching a client defined status mask.                                                                                                                                                                                                                                                             | M   | RNODTCBSM   |
| 02  | <b>reportDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client a list of DTC's and corresponding statuses matching a client defined status mask.                                                                                                                                                                                                                                              | M   | RDTCBsm     |
| 03  | <b>reportDTCSnapshotIdentification</b><br>This parameter specifies that the server shall transmit to the client all DTCSnapshot data record identifications (DTC number(s) and DTCSnapshot record number(s)).                                                                                                                                                                                                                        | U   | RDTCCSI     |
| 04  | <b>reportDTCSnapshotRecordByDTCNumber</b><br>This parameter specifies that the server shall transmit to the client the DTCSnapshot record(s) associated with a client defined DTC number and DTCSnapshot record number (FF hex for all records).                                                                                                                                                                                     | U   | RDTCCSBDTC  |
| 05  | <b>reportDTCSnapshotRecordByRecordNumber</b><br>This parameter specifies that the server shall transmit to the client the DTCSnapshot record(s) associated with a client defined DTCSnapshot record number (FF hex for all records). Note that this sub-function parameter can only be supported if the DTCSnapshotRecordNumber is unique to the server (each record number exists only once in the server) and not unique to a DTC. | U   | RDTCCSBRN   |
| 06  | <b>reportDTCExtendedDataRecordByDTCNumber</b><br>This parameter specifies that the server shall transmit to the client the DTCExtendedData record(s) associated with a client defined DTC number and DTCExtendedData record number (FF hex for all records, FE hex for all OBD records).                                                                                                                                             | U   | RDTCCEDRBDN |
| 07  | <b>reportNumberOfDTCBySeverityMaskRecord</b><br>This parameter specifies that the server shall transmit to the client the number of DTC's matching a client defined severity mask record.                                                                                                                                                                                                                                            | U   | RNODTCBSMR  |
| 08  | <b>reportDTCBySeverityMaskRecord</b><br>This parameter specifies that the server shall transmit to the client a list of DTC's and corresponding statuses matching a client defined severity mask record.                                                                                                                                                                                                                             | U   | RDTCBSMR    |
| 09  | <b>reportSeverityInformationOfDTC</b><br>This parameter specifies that the server shall transmit to the client the severity information of a specific DTC specified in the client request message.                                                                                                                                                                                                                                   | U   | RSIODTC     |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |   |             |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|-------------|
| 0A | <b>reportSupportedDTC</b><br>This parameter specifies that the server shall transmit to the client a list of all DTC's and corresponding statuses supported within the server.                                                                                                                                                                                                                                                                                                                                                                             | M | RSUPDTC     |
| 0B | <b>reportFirstTestFailedDTC</b><br>This parameter specifies that the server shall transmit to the client the first failed DTC to be detected by the server since the last clear of diagnostic information. Note that the information reported via this sub-function parameter shall be independent of whether or not the DTC was confirmed or aged.                                                                                                                                                                                                        | U | RFTFDTC     |
| 0C | <b>reportFirstConfirmedDTC</b><br>This parameter specifies that the server shall transmit to the client the first confirmed DTC to be detected by the server since the last clear of diagnostic information. The information reported via this sub-function parameter shall be independent of the aging process of confirmed DTC's (e.g. if a DTC ages such that its status is allowed to be reset, the first confirmed DTC record shall continue to be preserved by the server, regardless of any other DTC's that become confirmed afterwards).          | U | DFCDTC      |
| 0D | <b>reportMostRecentTestFailedDTC</b><br>This parameter specifies that the server shall transmit to the client the most recent failed DTC to be detected by the server since the last clear of diagnostic information. Note that the information reported via this sub-function parameter shall be independent of whether or not the DTC was confirmed or aged.                                                                                                                                                                                             | U | RMRVDTC     |
| 0E | <b>reportMostRecentConfirmedDTC</b><br>This parameter specifies that the server shall transmit to the client the most recent confirmed DTC to be detected by the server since the last clear of diagnostic information. Note that the information reported via this sub-function parameter shall be independent of the aging process of confirmed DTC's (e.g. if a DTC ages such that its status is allowed to be reset, the first confirmed DTC record shall continue to be preserved by the server assuming no other DTC's become confirmed afterwards). | U | RMRC DTC    |
| 0F | <b>reportMirrorMemoryDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client a list of DTC's out of the DTC mirror memory and corresponding statuses matching a client defined status mask.                                                                                                                                                                                                                                                                                                                           | U | RMMDTCBSM   |
| 10 | <b>reportMirrorMemoryDTCExtendedDataRecordByDTCNumber</b><br>This parameter specifies that the server shall transmit to the client the DTCExtendedData record(s) - out of the DTC mirror memory - associated with a client defined DTC number and DTCExtendedData record number (FF hex for all records) DTC's.                                                                                                                                                                                                                                            | U | RMMDED RBDN |
| 11 | <b>reportNumberOfMirrorMemoryDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client the number of DTC's out of mirror memory matching a client defined status mask.                                                                                                                                                                                                                                                                                                                                                  | U | RNOMMDTCBSM |

|       |                                                                                                                                                                                                                                                                                                                                                                    |   |               |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---------------|
| 12    | <b>reportNumberOfEmissionsRelatedOBDDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client the number of emissionsrelated OBD DTC's matching a client defined status mask. The number of OBD DTC's reported shall only be those which are required to be compatible with emissions-related legal requirements.               | U | RNOOBDDTCBSM  |
| 13    | <b>reportEmissionsRelatedOBDDTCByStatusMask</b><br>This parameter specifies that the server shall transmit to the client a list of emissions-related OBD DTC's and corresponding statuses matching a client defined status mask. The list of OBD DTC's reported shall only be those which are required to be compatible with emissions-related legal requirements. | U | ROBDDTCBSM    |
| 14~19 | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                    | U | RESRVD        |
| 1A    | <b>reportDTCExtendedDataRecordIdentification</b><br>This parameter specifies that the server shall transmit to the client the DTCs which supports a DTCExtendedDataRecord.                                                                                                                                                                                         | U |               |
| 1B~41 | <b>ISOSAEReserved</b>                                                                                                                                                                                                                                                                                                                                              | M | ISOSAERESRVD  |
| 42    | <b>reportWWHOBDDTCByMaskRecord</b><br>This parameter specifies that the server shall transmit to the client a list of WWH OBD DTCs and corresponding status and severity information matching a client defined status mask and severity mask record.                                                                                                               | U | RWWHOBDDTCBMR |
| 43~54 | <b>ISOSAEReserved</b>                                                                                                                                                                                                                                                                                                                                              | M | ISOSAERESRVD  |
| 55    | <b>reportWWHOBDDTCWithPermanentStatus</b><br>This parameter specifies that the server shall transmit to the client a list of WWH OBD DTCs with "permanent DTC" status                                                                                                                                                                                              | U | RWWHOBDDTCWPS |
| 56    | <b>reportDTCInformationByDTCReadinessGroupIdentifier</b><br>This parameter specifies that the server shall transmit to the client a list of OBD DTCs which matches the DTCReadiness Group Identifier.                                                                                                                                                              | U | RDTCBRGI      |
| 57~7F | <b>ISOSAEReserved</b>                                                                                                                                                                                                                                                                                                                                              | M | ISOSAERESRVD  |

Table6-13. Request message sub-function definition

### 6.2.2 Positive response message

| Data Byte | Parameter Name                                                                                                                                                                             | Cvt      | Hex Value                    | Mnemonic                                                       |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------|----------------------------------------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                                                                                                                                              | <b>M</b> | <b>#59h</b>                  | <b>RDTCIPLR</b>                                                |
| #2d       | reportType = [<br>reportNumberOfDTCByStatusMask<br>reportNumberOfDTCBySeverityMaskRecord<br>reportNumberOfMirrorMemoryDTCByStatusMask<br>reportNumberOfEmissionRelatedOBDDTCByStatusMask ] | M        | #01h<br>#07h<br>#11h<br>#12h | LEV_<br>RNODTCBSM<br>RNODTCBSMR<br>RNOMMDTCBSM<br>RNOOBDDTCBSM |
| #3d       | DTCStatusAvailabilityMask                                                                                                                                                                  | M        | #XXh                         | DTCSAM                                                         |
| #4d       | DTCFormatIdentifier = [<br>SAE_J2012-DA_DTCFormat_04]                                                                                                                                      | M        | #04h                         | DTCFID_<br>SAE_J2012-DA DTC<br>Format_04                       |

|     |                                                            |   |      |                 |
|-----|------------------------------------------------------------|---|------|-----------------|
| #5d | DTCCCount[] = [<br>DTCCCountHighByte<br>DTCCCountLowByte ] | M | #XXh | DTCC_<br>DTCCHB |
| #6d |                                                            | M | #XXh | DTCC_<br>DTCCLB |

Table6-14. Positive Response Message definition – sub-function = #01 / 07 / 11 / 12h

| Hex   | Description                | Mnemonic       |
|-------|----------------------------|----------------|
| 00    | SAE_J2012-DA_DTCFormat_00  | J2012-DADTCF00 |
| 01    | ISO 14229-1 DTC Format     | 14229-1DTCF    |
| 02    | SAE J1939-73 DTC Format    | J1939-73DTCF   |
| 03    | ISO 11992-4 DTC Format     | 11992-4DTCF    |
| 04    | SAE_J2012-DA DTC Format_04 | J2012-DADTCF04 |
| 05~FF | Reserved                   |                |

Table6-15. DTCFormat Identifier definition

| Data Byte | Parameter Name                                                                                                                                                                                                                                | Cvt      | Hex Value                                            | Mnemonic                                                                            |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------|-------------------------------------------------------------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                                                                                                                                                                                                 | <b>M</b> | <b>#59h</b>                                          | <b>RDTCIPLR</b>                                                                     |
| #2d       | reportType = [<br>reportDTCByStatusMask<br>reportSupportedDTCs<br>reportFirstTestFailedDTC<br>reportFirstConfirmedDTC<br>reportMostRecentTestFailedDTC<br>reportMostRecentConfirmedDTC<br>reportDTCWithPermanentStatus]                       | M        | #02h<br>#0Ah<br>#0Bh<br>#0Ch<br>#0Dh<br>#0Eh<br>#15h | LEV_<br>RDTCBMS<br>RSUPDTC<br>RFTFDTC<br>RFCDDTC<br>RMRTFDTC<br>RMRCDDTC<br>RDTCWPS |
| #3d       | DTCStatusAvailabilityMask                                                                                                                                                                                                                     | M        | #XXh                                                 | DTCSAM                                                                              |
| #4d       | DTCAndStatusRecord[] = [<br>DTCHighByte#1<br>DTCMiddleByte#1<br>DTCLowByte#1<br>statusOfDTC#1<br>DTCHighByte#2<br>DTCMiddleByte#2<br>DTCLowByte#2<br>statusOfDTC#2<br>..<br>DTCHighByte#m<br>DTCMiddleByte#m<br>DTCLowByte#m<br>statusOfDTC#m | C        | #XXh                                                 | DTCASR_<br>DTCHB                                                                    |
| #5d       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB                                                                      |
| #6d       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB                                                             |
| #7d       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #8d       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #9d       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #10d      |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #11d      |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| ..        |                                                                                                                                                                                                                                               | ..       | ..                                                   | ..                                                                                  |
| #n-3d     |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #n-2d     |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #n-1d     |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |
| #nd       |                                                                                                                                                                                                                                               | C        | #XXh                                                 | DTCHB<br>DTCMB<br>DTCLB<br>SODTC                                                    |

Table6-16. Positive Response Message definition – sub-function = #02 / 0A~0E / 15

| Data Byte | Parameter Name                                                         | Cvt      | Hex Value   | Mnemonic                |
|-----------|------------------------------------------------------------------------|----------|-------------|-------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                          | <b>M</b> | <b>#59h</b> | <b>RDTCIPLR</b>         |
| #2d       | reportType = [<br>reportDTCSnapshotIdentification ]                    | M        | #03h        | LEV_<br>RDTCCSI         |
| #3d       | DTCRecord[] #1 = [<br>DTCHighByte#1<br>DTCMiddleByte#1<br>DTCLowByte#1 | C        | #XXh        | DTCASR_<br>DTCHB        |
| #4d       |                                                                        | C        | #XXh        | DTCHB<br>DTCMB          |
| #5d       |                                                                        | C        | #XXh        | DTCHB<br>DTCMB<br>DTCLB |
| #6d       | DTCSnapshotRecordNumber #1                                             | C        | #XXh        | DTCCSRN                 |
| ..        | ..                                                                     | ..       | ..          | ..                      |

|      |                                                                          |   |      |         |
|------|--------------------------------------------------------------------------|---|------|---------|
| #9d  | DTCRecord[] #m = [<br>DTCHighByte#m<br>DTCMiddleByte#m<br>DTCLowByte#m ] | C | #XXh | DTCASR_ |
| #10d |                                                                          | C | #XXh | DTCHB   |
| #11d |                                                                          | C | #XXh | DTCMB   |
| #nd  | DTCSnapshotRecordNumber #m                                               | C | #XXh | DTCLB   |
|      |                                                                          |   |      | DTCSSRN |

Table6-17. Positive Response Message definition – sub-function = #03h

| Data Byte     | Parameter Name                                                                                                                                                                                                                                                                         | Cvt      | Hex Value   | Mnemonic           |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------------|--------------------|
| #1d           | <b>ReadDTCInformation Response Service ID</b>                                                                                                                                                                                                                                          | <b>M</b> | <b>#59h</b> | <b>RDTICIPR</b>    |
| #2d           | reportType = [<br>reportDTCSnapshotRecordByDTCNumber ]                                                                                                                                                                                                                                 | M        | #04h        | LEV_<br>RDTCSSBDTC |
| #3d           | DTCAndStatusRecord[] = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte<br>statusOfDTC ]                                                                                                                                                                                                | M        | #XXh        | DTCASR_            |
| #4d           |                                                                                                                                                                                                                                                                                        | M        | #XXh        | DTCHB              |
| #5d           |                                                                                                                                                                                                                                                                                        | M        | #XXh        | DTCMB              |
| #6d           |                                                                                                                                                                                                                                                                                        | M        | #XXh        | DTCLB              |
| #7d           | DTCSnapshotRecordNumber #1                                                                                                                                                                                                                                                             | C        | #XXh        | SODTC              |
| #8d           | DTCSnapshotRecordNumberOfIdentifiers #1                                                                                                                                                                                                                                                | C        | #XXh        | DTCSSRN            |
| #9d           | DTCSnapshotRecord[] #1 = [<br>dataIdentifier#1 byte#1 (MSB)<br>..<br>dataIdentifier#1 byte#k<br>snapshotData#1 byte#1<br>..<br>snapshotData#1 byte#p<br>..<br>dataIdentifier#w byte#1 (MSB)<br>..<br>dataIdentifier#2 byte#k<br>snapshotData#w byte#1<br>..<br>snapshotData#w byte#m ] | C        | #XXh        | DTCSSRN            |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | DTCSSR_<br>DIDB11  |
| #9+k-1d       |                                                                                                                                                                                                                                                                                        | C        | #XXh        | ..                 |
| #9+k-1d       |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDB1k             |
| #9+kd         |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSD11              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #9+k+(p-1)d   |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSD1p              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #r-(m-1)-2d   |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDBw1             |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #r-(m-1)-1d   |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDBwk             |
| #r-(m-1)d     |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSDw1              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #rd           |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSDwm              |
| ..            | ..                                                                                                                                                                                                                                                                                     | ..       | ..          | ..                 |
| #t            | DTCSnapshotRecordNumber#x                                                                                                                                                                                                                                                              | C        | #XXh        | DTCSSRN            |
| #t+1d         | DTCSnapshotRecordNumberOfIdentifiers #x                                                                                                                                                                                                                                                | C        | #XXh        | DTCSSRN            |
| #t+2d         | DTCSnapshotRecord[] #x = [<br>dataIdentifier#1 byte#1 (MSB)<br>..<br>dataIdentifier#1 byte#k<br>snapshotData#1 byte#1<br>..<br>snapshotData#1 byte#p<br>..<br>dataIdentifier#w byte#1 (MSB)<br>..<br>dataIdentifier#2 byte#k<br>snapshotData#w byte#1<br>..<br>snapshotData#w byte#u ] | C        | #XXh        | DTCSSR_<br>DIDB11  |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #t+2+k-1d     |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDB1k             |
| #t+2+kd       |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSD11              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #t+2+k+(p-1)d |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSD1p              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #n-(u-1)-2d   |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDBw1             |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #n-(u-1)-1d   |                                                                                                                                                                                                                                                                                        | C        | #XXh        | DIDBwk             |
| #n-(u-1)d     |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSDw1              |
| ..            |                                                                                                                                                                                                                                                                                        | ..       | ..          | ..                 |
| #nd           |                                                                                                                                                                                                                                                                                        | C        | #XXh        | SSDwu              |

Table6-18. Positive Response Message definition – sub-function = #04h

| Data Byte | Parameter Name                                | Cvt      | Hex Value   | Mnemonic        |
|-----------|-----------------------------------------------|----------|-------------|-----------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b> | <b>M</b> | <b>#59h</b> | <b>RDTICIPR</b> |

|               |                                                                                                                                                                                                                                                                                        |    |      |                   |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------|-------------------|
| #2d           | reportType = [<br>reportDTCSnapshotRecordByRecordNumber ]                                                                                                                                                                                                                              | M  | #05h | LEV_<br>RDTCSSBRN |
| #3d           | DTCSnapshotRecordNumber#1                                                                                                                                                                                                                                                              | M  | #XXh | DTCEDRN           |
| #4d           | DTCAndStatusRecord[] #1 = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte<br>statusOfDTC ]                                                                                                                                                                                             | M  | #XXh | DTCASR_<br>DTCHB  |
| #5d           |                                                                                                                                                                                                                                                                                        | M  | #XXh | DTCMB             |
| #6d           |                                                                                                                                                                                                                                                                                        | M  | #XXh | DTCLB             |
| #7d           |                                                                                                                                                                                                                                                                                        | M  | #XXh | SODTC             |
| #8d           | DTCSnapshotRecordNumberOfIdentifiers #1                                                                                                                                                                                                                                                | C  | #XXh | DTCSSRNI          |
| #9d           | DTCSnapshotRecord[] #1 = [<br>dataIdentifier#1 byte#1 (MSB)<br>..<br>dataIdentifier#1 byte#k<br>snapshotData#1 byte#1<br>..<br>snapshotData#1 byte#p<br>..<br>dataIdentifier#w byte#1 (MSB)<br>..<br>dataIdentifier#2 byte#k<br>snapshotData#w byte#1<br>..<br>snapshotData#w byte#m ] | C  | #XXh | DTCSSR_<br>DIDB11 |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #9+k-1d       |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDB1k            |
| #9+kd         |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSD11             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #9+k+(p-1)d   |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSD1p             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #r-(m-1)-2d   |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDBw1            |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #r-(m-1)-1d   |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDBwk            |
| #r-(m-1)d     |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSDw1             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #rd           |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSDwm             |
| ..            | ..                                                                                                                                                                                                                                                                                     | .. | ..   | ..                |
| #t            | DTCSnapshotRecordNumber#x                                                                                                                                                                                                                                                              | C  | #XXh | DTCSSRN           |
| #t+1d         | DTCAndStatusRecord[] #x = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte<br>statusOfDTC ]                                                                                                                                                                                             | M  | #XXh | DTCASR_<br>DTCHB  |
| #t+2d         |                                                                                                                                                                                                                                                                                        | M  | #XXh | DTCMB             |
| #t+3d         |                                                                                                                                                                                                                                                                                        | M  | #XXh | DTCLB             |
| #t+4d         |                                                                                                                                                                                                                                                                                        | M  | #XXh | SODTC             |
| #t+5d         | DTCSnapshotRecordNumberOfIdentifiers #x                                                                                                                                                                                                                                                | C  | #XXh | DTCSSRNI          |
| #t+6d         | DTCSnapshotRecord[] #x = [<br>dataIdentifier#1 byte#1 (MSB)<br>..<br>dataIdentifier#1 byte#k<br>snapshotData#1 byte#1<br>..<br>snapshotData#1 byte#p<br>..<br>dataIdentifier#w byte#1 (MSB)<br>..<br>dataIdentifier#2 byte#k<br>snapshotData#w byte#1<br>..<br>snapshotData#w byte#u ] | C  | #XXh | DTCSSR_<br>DIDB11 |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #t+6+k-1d     |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDB1k            |
| #t+6+kd       |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSD11             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #t+6+k+(p-1)d |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSD1p             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| ???           |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDBw1            |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #n-(u-1)-1d   |                                                                                                                                                                                                                                                                                        | C  | #XXh | DIDBwk            |
| #n-(u-1)d     |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSDw1             |
| ..            |                                                                                                                                                                                                                                                                                        | .. | ..   | ..                |
| #nd           |                                                                                                                                                                                                                                                                                        | C  | #XXh | SSDwu             |

Table6-19. Positive Response Message definition – sub-function = #05h

| Data Byte | Parameter Name                                                                                                  | Cvt      | Hex Value    | Mnemonic                         |
|-----------|-----------------------------------------------------------------------------------------------------------------|----------|--------------|----------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                                                                   | <b>M</b> | <b>#59h</b>  | <b>RDTICIPR</b>                  |
| #2d       | reportType = [<br>reportDTCExtendedDataRecordByDTCNumber<br>reportMirrorMemoryDTCExtendedDataReordByDTCNumber ] | M        | #06h<br>#10h | LEV_<br>RDTCEDRBDN<br>RMMDEDRBDN |

|             |                                                                                          |    |      |         |
|-------------|------------------------------------------------------------------------------------------|----|------|---------|
| #3d         | DTCAndStatusRecord[] = [<br>DTCHighByte<br>DTCMiddleByte<br>DTCLowByte<br>statusOfDTC ]  | M  | #XXh | DTCASR_ |
| #4d         |                                                                                          | M  | #XXh | DTCHB   |
| #5d         |                                                                                          | M  | #XXh | DTCMB   |
| #6d         |                                                                                          | M  | #XXh | DTCLB   |
| #7d         | DTCExtendedDataRecordNumber#1                                                            | C  | #XXh | SODTC   |
| #8d         | DTCExtendedDataRecord[] #1 = [<br>extendedData#1 byte#1<br>..<br>extendedData#1 byte#p ] | C  | #XXh | DTCEDRN |
| ..          |                                                                                          | .. | ..   | DTCSSR_ |
| #8+(p-1)d   |                                                                                          | C  | #XXh | EDD11   |
| ..          | ..                                                                                       | .. | ..   | EDD1p   |
| #td         | DTCExtendedDataRecord#x                                                                  | C  | #XXh | ..      |
| #t+1d       | DTCExtendedDataRecord[] #x = [<br>extendedData#1 byte#1<br>..<br>extendedData#1 byte#q ] | C  | #XXh | DTCEDRN |
| ..          |                                                                                          | .. | ..   | DTCSSR_ |
| #t+1+(q-1)d |                                                                                          | C  | #XXh | EDDx1   |
| ..          | ..                                                                                       | .. | ..   | EDDxq   |

Table6-20. Positive Response Message definition – sub-function = #06 / 10h

| Data Byte | Parameter Name                                                                                                                                                                                                                                                         | Cvt      | Hex Value    | Mnemonic                     |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------|------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                                                                                                                                                                                                                          | <b>M</b> | <b>#59h</b>  | <b>RDTICIPR</b>              |
| #2d       | reportType = [<br>reportDTCBySeverityMaskRecord<br>reportSeverityInformationOfDTC ]                                                                                                                                                                                    | M        | #08h<br>#09h | LEV_<br>RDTCBMSMR<br>RSIODTC |
| #3d       | DTCStatusAvailabilityMask                                                                                                                                                                                                                                              | C        | #XXh         | DTCSSAM                      |
| #4d       | DTCAndSeverityRecord[] = [<br>DTCSeverity #1<br>DTCFunctionalUnit #1<br>DTCHighByte #1<br>DTCMiddleByte #1<br>DTCLowByte #1<br>statusOfDTC #1<br>..<br>DTCSeverity #m<br>DTCFunctionalUnit #m<br>DTCHighByte #m<br>DTCMiddleByte #m<br>DTCLowByte #m<br>statusOfDTC #m | C        | #XXh         | DTCASR_                      |
| #5d       |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCS                         |
| #6d       |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCSFU                       |
| #7d       |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCHB                        |
| #8d       |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCMB                        |
| #9d       |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCLB                        |
| ..        |                                                                                                                                                                                                                                                                        | ..       | ..           | SODTC                        |
| ..        |                                                                                                                                                                                                                                                                        | ..       | ..           | ..                           |
| #n-5d     |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCS                         |
| #n-4d     |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCSFU                       |
| #n-3d     |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCHB                        |
| #n-2d     |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCMB                        |
| #n-1d     |                                                                                                                                                                                                                                                                        | C        | #XXh         | DTCLB                        |
| #nd       |                                                                                                                                                                                                                                                                        | C        | #XXh         | SODTC                        |

Table6-21. Positive Response Message definition – sub-function = #08 / 09h

| Data Byte | Parameter Name                                                                                       | Cvt      | Hex Value   | Mnemonic        |
|-----------|------------------------------------------------------------------------------------------------------|----------|-------------|-----------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>                                                        | <b>M</b> | <b>#59h</b> | <b>RDTICIPR</b> |
| #2d       | reportType = [<br>reportSupportedDTCExtDataRecord]                                                   | M        | #1Ah        | LEV_<br>RDTCEDI |
| #3d       | DTCStatusAvailabilityMask                                                                            | M        | #XXh        | DTCSSAM         |
| #4d       | DTCExtDataRecordNumber                                                                               | C1       | #01~FD      | DTCEDRN         |
| #5d       | DTCAndStatusRecord[]#1 = [<br>DTCHighByte #1<br>DTCMiddleByte #1<br>DTCLowByte #1<br>statusOfDTC #1] | C1       | #XXh        | DTCASR_         |
| #6d       |                                                                                                      | C1       | #XXh        | DTCHB           |
| #7d       |                                                                                                      | C1       | #XXh        | DTCMB           |
| #8d       |                                                                                                      | C1       | #XXh        | DTCLB           |
| ..        | ..                                                                                                   | ..       | ..          | SODTC           |

|       |                            |    |      |         |
|-------|----------------------------|----|------|---------|
| #n-3d | DTCAndStatusRecord[]#m = [ | C2 | #XXh | DTCASR_ |
| #n-2d | DTCHighByte #m             | C2 | #XXh | DTCHB   |
| #n-1d | DTCMiddleByte #m           | C2 | #XXh | DTCMB   |
| #nd   | DTCLowByte #m              | C2 | #XXh | DTCLB   |
|       | statusOfDTC #m]            | C2 | #XXh | SODTC   |

C1: This parameter shall be present if at least one DTC supports the DTCEXtendedDataRecord  
C2: This parameter record is only present if more than one DTC support the DTCEXtendedDataRecord

표 6-28. Positive Response Message definition – sub-function = #1Ah

| Data Byte | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic                      |
|-----------|-------------------------------------------------------|----------|-------------|-------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>         | <b>M</b> | <b>#59h</b> | <b>RDTCI PR</b>               |
| #2d       | reportType = [<br>reportWWHOBDDTCByMaskRecord]        | M        | #42h        | LEV_<br>RWWHOBDDT<br>CBSMR    |
| #3d       | FunctionalGroupIdentifier                             | M        | #XXh        | FGID                          |
| #4d       | DTCStatusAvailabilityMask                             | M        | #XXh        | DTCSAM                        |
| #5d       | DTCSeverityAvailabilityMask                           | M        | #XXh        | DTCSVAM                       |
| #6d       | DTCFormatIdentifier = [<br>SAE_J2012-DA_DTCFormat_04] | M        | #04h        | DTCFID_<br>J2012-<br>DADTCF04 |
| #7d       | DTCAndSeverityRecord[] = [<br>DTCSeverity#1           | C1       | #XXh        | DTCASR_<br>DTCS               |
| #8d       | DTCHighByte#1 (MSB)                                   | C1       | #XXh        | DTCHB                         |
| #9d       | DTCMiddleByte#1                                       | C1       | #XXh        | DTCMB                         |
| #10d      | DTCLowByte#1                                          | C1       | #XXh        | DTCLB                         |
| #11d      | statusOfDTC#1                                         | C1       | #XXh        | SODTC                         |
| ..        | ..                                                    | ..       | ..          | ..                            |
| #n-4d     | DTCSeverity#m                                         | C2       | #XXh        | DTCS                          |
| #n-3d     | DTCHighByte#m (MSB)                                   | C2       | #XXh        | DTCHB                         |
| #n-2d     | DTCMiddleByte#m                                       | C2       | #XXh        | DTCMB                         |
| #n-1d     | DTCLowByte#m                                          | C2       | #XXh        | DTCLB                         |
| #nd       | statusOfDTC#m ]                                       | C2       | #XXh        | SODTC                         |

C1: This parameter is only present if DTC information is available to be reported.  
C2: This parameter is only present if more than one DTC information is available to be reported.

표 6-29. Positive Response Message definition – sub-function = #42h

| Data Byte | Parameter Name                                        | Cvt      | Hex Value   | Mnemonic                      |
|-----------|-------------------------------------------------------|----------|-------------|-------------------------------|
| #1d       | <b>ReadDTCInformation Response Service ID</b>         | <b>M</b> | <b>#59h</b> | <b>RDTCI PR</b>               |
| #2d       | reportType = [<br>reportWWHOBDDTCWithPermanentStatus] | M        | #55h        | LEV_<br>RWWHOBDDT<br>CWPS     |
| #3d       | FunctionalGroupIdentifier                             | M        | #XXh        | FGID                          |
| #4d       | DTCStatusAvailabilityMask                             | M        | #XXh        | DTCSAM                        |
| #5d       | DTCFormatIdentifier = [<br>SAE_J2012-DA_DTCFormat_04] | M        | #04h        | DTCFID_<br>J2012-<br>DADTCF04 |

|       |                          |                     |    |      |         |
|-------|--------------------------|---------------------|----|------|---------|
| #6d   | DTCAndStatusRecord[] = [ | DTCHighByte#1 (MSB) | C1 | #XXh | DTCASR_ |
| #7d   |                          | DTCMiddleByte#1     | C1 | #XXh | DTCHB   |
| #8d   |                          | DTCLowByte#1        | C1 | #XXh | DTCMB   |
| #9d   |                          | statusOfDTC#1       | C1 | #XXh | DTCLB   |
| ..    |                          | ..                  | .. | ..   | SODTC   |
| #n-3d |                          | DTCHighByte#m (MSB) | C2 | #XXh | ..      |
| #n-2d |                          | DTCMiddleByte#m     | C2 | #XXh | DTCHB   |
| #n-1d |                          | DTCLowByte#m        | C2 | #XXh | DTCMB   |
| #nd   |                          | statusOfDTC#m ]     | C2 | #XXh | DTCLB   |
|       |                          |                     |    |      | SODTC   |

C1: This parameter is only present if DTC information is available to be reported.  
C2: This parameter is only present if more than one DTC information is available to be reported.

표 6-30. Positive Response Message definition – sub-function = #55h

| Data Byte                                                                                                            | Parameter Name                                                                                                                                                                     | Cvt | Hex Value | Mnemonic                  |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----------|---------------------------|
| #1d                                                                                                                  | ReadDTCInformation Response Service ID                                                                                                                                             | M   | #59h      | RDTCIPR                   |
| #2d                                                                                                                  | reportType = [reportDTCByReadinessGroupIdentifier]                                                                                                                                 | M   | #56h      | LEV_RWWHOBDT<br>CWPS      |
| #3d                                                                                                                  | FunctionalGroupIdentifier                                                                                                                                                          | M   | #XXh      | FGID                      |
| #4d                                                                                                                  | DTCStatusAvailabilityMask                                                                                                                                                          | M   | #XXh      | DTCSAM                    |
| #5d                                                                                                                  | DTCFormatIdentifier = [SAE_J2012-DA_DTCFormat_04]                                                                                                                                  | M   | #04h      | DTCFID_J2012-<br>DADTCF04 |
| #6d                                                                                                                  | DTCReadinessGroupIdentifier                                                                                                                                                        | M   | #XXh      | RGI                       |
| #7d                                                                                                                  | DTCAndStatusRecord[] = [DTCHighByte#1 (MSB)<br>DTCMiddleByte#1<br>DTCLowByte#1<br>statusOfDTC#1<br>..<br>DTCHighByte#m (MSB)<br>DTCMiddleByte#m<br>DTCLowByte#m<br>statusOfDTC#m ] | C1  | #XXh      | DTCASR_                   |
| #8d                                                                                                                  |                                                                                                                                                                                    | C1  | #XXh      | DTCHB                     |
| #9d                                                                                                                  |                                                                                                                                                                                    | C1  | #XXh      | DTCMB                     |
| #10d                                                                                                                 |                                                                                                                                                                                    | C1  | #XXh      | DTCLB                     |
| ..                                                                                                                   |                                                                                                                                                                                    | ..  | ..        | ..                        |
| #n-3d                                                                                                                |                                                                                                                                                                                    | C2  | #XXh      | DTCHB                     |
| #n-2d                                                                                                                |                                                                                                                                                                                    | C2  | #XXh      | DTCMB                     |
| #n-1d                                                                                                                |                                                                                                                                                                                    | C2  | #XXh      | DTCLB                     |
| #nd                                                                                                                  |                                                                                                                                                                                    | C2  | #XXh      | SODTC                     |
| C1: This parameter shall be present if at least one DTC matches the client defined Readiness Group Identifier.       |                                                                                                                                                                                    |     |           |                           |
| C2: This parameter record is only present if more than one DTC matches the client defined Readiness Group Identifier |                                                                                                                                                                                    |     |           |                           |

표 6-31. Positive Response Message definition – sub-function = #56h

### 6.2.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                  |
|-----|---------------------------------------|-----|----------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | This code is returned if the requested sub-function is not supported |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                   |



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|    |                   |   |                                                                                                                                                                                                                                               |
|----|-------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | requestOutOfRange | M | This code is returned if :<br>1. The client specified a DTCMaskRecord /<br>DTCSeverityMaskRecord that was not recognized<br>by the server.<br>2. The client specified an invalid<br>DTCSnapshotRecordNumber /<br>DTCExtendedDataRecordNumber. |
|----|-------------------|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

Table6-22. Supported Negative Response Codes

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㈜모비스전자 / 김의진

## 7. InputOutput Control functional unit

### 7.1 InputOutputControlByIdentifier (2F hex) service

The InputOutputControlByIdentifier service is used by the client to substitute a value for an input signal, internal server function and/or control an output (actuator) of an electronic system.  
The server shall send a positive response message to a request message with an inputOutputControlParameter of returnControlToECU even if the dataIdentifier is currently not under tester control.

#### 7.1.1 Request message

| Data Byte   | Parameter Name                                                             | Cvt      | Hex Value   | Mnemonic          |
|-------------|----------------------------------------------------------------------------|----------|-------------|-------------------|
| #1d         | <b>InputOutputControlByIdentifier Request Service ID</b>                   | <b>M</b> | <b>#2Fh</b> | <b>IOCB</b>       |
| #2d         | dataIdentifier #1[] = [<br>byte #1 (MSB)<br>byte #2 (LSB) ]                | M        | #XXh        | IOI_<br>B1        |
| #3d         |                                                                            | M        | #XXh        | B2                |
| #4d         | controlOptionRecord #1[] = [<br>inputOutputControlParameter                | M        | #XXh        | CSR_<br>IOCP      |
| #5d         | controlOptionRecord #2[] = [<br>controlState#1<br>..<br>controlState#m     | C1       | #XXh        | CSR_<br>CS_<br>.. |
| ..          |                                                                            | ..       | ..          | ..                |
| #5+(m-1)d   | controlState#m                                                             | C1       | #XXh        | CS_<br>..         |
| #5+md       | controlEnableMaskRecord #1[] = [<br>controlMask#1<br>..<br>controlMask#r ] | C2       | #XXh        | CEM_<br>CM_<br>.. |
| ..          |                                                                            | ..       | ..          | ..                |
| #5+m+(r-1)d | controlMask#r ]                                                            | C2       | #XXh        | CM_<br>..         |

C1 : The presence of this parameter depends on the dataIdentifier#1 and the inputOutputControlParameter of controlOptionRecord#1. (See Table 7-2)

C2 : The presence of this parameter depends on the dataIdentifier#1.

dataIdentifier : see Annex C.1

controlEnableMaskRecord : see Annex C.1.1

The value of each bit shall determine whether the corresponding parameter in the dataIdentifier will be affected by the request. A bit value of '0' in the ControlEnableMask shall represent that the corresponding parameter is not affected by this request and a bit value of '1' shall represent that the corresponding parameter is affected by this request. The most significant bit of ControlMask#1 shall correspond to the first parameter in the ControlState starting at the most significant bit of ControlState#1, the second most significant bit of ControlMask#1 shall correspond to the second parameter in the ControlState, and continuing on in this fashion utilising as many ControlMask bytes as necessary to mask all parameters. For example, the least significant bit of ControlMask#2 would correspond to the 16th parameter in the controlState. For bitmapped dataIdentifiers, unsupported bits shall also have a corresponding bit in the ControlEnableMask so that the position of the mask bit of every parameter in the ControlEnableMask shall exactly match the position of the corresponding parameter in the controlState.

Table7-1. Request Message

| Hex | Description | Cvt | Mnemonic |
|-----|-------------|-----|----------|
|-----|-------------|-----|----------|

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |   |        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|--------|
| 00 | <b>returnControlToECU</b><br>This value shall indicate to the server that the client does no longer have control about the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br><br>Number of controlState bytes in request: 0<br>Number of controlState bytes in pos. response: if the dataIdentifier consists of more than one parameter                                                                                                                                                                       | U | RCTECU |
| 01 | <b>resetToDefault</b><br>This value shall indicate to the server that it is requested to reset the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier to its default state.<br><br>Number of controlState bytes in request: 0<br>Number of controlState bytes in pos. response: depends on the dataIdentifier                                                                                                                                                                                                      | U | RTD    |
| 02 | <b>freezeCurrentState</b><br>This value shall indicate to the server that it is requested to freeze the current state of the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br><br>Number of controlState bytes in request: 0<br>Number of controlState bytes in pos. response: depends on the dataIdentifier                                                                                                                                                                                                 | U | FCS    |
| 03 | <b>shortTermAdjustment</b><br>This value shall indicate to the server that it is requested to adjust the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier in RAM to the value(s) included in the controlOption parameter(s). (e.g. set Idle Air Control Valve to a specific step number, set pulse width of valve to a specific value/duty cycle).<br><br>Number of controlState bytes in request: depends on the dataIdentifier<br>Number of controlState bytes in pos. response: depends on the dataIdentifier | U | STA    |

Table7-2. inputOutputControlParameter definition

## 7.1.2 Positive response message

| Data Byte | Parameter Name                                              | Cvt | Hex Value | Mnemonic     |
|-----------|-------------------------------------------------------------|-----|-----------|--------------|
| #1d       | InputOutputControlByIdentifier Response Service ID          | M   | #6Fh      | IOCBIPR      |
| #2d       | dataIdentifier #1[] = [<br>byte #1 (MSB)<br>byte #2 (LSB) ] | M   | #XXh      | IOI_<br>B1   |
| #3d       |                                                             | M   | #XXh      | B2           |
| #4d       | controlOptionRecord #1[] = [<br>inputOutputControlParameter | M   | #XXh      | CSR_<br>IOCP |
| #5d       | controlOptionRecord #2[] = [<br>controlState#1              | C   | #XXh      | CSR_<br>CS_  |
| ..        | ..                                                          | ..  | ..        | ..           |
| #5+(m-1)d | controlState#m                                              | C   | #XXh      | CS_          |

Table7-3. Positive Response Message definition

## 7.1.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                     |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                      |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the request InputOutputControlByIdentifier are not met. |



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|    |                      |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----|----------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 31 | requestOutOfRange    | M | This code shall be returned if :<br>1. the requested dataIdentifier value is not supported by the device.<br>2. the dataIdentifier uses the controlState#1 parameter as an inputOutputControlParameter and the value contained in this parameter is invalid.<br>3. the one or multiple of the applicable controlStates of the controlOptionRecord record are invalid.<br>4. the combination of bits enabling control in the ControlEnableMaskRecord is not supported by the device. |
| 33 | securityAccessDenied | M | This code shall be returned if a client sends a request with a valid secure dataIdentifier and the server's security feature is currently active.                                                                                                                                                                                                                                                                                                                                   |

Table7-4. Supported Negative Response Codes

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## 8. Remote Activation of Routine functional unit

The following section describes two (2) different methods of implementation (Methods “A” and “B”).

### - Method “A” ;

- This method is based on the assumption that after a routine has been started by the client in the server's memory the client shall be responsible to stop the routine.
- The server routine shall be started in the server's memory some time between the completion of the RoutineControl request message that starts the routine and the completion of the first response message (if “positive” based on the server's conditions).
- The server routine shall be stopped in the server's memory some time after the completion of the StopRoutine request message and the completion of the first response message (if “positive” based on the server's conditions).
- The client may request routine results after the routine has been stopped.

### - Method “B” ;

- This method is based on the assumption that after a routine has been started by the client in the server's memory that the server shall be responsible to stop the routine.
- The server routine shall be started in the server's memory some time between the completion of the RoutineControl request message that starts the routine and the completion of the first response message (if “positive” based on the server's conditions).
- The server routine shall be stopped any time as programmed or previously initialized in the server's memory.

## 8.1 RoutineControl (31 hex) service

The RoutineControl service is used by the client to ;

- start a routine,
- stop a routine, and
- request routine results

A routine is referenced by a 2-byte routineIdentifier.

Each routine has security option. (e.g. routineIdentifier XXXX : Security is not required, routineIdentifier YYYY : ASK required)

Reprogram routine (eraseMemory, checkProgrammingDependencies, checkmemory, ..., etc) has to be secured by respective ECU security option(ASK, C-SAC, NormalSeedKey and so on).

For other routines, If there is no special purpose, it is necessary to apply security options to routines.

Please refer to each functional specification or ECU diagnostic specification.

### 8.1.1 Request message

| Data Byte | Parameter Name                                                                                  | Cvt      | Hex Value   | Mnemonic       |
|-----------|-------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl request Service ID</b>                                                        | <b>M</b> | <b>#31h</b> | <b>RC</b>      |
| #2d       | Sub-function = [<br>routineControlType ]                                                        | M        | #XXh        | LEV_<br>RCTP_  |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                          | M        | #XXh        | RI_<br>B1      |
| #4d       |                                                                                                 | M        | #XXh        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>routineControlOption #1<br>..<br>routineControlOption #m ] | C/U      | #XXh        | RCEOR_<br>RCO_ |
| ..        |                                                                                                 | ..       | ..          | ..             |
| #nd       |                                                                                                 | C/U      | #XXh        | RCO_           |

routineControlOptionRecord : This parameter record contains either

- Routine entry option parameters, which optionally specify start conditions of the routine, or
- Routine exit option parameters, which optionally specify stop conditions of the routine, or
- requestRoutineResults option parameters.

Table8-1. Request Message

| Hex | Description           | Cvt | Mnemonic |
|-----|-----------------------|-----|----------|
| 01  | startRoutine          | M   | STR      |
| 02  | stopRoutine           | U   | STPR     |
| 03  | requestRoutineResults | U   | RRR      |

Table8-2. Request message sub-function definition

| Hex         | Description                                                                       | Cvt | Mnemonic     |
|-------------|-----------------------------------------------------------------------------------|-----|--------------|
| 0000 – 00FF | Reserved (ISOSAE Reserved)                                                        | M   | ISOSAERESRVD |
| 0100 – 01FF | Reserved (Tachograph TestIds)                                                     | M   | TACHORI_     |
| 0200        | CheckMemory                                                                       | M   | CM           |
| 0201 – 020F | Reserved (vehicleManufacturerSpecific)                                            | M   | VMS_         |
| 0210        | ReadActiveArea                                                                    | U   | RAA          |
| 0211        | FinishUpdate / (OLD)CopyActiveareaTo                                              | U   | CAT          |
| 0212        | EraseTargetArea                                                                   | U   | ETA          |
| 0213        | SwapActiveArea                                                                    | U   | SAA          |
| 0214 – 021F | Reserved (vehicleManufacturerSpecific)                                            | M   | VMS_         |
| 0220        | WriteDeltaPatch                                                                   | U   | WDP          |
| 0221 – 02FF | Reserved (vehicleManufacturerSpecific)                                            | M   | VMS_         |
| 0300        | OtaReady                                                                          | U   | OR           |
| 0301        | EnableUpdateCondition                                                             | U   | EUC          |
| 0302 – DFFF | Reserved (vehicleManufacturerSpecific)                                            | M   | VMS_         |
| 0400        | ReadFoDStatus                                                                     | U   | RFS          |
| 0410        | PrepareFoDActivation                                                              | U   | RFA          |
| 0411        | ExecuteFoDActivation                                                              | U   | EFA          |
| 0412        | ConfirmFoDActivation                                                              | U   | CFA          |
| 0420        | PrepareFoDDeactivation                                                            | U   | PFD          |
| 0421        | ExecuteFoDDeactivation                                                            | U   | EFD          |
| 0422        | ConfirmFoDDeactivation                                                            | U   | CFD          |
| 0430        | FoDReady                                                                          | U   | FR           |
| 0440        | DeleteFoDCertificates                                                             | U   | DFC          |
| 0441 – DFFF | Reserved (vehicleManufacturerSpecific)                                            | M   | VMS_         |
| E000 – E1FF | Reserved (OBDTestIds)<br><b>IMPORTANT : Do not use other than legislative ECU</b> | M   | OBDRI_       |
| E200 – EFFF | Reserved (ISOSAE Reserved)                                                        | M   | ISOSAERESRVD |
| F000 – FFFF | Reserved (SystemSupplierSpecific)                                                 | M   | SSS_         |
| FF00        | eraseMemory                                                                       | U   | EM_          |
| FF01        | checkProgrammingDependencies                                                      | U   | CPD_         |
| FF02 – FFFF | Reserved (ISOSAE Reserved)                                                        | M   | ISOSAERESRVD |

Table8-3. Request message routineIdentifier definition

### 8.1.2 Positive response message

| Data Byte | Parameter Name                                                             | Cvt      | Hex Value   | Mnemonic    |
|-----------|----------------------------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>                                  | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                                         | M        | #XXh        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                     | M        | #XXh        | RI_<br>B1   |
| #4d       |                                                                            | M        | #XXh        | B2          |
| #5d       | routineStatusRecord [] = [<br>routineStatus #1<br>..<br>routineStatus #m ] | U        | #XXh        | RSR_<br>RS_ |
| ..        |                                                                            | ..       | ..          | ..          |
| #nd       |                                                                            | U        | #XXh        | RS_         |

Table8-4. Positive Response Message definition

### 8.1.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                     |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12  | subFunctionNotSupported               | M   | This code is returned if the requested sub-function is not supported                                                                                                                                                                                                                                    |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                                                                                                      |
| 22  | conditionNotCorrect                   | M   | This code shall be returned if the criteria for the request RoutineControl are not met.                                                                                                                                                                                                                 |
| 24  | requestSequenceError                  | M   | This code shall be returned if the 'stopRoutine' or 'requestRoutineResults' sub-function is received without first receiving a 'startRoutine' for the requested routineIdentifier.                                                                                                                      |
| 31  | requestOutOfRange                     | M   | The code shall be returned if ;<br>1. The server does not supported the requested routineIdentifier.<br>2. The user optional routineControlOptionRecord contains invalid data for the requested routineIdentifier.                                                                                      |
| 33  | seurityAccessDenied                   | M   | This NRC shall be sent if a client sends a requested with a valid secure routineIdentifier and the server's security feature is currently active.                                                                                                                                                       |
| 72  | GeneralProgrammingFailure             | M   | This NRC shall be returned if the server detects an error when performing a routine, which accesses server internal memory. An example is when the routine erases or prgrams a certain memory location in the permanent memory device (e.g. Flash Memory) and the access to that memory location fails. |

Table8-5. Supported Negative Response Codes

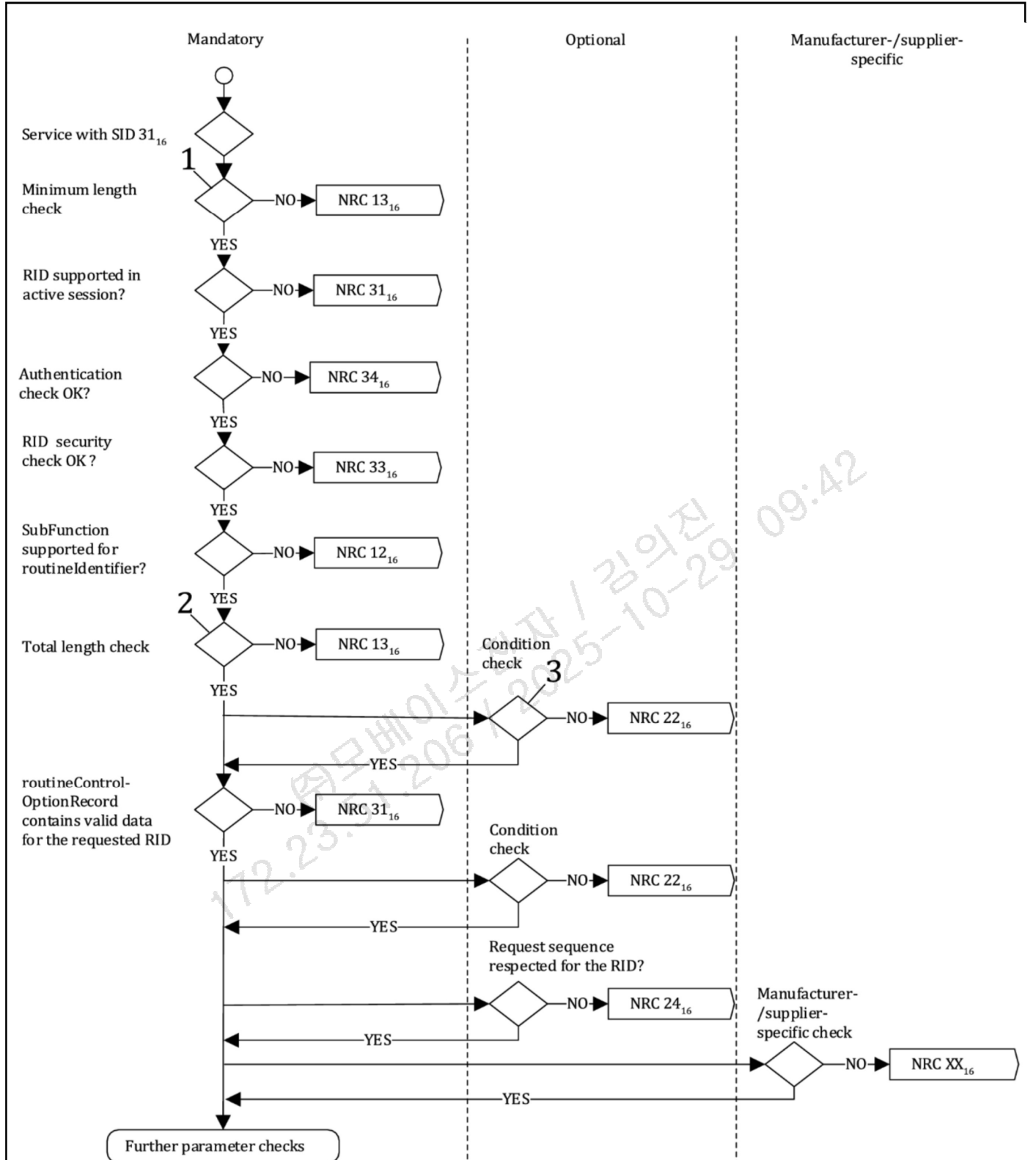


Figure8-1. NRC handling for RoutineControlService

### Key

- 1 at least 4 (SI+SubFunction+RID Parameter)
- 2 1 byte SI + 1 byte SF + 2 byte RID + nth byte routineControlOptionRecord required for the specific RID
- 3 optional condition check independent from data content

### 8.1.4 Implementation rules for legacy update / OTA 1.0

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### 8.1.4.1 OTA Ready

Session : Default and Non-default session

Security level : No

#### 8.1.4.1.1 Request message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic      |
|-----------|--------------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b> | <b>RC</b>     |
| #2d       | Sub-function = [<br>routineControlType ]               | M        | #01h        | LEV_<br>RCTP_ |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #03h        | RI_<br>B1     |
| #4d       |                                                        | M        | #00h        | B2            |

Table8-6. Request Message

#### 8.1.4.1.2 Positive response message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic    |
|-----------|--------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                     | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #03h        | RI_<br>B1   |
| #4d       |                                                        | M        | #00h        | B2          |
| #5d       | routineControlOptionRecord [] = [<br>routineStatus ]   | M        | #XXh        | RCOR_RS     |

Table8-7. Positive Response Message definition

| Hex   | Description                                    | Cvt | Mnemonic |
|-------|------------------------------------------------|-----|----------|
| 00    | (Reserved)                                     | M   |          |
| 01    | OK<br>(Server can be reprogrammed immediately) | M   |          |
| 02~FF | (Reserved)                                     | M   |          |

Table8-8. Routine status definition

#### 8.1.4.1.3 Supported negative response codes

| Hex | NegativeResponseCode                         | Cvt | Cause of Occurence |
|-----|----------------------------------------------|-----|--------------------|
| 10  | generalReject                                | M   |                    |
| 13  | incorrectMessageLengthOrInvalidFormat        | M   |                    |
| 31  | requestOutOfRange                            | M   |                    |
| 33  | securityAccessDenied                         | M   |                    |
| 78  | requestCorrectlyReceived-<br>ResponsePending | M   |                    |

Table8-9. Supported Negative Response Codes

### 8.1.4.2 (OLD)EraseTargetArea (ECUs without memory redundancy)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.2.1 Request message

| Data Byte | Parameter Name                                                                                 | Cvt      | Hex Value   | Mnemonic       |
|-----------|------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b> | <b>RC</b>      |
| #2d       | Sub-function = [<br>routineControlType ]                                                       | M        | #01h        | LEV_<br>RCTP_  |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #FFh        | RI_<br>B1      |
| #4d       |                                                                                                | M        | #00h        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh        | RCOR_SW<br>UMA |
| #6d       |                                                                                                |          |             |                |
| #7d       | routineControlOptionRecord [] = [<br>Memory block ]                                            | U        | #XXh        | RCOR_MB        |
| #8d       |                                                                                                |          | #XXh        |                |

The SWUnitMemoryArea represents the memory area to be erased.  
Memory block would be used as ECU specific memory index.

Table8-10. Request Message

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-11. SWUnitMemoryArea definition

#### 8.1.4.2.2 Positive response message

| Data Byte | Parameter Name                                                                                 | Cvt      | Hex Value   | Mnemonic       |
|-----------|------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl Response Service ID</b>                                                      | <b>M</b> | <b>#71h</b> | <b>RCPR</b>    |
| #2d       | routineControlType                                                                             | M        | #01h        | RCTP_          |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #FFh        | RI_<br>B1      |
|           |                                                                                                | M        | #00h        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh        | RCOR_SW<br>UMA |
| #6d       |                                                                                                |          |             |                |

Table8-12. Positive Response Message definition



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| Hex                               | Description                   | Cvt | Mnemonic   |
|-----------------------------------|-------------------------------|-----|------------|
| F1B0                              | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1                              | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2                              | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3                              | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4                              | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5                              | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6                              | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7                              | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8                              | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9                              | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA                              | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |
| C: More than 1 SWUnit is defined. |                               |     |            |

Table8-13. SWUnitMemoryArea definition

### 8.1.4.2.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurrence |
|-----|------------------------------------------|-----|---------------------|
| 10  | generalReject                            | M   |                     |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   |                     |
| 31  | requestOutOfRange                        | M   |                     |
| 33  | securityAccessDenied                     | M   |                     |
| 78  | requestCorrectlyReceived-ResponsePending | M   |                     |

Table8-14. Supported Negative Response Codes

### 8.1.4.3 (OLD)ReadActiveArea (ECU with memory redundancy)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.3.1 Request message

| Data Byte  | Parameter Name                                                                                 | Cvt      | Hex Value    | Mnemonic       |
|------------|------------------------------------------------------------------------------------------------|----------|--------------|----------------|
| #1d        | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b>  | <b>RC</b>      |
| #2d        | Sub-function = [<br>routineControlType ]                                                       | M        | #01h         | LEV_<br>RCTP_  |
| #3d        | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #02h         | RI_<br>B1      |
| #4d        |                                                                                                | M        | #10h         | B2             |
| #5d<br>#6d | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | C        | #F1h<br>#BXh | RCOR_SW<br>UMA |

C : ECU supports OTA and More than 1 SWUnit is defined.

Table8-15. Request Message

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-16. SWUnitMemoryArea definition

#### 8.1.4.3.2 Positive response message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic    |
|-----------|--------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                     | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #02h        | RI_<br>B1   |
| #4d       |                                                        | M        | #10h        | B2          |
| #5d       | routineControlOptionRecord [] = [<br>MemoryArea]       | M        | #XXh        | RCOR_MA     |

Table8-17. Positive Response Message definition

| Hex | Description                | Cvt | Mnemonic |
|-----|----------------------------|-----|----------|
| 0A  | The current boot area is A | M   |          |
| 0B  | The current boot area is B | M   |          |

Table8-18. MemoryArea definition



#### 8.1.4.3.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurrence |
|-----|------------------------------------------|-----|---------------------|
| 10  | generalReject                            | M   |                     |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   |                     |
| 31  | requestOutOfRange                        | M   |                     |
| 33  | securityAccessDenied                     | M   |                     |
| 78  | requestCorrectlyReceived-ResponsePending | M   |                     |

Table8-19. Supported Negative Response Codes

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주모베이스전자 / 김의진

### 8.1.4.4 (OLD)CopyActiveareaTo (ECUs with memory redundancy)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.4.1 Request message

| Data Byte | Parameter Name                                                                                 | Cvt      | Hex Value   | Mnemonic       |
|-----------|------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b> | <b>RC</b>      |
| #2d       | Sub-function = [<br>routineControlType ]                                                       | M        | #01h        | LEV_<br>RCTP_  |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #02h        | RI_<br>B1      |
| #4d       |                                                                                                | M        | #11h        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>MemoryArea ]                                              | M        | #XXh        | RCOR_MA        |
| #6d       | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh        | RCOR_SW<br>UMA |
| #7d       |                                                                                                |          | #XXh        |                |

The MemoryArea represents the target memory area to write after copying data.  
The SWUnitMemoryArea represents the memory area of the S/W unit to be designated as write.

Table8-20. Request Message

| Hex | Description                  | Cvt | Mnemonic |
|-----|------------------------------|-----|----------|
| 00  | No specifying                | M   |          |
| 0A  | Copy active area B to area A | M   |          |
| 0B  | Copy active area A to area B | M   |          |

Table8-21. MemoryArea definition

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-22. SWUnitMemoryArea definition

#### 8.1.4.4.2 Positive response message

| Data Byte | Parameter Name                            | Cvt      | Hex Value   | Mnemonic    |
|-----------|-------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b> | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |

|            |                                                                                                |        |              |                 |
|------------|------------------------------------------------------------------------------------------------|--------|--------------|-----------------|
| #2d        | routineControlType                                                                             | M      | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M<br>M | #02h<br>#11h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [<br>MemoryArea]                                               | M      | #XXh         | RCOR_MA         |
| #6d<br>#7d | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M      | #XXh         | RCOR_SW<br>UMA  |

Table8-23. Positive Response Message definition

| Hex | Description                  | Cvt | Mnemonic |
|-----|------------------------------|-----|----------|
| 0A  | Copy active area B to area A | M   |          |
| 0B  | Copy active area A to area B | M   |          |

Table8-24. MemoryArea definition

| Hex                              | Description                   | Cvt | Mnemonic   |
|----------------------------------|-------------------------------|-----|------------|
| F1B0                             | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1                             | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2                             | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3                             | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4                             | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5                             | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6                             | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7                             | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8                             | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9                             | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA                             | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |
| C: More than 1 SWUnit is defined |                               |     |            |

Table8-25. S/W Unit MemoryArea definition

### 8.1.4.4.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurence |
|-----|------------------------------------------|-----|--------------------|
| 10  | generalReject                            | M   |                    |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   |                    |
| 31  | requestOutOfRange                        | M   |                    |
| 33  | securityAccessDenied                     | M   |                    |
| 78  | requestCorrectlyReceived-ResponsePending | M   |                    |

Table8-26. Supported Negative Response Codes

### 8.1.4.5 (OLD)EraseTargetArea (ECU with memory redundancy)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.5.1 Request message

| Data Byte | Parameter Name                                                                                 | Cvt      | Hex Value   | Mnemonic       |
|-----------|------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b> | <b>RC</b>      |
| #2d       | Sub-function = [<br>routineControlType ]                                                       | M        | #01h        | LEV_<br>RCTP_  |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #02h        | RI_<br>B1      |
| #4d       |                                                                                                | M        | #12h        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>MemoryArea ]                                              | M        | #XXh        | RCOR_MA        |
| #6d       | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh        | RCOR_SW<br>UMA |
| #7d       |                                                                                                |          | #XXh        |                |

The MemoryArea represents the memory area to be designated as erase.

Table8-27. Request Message

| Hex | Description   | Cvt | Mnemonic |
|-----|---------------|-----|----------|
| 00  | No specifying | M   |          |
| 0A  | Erase area A  | M   |          |
| 0B  | Erase area B  | M   |          |

Table8-28. MemoryArea definition

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined

Table8-29. SWUnitMemoryArea definition

#### 8.1.4.5.2 Positive response message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic    |
|-----------|--------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                     | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #02h        | RI_<br>B1   |
| #4d       |                                                        | M        | #12h        | B2          |

|     |                                                   |   |      |         |
|-----|---------------------------------------------------|---|------|---------|
| #5d | routineControlOptionRecord [] = [<br>MemoryArea ] | M | #XXh | RCOR_MA |
|-----|---------------------------------------------------|---|------|---------|

Table8-30. Positive Response Message definition

| Hex | Description  | Cvt | Mnemonic |
|-----|--------------|-----|----------|
| 0A  | Erase area A | M   |          |
| 0B  | Erase area B | M   |          |

Table8-31. MemoryArea definition

| Hex                               | Description                   | Cvt | Mnemonic   |
|-----------------------------------|-------------------------------|-----|------------|
| F1B0                              | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1                              | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2                              | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3                              | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4                              | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5                              | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6                              | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7                              | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8                              | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9                              | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA                              | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |
| C: More than 1 SWUnit is defined. |                               |     |            |

Table8-32. SWUnitMemoryArea definition

### 8.1.4.5.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurence |
|-----|------------------------------------------|-----|--------------------|
| 10  | generalReject                            | M   |                    |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   |                    |
| 31  | requestOutOfRange                        | M   |                    |
| 33  | securityAccessDenied                     | M   |                    |
| 78  | requestCorrectlyReceived-ResponsePending | M   |                    |

Table8-33. Supported Negative Response Codes

### 8.1.4.6 (OLD)SwapActiveArea (ECU with memory redundancy)

This routine can cause system reset. System reset should be performed according to the ECU reset service. For example, the Ethdiag TCP socket shall be closed(FIN or RST) after a positive response is transmitted and the Server shall perform a reset thereafter. And the engine conditions of the ECU reset service must be complied with.. See also OTA requirement(ES98765-01) indetailed.

Session : Non-default session

Security level : Required (ASK / OEUK)

### 8.1.4.6.1 Request message

| Data Byte | Parameter Name                    | Cvt | Hex Value | Mnemonic |
|-----------|-----------------------------------|-----|-----------|----------|
| #1d       | RoutineControl request Service ID | M   | #31h      | RC       |

|                                                                                                                                                                                                                          |                                                                                             |        |              |                 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------|--------------|-----------------|
| #2d                                                                                                                                                                                                                      | Sub-function = [ routineControlType ]                                                       | M      | #01h         | LEV_RCTP_       |
| #3d<br>#4d                                                                                                                                                                                                               | routineIdentifier [] = [ byte #1 (MSB)<br>byte #2 ]                                         | M<br>M | #02h<br>#13h | RI_<br>B1<br>B2 |
| #5d                                                                                                                                                                                                                      | routineControlOptionRecord [] = [ MemoryArea ]                                              | U      | #XXh         | RCOR_MA         |
| #6d<br>#7d                                                                                                                                                                                                               | routineControlOptionRecord [] = [ SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | C      | #F1h<br>#BXh | RCOR_SW<br>UMA  |
| <p>The MemoryArea represents the memory area specified as the boot(operation) area for ECU with memory redundancy.</p> <p>C: If MemoryArea which is placed in byte field #5d is used, this field would be available.</p> |                                                                                             |        |              |                 |

Table8-34. Request Message

| Hex                                                                     | Description                            | Cvt | Mnemonic |
|-------------------------------------------------------------------------|----------------------------------------|-----|----------|
| 00                                                                      | No specifying                          | U   |          |
| 0A                                                                      | Specify area A as boot(operation) area | M   |          |
| 0B                                                                      | Specify area B as boot(operation) area | M   |          |
| 'No specifying' would be used if ECU can decide swap area on their own. |                                        |     |          |

Table8-35. MemoryArea definition

| Hex                               | Description                   | Cvt | Mnemonic   |
|-----------------------------------|-------------------------------|-----|------------|
| F1B0                              | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1                              | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2                              | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3                              | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4                              | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5                              | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6                              | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7                              | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8                              | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9                              | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA                              | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |
| C: More than 1 SWUnit is defined. |                               |     |            |

Table8-36. SWUnitMemoryArea definition

### 8.1.4.6.2 Positive response message

| Data Byte  | Parameter Name                                                                              | Cvt      | Hex Value    | Mnemonic        |
|------------|---------------------------------------------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>                                                   | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                                                          | M        | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [ byte #1 (MSB)<br>byte #2 ]                                         | M<br>M   | #02h<br>#13h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [ MemoryArea ]                                              | U        | #XXh         | RCOR_MA         |
| #6d<br>#7d | routineControlOptionRecord [] = [ SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | C        | #F1h<br>#BXh | RCOR_SW<br>UMA  |



# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)

ES95486-02

페이지

(SHT/SHTS) 141/244

|                                                                                                                                                                                                                                                        |                                                                                  |   |              |         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---|--------------|---------|
| #8d<br>#9d                                                                                                                                                                                                                                             | routineControlOptionRecord [] = [<br>DelayTime byte#1(MSB)<br>DelayTime byte#2 ] | U | #XXh<br>#XXh | RCOR_DT |
| <p>The MemoryArea represents the memory area specified as the boot(operation) area for ECU with memory redundancy.<br/>Unit of DelayTime is Second.<br/>C: If MemoryArea which is placed in byte field #5d is used, this field would be available.</p> |                                                                                  |   |              |         |

Table8-37. Positive Response Message definition

| Hex         | Description                | Cvt | Mnemonic |
|-------------|----------------------------|-----|----------|
| 0000        | (Reserved)                 | M   |          |
| 0001 – 012C | Time Second (1 ~ 300 sec ) | M   |          |
| 012D – FFFF | (Reserved)                 | M   |          |

Table8-38. DelayTime definition

### 8.1.4.6.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                                                          |
|-----|------------------------------------------|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | generalReject                            | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                             |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                             |
| 22  | conditionsNotCorrect                     | M   | ECU Specific Conditions are not met                                                                                                                                                                                                                                                                                                                                          |
| 24  | requestSequenceError                     | M   | CheckProgramming dependencies were not successfully performed before the Swap Active Area request                                                                                                                                                                                                                                                                            |
| 31  | requestOutOfRange                        | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                             |
| 33  | securityAccessDenied                     | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                             |
| 78  | requestCorrectlyReceived-ResponsePending | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                             |
| F4  | LowerVersionDownloadAttempt              | M   | <p>This response code indicates that the requested action will not be taken because the downloaded software is a lower version than the existing version. <b>e.g. ) ECU downgrade attempts</b></p> <p><b>IMPORTANT) The memory swap ECU supporting SwapActiveArea should determine the downgrade in SwapActiveArea instead of the CheckProgrammingdependencies step.</b></p> |

Table8-39. Supported Negative Response Codes

### 8.1.4.7 (OLD)WriteDeltaPatch (ECU supporting differential data patch)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.7.1 Request message

| Data Byte  | Parameter Name                                                                                 | Cvt      | Hex Value    | Mnemonic        |
|------------|------------------------------------------------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b>  | <b>RC</b>       |
| #2d        | Sub-function = [<br>routineControlType ]                                                       | M        | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M<br>M   | #02h<br>#20h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [<br>MemoryArea ]                                              | M        | #XXh         | RCOR_MA         |
| #6d<br>#7d | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh         | RCOR_SW<br>UMA  |

The MemoryArea represents the memory area specified as the boot(operation) area for ECU with memory redundancy.

Table8-40. Request Message

| Hex | Description                                                          | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------|-----|----------|
| 00  | Write differential data patch (ECU without memory redundancy)        | M   |          |
| 0A  | Write differential data patch to area A (ECU with memory redundancy) | M   |          |
| 0B  | Write differential data patch to area B (ECU with memory redundancy) | M   |          |

Table8-41. MemoryArea definition

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-42. SWUnitMemoryArea definition

#### 8.1.4.7.2 Positive response message

| Data Byte | Parameter Name                            | Cvt      | Hex Value   | Mnemonic    |
|-----------|-------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b> | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |

|     |                                                        |   |      |           |
|-----|--------------------------------------------------------|---|------|-----------|
| #2d | routineControlType                                     | M | #01h | RCTP_     |
| #3d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M | #02h | RI_<br>B1 |
| #4d |                                                        | M | #20h | B2        |
| #5d | routineControlOptionRecord [] = [<br>MemoryArea ]      | M | #XXh | RCOR_MA   |

Table8-43. Positive Response Message definition

| Hex | Description                                                          | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------|-----|----------|
| 00  | Write differential data data (ECU without memory redundancy)         | M   |          |
| 0A  | Write differential data patch to area A (ECU with memory redundancy) | M   |          |
| 0B  | Write differential data patch to area B (ECU with memory redundancy) | M   |          |

Table8-44. MemoryArea definition

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-45. SWUnitMemoryArea definition

### 8.1.4.7.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurence |
|-----|------------------------------------------|-----|--------------------|
| 10  | generalReject                            | M   |                    |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   |                    |
| 31  | requestOutOfRange                        | M   |                    |
| 33  | securityAccessDenied                     | M   |                    |
| 78  | requestCorrectlyReceived-ResponsePending | M   |                    |

Table8-46. Supported Negative Response Codes

### 8.1.4.8 (OLD)CheckProgrammingDependancy(Not only OTA but also reprogrammable ECU)

Session : Non-default session

Security level : Required (ASK / OEUK)

#### 8.1.4.8.1 Request message

| Data Byte | Parameter Name                    | Cvt | Hex Value | Mnemonic |
|-----------|-----------------------------------|-----|-----------|----------|
| #1d       | RoutineControl request Service ID | M   | #31h      | RC       |

|            |                                                                                             |        |              |                 |
|------------|---------------------------------------------------------------------------------------------|--------|--------------|-----------------|
| #2d        | Sub-function = [ routineControlType ]                                                       | M      | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d | routineIdentifier [] = [ byte #1 (MSB)<br>byte #2 ]                                         | M<br>M | #FFh<br>#01h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [ MemoryArea]                                               | C      | #XXh         | RCOR_MA         |
| #6d<br>#7d | routineControlOptionRecord [] = [ SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | C      | #F1h<br>#BXh | RCOR_SW<br>UMA  |

The MemoryArea represents the memory area specified as the boot(operation) area for ECU with memory redundancy.

C : ECU supports OTA and More than 1 SWUnit is defined.

Table8-47. Request Message

| Hex | Description               | Cvt | Mnemonic |
|-----|---------------------------|-----|----------|
| 00  | No specifying             | U   |          |
| 0A  | Action request for area A | M   |          |
| 0B  | Action request for area B | M   |          |

'No specifying' would be used if ECU can recognize MemoryArea on their own.

Table8-48. MemoryArea definition

| Hex  | Description                   | Cvt | Mnemonic   |
|------|-------------------------------|-----|------------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2 | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3 | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4 | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5 | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6 | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7 | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8 | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9 | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |

C: More than 1 SWUnit is defined.

Table8-49. SWUnitMemoryArea definition

### 8.1.4.8.2 Positive response message

The positive response message has to be sent when the examination proves normal operation of the ECU is successful.

| Data Byte  | Parameter Name                                      | Cvt      | Hex Value    | Mnemonic        |
|------------|-----------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>           | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                  | M        | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [ byte #1 (MSB)<br>byte #2 ] | M<br>M   | #FFh<br>#01h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [ MemoryArea ]      | C        | #XXh         | RCOR_MA         |

|            |                                                                                                |   |              |                |
|------------|------------------------------------------------------------------------------------------------|---|--------------|----------------|
| #6d<br>#7d | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | C | #F1h<br>#BXh | RCOR_SW<br>UMA |
|------------|------------------------------------------------------------------------------------------------|---|--------------|----------------|

Table8-50. Positive Response Message definition

#### 8.1.4.8.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurence                                                                                                                                                                                                                                                                                                                                                |
|-----|------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | generalReject                            | M   | This response code indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation.                                                                                             |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                  |
| 24  | requestSequenceError                     | M   | The requested action will not be taken because the server expects a different sequence of request messages or message as sent by the client. This may occur when sequence sensitive requests are issued in the wrong order.<br>e.g.) expected reprogramming requests are not issued.                                                                              |
| 31  | requestOutOfRange                        | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                  |
| 33  | securityAccessDenied                     | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                  |
| 72  | generalProgrammingFailure                | M   | This NRC shall be returned if the server detects an error when performing a routine, which accesses server internal memory. An example is when the routine erases or prgrams a certain memory location in the permanent memory device (e.g. Flash Memory) and the access to that memory location fails.<br><b>e.g. ) SW integrity fails</b>                       |
| 78  | requestCorrectlyReceived-ResponsePending | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                                                  |
| F4  | LowerVersionDownloadAttempt              | M   | This response code indicates that the requested action will not be taken because the downloaded software is a lower version than the existing version. <b>e.g. ) ECU downgrade attempts</b><br><b>IMPORTANT) The memory swap ECU supporting SwapActiveArea should determine the downgrade in SwapActiveArea instead of the CheckProgrammingdependencies step.</b> |

Table8-51. Supported Negative Response Codes

#### 8.1.4.9 (OLD)FinishUpdate

Note : The Identifier(0211) is identical with CopyActiveAreaTo which is legacy routine identifier. But

request/response message format is different from CopyActiveAreaTo  
When transmitting a FinishUpdate through functional addressing, copyActiveAreaTo with a different total length of the routine should not operate.  
Session : Default(01), Extended(03)  
Security level : No

### 8.1.4.9.1 Request message

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b>  | <b>RC</b>       |
| #2d        | Sub-function = [<br>routineControlType ]               | M        | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #02h<br>#11h | RI_<br>B1<br>B2 |

Table8-113. Request Message

### 8.1.4.9.2 Positive response message

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                     | M        | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #02h<br>#11h | RI_<br>B1<br>B2 |

Table8-114. Positive Response Message definition

### 8.1.4.9.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence |
|-----|---------------------------------------|-----|--------------------|
| 10  | generalReject                         | M   | Refer to Annex A   |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Refer to Annex A   |
| 31  | requestOutOfRange                     | M   | Refer to Annex A   |

Table8-115. Supported Negative Response Codes

## 8.1.5 Implementation rules (regarding ES98765-02)

### 8.1.5.1 OTA Ready

Session : Default and Non-default session  
Security level : No

#### 8.1.5.1.1 Request message

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b>  | <b>RC</b>       |
| #2d        | Sub-function = [<br>routineControlType ]               | M        | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #03h<br>#00h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [<br>InitParam]        | C        | #01h         | RCOR_IP         |

**C : The only adaptive platform that includes UCM can support this parameter. The others shall not be used.**  
**UCM ECU shall support both format. One is 31 01 03 00, the other is 31 01 03 00 01**

Table8-91. Request Message

#### 8.1.5.1.2 Positive response message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic    |
|-----------|--------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                     | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #03h        | RI_<br>B1   |
| #4d       |                                                        | M        | #00h        | B2          |
| #5d       | routineControlOptionRecord [] = [<br>routineStatus ]   | M        | #XXh        | RCOR_RS     |

Table8-92. Positive Response Message definition

| Hex   | Description                                    | Cvt | Mnemonic |
|-------|------------------------------------------------|-----|----------|
| 00    | (Reserved)                                     | M   |          |
| 01    | OK<br>(Server can be reprogrammed immediately) | M   |          |
| 02~FF | (Reserved)                                     | M   |          |

Table8-93. Routine status definition

#### 8.1.5.1.3 Supported negative response codes

| Hex | NegativeResponseCode                     | Cvt | Cause of Occurence                                                                                                                                      |
|-----|------------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | generalReject                            | M   | Unable to reprogram.<br>Negative response during pending.<br>If operation is required for more than 3 minutes while IG is off.<br>IG on during Pending. |
| 13  | incorrectMessageLengthOrInvalidFormat    | M   | Message format or Length error                                                                                                                          |
| 31  | requestOutOfRange                        | M   | Routine identifier out of range.<br>Routine identifier is not supported in active session                                                               |
| 78  | requestCorrectlyReceived-ResponsePending | M   | Essential functions that occur only in special driving and use environments (less than 5 out of 100 drives) are in operation.                           |

Table8-94. Supported Negative Response Codes

#### 8.1.5.2 EnableUpdateCondition

Session : Default and Non-default session

Security level : No

##### 8.1.5.2.1 Request message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic      |
|-----------|--------------------------------------------------------|----------|-------------|---------------|
| #1d       | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b> | <b>RC</b>     |
| #2d       | Sub-function = [<br>routineControlType ]               | M        | #01h        | LEV_<br>RCTP_ |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #03h        | RI_<br>B1     |
| #4d       |                                                        | M        | #01h        | B2            |

Table8-110. Request Message

### 8.1.5.2.2 Positive response message

| Data Byte | Parameter Name                                         | Cvt      | Hex Value   | Mnemonic    |
|-----------|--------------------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                                     | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M        | #03h        | RI_<br>B1   |
| #4d       |                                                        | M        | #01h        | B2          |

Table8-111. Positive Response Message definition

### 8.1.5.2.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence                                                                        |
|-----|---------------------------------------|-----|-------------------------------------------------------------------------------------------|
| 10  | generalReject                         | M   | Refer to Annex. A                                                                         |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Message format or Length error                                                            |
| 31  | requestOutOfRange                     | M   | Routine identifier out of range.<br>Routine identifier is not supported in active session |

Table8-112. Supported Negative Response Codes

### 8.1.5.3 EraseMemory

Session : Programming(02) – Mandaroty, Extended(03) – Conditional

※Conditional : Background Transfer & Memory redundancy ECU

Security level : Required (ASK / OEUK)

#### 8.1.5.3.1 Request message

| Data Byte | Parameter Name                                                                                 | Cvt      | Hex Value   | Mnemonic       |
|-----------|------------------------------------------------------------------------------------------------|----------|-------------|----------------|
| #1d       | <b>RoutineControl request Service ID</b>                                                       | <b>M</b> | <b>#31h</b> | <b>RC</b>      |
| #2d       | Sub-function = [<br>routineControlType ]                                                       | M        | #01h        | LEV_<br>RCTP_  |
| #3d       | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M        | #FFh        | RI_<br>B1      |
| #4d       |                                                                                                | M        | #00h        | B2             |
| #5d       | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh        | RCOR_SW<br>UMA |
| #6d       |                                                                                                |          | #XXh        |                |

The SWUnitMemoryArea represents the memory area to be erased.

Table8-95. Request Message

| Hex  | Description                   | Cvt | Mnemonic  |
|------|-------------------------------|-----|-----------|
| F1B0 | ECUSoftwareUNIT's entire area | M   | ECUSUNIT  |
| F1B1 | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1 |
| F1B2 | ECUSoftwareUNIT2 area         | C1  | ECUSUNIT2 |
| F1B3 | ECUSoftwareUNIT3 area         | C1  | ECUSUNIT3 |
| F1B4 | ECUSoftwareUNIT4 area         | C1  | ECUSUNIT4 |
| F1B5 | ECUSoftwareUNIT5 area         | C1  | ECUSUNIT5 |
| F1B6 | ECUSoftwareUNIT6 area         | C1  | ECUSUNIT6 |
| F1B7 | ECUSoftwareUNIT7 area         | C1  | ECUSUNIT7 |
| F1B8 | ECUSoftwareUNIT8 area         | C1  | ECUSUNIT8 |

|                                                                           |                                 |    |            |
|---------------------------------------------------------------------------|---------------------------------|----|------------|
| F1B9                                                                      | ECUSoftwareUNIT9 area           | C1 | ECUSUNIT9  |
| F1BA                                                                      | ECUSoftwareUNIT10 area          | C1 | ECUSUNIT10 |
| FFFF                                                                      | initialize Adaptive AUTOSAR UCM | C2 | IAASARU    |
| C1: More than 1 SWUnit is defined.<br>C2 :In case of Adaptive Autosar UCM |                                 |    |            |

Table8-96. SWUnitMemoryArea definition

### 8.1.5.3.2 Positive response message

| Data Byte  | Parameter Name                                                                                 | Cvt      | Hex Value    | Mnemonic        |
|------------|------------------------------------------------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>                                                      | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                                                             | M        | #01h         | RCTP_           |
| #3d        | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ]                                         | M<br>M   | #FFh<br>#00h | RI_<br>B1<br>B2 |
| #5d<br>#6d | routineControlOptionRecord [] = [<br>SWUnitMemoryArea byte#1(MSB)<br>SWUnitMemoryArea byte#2 ] | M        | #XXh<br>#XXh | RCOR_SW<br>UMA  |
| #7d        | routineStatus [] = [<br>ErasedPartitionArea #1]                                                | M        | #XXh         | RCOR_EP<br>A    |

Table8-97. Positive Response Message definition

| Hex                               | Description                   | Cvt | Mnemonic   |
|-----------------------------------|-------------------------------|-----|------------|
| F1B0                              | ECUSoftwareUNIT's entire area | M   | ECUSUNIT   |
| F1B1                              | ECUSoftwareUNIT1 area         | M   | ECUSUNIT1  |
| F1B2                              | ECUSoftwareUNIT2 area         | C   | ECUSUNIT2  |
| F1B3                              | ECUSoftwareUNIT3 area         | C   | ECUSUNIT3  |
| F1B4                              | ECUSoftwareUNIT4 area         | C   | ECUSUNIT4  |
| F1B5                              | ECUSoftwareUNIT5 area         | C   | ECUSUNIT5  |
| F1B6                              | ECUSoftwareUNIT6 area         | C   | ECUSUNIT6  |
| F1B7                              | ECUSoftwareUNIT7 area         | C   | ECUSUNIT7  |
| F1B8                              | ECUSoftwareUNIT8 area         | C   | ECUSUNIT8  |
| F1B9                              | ECUSoftwareUNIT9 area         | C   | ECUSUNIT9  |
| F1BA                              | ECUSoftwareUNIT10 area        | C   | ECUSUNIT10 |
| C: More than 1 SWUnit is defined. |                               |     |            |

Table8-98. SWUnitMemoryArea definition

| Hex | Description                                     | Cvt | Mnemonic |
|-----|-------------------------------------------------|-----|----------|
| 00  | Erased (if not memory redundant ECU or MMU ECU) | M   | ER       |
| 0A  | Erased area A                                   | M   | ERA      |
| 0B  | Erased area B                                   | M   | ERB      |

Table8-99. ErasedPartitionArea definition

### 8.1.5.3.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence |
|-----|---------------------------------------|-----|--------------------|
| 10  | generalReject                         | M   | Refer to Annex A   |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Refer to Annex A   |
| 31  | requestOutOfRange                     | M   | Refer to Annex A   |
| 33  | securityAccessDenied                  | M   | Refer to Annex A   |

Table8-100. Supported Negative Response Codes

#### 8.1.5.4 CheckMemory

Session : Programming(02) – Mandaroty, Extended(03) – Conditional

※Conditional : Background Transfer & Memory redundancy ECU

Security level : Required (ASK / OEUK)

##### 8.1.5.4.1 Request message

| Data Byte                                                     | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|---------------------------------------------------------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d                                                           | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b>  | <b>RC</b>       |
| #2d                                                           | Sub-function = [<br>routineControlType ]               | M        | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d                                                    | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #02h<br>#00h | RI_<br>B1<br>B2 |
| The SWUnitMemoryArea represents the memory area to be erased. |                                                        |          |              |                 |

Table8-101. Request Message

##### 8.1.5.4.2 Positive response message

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                     | M        | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #02h<br>#00h | RI_<br>B1<br>B2 |

Table8-102. Positive Response Message definition

##### 8.1.5.4.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence                                                                                                                                                                                                                                                                                                                          |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | generalReject                         | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                            |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                            |
| 31  | requestOutOfRange                     | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                            |
| 33  | securityAccessDenied                  | M   | Refer to Annex A                                                                                                                                                                                                                                                                                                                            |
| 72  | generalProgrammingFailure             | M   | This NRC shall be returned if the server detects an error when performing a routine, which accesses server internal memory. An example is when the routine erases or prgrams a certain memory location in the permanent memory device (e.g. Flash Memory) and the access to that memory location fails.<br><b>e.g. ) SW integrity fails</b> |

Table8-103. Supported Negative Response Codes

### 8.1.5.5 CheckProgrammingDependancies

Session : Programming(02) - Mandaroty, Extended(03) - Optional  
Security level : Required (ASK / OEUK)

#### 8.1.5.5.1 Request message

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl request Service ID</b>               | <b>M</b> | <b>#31h</b>  | <b>RC</b>       |
| #2d        | Sub-function = [<br>routineControlType ]               | M        | #01h         | LEV_<br>RCTP_   |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #FFh<br>#01h | RI_<br>B1<br>B2 |
| #5d        | routineControlOptionRecord [] = [<br>UpdateTarget]     | C        | #01h         | RCOR_UT         |

**C: The only adaptive platform that includes UCM can support this parameter. The others should not use this parameter. App update: 01h**  
**ECU that uses this parameter shall support both format. One is 31 01 FF 01, the other is 31 01 FF 01 01 can be used.**

Table8-104. Request Message

#### 8.1.5.5.2 Positive response message

The positive response message has to be sent when the examination proves normal operation of the ECU is successful.

| Data Byte  | Parameter Name                                         | Cvt      | Hex Value    | Mnemonic        |
|------------|--------------------------------------------------------|----------|--------------|-----------------|
| #1d        | <b>RoutineControl Response Service ID</b>              | <b>M</b> | <b>#71h</b>  | <b>RCPR</b>     |
| #2d        | routineControlType                                     | M        | #01h         | RCTP_           |
| #3d<br>#4d | routineIdentifier [] = [<br>byte #1 (MSB)<br>byte #2 ] | M<br>M   | #FFh<br>#01h | RI_<br>B1<br>B2 |
| #5d        | routineStatusRecord [] = [<br>UpdateTarget]            | C        | #01h         | RS_UT           |

**C: The only adaptive platform that includes UCM can support this parameter. The others should not use this parameter. App update: 01h**  
**ECU that uses this parameter shall support both format. One is 31 01 FF 01, the other is 31 01 FF 01 01 can be used.**

Table8-105. Positive Response Message definition

#### 8.1.5.5.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence                                                                                                                                                                                                                                                                |
|-----|---------------------------------------|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10  | generalReject                         | M   | Refer to Annex A                                                                                                                                                                                                                                                                  |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Refer to Annex A                                                                                                                                                                                                                                                                  |
| 24  | requestSequenceError                  | M   | The requested action will not be taken because the server expects a different sequence of request messages or message as sent by the client. This may occur when sequence sensitive requests are issued in the wrong order. e.g.) expected reprogramming requests are not issued. |
| 31  | requestOutOfRange                     | M   | Refer to Annex A                                                                                                                                                                                                                                                                  |

|    |                             |   |                                                                                                                                                                                                                                                                                                          |
|----|-----------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 33 | securityAccessDenied        | M | Refer to Annex A                                                                                                                                                                                                                                                                                         |
| 72 | generalProgrammingFailure   | M | This NRC shall be returned if the server detects an error when performing a routine, which accesses server internal memory. An example is when the routine erases or programs a certain memory location in the permanent memory device (e.g. Flash Memory) and the access to that memory location fails. |
| F4 | LowerVersionDownloadAttempt | M | This response code indicates that the requested action will not be taken because the downloaded software is a lower version than the existing version. <b>e.g. ) ECU downgrade attempts</b>                                                                                                              |

Table8-106. Supported Negative Response Codes

### 8.1.5.6 FinishUpdate

Note : The Identifier(0211) is identical with CopyActiveAreaTo which is legacy routine identifier. But request/response message format is different from CopyActiveAreaTo

When transmitting a FinishUpdate through functional addressing, copyActiveAreaTo with a different total length of the routine should not operate.

Session : Default(01), Extended(03)

Security level : No

#### 8.1.5.6.1 Request message

| Data Byte | Parameter Name                           | Cvt      | Hex Value   | Mnemonic  |
|-----------|------------------------------------------|----------|-------------|-----------|
| #1d       | <b>RoutineControl request Service ID</b> | <b>M</b> | <b>#31h</b> | <b>RC</b> |
| #2d       | Sub-function = [ routineControlType ]    | M        | #01h        | LEV_RCTP_ |
| #3d       | routineIdentifier [] = [ byte #1 (MSB)   | M        | #02h        | RI_B1     |
| #4d       | byte #2 ]                                | M        | #11h        | B2        |

Table8-107. Request Message

#### 8.1.5.6.2 Positive response message

| Data Byte | Parameter Name                            | Cvt      | Hex Value   | Mnemonic    |
|-----------|-------------------------------------------|----------|-------------|-------------|
| #1d       | <b>RoutineControl Response Service ID</b> | <b>M</b> | <b>#71h</b> | <b>RCPR</b> |
| #2d       | routineControlType                        | M        | #01h        | RCTP_       |
| #3d       | routineIdentifier [] = [ byte #1 (MSB)    | M        | #02h        | RI_B1       |
| #4d       | byte #2 ]                                 | M        | #11h        | B2          |

Table8-108. Positive Response Message definition

#### 8.1.5.6.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurence |
|-----|---------------------------------------|-----|--------------------|
| 10  | generalReject                         | M   | Refer to Annex A   |
| 13  | incorrectMessageLengthOrInvalidFormat | M   | Refer to Annex A   |
| 31  | requestOutOfRange                     | M   | Refer to Annex A   |

Table8-109. Supported Negative Response Codes



| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                   |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                                                                                    |
| 22  | conditionNotCorrect                   | M   | This return code shall be sent if a server receives a request for this service while in the process of receiving a download of a software or calibration module. This could occur if there is a data size mismatch between the server and the client during the download of a module. |
| 24  | requestSequenceError                  | M   | The server shall return this response code in the event that the requested memory area includes a reprogramming zone, and the 'Erase Memory' operation for the requested memory area has not been successfully completed prior to this request.                                       |
| 31  | requestOutOfRange                     | M   | This return code shall be sent if ;<br>1. The specified dataFormatIdentifier is not valid.<br>2. The specified addressAndLengthFormatIdentifier is not valid.<br>3. The specified memoryAddress / memorySize is not valid.                                                            |
| 33  | securityAccessDenied                  | M   | This return code shall be sent if the server is secure (for server's that support the SecurityAccess service) when a request for this service has been received.                                                                                                                      |
| 70  | uploadDownloadNotAccepted             | M   | This response code indicates that an attempt to download to a server's memory cannot be accomplished due to some fault conditions.                                                                                                                                                    |

Table9-3. Supported Negative Response Codes

### 9.1.4 Implementation rules

The address may represent a physical address location within the ECU, but addresses may also be mapped (e.g. to implement a memory selector in upper bits). However, if the address parameter does not correspond to the physical location, the mapping shall be documented within the ECU diagnostic specification.

The DataFormatIdentifier consists of 2 nibbles, the left one containing the compression method, the right one containing the encryption method. If no compression or encryption applies, the corresponding parameter shall be set to 0.

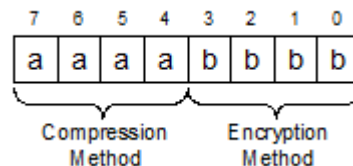


Figure9-1. Structure of DataFormatIdentifier

At least a DataFormatIdentifier of 00 shall be supported.

| HEX | Description                               | Cvt | Mnemonic |
|-----|-------------------------------------------|-----|----------|
| 0   | No Compression Method                     | M   |          |
| 1~9 | Reserved (System Supplier Specific Range) |     |          |
| A   | LZMA                                      | U   |          |
| B~F | Reserved (Manufacturer Specific Range)    |     |          |

Table9-4. Compression Method



| HEX | Description                               | Cvt | Mnemonic |
|-----|-------------------------------------------|-----|----------|
| 0   | No Encryption Method                      | M   |          |
| 1~9 | Reserved (System Supplier Specific Range) |     |          |
| A~F | Reserved (Manufacturer Specific Range)    |     |          |

Table9-5. Encryption Method

The addressAndLengthFormatIdentifier is a one byte value with each nibble coded separately

- bit 7 – 4 : Length (number of bytes) of the memorySize parameter
- bit 3 – 0 : Length (number of bytes) of the momoryAddress parameter

The parameter memoryAddress is the starting address of the server memory where the data is to be written to.

The momorySize parameter shall be used by the server to compare the memory size with the total amount of data transferred during the TransferData service. The number of bytes used for the this size is defined by the high nibble (bit 7 – 4) of the addressFormatIdentifier.

The LengthFormatIdentifier is a one byte value with each nibble coded separately:

- bit 7 – 4 : Length (number of bytes) of the maxNumberOfBlockLength parameter
- bit 3 – 0 : reserved by document to be set 0 hex

The format of this parameter is compatible to the format of the addressAndLengthFormatIdentifier parameter contained in the request message, except that the lower nibble has to be set to 0 hex.

The parameter maxnumberOfBlockLength is used by the requestDownload positive response message to inform the client how many data bytes shall be included in each TransferData request message from the client. This length reflects the complete message length, including the service identifier and the data parameters present in the TransferData request message. This parameter allows the client to adapt to the receive buffer size of the server before it starts transferring to the server.

## 9.2 RequestUpload (35 hex) service

The RequestUpload service is used by the client to initiate a data transfer from the server to the client (upload). After the server has received the requestUpload request message the server shall take all necessary actions to send data before it sends a positive response message.

### 9.2.1 Request message

| Data Byte | Parameter Name                          | Cvt      | Hex Value   | Mnemonic  |
|-----------|-----------------------------------------|----------|-------------|-----------|
| #1d       | <b>RequestUpload Request Service ID</b> | <b>M</b> | <b>#35h</b> | <b>RU</b> |
| #2d       | dataFormatIdentifier                    | M        | #00h        | DFI_      |
| #3d       | addressAndLengthFormatIdentifier        | M        | #XXh        | ALFID     |
| #4d       | memoryAddress[] = [<br>byte#1 (MSB)     | M        | #XXh        | MA_<br>B1 |
| ..        | ..                                      | ..       | ..          | ..        |
| #(m-1)+4d | byte#m ]                                | C        | #XXh        | Bm        |
| #n-(k-1)d | memorySize[] = [<br>byte#1 (MSB)        | M        | #XXh        | MS_<br>B1 |
| ..        | ..                                      | ..       | ..          | ..        |
| #nd       | byte#k ]                                | C        | #XXh        | Bk        |

Table9-6. Request Message definition

### 9.2.2 Positive response message

| Data Byte | Parameter Name                           | Cvt      | Hex Value   | Mnemonic     |
|-----------|------------------------------------------|----------|-------------|--------------|
| #1d       | <b>RequestUpload Response Service ID</b> | <b>M</b> | <b>#75h</b> | <b>RUPR</b>  |
| #2d       | lengthFormatIdentifier                   | M        | #XXh        | LFID         |
| #3d       | maxNumberOfBlockLength [<br>byte#1 (MSB) | M        | #XXh        | MNROB_<br>B1 |
| ..        | ..                                       | ..       | ..          | ..           |
| #nd       | byte#m ]                                 | M        | #XXh        | Bm           |

Table9-7. Positive Response Message definition

### 9.2.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                                                        |
|-----|---------------------------------------|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                                         |
| 22  | conditionNotCorrect                   | M   | This return code shall be sent if the criteria for the requestUpload are not met. This could occur if a server receives a request for this service while a requestUpload is already active, but not yet completed.         |
| 31  | requestOutOfRange                     | M   | This return code shall be sent if ;<br>1. The specified dataFormatIdentifier is not valid.<br>2. The specified addressAndLengthFormatIdentifier is not valid.<br>3. The specified memoryAddress / memorySize is not valid. |
| 33  | securityAccessDenied                  | M   | This return code shall be sent if the server is secure (for server's that support the SecurityAccess service) when a request for this service has been received.                                                           |



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|    |                           |   |                                                                                                                                  |
|----|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------|
| 70 | uploadDownloadNotAccepted | M | This response code indicates that an attempt to upload to a server's memory cannot be accomplished due to some fault conditions. |
|----|---------------------------|---|----------------------------------------------------------------------------------------------------------------------------------|

Table9-8. Supported Negative Response Codes

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### 9.3 TransferData (36 hex) service

The TransferData service is used by the client to transfer data either from the client to the server (download) or from the server to the client (upload).

The data transfer direction is defined by the preceding RequestDownload or RequestUpload service. If the client initiated a RequestDownload the data to be downloaded is included in the parameter(s) transferRequestParameter in the TransferData request message(s). If the client initiated a RequestUpload the data to be uploaded is included in the parameter(s) transferResponseParameter in the TransferData response message(s).

The TransferData service request includes a blockSequenceCounter to allow for an improved error handling in case a TransferData service fails during a sequence of multiple TransferData requests. The blockSequenceCounter of the server shall be initialized to one (1) when receiving a RequestDownload (34 hex) or RequestUpload (35 hex) request message. This means that the first TransferData (36 hex) request message following the RequestDownload (34 hex) or RequestUpload (35 hex) request message starts with a blockSequenceCounter of one (1).

Example use cases:

- If a TransferData request to download data is correctly received and processed in the server but the positive response message does not reach the client then the client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this TransferData request is repeated. The server would send the positive response message immediately without writing the data once again into its memory.
- If the TransferData request to download data is not received correctly in the server then the server would not send a positive response message. The client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this is a new TransferData. The server would process the service and would send the positive response message.
- If a TransferData request to upload data is correctly received and processed in the server but the positive response message does not reach the client then the client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this TransferData request is repeated. The server would send the positive response message immediately accessing the previously provided data once again in its memory.
- If the TransferData request to upload data is not received correctly in the server then the server would not send a positive response message. The client would determine an application layer timeout and would repeat the same request (including the same blockSequenceCounter). The server would receive the repeated TransferData request and could determine based on the included blockSequenceCounter that this is a new TransferData. The server would process the service and would send the positive response message.

#### 9.3.1 Request message

| Data Byte | Parameter Name                  | Cvt | Hex Value | Mnemonic |
|-----------|---------------------------------|-----|-----------|----------|
| #1d       | TransferData Request Service ID | M   | #36h      | TD       |
| #2d       | blockSequenceCounter            | M   | #XXh      | BSC      |

|     |                                      |    |      |       |
|-----|--------------------------------------|----|------|-------|
| #3d | transferRequestParameterRecord[] = [ | C  | #XXh | TRPR_ |
| ..  | transferRequestParameter#1           | .. | ..   | TRTP_ |
| #nd | transferRequestParameter#m ]         | U  | #XXh | TRTP_ |

C = Conditional : this parameter is mandatory if a download is in progress.

**blockSequenceCounter**  
The blockSequenceCounter parameter value starts at 01 hex with the first TransferData request that follows the RequestDownload (34 hex) or RequestUpload (35 hex) service. Its value is incremented by 1 for each subsequent TransferData request. At the value of FF hex the blockSequenceCounter rolls over and starts at 00 hex with the next TransferData request message.

Table9-9. Request Message definition

## 9.3.2 Positive response message

| Data Byte | Parameter Name                        | Cvt | Hex Value | Mnemonic |
|-----------|---------------------------------------|-----|-----------|----------|
| #1d       | TransferData Response Service ID      | M   | #76h      | TDPR     |
| #2d       | blockSequenceCounter                  | M   | #XXh      | BSC      |
| #3d       | transferResponseParameterRecord[] = [ | C   | #XXh      | TREPR_   |
| ..        | transferResponseParameter#1           | ..  | ..        | TREP_    |
| #nd       | transferResponseParameter#m ]         | U   | #XXh      | TREP_    |

C = Conditional : this parameter is mandatory if an upload is in progress.

Table9-10. Positive Response Message definition

## 9.3.3 Supported negative response codes

| Hex | NegativeResponseCode                 | Cvt | Cause of Occurrence                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----|--------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | icorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong (e.g., message length does not meet requirements of maxNumberOfBlockLength parameter returned in the positive response to requestDownload).                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 24  | requestSequenceError                 | M   | <p>The server shall use this response code</p> <ol style="list-style-type: none"> <li>1. If the RequestDownload or RequestUpload service is not active when a request for this service is received.</li> <li>2. If the RequestDownload or RequestUpload service is active, but the server has already received all data as determined by the memorySize parameter in the active RequestDownlod or RequestUpload service.</li> </ol> <p>NOTE) The repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the server.</p> |
| 31  | requestOutOfRange                    | M   | This return code shall be sent if the transferRequestParameterRecord contains additional control parameters (e.g. additional address information) and this control information is invalid.                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|    |                           |   |                                                                                                                                                                                                                                                                                                              |
|----|---------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 71 | transferDataSuspended     | M | This return code shall be sent if<br>1. The response code indicates that a data transfer operation was halted due to some fault.<br>2. The download module length does not meet the requirements of the memorySize parameter sent in the request message of the requestDownload service.                     |
| 72 | generalProgrammingFailure | M | This return code shall be sent if the server detects an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory) during the download of data.                                                                                                                  |
| 73 | wrongBlockSequenceCounter | M | This return code shall be sent if the server detects an error in the sequence of the blockSequenceCounter.<br><br>NOTE) The repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the server. |
| 92 | voltageTooHigh            | M | This return code shall be sent as applicable if the voltage measured at the primary power pin of the server is out of the acceptable range for downloading data into the server's permanent memory (e.g. Flash Memory).                                                                                      |
| 93 | voltageTooLow             | M |                                                                                                                                                                                                                                                                                                              |

Table9-11. Supported Negative Response Codes

### 9.3.4 Implementation rules

The BlockSequenceCounter allows the diagnostic server to detect and handle transmission errors. It shall start at 1. Its value is incremented by 1 for each subsequent request of this service. When the block number 255 (0xFF) is transmitted, it shall continue with block number 0.

## 9.4 RequestTransferExit (37 hex) service

This service is used by the client to terminate a data transfer between client and server (upload or download).

### 9.4.1 Request message

| Data Byte                                                                                                                                                                                                                    | Parameter Name                                                     | Cvt      | Hex Value   | Mnemonic       |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|----------|-------------|----------------|
| #1d                                                                                                                                                                                                                          | <b>RequestTransferExit Request Service ID</b>                      | <b>M</b> | <b>#37h</b> | <b>RTE</b>     |
| #2d                                                                                                                                                                                                                          | transferRequestParameterRecord[] = [<br>transferRequestParameter#1 | U        | #XXh        | TRPR_<br>TRTP_ |
| ..                                                                                                                                                                                                                           | ..                                                                 | ..       | ..          | ..             |
| #nd                                                                                                                                                                                                                          | trans ferRequestParameter#m]                                       | U        | #XXh        | TRTP_          |
| transferReqeustParameterRecord<br>- This parameter record contains parameter(s), which are required by the server to support the transfer of data. Format and length of this parameter(s) are vehicle manufacturer specific. |                                                                    |          |             |                |

Table9-12. Request Message

### 9.4.2 Positive response message

| Data Byte                                                                                                                                                                                                                  | Parameter Name                                                       | Cvt      | Hex Value   | Mnemonic        |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|----------|-------------|-----------------|
| #1d                                                                                                                                                                                                                        | <b>RequestTransferExit Response Service ID</b>                       | <b>M</b> | <b>#77h</b> | <b>RTEPR</b>    |
| #2d                                                                                                                                                                                                                        | transferResponseParameterRecord[] = [<br>transferResponseParameter#1 | U        | #XXh        | TREPR_<br>TREP_ |
| ..                                                                                                                                                                                                                         | ..                                                                   | ..       | ..          | ..              |
| #nd                                                                                                                                                                                                                        | trans ferResponseParameter#m]                                        | U        | #XXh        | TREP_           |
| transferResponseParameterRecord<br>- This parameter shall contain parameter(s) which are required by the client to support the transfer of data. Format and length of this parameter(s) are vehicle manufacturer specific. |                                                                      |          |             |                 |

Table9-13. Positive Response Message definition

### 9.4.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                                                                                                                                                                               |
|-----|---------------------------------------|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong                                                                                                                                                                |
| 24  | requestSequenceError                  | M   | This return code shall be sent if ;<br>1. The programming process is not completed when a request for this service is received.<br>2. The RequestDownload or ReqeustUpload service is not active. |
| 31  | requestOutOfRange                     | U   | This NRC shall be returned if the transferRequestParameterRecord contains invalid data.                                                                                                           |
| 72  | GeneralProgrammingFailure             | U   | This NRC shall be returned if the server detects an error when finalizing the data transfer between the client and server (e.g., via an integrity check).                                         |

Table9-14. Supported Negative Response Codes

## 9.5 RequestFileTransfer (38hex) service

This service is used for initialization to transmit file data and is used only for controllers with necessary use cases. As an alternative solution to the existing Request download 34 and Request upload 35, this service can be used if the controller has a file system or data storage space. This service can be used for file system search. It may be used for download to ECU, deletion, replacement, upload to client, and directory read. IMPORTANT : This service shall restrict access to only permitted areas of the controller.

### 9.5.1 Request message

| Data Byte                     | Parameter Name                                                | Cvt              | Hex Value          | Mnemonic                 |
|-------------------------------|---------------------------------------------------------------|------------------|--------------------|--------------------------|
| #1d                           | <b>RequestFileTransfer Request SID</b>                        | <b>M</b>         | <b>#38h</b>        | <b>RFT</b>               |
| #2d                           | modeOfOperation                                               | M                | #01~06h            | MOOP                     |
| #3d<br>#4d                    | filePathAndNameLength[] = [<br>byte#1 (MSB)<br>byte#2 (LSB) ] | M                | #XXh<br>#XXh       | FPL_B1<br>FPL_B2         |
| #5d<br>..<br>#5+n-1d          | filePathAndName[] = [<br>byte#1 (MSB)<br>..<br>byte#n ]       | M<br>..<br>C1    | #XXh<br>..<br>#XXh | FP_B1<br>..<br>FP_Bn     |
| #5+nd                         | dataFormatIdentifier                                          | C2               | #00h               | DFI_                     |
| #5+n+1d                       | fileSizeParameterLength                                       | C2               | #XXh               | FSL                      |
| #5+n+2d<br>..<br>#5+n+2+k-1d  | fileSizeUnCcompressed[] = [<br>byte#1 (MSB)<br>..<br>byte#k ] | C2<br>..<br>C2,3 | #XXh<br>..<br>#XXh | FSUC_B1<br>..<br>FSUC_Bk |
| #5+n+2+kd<br>..<br>#5+n+1+2kd | fileSizeCompressed[] = [<br>byte#1 (MSB)<br>..<br>byte#k ]    | C2<br>..<br>C2,3 | #XXh<br>..<br>#XXh | FSUC_B1<br>..<br>FSUC_Bk |

C1 : The length (number of bytes) of this message parameter is defined by the filePathAndNameLength parameter.  
C2 : The presence of these parameters depends on the modeOfOperation parameter.  
C3 : The length (number of bytes) of this message parameter is defined by the fileSizeParameterLength

Table9-15. Request message definition

| Parameter Name Definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>modeOfOperation</b><br>This data-parameter specifies the type of operation to be applied to the file or directory indicated in the filePathAndName parameter.<br>The values of the data-parameter shall be defined as specified in Annex G.                                                                                                                                                                                                                                                                                       |
| <b>filePathAndNameLength</b><br>Specifies the length in byte for the parameter filePath.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>filePathAndName</b><br>Specifies the file system location of the server where the file which shall be added, deleted, replaced or read from depending on the parameter modeOfOperation parameter.<br>In addition this parameter includes the file name of the file which shall be added, deleted, replaced or read as part of the file path.<br>If the modeOfOperation parameter equals <b>05 (ReadDir)</b> , this parameter indicates the directory to be read.<br>Each byte of this parameter shall be encoded in ASCII format. |

### dataFormatIdentifier

This data-parameter is a one Byte value with each nibble encoded separately. The high nibble specifies the "compressionMethod, and the low nibble specifies the "encryptingMethod". The value 00 specifies that neither compressionMethod nor encryptingMethod is used. Values other than 00 are vehicle manufacturer specific.

If the modeOfOperation parameter equals to **02 (DeleteFile)** and **05 (ReadDir)** this parameter shall not be included in the request message.

### fileSizeParameterLength

Specifies the length in bytes for both parameters fileSizeUncompressed and fileSizeCompressed.

If the modeOfOperation parameter equals to **02 (DeleteFile)**, **04 (ReadFile)** or **05 (ReadDir)** this parameter shall not be included in the request message.

### fileSizeUncompressed

Specifies the size of the uncompressed file in bytes.

If the modeOfOperation parameter equals **02 (DeleteFile)**, **04 (ReadFile)** or **05 (ReadDir)** this parameter shall not be included in the request message.

### fileSizeCompressed

Specifies the size of the compressed file in bytes.

If an uncompressed file is transferred all bytes of this parameter shall be set to the size information used in the parameter fileSizeUncompressed.

If the modeOfOperation parameter equals to **02 (DeleteFile)**, **04 (ReadFile)** or **05 (ReadDir)** this parameter shall not be included in the request message.

Table9-16. Request parameter definition

| Hex   | Description                                                                                                                                                                                                                                          | Cvt | Mnemonic     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------------|
| 00    | <b>ISO/SAE reserved</b><br>This value is reserved by this document for future definition.                                                                                                                                                            | M   | ISOSAERESRVD |
| 01    | <b>AddFile</b><br>This value shall be used to add the file (download) defined in the filePathAndName parameter                                                                                                                                       | U   | ADDFILE      |
| 02    | <b>DeleteFile</b><br>This value shall be used to delete the file defined in the filePathAndName parameter.                                                                                                                                           | U   | DELFILE      |
| 03    | <b>ReplaceFile</b><br>This value shall be used to replace the file (download) defined in the filePathAndName parameter. If the file is not stored at the location the file shall be added.                                                           | U   | REPLFILE     |
| 04    | <b>ReadFile</b><br>This value shall be used to read the file (upload) at the location defined by the filePathAndName parameter.                                                                                                                      | U   | RDFILE       |
| 05    | <b>ReadDir</b><br>This value shall be used to read the directory defined in the filePathAndName parameter. This value implies that the request does not include a fileName.                                                                          | U   | RDDIR        |
| 06    | <b>ResumeFile</b><br>This value shall be used to resume downloading the file defined in the filePathAndName parameter at the returned filePosition indicator. The file specified in the filePathAndName must already exist in the ECU's file system. | U   | RSFILE       |
| 07~FF | <b>ISO/SAE reserved</b><br>This value is reserved by this document for future definition.                                                                                                                                                            | M   | ISOSAERESRVD |

Table9-17. modeOfOperation definition

## 9.5.2 Positive response message

| Data Byte        | Parameter Name                                             | Cvt      | Hex Value   | Mnemonic      |
|------------------|------------------------------------------------------------|----------|-------------|---------------|
| #1d              | <b>RequestFileTransfer Response SID</b>                    | <b>M</b> | <b>#78h</b> | <b>RRFT</b>   |
| #2d              | modeOfOperation                                            | M        | #01~06h     | MOOP          |
| #3d              | lengthFormatIdentifier                                     | C1       | #XXh        | LFID          |
| #4d              | maxNumberOfBlockLength[] = [<br>byte#1 (MSB)               | C1,2     | #XXh        | MNROB<br>B1   |
| ..               | ..                                                         | ..       | ..          | ..            |
| #4+(m-1)d        | byte#m ]                                                   | C1,2     | #XXh        | Bm            |
| #4+md            | dataFormatIdentifier                                       | C1       | #00h        | DFI_          |
| #4+m+1d          | fileSizeOrDirInfoParameterLength [<br>byte#1 (MSB)         | C1       | #XXh        | FSDIL_B1      |
| #4+m+2d          | byte#2 (LSB) ]                                             | C1       |             | FSDIL_B2      |
| #4+m+3d          | fileSizeUncompressedOrDirInfoLength [] = [<br>byte#1 (MSB) | C1,3     | #XXh        | FSUDIL_B<br>1 |
| ..               | ..                                                         | ..       | ..          | ..            |
| #4+m+3+k-1d      | byte#k ]                                                   | C1,3     | #XXh        | FSUDIL_B<br>k |
| #4+m+3+k<br>d    | fileSizeCompressed[] = [<br>byte#1 (MSB)                   | C1,3     | #XXh        | FSC_B1        |
| ..               | ..                                                         | ..       | ..          | ..            |
| #4+m+3+2<br>k-1d | byte#k ]                                                   | C1,3     | #XXh        | FSC_Bk        |
| #4+m+4+2<br>k    | filePosition [] = [<br>byte#1 (MSB)                        | C1       | #XXh        | FPOS_B1       |
| ..               | ..                                                         | ..       | ..          | ..            |
| #4+m+11+<br>2k   | byte#8 (LSB) ]                                             | C1       | #XXh        | FPOS_B8       |

C1: The presence of these parameters depends on the modeOfOperation parameter.  
C2: The length (number of bytes) of this message parameter is defined by the lengthFormatIdentifier parameter.  
C3: The length (number of bytes) of this message parameter is defined by the fileSizeOrDirInfoParameterLength parameter.

Table9-18. Positive response message definition

| Parameter Name Definition                                                                                                                                                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>modeOfOperation</b><br>This is parameter echoes the value of the request                                                                                                                                                                         |
| <b>lengthFormatIdentifier</b><br>Specifies the length (number of bytes) of the maxNumberOfBlockLength parameter.<br>If the modeOfOperation parameter equals to <b>02 (DeleteFile)</b> this parameter shall not be included in the response message. |

### maxNumberOfBlockLength

This parameter is used by the requestFileTransfer positive response message to inform the client how many data bytes (maxNumberOfBlockLength) to include in each TransferData request message from the client or how many data bytes the server will include in a TransferData positive response when uploading data. This length reflects the complete message length, including the service identifier and the data parameters present in the TransferData request message or positive response message. This parameter allows either the client to adapt to the receive buffer size of the server before it starts transferring data to the server or to indicate how many data bytes will be included in each TransferData positive response in the event that data is uploaded. A server is required to accept transferData requests that are equal in length to its reported maxNumberOfBlockLength. It is server specific what transferData request lengths less than maxNumberOfBlockLength are accepted (if any).

NOTE The last transferData request within a given block can be required to be less than maxNumberOfBlockLength. It is not allowed for a server to write additional data bytes (i.e. pad bytes) not contained within the transferData message (either in a compressed or uncompressed format), as this would affect the memory address of where the subsequent transferData request data would be written. If the modeOfOperation parameter equals to **02 (DeleteFile)** this parameter shall not be included in the response message.

### dataFormatIdentifier

This parameter echoes the value of the request.

If the modeOfOperation parameter equals to **02 (DeleteFile)** this parameter shall not be included in the response message.)

If the modeOfOperation parameter equals to **05 (ReadDir)** the value of this parameter shall be equal to 00.

### fileSizeOrDirInfoParameterLength

Specifies the length in bytes for both parameters fileSizeUncompressedOrDirInfoLength and fileSizeCompressed.

If the modeOfOperation parameter equals to **01 (AddFile)**, **02 (DeleteFile)**, **03 (ReplaceFile)** or **06 (ResumeFile)** this parameter shall not be included in the response message.

### fileSizeUncompressedOrDirInfoLength

Specifies the size of the uncompressed file to be uploaded or the length of the directory information to be read in bytes.

If the modeOfOperation parameter equals to **01 (AddFile)**, **02 (DeleteFile)**, **03 (ReplaceFile)** or **06 (ResumeFile)**, this parameter shall not be included in the response message.

### fileSizeCompressed

Specifies the size of the compressed file in bytes.

If the modeOfOperation parameter equals to 01 (AddFile), 02 (DeleteFile), 03 (ReplaceFile), 05 (ReadDir) or 06 (ResumeFile) this parameter shall not be included in the response message.

### filePosition

Specifies the byte position within the file at which the Tester will resume downloading after an initial download is suspended. A download is suspended when the ECU stops receiving TransferData requests and does not receive the RequestTransferExit request before returning to the defaultSession.

The filePosition is relative to the compressed size or uncompressed size, depending if the file was originally sent compressed or uncompressed during the initiating ModeOfOperation = AddFile or ReplaceFile.

If the modeOfOperation parameter equals to **01 (AddFile)**, **02 (DeleteFile)**, **03 (ReplaceFile)**, **04 (ReadFile)**, or **05 (ReadDir)** this parameter shall not be included in the request.

Table9-19. Response parameter definition

## 9.5.3 Supported negative response codes

| Hex | NegativeResponseCode                  | Cvt | Cause of Occurrence                |
|-----|---------------------------------------|-----|------------------------------------|
| 13  | incorrectMessageLengthOrInvalidFormat | M   | The length of the message is wrong |



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|    |                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----|---------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22 | conditionsNotCorrect      | M | This NRC shall be returned if a server receives a request for this service while in the process of downloading or uploading data or other conditions to be able to execute this service are not met.                                                                                                                                                                                                                                                                                       |
| 24 | requestSequenceError      | M | This NRC shall be returned when modeOfOperation is 0616 (ResumeFile) and the requested file has already been completely transferred and therefore is not in a state to be resumed.                                                                                                                                                                                                                                                                                                         |
| 31 | requestOutOfRange         | M | This NRC shall be returned if: <ul style="list-style-type: none"><li>- The specified dataFormatIdentifier is not valid.</li><li>- The specified modeOfOperation is not valid.</li><li>- The specified fileSizeParameterLength is not valid.</li><li>- The specified filePathAndNameLength is not valid.</li><li>- The specified fileSizeUncompressed is not valid.</li><li>- The specified fileSizeCompressed is not valid.</li><li>-The specified filePathAndName is not valid.</li></ul> |
| 33 | securityAccessDenied      | M | This NRC shall be returned if the server is secure (for server's that support the SecurityAccess service) when a request for this service has been received.                                                                                                                                                                                                                                                                                                                               |
| 70 | uploadDownloadNotAccepted | M | This NRC indicates that an attempt to download to a server's memory cannot be accomplished due to some fault conditions.                                                                                                                                                                                                                                                                                                                                                                   |

Table9-20. Supported Negative Response Codes

## 10. Non-volatile server memory programming process

### 10.1 Requirements for Flash Reprogramming

**NOTE :** This specification describes Legacy reprogramming.  
Refer to the ES98765-02 specification for reprogramming standards

#### 10.1.1 General Requirement

Identification information<sup>1)</sup>, which would be compared with tool's one during the programming process or have to be always available, should be located in a non-erasable area. It will enable reprogramming process even if the server has incomplete set of application software and application data caused by failure of Flash Reprogramming (erasing identification information).

#### 10.1.2 Boot Area

- 1) All programmable servers that support programming of the application software shall contain boot software in a boot memory region.
- 2) The boot memory shall be protected against inadvertent erasure.
- 3) The boot software can be protected via hardware or software.
- 4) Servers that support boot software shall continue to execute out of the boot until a complete set of application software and/or application data is programmed.
- 5) The server shall be able to recover and be reprogrammed if any of the following error conditions occur during the programming process:
  - ① Loss of supplied power connection
  - ② Loss of the ground connection
  - ③ Disruption of CAN communication
  - ④ Over- or under-voltage conditions
- 6) No programmable server operating out of boot memory shall transmit a non-diagnostic communication message or unsolicited diagnostic message.
- 7) All servers operation out of boot memory shall be able to receive diagnostic messages.
- 8) A server shall be capable of using the same diagnostic CAN IDs for the duration of a programming event.
- 9) The Boot software shall support 'Extended Diagnostic Mode' with the 'disable normal message transmission', 'ECU reset', 'Tester Present' and 'Start Diagnostic Session' services to enable to restart reprogramming procedure when a sever exits the programming process unsuccessfully.

#### 10.1.3 Server reprogramming requirements

##### 10.1.3.1 Requirements for all servers to support programming

The DiagnosticSessionControl (10 hex) service, with sessionType equal to extendedDiagnosticSession followed by a CommunicationControl (28 hex) service that disables non-diagnostic messages and a ControlDTCSetting (85 hex) service that disables setting of DTCs, shall be used by the server to recognize the disabling of normal communication and to ensure that a server does not set DTCs while



another server is being programmed. The extendedDiagnosticSession shall be kept active by the client for the duration of the programming event.

#### 10.1.3.2 Hardware requirements for programmable servers to support programming

All servers that are programmable shall be able to interface with the programming tools used by development, the assembly plant and by service via the appropriate pins of the vehicle diagnostic connector.

The only power required at the vehicle diagnostic connector for programming shall be vehicle battery power.

Any server that is properly installed in the vehicle and is programmable shall be able to be programmed via the vehicle diagnostic connector. Removal of the server from the vehicle in order to perform programming shall not be required.

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## 10.2 Non-volatile server memory programming sequence

### 10.2.1 Programming Phase #1

| Service Name         | SID (hex) | Sub-function Name                                                 | Sub-function ID (hex) | Identifiers | Request Type | Remark |
|----------------------|-----------|-------------------------------------------------------------------|-----------------------|-------------|--------------|--------|
| ReadDataByIdentifier | 22        | VehicleManufacturerECU<br>SoftwareVersionNumber<br>DataIdentifier | F189                  | -           | Physical     |        |

Table10-1. Programming Phase #1 Sequence

#### 10.2.1.1 ReadDataByIdentifier – ECUSoftwareVersion

The tester will send request message to compare the software of ECU in the vehicle with the software in the tester. If the software in the vehicle is later version than the software in the tester or the same version, the programming sequence will be suspended by the tester.

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## 10.2.2 Programming Phase #2

※ The server[s] should be reprogrammed by the following service procedures.

| Service Name             | SID (hex) | Sub-function Name      | Sub-function ID (hex) | Identifiers                                              | Request Type | Remark                                                                                               |
|--------------------------|-----------|------------------------|-----------------------|----------------------------------------------------------|--------------|------------------------------------------------------------------------------------------------------|
| DiagnosticSessionControl | 10        | default Session        | 81                    | -                                                        | Functional   | Initialization Step                                                                                  |
| DiagnosticSessionControl | 10        | Extended Session       | 83                    | -                                                        | Functional   | Pre-Programming Step<br><br>※ In case of Secure Access feature in CGW, refer to ES95486-01 sequence. |
| ControlDTCSetting        | 85        | OFF                    | 82                    | -                                                        | Functional   |                                                                                                      |
| CommunicationControl     | 28        | disableRxAndTx         | 83                    | Communication Type = 03 hex                              | Functional   |                                                                                                      |
| DiagnosticSessionControl | 10        | Programming Session    | 02                    | -                                                        | Physical     | -                                                                                                    |
| SecurityAccess           | 27        | Request Seed           | 01                    | -                                                        | Physical     | -                                                                                                    |
| SecurityAccess           | 27        | Send Key               | 02                    | -                                                        | Physical     | -                                                                                                    |
| RequestDownload          | 34        | Data Format Identifier | -                     | -                                                        | Physical     | Download the Boot Loader Software (Optional)                                                         |
| TransferData             | 36        | Block Sequence Counter | -                     | -                                                        | Physical     |                                                                                                      |
| RequestTransferExit      | 37        | -                      | -                     | -                                                        | Physical     |                                                                                                      |
| RoutineControl           | 31        | Start Routine          | 01                    | routineIdentifier = checkProgrammingDependencies (#FF01) | Physical     |                                                                                                      |
| RoutineControl           | 31        | Start Routine          | 01                    | routineIdentifier = eraseMemory                          | Physical     | Download the Application Software                                                                    |
| RequestDownload          | 34        | -                      | -                     | Data Format Identifier                                   | Physical     |                                                                                                      |
| TransferData             | 36        | -                      | -                     | Block Sequence Counter                                   | Physical     |                                                                                                      |
| RequestTransferExit      | 37        | -                      | -                     | -                                                        | Physical     |                                                                                                      |
| RoutineControl           | 31        | Start Routine          | 01                    | routineIdentifier = checkProgrammingDependencies (#FF01) | Physical     |                                                                                                      |
| ECUReset                 | 11        | Hard Reset             | 01                    | -                                                        | Physical     | -                                                                                                    |
| CommunicationControl     | 28        | enableRxAndTx          | 80                    | Communication Type = 03 hex                              | Functional   | within 15ms after ECUReset Pos. Rsp.                                                                 |

Table10-2. Programming Phase #2 Sequence

### 10.2.2.1 DiagnosticSessionControl – extended diagnostic session

In order to be able to disable the normal communication between the servers and the setting of DTCs, it is required to start a non-defaultSession in each server where normal communication and DTCs shall be disabled. This is achieved via a DiagnosticSessionControl (10 hex) service with sessionType equal to extendedDiagnosticSession. The request shall be transmitted functionally addressed to all servers. Response messages are not required in this step.

### 10.2.2.2 ControlDTCSetting – DTC setting type = off

The client disables the setting of DTCs in each server using the ControlDTCSetting (85 hex) service with DTCSettingType equal to “off”. The request shall be transmitted functionally addressed to all servers.

Response messages are not required in this step.

#### 10.2.2.3 CommunicationControl – disable non-diagnostic communication

The client disables the transmission and reception of non-diagnostic messages using the CommunicationControl (28 hex) service. The request shall be transmitted functionally addressed to all servers. Response messages are not required in this step.

#### 10.2.2.4 DiagnosticSessionControl – programming session

The programming event is started in the server(s) via a physically addressed request of the DiagnosticSessionControl (10 hex) service with sessionType equal to programmingSession. When the server receives the request, it shall allocate all necessary resources required for programming. It is implementation specific whether the server starts executing out of boot memory.

#### 10.2.2.5 SecurityAccess – read seed / send key

A programming event should be secured. The SecurityAccess (27 hex) service shall be mandatory for all systems.

#### 10.2.2.6 RoutineControl – erase memory

The memory of the server shall be erased in order to allow an application software/data download. This is achieved via a routine, using the RoutineControl (31 hex) service to execute the erase routine.

#### 10.2.2.7 Request Download / TransferData / RequestTransferExit

Each download of a contiguous block of application software/data to a non-volatile server memory location (either a complete application software/data module or part of a software/data module) shall always follow the general data transfer method using the following service sequence:

- RequestDownload (34 hex)
- TransferData (36 hex)
- RequestTransferExit (37 hex)

#### 10.2.2.8 RoutineControl – check reprogramming dependencies

Once all application software/data blocks/modules are completely downloaded, the client shall verify if the download has been performed successfully by initiating a routine in the server using the RoutineControl (31 hex) service. This routine either triggers the server to check the reprogramming dependencies and to perform all necessary action to proof that the download and programming into non-volatile memory was successful, or it requests the server to calculate the checksum and submit the checksum to the client via a RoutineControl positive response message (requestRoutineResults) for a comparison with a checksum contained in the client.

#### 10.2.2.9 ECUReset – hard reset

The client transmits an ECUReset (11 hex) service request message onto the CAN network with resetType equal to hardReset. This shall be done physically addressed. The reception of the ECUReset (11 hex) request message causes the server(s) to perform a reset and to start the defaultSession.



#### 10.2.2.10 CommunicationControl – enable non-diagnostic communication

The client enables the transmission and reception of non-diagnostic messages using the CommunicationControl (28 hex) service. This service must be requested within 15ms after ECU Reset positive response to prevent Time-Out Errors. When the Gateway is reprogramming by itself, the client transmits the CommunicationControl service in 400ms ~ 500ms after ECU Reset positive response. The request shall be transmitted functionally addressed to all servers. Response messages are not required in this step.

#### 10.2.3 Programming Phase #3

| Service Name         | SID (hex) | Sub-function Name                                                 | Sub-function ID (hex) | Identifiers | Request Type | Remark |
|----------------------|-----------|-------------------------------------------------------------------|-----------------------|-------------|--------------|--------|
| ReadDataByIdentifier | 22        | VehicleManufacturerECU<br>SoftwareVersionNumber<br>DataIdentifier | F189                  | -           | Physical     |        |

Table10-3. Programming Phase #3 Sequence

##### 10.2.3.1 ReadDataByIdentifier – ECUSoftwareVersion

After the completion of reprogramming, the tester will send request message ReadDataByIdentifier to the ECU. If the ECU send a positive response message, it shows that the reprogramming procedure is successfully completed and the ECU has complete set of updated application software and application data.

## 10.3 Non-volatile server memory programming message flow examples

### 10.3.1 General Information

The following presents the message for a non-volatile server memory-programming event of a single server.

Assume that,

- Two ECUs (07E0/07E8 & 07E1/07E9) and Tester are in CAN or CAN-FD network
- Transfer of four (4) modules, where each module has a length of 511 bytes (data bytes 02 hex through FE hex)
- The network layer buffer size of the server that is re-programmed is 255 bytes (reported in the RequestDownload positive response message)

### 10.3.2 Programming Phase #1

Assume that,

- BS = 4, STmin = 2 (in Default Session)
- Vehicle Manufacturer ECU Software Version Number = 123456789 (ASCII)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 03         | 22 | F1 | 89 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E8   | Response           | 8   | 10         | 0D | 62 | F1 | 89 | 31 | 32 | 33 | First Frame          |
| 07E0   | Request            | 8   | 30         | 04 | 02 | 55 | 55 | 55 | 55 | 55 | Flow Control         |
| 07E8   | Response           | 8   | 21         | 34 | 35 | 36 | 37 | 38 | 39 | AA | Consecutive Frame #1 |

Table10-4. ReadDataByIdentifier (VehicleManufacturerECUSoftwareVersionNumber)

### 10.3.3 Programming Phase #2

Assume that,

- BS = 0, STmin = 0 (in ProgrammingSession)
- seed of the server (2 bytes) : 3657 hex
- key of the server (2 bytes) : C9A9 hex

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07DF   | Func. Request      | 8   | 02         | 10 | 83 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-5. DiagnosticSessionControl (extendedSession)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07DF   | Func. Request      | 8   | 02         | 85 | 82 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-6. ControlDTCSetting (off)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07DF   | Func. Request      | 8   | 03         | 28 | 83 | 03 | 55 | 55 | 55 | 55 | Single Frame |

Table10-7. CommunicationControl (disableRxAndTx)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 02         | 10 | 02 | 55 | 55 | 55 | 55 | 55 | Single Frame |



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|      |               |   |    |    |    |    |    |    |    |    |              |
|------|---------------|---|----|----|----|----|----|----|----|----|--------------|
| 07E8 | Response      | 8 | 02 | 50 | 02 | AA | AA | AA | AA | AA | Single Frame |
| 07DF | Func. Request | 8 | 02 | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-8. DiagnosticSessionControl (programmingSession)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 02         | 27 | 01 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 02         | 67 | 01 | 36 | 57 | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-9. SecurityAccess (requestSeed)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 02         | 27 | 02 | C9 | A9 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 02         | 67 | 02 | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-10. SecurityAccess (sendKey)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 04         | 31 | 01 | FF | 00 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 03         | 7F | 31 | 78 | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 03         | 7F | 31 | 78 | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 04         | 71 | 01 | FF | 00 | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-11. RoutineControl (eraseMemory)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 10         | 09 | 34 | 00 | 33 | 00 | 1B | 67 | First Frame          |
| 07E8   | Response           | 8   | 30         | 00 | 00 | AA | AA | AA | AA | AA | Flow Control         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E0   | Phys. Request      | 8   | 21         | 00 | 01 | FF | 55 | 55 | 55 | 55 | Consecutive Frame #1 |
| 07E8   | Response           | 8   | 04         | 74 | 20 | 00 | FF | AA | AA | AA | Single Frame         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |

Table10-12. RequestDownload – Module #1

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 01  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 01  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-13. TransferData – Module #1 (block #1)

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 02  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 02  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-14. TransferData – Module #1 (block #2)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 07         | 36 | 01 | 02 | 03 | 04 | 05 | 06 | Single Frame |
| 07E8   | Response           | 8   | 02         | 76 | 03 | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-15. TransferData – Module #1 (block #3)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 01         | 37 | 55 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 01         | 77 | AA | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-16. RequestTransferExit – Module #1

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 10         | 09 | 34 | 00 | 33 | 00 | 1B | 67 | First Frame          |
| 07E8   | Response           | 8   | 30         | 00 | 00 | AA | AA | AA | AA | AA | Flow Control         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E0   | Phys. Request      | 8   | 21         | 00 | 01 | FF | 55 | 55 | 55 | 55 | Consecutive Frame #1 |
| 07E8   | Response           | 8   | 04         | 74 | 20 | 00 | FF | AA | AA | AA | Single Frame         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |

Table10-17. RequestDownload – Module #2

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 01  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 01  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-18. TransferData – Module #2 (block #1)

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 02  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 02  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-19. TransferData – Module #2 (block #2)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 07         | 36 | 01 | 02 | 03 | 04 | 05 | 06 | Single Frame |
| 07E8   | Response           | 8   | 02         | 76 | 03 | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-20. TransferData – Module #2 (block #3)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 01         | 37 | 55 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 01         | 77 | AA | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-21. RequestTransferExit – Module #2

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 10         | 09 | 34 | 00 | 33 | 00 | 1B | 67 | First Frame          |
| 07E8   | Response           | 8   | 30         | 00 | 00 | AA | AA | AA | AA | AA | Flow Control         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E0   | Phys. Request      | 8   | 21         | 00 | 01 | FF | 55 | 55 | 55 | 55 | Consecutive Frame #1 |
| 07E8   | Response           | 8   | 04         | 74 | 20 | 00 | FF | AA | AA | AA | Single Frame         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |

Table10-22. RequestDownload – Module #3

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 01  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 01  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-23. TransferData – Module #3 (block #1)

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 02  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 02  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-24. TransferData – Module #3 (block #2)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 07         | 36 | 01 | 02 | 03 | 04 | 05 | 06 | Single Frame |
| 07E8   | Response           | 8   | 02         | 76 | 03 | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-25. TransferData – Module #3 (block #3)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 01         | 37 | 55 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 01         | 77 | AA | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-26. RequestTransferExit – Module #3

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 10         | 09 | 34 | 00 | 33 | 00 | 1B | 67 | First Frame          |
| 07E8   | Response           | 8   | 30         | 00 | 00 | AA | AA | AA | AA | AA | Flow Control         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E0   | Phys. Request      | 8   | 21         | 00 | 01 | FF | 55 | 55 | 55 | 55 | Consecutive Frame #1 |
| 07E8   | Response           | 8   | 04         | 74 | 20 | 00 | FF | AA | AA | AA | Single Frame         |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame         |

Table10-27. RequestDownload – Module #4

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 01  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 01  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-28. TransferData – Module #4 (block #1)

| CAN ID | Request / Response | DLC | Data Bytes |     |     |     |     |     |     |     | Comments              |
|--------|--------------------|-----|------------|-----|-----|-----|-----|-----|-----|-----|-----------------------|
| 07E0   | Phys. Request      | 8   | 10         | FF  | 36  | 02  | 02  | 03  | 04  | 05  | First Frame           |
| 07E8   | Response           | 8   | 30         | 00  | 00  | AA  | AA  | AA  | AA  | AA  | Flow Control          |
| 07E0   | Phys. Request      | 8   | 21         | 06  | 07  | 08  | 09  | 0A  | 0B  | 0C  | Consecutive Frame #1  |
| 07E0   | Phys. Request      | 8   | 22         | 0D  | 0E  | 0F  | 10  | 11  | 12  | 13  | Consecutive Frame #2  |
| 07E0   | Phys. Request      | 8   | 23         | 14  | 15  | 16  | 17  | 18  | 19  | 1A  | Consecutive Frame #3  |
| ...    | ...                | ... | ...        | ... | ... | ... | ... | ... | ... | ... | ...                   |
| 07E0   | Phys. Request      | 8   | 23         | F4  | F5  | F6  | F7  | F8  | F9  | FA  | Consecutive Frame #35 |
| 07E0   | Phys. Request      | 8   | 24         | FB  | FC  | FD  | FE  | 55  | 55  | 55  | Consecutive Frame #36 |
| 07E8   | Response           | 8   | 02         | 76  | 02  | AA  | AA  | AA  | AA  | AA  | Single Frame          |
| 07DF   | Func. Request      | 8   | 02         | 3E  | 80  | 55  | 55  | 55  | 55  | 55  | Single Frame          |

Table10-29. TransferData – Module #4 (block #2)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 07         | 36 | 01 | 02 | 03 | 04 | 05 | 06 | Single Frame |
| 07E8   | Response           | 8   | 02         | 76 | 03 | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-30. TransferData – Module #4 (block #3)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 01         | 37 | 55 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 01         | 77 | AA | AA | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-31. RequestTransferExit – Module #4

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07E0   | Phys. Request      | 8   | 04         | 31 | 01 | FF | 01 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 03         | 7F | 31 | 78 | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 03         | 7F | 31 | 78 | AA | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 04         | 71 | 01 | FF | 01 | AA | AA | AA | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07DF   | Func. Request      | 8   | 02         | 3E | 80 | 55 | 55 | 55 | 55 | 55 | Single Frame |

Table10-32. RoutineControl (checkProgrammingDependencies)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07DF   | Phys. Request      | 8   | 02         | 11 | 01 | 55 | 55 | 55 | 55 | 55 | Single Frame |
| 07E8   | Response           | 8   | 02         | 51 | 01 | AA | AA | AA | AA | AA | Single Frame |

Table10-33. ECUReset (hardReset)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments     |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|--------------|
| 07DF   | Func. Request      | 8   | 03         | 28 | 80 | 03 | 55 | 55 | 55 | 55 | Single Frame |



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Table10-34. CommnunicationControl (enableRxAndTx)

10.3.4 Programming Phase #3

- Assume that,
- BS = 4, STmin = 2 (in Default Session)
  - Vehicle Manufacturer ECU Software Version Number = 12345678A (ASCII)

| CAN ID | Request / Response | DLC | Data Bytes |    |    |    |    |    |    |    | Comments             |
|--------|--------------------|-----|------------|----|----|----|----|----|----|----|----------------------|
| 07E0   | Phys. Request      | 8   | 03         | 22 | F1 | 89 | 55 | 55 | 55 | 55 | Single Frame         |
| 07E8   | Response           | 8   | 10         | 0D | 62 | F1 | 89 | 31 | 32 | 33 | First Frame          |
| 07E0   | Request            | 8   | 30         | 04 | 02 | 55 | 55 | 55 | 55 | 55 | Flow Control         |
| 07E8   | Response           | 8   | 21         | 34 | 35 | 36 | 37 | 38 | 41 | AA | Consecutive Frame #1 |

Table10-35. ReadDataByIdentifier (VehicleManufacturerECUSoftwareVersionNumber)

### 10.4 Servers status in programming procedure

ECU #1 is the server actually being programmed.

ECU #2 is the server which is not actually programmed (programmable or non-programmable).

ECU #3 is the server which is participating in a CAN or CAN-FD network but the server itself is diagnosed over another vehicle bus (e.g. KWP2000 ...) or does not support diagnostic.

| Tester                                        |           |       | ECU #1                                                                       | ECU #2                                       | ECU #3                          |
|-----------------------------------------------|-----------|-------|------------------------------------------------------------------------------|----------------------------------------------|---------------------------------|
| Service Name                                  | SID (hex) |       | DiagnosticSession / CommunicationControl setting / DTCSetting                |                                              |                                 |
| ReadDataByIdentifier                          | 22        | Phys. | defaultSession / default (enableRxAndTx) / default (ON)                      |                                              | - / default (enableRxAndTx) / - |
| DiagnosticSessionControl - extendedSession    | 10        | Func. | <u>extendedSession</u> / default (enableRxAndTx) / default (ON)              |                                              |                                 |
| ControlDTCSetting                             | 85        | Func. | extendedSession / default (enableRxAndTx) / <u>OFF</u>                       |                                              |                                 |
| CommunicationControl - disableRxAndTx         | 28        | Func. | extendedSession / <u>disableRxAndTx</u> / OFF                                |                                              |                                 |
| DiagnosticSessionControl - programmingSession | 10        | Phys. | <u>programmingSession</u> / disableRxAndTx / OFF                             |                                              | - / EnableRxAndDisableTx / -    |
| SecurityAccess                                | 27        | Phys. |                                                                              |                                              |                                 |
| SecurityAccess                                | 27        | Phys. |                                                                              |                                              |                                 |
| RequestDownload                               | 34        | Phys. |                                                                              |                                              |                                 |
| TransferData                                  | 36        | Phys. |                                                                              |                                              |                                 |
| RequestTransferExit                           | 37        | Phys. |                                                                              |                                              |                                 |
| RoutineControl                                | 31        | Phys. |                                                                              |                                              |                                 |
| RoutineControl                                | 31        | Phys. |                                                                              |                                              |                                 |
| RequestDownload                               | 34        | Phys. |                                                                              |                                              |                                 |
| TransferData                                  | 36        | Phys. |                                                                              |                                              |                                 |
| RequestTransferExit                           | 37        | Phys. |                                                                              |                                              |                                 |
| RoutineControl                                | 31        | Phys. |                                                                              |                                              |                                 |
| ECUReset - hardReset                          | 11        | Phys. | <u>defaultSession</u> / <u>default (enableRxAndTx)</u> / <u>default (ON)</u> |                                              |                                 |
| CommunicationControl - enableRxAndTx          | 28        | Func. |                                                                              | extendedSession / <u>enableRxAndTx</u> / OFF | - / enableRxAndTx / -           |
| ReadDataByIdentifier                          | 22        | Phys. |                                                                              |                                              |                                 |

## Annex. A Negative Response Codes

When an ECU detects any error condition on communication or have internal problem, it cannot respond or react to the request normally. In this case the ECU should respond to tester with negative response code described in table A-1.

\*) These codes are general negative response codes. These NRCs are in general supported by each diagnostic service and not listed in the list of applicable response codes of the diagnostic services. The occurrence depends on the system configuration (Hex value – 10, 11, 33, 78, 7F, 21, 7E (with a sub-function))

| Hex value | responseCode                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Mnemonic |
|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 10        | <b>generalReject *)</b><br><br>This response code indicates that the requested action has been rejected by the server. The generalReject response code shall only be implemented in the server if none of the negative response codes defined in this document meet the needs of the implementation.                                                                                                                                                                                                                                                                 | GR       |
| 11        | <b>serviceNotSupported *)</b><br><br>This response code indicates that the requested action will not be taken because the server does not support the requested service. The server shall send this response code in case the client has sent a request message with a service identifier, which is either unknown or not supported by the server. Therefore this negative response code is not shown in the list of negative response codes to be supported for a diagnostic service, because this negative response code is not applicable for supported services. | SNS      |
| 12        | <b>subFunctionNotSupported</b><br><br>This response code indicates that the requested action will not be taken because the server does not support the service specific parameters of the request message. The server shall send this response code in case the client has sent a request message with a known and supported service identifier but with "sub function" which is either unknown or not supported.                                                                                                                                                    | SFNS     |
| 13        | <b>incorrectMessageLengthOrInvalidFormat</b><br><br>This response code indicates that the requested action will not be taken because the length of the received request message does not match the prescribed length for the specified service or the format of the paramters do not match the prescribed format for the specified service.                                                                                                                                                                                                                          | IMLOIF   |
| 14        | <b>responseTooLong</b><br><br>This response code shall be reported by the server if the response to be generated exceeds the maximum number of bytes available by the underlying network layer.<br><br>- EXAMPLE This problem may occur when several DIDs at a time are requested and the combination of all DIDs in the response exceeds the limit of the underlying transport protocol.                                                                                                                                                                            | RTL      |
| 21        | <b>busyRepeatRequest</b><br><br>This response code indicates that the server is temporarily too busy to perform the requested operation. In this circumstance the client shall perform repetition of the "identical request message" or "another request message". The repetition of the request shall be delayed by a time specified in the respective implementation documents.                                                                                                                                                                                    | BRR      |



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Example: In a multi-client environment the diagnostic request of one client might be blocked temporarily by a NRC \$21 while a different client finishes a diagnostic task.

NOTE) If the server is able to perform the diagnostic task but needs additional time to finish the task and prepare the response, the NRC \$78 shall be used instead of NRC \$21.  
This response code is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.

22

**conditionNotCorrect**

CNC

This response code indicates that the requested action will not be taken because the server prerequisite conditions are not met.

24

**requestSequenceError**

RSE

This response code indicates that the requested action will not be taken because the server expects a different sequence of request messages or message as sent by the client. This may occur when sequence sensitive requests are issued in the wrong order.

EXAMPLE A successful SecurityAccess service specifies a sequence of requestSeed and sendKey as sub-functions in the request messages. If the sequence is sent different by the client the server shall send a negative response message with the negative response code 24 hex - . requestSequenceError.

31

**requestOutOfRange**

ROOR

This response code indicates that the requested action will not be taken because the server has detected that the request message contains a parameter which attempts to substitute a value beyond its range of authority (e.g. attempting to substitute a data byte of 111 when the data is only defined to 100), or which attempts to access a dataIdentifier/routineIdentifier that is not supported or not supported in active session. This response code shall be implemented for all services, which allow the client to read data, write data or adjust functions by data in the server.

33

**securityAccessDenied \*)**

SAD

This response code indicates that the requested action will not be taken because the server's security strategy has not been satisfied by the client. The server shall send this response code if one of the following cases occur:

- the test conditions of the server are not met,
- the required message sequence e.g. DiagnosticSessionControl, securityAccess is not met,
- the client has sent a request message which requires an unlocked server.

Beside the mandatory use of this negative response code as specified in the applicable services within this standard, this negative response code can also be used for any case where security is required and is not yet granted to perform the required service.



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|    |                                                                                                                                                                                                                                                                                                                               |        |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 35 | <b>invalidKey</b><br><br>This response code indicates that the server has not given security access because the key sent by the client did not match with the key in the server's memory. This counts as an attempt to gain security. The server shall remain locked and increment its internal securityAccessFailed counter. | IK     |
| 36 | <b>exceedNumberOfAttempts</b><br><br>This response code indicates that the requested action will not be taken because the client has unsuccessfully attempted to gain security access more times than the server's security strategy will allow.                                                                              | ENOA   |
| 37 | <b>requiredTimeDelayNotExpired</b><br><br>This response code indicates that the requested action will not be taken because the client's latest attempt to gain security access was initiated before the server's required timeout period had elapsed.                                                                         | RTDNE  |
| 50 | <b>Certificate verification failed – Invalid Time Period</b><br><br>Date and time of the server does not match the validity period of the Certificate.                                                                                                                                                                        | CVFITP |
| 51 | <b>Certificate verification failed - Invalid Signature</b><br><br>Signature of the Certificate could not be verified.                                                                                                                                                                                                         | CVFIS  |
| 54 | <b>Certificate verification failed - Invalid Format</b><br><br>Certificate could not be evaluated because the format requirement has not been met.                                                                                                                                                                            | CVFIF  |
| 56 | <b>Certificate verification failed - Invalid Scope</b><br><br>The scope of the Certificate does not match the contents of the server.                                                                                                                                                                                         | CVFIS  |
| 57 | <b>Certificate verification failed – Invalid Certificate (revoked)</b><br><br>Certificate received from client is invalid, because the server has revoked access for some reason.                                                                                                                                             | CVFIC  |
| 58 | <b>Ownership verification failed</b><br><br>Delivered Ownership does not match the provided challenge or could not be verified with the own private key.                                                                                                                                                                      | OVF    |
| 59 | <b>Challenge calculation failed</b><br><br>The challenge could not be calculated on the server side.                                                                                                                                                                                                                          | CCF    |
| 5A | <b>Setting Access Rights failed</b><br><br>The server could not set the access rights.                                                                                                                                                                                                                                        | SARF   |
| 5C | <b>Configuration data usage failed</b><br><br>The server could not work with the provided configuration data.                                                                                                                                                                                                                 | CDUF   |
| 5D | <b>DeAuthentication failed</b><br><br>DeAuthentication was not successful, server could still be unprotected.                                                                                                                                                                                                                 | DAF    |
| 70 | <b>uploadDownloadNotAccepted</b><br><br>This response code indicates that an attempt to upload/download to a server's memory cannot be accomplished due to some fault conditions.                                                                                                                                             | UDNA   |
| 71 | <b>transferDataSuspended</b><br><br>This response code indicates that a data transfer operation was halted due to some fault. The active transferData sequence shall be aborted.                                                                                                                                              | TDS    |



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|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 72 | <p><b>generalProgrammingFailure</b></p> <p>This response code indicates that the server detected an error when erasing or programming a memory location in the permanent memory device (e.g. Flash Memory).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | GPF     |
| 73 | <p><b>wrongBlockSequenceCounter</b></p> <p>This response code indicates that the server detected an error in the sequence of blockSequenceCounter values. Note that the repetition of a TransferData request message with a blockSequenceCounter equal to the one included in the previous TransferData request message shall be accepted by the server.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | WBSC    |
| 78 | <p><b>requestCorrectlyReceived-ResponsePending *)</b></p> <p>This response code indicates that the request message was received correctly, and that all parameters in the request message were valid, but the action to be performed is not yet completed and the server is not yet ready to receive another request. As soon as the requested service has been completed, the server shall send a positive response message or negative response message with a response code different from this.</p> <p>The negative response message with this response code may be repeated by the server until the requested service is completed and the final response message is sent. This response code might impact the application layer timing parameter values. The detailed specification shall be included in the data link specific implementation document.</p> <p><b>This response code shall only be used in a negative response message if the server will not be able to receive further request messages from the client while completing the requested diagnostic service. When this response code is used, the server shall always send a final response (positive or negative) independent of the suppressPosRspMsgIndicationBit value.</b></p> <p>A typical example where this response code may be used is when the client has sent a request message, which includes data to be programmed or erased in flash memory of the server. If the programming/erasing routine (usually executed out of RAM) is not able to support serial communication while writing to the flash memory the server shall send a negative response message with this response code.</p> <p>This response code is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p> | RCRRP   |
| 7E | <p><b>subFunctionNotSupportedInActiveSession</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested sub-function in the session currently active. Within the programmingSession negative response code SFNS (subFunctionNotSupported) may optionally be reported instead of negative response code SNFSIAS (subFunctionNotSupportedInActiveSession). This response code shall only be used when the requested sub-function is known to be supported in another session, otherwise response code SFNS (subFunctionNotSupported) shall be used. This response code shall be supported by each diagnostic service with a sub-function parameter, if not otherwise stated in</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SFNSIAS |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
|    | the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |        |
| 7F | <p><b>serviceNotSupportedInActiveSession *)</b></p> <p>This response code indicates that the requested action will not be taken because the server does not support the requested service in the session currently active. This response code shall only be used when the requested service is known to be supported in another session, otherwise response code SNS (serviceNotSupported) shall be used. This response code is in general supported by each diagnostic service, as not otherwise stated in the data link specific implementation document, therefore it is not listed in the list of applicable response codes of the diagnostic services.</p> | SNSIAS |
| 81 | <p><b>rpmTooHigh</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for RPM is not met (current RPM is above a pre-programmed maximum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                | RPMTH  |
| 82 | <p><b>rpmTooLow</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for RPM is not met (current RPM is below a pre-programmed minimum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                                                 | RPMTL  |
| 83 | <p><b>enginesRunning</b></p> <p>This is required for those actuator tests which cannot be actuated while the Engine is running. This is different from RPM too high negative response, and needs to be allowed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                             | EIR    |
| 84 | <p><b>enginesNotRunning</b></p> <p>This is required for those actuator tests which cannot be actuated unless the Engine is running. This is different from RPM too low negative response, and needs to be allowed.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                          | EINR   |
| 85 | <p><b>engineRunTimeTooLow</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for engine run time is not met (current engine run time is below a preprogrammed limit).</p>                                                                                                                                                                                                                                                                                                                                                                                                            | ERTTL  |
| 86 | <p><b>temperatureTooHigh</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for temperature is not met (current temperature is above a preprogrammed maximum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                         | TEMPTH |
| 87 | <p><b>temperatureTooLow</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for temperature is not met (current temperature is below a preprogrammed minimum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                          | TEMPTL |
| 88 | <p><b>vehicleSpeedTooHigh</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for vehicle speed is not met (current VS is above a pre-programmed maximum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                              | VSTH   |
| 89 | <p><b>vehicleSpeedTooLow</b></p> <p>This response code indicates that the requested action will not be taken because the server prerequisite condition for vehicle speed is not met (current VS is below a pre-programmed minimum threshold).</p>                                                                                                                                                                                                                                                                                                                                                                                                               | VSTL   |



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|    |                                                                                                                                                                                                                                                                              |       |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 8A | <b>throttle/PedalTooHigh</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for throttle/pedal position is not met (current TP/APP is above a preprogrammed maximum threshold).                   | TPTH  |
| 8B | <b>throttle/PedalTooLow</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for throttle/pedal position is not met (current TP/APP is below a preprogrammed minimum threshold).                    | TPTL  |
| 8C | <b>transmissionRangeNotInNeutral</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for being in neutral is not met (current transmission range is not in neutral).                               | TRNIN |
| 8D | <b>transmissionRangeNotInGear</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for being in gear is not met (current transmission range is not in gear).                                        | TRNIG |
| 8F | <b>brakeSwitch(es)NotClosed</b> (Brake Pedal not pressed or not applied)<br><br>For saftey reasons, this is required for certain tests before it begins, and must be maintained for the entire duration of the test.                                                         | BSNC  |
| 90 | <b>shiftLeverNotInPark</b><br><br>For saftey reasons, this is required for certain tests before it begins, and must be maintained for the entire duration of the test.                                                                                                       | SLNIP |
| 91 | <b>torqueConverterClutchLocked</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for torque converter clutch is not met (current TCC status above a preprogrammed limit or locked).              | TCCL  |
| 92 | <b>voltageTooHigh</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for voltage at the primary pin of the server (ECU) is not met (current voltage is above a pre-programmed maximum threshold). | VTH   |
| 93 | <b>voltageTooLow</b><br><br>This response code indicates that the requested action will not be taken because the server prerequisite condition for voltage at the primary pin of the server (ECU) is not met (current voltage is below a pre-programmed maximum threshold).  | VTL   |
| F0 | <b>CRLintegrityFailed</b><br><br>CRL integrity check (Verification of CRL Signature) is failed. The hash value of SHA-256 of Certificate authority's public key and the CRL Issuer Public Key Identifier value are not matched                                               | CRLF  |
| F1 | <b>CRLvalidityPeriodFailed</b><br><br>The current time is not included between CRL effective date and CRL expiration date                                                                                                                                                    | CRLPD |
| F2 | <b>RoleandRightofCertificateDenied</b><br><br>Role and Right of Certificate verification failed. The request service or the target from the external diagnostic tool are not matched with Certificate Holder Reference : Authorization                                       | RRCD  |



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|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| F3 | <b>securityAccessDeniedfromCGW</b><br><br>This response code indicates that the requested action will not be taken from the CGW because the CGW's security strategy (Secure Access) has not been satisfied by the client. | SADCGW |
| F4 | <b>LowerVersionDownloadAttempt</b><br><br>This response code indicates that the requested action will not be taken because the downloaded software is lower version than the existing version.                            | LVDA   |
| F5 | <b>ServiceNotSupportedInActiveConnection</b><br><br>This response code indicates that the requested action will not be taken because the service is not allowed in active connection(CAN or Ethernet).                    | SNSIAC |

Table A-1. Definition of responseCode values

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## Annex. B Diagnostic and Communication Management Functional Unit Data Parameter Definitions

### B.1 CommnuicationType Parameter Definition

The communicationType is a 1-byte value. The bit-encoded low nibble of this byte represents the communicationTypes, which can be controlled via the CommunicationControl (28 hex) service.

For example, a communicationType with a bit combination (Bits 1-0) of "11b" is valid and disables both "normalCommunicationMessages" and "networkManagementCommunicationMessages" messages.

The high nibble of the communicationType 1-byte value defines which of the subnets connected to the receiving node shall be disabled/enabled when an appropriate CommunicationControl service is received.

| Encoding of Bit | Value | Description                                                                                                                                                                                                                                                                                                                        | Cvt | Mnemonic |
|-----------------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 0 - 1           | 0     | Reserved                                                                                                                                                                                                                                                                                                                           | M   |          |
|                 | 1     | <b>NormalCommunicationMessages</b><br>This value references all application-related communication (inter-application signal exchange between multiple in-vehicle servers).                                                                                                                                                         | M   | NCM      |
|                 | 2     | <b>NetworkManagementCommunicationMessages</b><br>This value references all network management related communication.                                                                                                                                                                                                               | M   | NWMCM    |
|                 | 3     | <b>NetworkManagementCommunicationMessage and NormalCommunicationMessages</b><br>This value references all network management and application-related communication.                                                                                                                                                                | M   |          |
| 2 - 3           | 0~3   | Reserved                                                                                                                                                                                                                                                                                                                           |     |          |
| 4 - 7           | 0     | Disable/Enable specified communicationType (see encoding of bit 0-1) in the receiving node and all connected subnets. This only disables the node's communication into the subnets but not the communication of other nodes on the subnet (receiving node is not responsible to disable communication in each node of the subnet). | M   |          |
|                 | 1~E   | Disable/Enable specific subnet identified by subnet number.                                                                                                                                                                                                                                                                        | U   |          |
|                 | F     | Disable/Enable network which request is received on (Receiving node (server))                                                                                                                                                                                                                                                      | U   |          |

Table B-1. Definition of CommunicationType and SubnetNumber Byte

### B.2 EventWindowTime Parameter Definition

| HEX     | Description                                                                                                                                                      | Cvt | Mnemonic |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00 - 01 | <b>Reserved</b>                                                                                                                                                  | M   |          |
| 02      | <b>InfiniteTimeToResponse</b><br>This value specifies that the event window shall stay active for an infinite amount of time. (e.g. open window until power off) | U   | ITTR     |
| 03 - 7F | <b>Reserved</b> (vehicle manufacture specific range)                                                                                                             |     |          |
| 80 - FF | <b>Reserved</b>                                                                                                                                                  | M   |          |

Table B-2. Definition of EventWindowTime Parameter



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### B.3 BaudratelIdentifier Parameter Definition

| HEX     | Description                                                                          | Cvt | Mnemonic   |
|---------|--------------------------------------------------------------------------------------|-----|------------|
| 00      | <b>Reserved</b>                                                                      | M   |            |
| 01      | <b>PC9600Baud</b><br>This value specifies the standard PC baudrate of 9.6 KBaud.     | U   | PC9600     |
| 02      | <b>PC19200Baud</b><br>This value specifies the standard PC baudrate of 19.2 KBaud.   | U   | PC19200    |
| 03      | <b>PC38400Baud</b><br>This value specifies the standard PC baudrate of 38.4 KBaud.   | U   | PC38400    |
| 04      | <b>PC57600Baud</b><br>This value specifies the standard PC baudrate of 57.6 KBaud.   | U   | PC57600    |
| 05      | <b>PC115200Baud</b><br>This value specifies the standard PC baudrate of 115.2 KBaud. | U   | PC115200   |
| 06 - 0F | <b>Reserved</b>                                                                      | M   |            |
| 10      | <b>CAN125000Baud</b><br>This value specifies the standard CAN baudrate of 125 KBaud. | U   | CAN125000  |
| 11      | <b>CAN250000Baud</b><br>This value specifies the standard CAN baudrate of 250 KBaud. | U   | CAN250000  |
| 12      | <b>CAN500000Baud</b><br>This value specifies the standard CAN baudrate of 500 KBaud. | U   | CAN500000  |
| 13      | <b>CAN1000000Baud</b><br>This value specifies the standard CAN baudrate of 1 MBaud.  | U   | CAN1000000 |
| 14 - FF | <b>Reserved</b>                                                                      |     |            |

Table B-3. Definition of BaudratelIdentifier Parameter

## Annex. C Data Transmission Functional Unit Data Parameter Definitions

### C.1 dataIdentifier parameter definitions

The parameter dataIdentifier (DID) is to identify a server specific local data record. This parameter shall be available in the server's memory. The dataIdentifier value shall either exist in fixed memory or temporarily stored in RAM if defined dynamically by the service dynamicallyDefineDataIdentifier.

The example of DataIdentifier Parameter Table is described in following table.

Example#1]

| Data Identifier (HEX) | Byte    | PID  | Description Text                                                                                                                                                  | Unit                                                                                             | HEX Range               | Conversion             |
|-----------------------|---------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-------------------------|------------------------|
|                       |         |      |                                                                                                                                                                   |                                                                                                  | Phy. Range              |                        |
| 0xZZZZ                | #1 ~ #4 | 0x00 | Supported PID : 0x01~0x20<br>Response : 4 Byte (A,B,C,D)<br>Byte #1 : PID 0x01-0x08, Byte #2 : PID 0x09-0x10,<br>Byte #3 : PID 0x11-0x18, Byte #4 : PID 0x19-0x20 |                                                                                                  |                         |                        |
|                       | ...     | ...  | ...                                                                                                                                                               | ...                                                                                              | ...                     | ...                    |
|                       | #       | 0xX1 | Intake Manifold Absolute Pressure                                                                                                                                 | kPa                                                                                              | 00 ~ FF<br>0 ~ 255      | X*1                    |
|                       | #       | 0xX2 | Short Term Fuel Trim                                                                                                                                              | %                                                                                                | 80 ~ 7F<br>-100 ~ 99.22 | (singled)<br>X*200/255 |
|                       | #       | 0xX3 | Intake Air Temperature                                                                                                                                            | °C                                                                                               | 00 ~ FF<br>-40 ~ 215    | X*1-40                 |
|                       | #       | 0xX4 | Location of Oxygen Sensor                                                                                                                                         | Bit 0 = 1 : Bank1 / Sensor1<br>Bit 1 = 1 : Bank1 / Sensor2<br>...<br>Bit 7 = 1 : Bank2 / Sensor4 |                         |                        |
|                       | #       | 0xX5 | OBD requirements to which vehicle is designed                                                                                                                     | 01 = OBD II<br>02 = OBD<br>...<br>0C~FF = Reserved                                               |                         |                        |
|                       | #       | 0xX6 | Bit 0 : Variant Coding                                                                                                                                            | 0 = MT<br>1 = AT                                                                                 |                         |                        |
|                       |         | 0xX7 | Bit 2 : Drive Position                                                                                                                                            | 0 = R<br>1 = D                                                                                   |                         |                        |
|                       |         | 0xX8 | Bit 3 : Aircon Switch                                                                                                                                             | 0 = OFF<br>1 = ON                                                                                |                         |                        |
|                       |         | ...  | ...                                                                                                                                                               | ...                                                                                              | ...                     | ...                    |
|                       |         | 0xXC | Bit 7 : Reserved                                                                                                                                                  | -                                                                                                | -                       | -                      |
|                       | ...     | ...  | ...                                                                                                                                                               | ...                                                                                              | ...                     | ...                    |

The definition of Supported PID Byte (byte #0~byte#3) is as below.

| Byte \ Bit | Bit 7    | Bit 6    | Bit 5    | Bit 4    | Bit 3    | Bit 2    | Bit 1    | Bit 0    |
|------------|----------|----------|----------|----------|----------|----------|----------|----------|
| Byte #0    | PID 0x01 | PID 0x02 | PID 0x03 | PID 0x04 | PID 0x05 | PID 0x06 | PID 0x07 | PID 0x08 |
| Byte #1    | PID 0x09 | PID 0x0A | PID 0x0B | PID 0x0C | PID 0x0D | PID 0x0E | PID 0x0F | PID 0x10 |
| Byte #2    | PID 0x11 | PID 0x12 | PID 0x13 | PID 0x14 | PID 0x15 | PID 0x16 | PID 0x17 | PID 0x18 |
| Byte #3    | PID 0x19 | PID 0x1A | PID 0x1B | PID 0x1C | PID 0x1D | PID 0x1E | PID 0x1F | PID 0x20 |

\*. Each bit would be set to '0' (if the assigned PID is not supported) or '1' (if the assigned PID is supported).

Table C-1. Example#1 of Record DataIdentifier Parameter Table



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### Example#2]

| Data Identifier (HEX) | Byte   | Description Text                    | Data Size | Data Format |
|-----------------------|--------|-------------------------------------|-----------|-------------|
| 0xZZZZ                | #1~#17 | VIN (Vehicle Identification Number) | 17        | ASCII       |

Table C-2. Example#2 of Identification Option DataIdentifier Parameter Table

### Example#3]

| Data Identifier (HEX) | Byte#       | Control Enable Mask Record | Description Text        | Input Output Control Parameter | Control State               | Control Type       | Test Condition       |
|-----------------------|-------------|----------------------------|-------------------------|--------------------------------|-----------------------------|--------------------|----------------------|
| 0xAAAA                | -           | -                          | Desired Idle Adjustment | 00, 01, 02, 03                 | (Hex) * 10 [r/min]          | Until Stop Request | IG ON & Eng. Running |
| 0xB BBBB              | -           | -                          | MIL                     | 00, 01, 02, 03                 | 0x00 = OFF<br>0xFF = ON     | Until Stop Request | IG ON & Eng. Stop    |
| 0xC CCCC              | -           | -                          | Fuel Pump Relay         | 00, 01, 02, 03                 | 0x00 = Stop<br>0xFF = Start | For 3 sec          | IG ON & Eng. Stop    |
| 0xD DDDD              | #1          | Bit 7                      | IAC Pintle Position     | 00, 01, 02, 03                 | X [count]                   | Until Stop Request | IG ON & Eng. Running |
|                       | #2          | Bit 6                      | RPM (High Byte)         |                                | X [U/min]                   |                    |                      |
|                       | #3          |                            | RPM (Low Byte)          |                                |                             |                    |                      |
|                       | #4 [bit7-4] | Bit 5                      | Pedal Position A        |                                | X*8 [%]                     |                    |                      |
|                       | #4 [bit3-0] | Bit 4                      | Pedal Position B        |                                | X*8 [%]                     |                    |                      |
|                       | #5          | Bit 3                      | EGR Duty Cycle          |                                | X*100/255 [%]               |                    |                      |
| ...                   | ...         | ...                        | ...                     | ...                            | ...                         | ...                | ...                  |

Table C-3. Example#3 of Input Output DataIdentifier Parameter Table

| Hex       | Description                                                                                                                                                                                                                                                                                               | Cvt | Mnemonic |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 0100-A5FF | Reserved for HMCKIA Specific – Record Data Identifiers                                                                                                                                                                                                                                                    | U   |          |
| A600-A7FF | Reserved for Legislative Use – IMPORTANT : Do not use other than legislative ECU                                                                                                                                                                                                                          | M   |          |
| A800-ACFF | Reserved for HMCKIA Specific – Record Data Identifiers                                                                                                                                                                                                                                                    | U   |          |
| AD00-AFFF | Reserved for Legislative Use – IMPORTANT : Do not use other than legislative ECU                                                                                                                                                                                                                          | M   |          |
| B000-B1FF | Reserved for HMCKIA Specific – Record Data Identifiers                                                                                                                                                                                                                                                    | U   |          |
| B200-BFFF | Reserved for Legislative Use – IMPORTANT : Do not use other than legislative ECU                                                                                                                                                                                                                          | M   |          |
| C000-C2FF | Reserved for HMCKIA Specific – Record Data Identifiers                                                                                                                                                                                                                                                    | U   |          |
| C300-CEFF | Reserved for Legislative Use – IMPORTANT : Do not use other than legislative ECU                                                                                                                                                                                                                          | M   |          |
| CF00-EFFF | Reserved for HMCKIA Specific – Record Data Identifiers                                                                                                                                                                                                                                                    | U   |          |
| C0DE      | <b>EOLCodingNumber</b><br><b>Notation – EOL Coding Number :</b><br>This value shall be used to reference the vehicle variant coding i.e)EOL Coding<br>Record data size shall be 4 or 8bytes(extended)                                                                                                     | U   | EOLCN    |
| DE00-DE50 | <b>EthernetPhyDiagnosticRecordData</b><br>These values are available for EthernetPHYDiagnosticsData, used to provide the health information/statistics of Ethernet PHY. Further details e.g. Ethernet PHY diagnostic features, protocols, and procedures, shall be referred to the relevant specification | U   | EPDRD    |



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|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------|
| ED90      | <b>NonSleepDataCollectedEachEcuDataIdentifier</b><br>This value shall be used to reference non-sleep log data collecting from each ECU. Each ECU shall response the data to CCU by RDBI(ReadDataByIdentifier) when the ECU can't enter to the sleep mode abnormally and CCU sends diagnostic request as ED90 DID. CCU shall collect the data from the ECU and Tester can get the all collected data by another diagnostic RDBI from CCU.<br>Each ECU shall response non-sleep log data in Max 100 Bytes. Specific data information responded shall be managed in each ECU's specification. For example, following data can be included in the data.<br>- Network Message and PNC bit info related to sleep and wakeup<br>- HW Input / Output pin info related to wakeup.<br>- internal specific data and flag value in ECU to cause non-sleep.<br>- Each application session info to hold entering sleep mode | M | NSDCEEDID  |
| ED91-ED95 | <b>Reserved for NonSleepDataCollectedEachEcuDataIdentifier</b><br>ED91-ED95 (Reserved for future non sleep data requirement)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | M | RNSDCEEDID |
| ED96-ED99 | <b>VehicleLoggerEcuLoggingDataIdentifier</b><br>This value shall be used to reference ECU internal logging information to be collected for vehicle logger. Vehicle logger in CCU references data in each ECU periodically at predefined interval, and the ECUs shall respond with their own internal logging information. Each response shall contain less than 1KB data, unless the exception is agreed with CCU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | U | VLELDID    |
| EDC0      | <b>ECUPlatformLoggingData 1 – Network Mode Log Data</b><br>This value shall be used to reference ECU internal logging information from the SW platform. This feature collect information of ComM and CanNm module in BSW such as ComM mode and also the state and transition value of CanNm.<br><br>This functionality is provided by SW Platform itself and does not including any security and personal data at all.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | U | ECUPLD1    |
| EDC1      | <b>ECUPlatformLoggingData 2 – Wakeup/Sleep Log Data</b><br>This value shall be used to reference ECU internal logging information from the SW platform. This feature collect information during the Sleep and Wakeup processes run on SW platform such as ComM Mode and NvM running state at requesting sleep and Wakeup delay at requesting wakeup.<br><br>This functionality is provided by SW Platform itself and does not including any security and personal data at all.                                                                                                                                                                                                                                                                                                                                                                                                                                | U | ECUPLD2    |
| EFF0-EFDF | <b>FaultDiagnosticRecordData</b><br>Please refer to Requirements document for this DIDs                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | U | FDRD       |
| F010-F0FF | <b>Reserved for HMCKIA Specific – Input Output Data Identifiers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | U |            |
| F100-F17F | <b>Reserved for HMCKIA Specific – Identification Options Data Identifiers</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | U |            |
| F180      | <b>BootSoftwareIdentificationDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU boot software identification record. The first data byte of the record data shall be the numberOfModules that are reported. Following the numberOfModules the boot software identification(s) are reported. The format of the boot software identification structure shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                                                                                                                                                                                                                                                                                                            | U | BSIDID     |
| F181      | <b>ApplicationSoftwareIdentificationDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU application software number(s). The first data byte of the record data shall be the numberOfModules that are reported. Following the numberOfModules the application software identification(s) are reported. The format of the application software identification structure shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                                                                                                                                                                                                                                                                                            | U | ASIDID     |
| F182      | <b>ApplicationDataIdentificationDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU application data identification record. The first data byte of the record data shall be the numberOfModules that are reported. Following the numberOfModules the application data identification(s) are reported. The format of the application data identification structure shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                                                                                                                                                                                                                                                                                                | U | ADIDID     |
| F183      | <b>BootSoftwareFingerprintDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU boot software fingerprint identification record. Record data content and format shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | U | BSFPDID    |



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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-------------|
| F184 | <b>ApplicationSoftwareFingerprintDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU application software fingerprint identification record. Record data content and format shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                     | U  | ASFPDID     |
| F185 | <b>ApplicationDataFingerprintDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer specific ECU application data fingerprint identification record. Record data content and format shall be ECU specific and defined by the vehicle manufacturer.                                                                                                                                                             | U  | ADFPDID     |
| F186 | <b>ActiveDiagnosticSessionDataIdentifier</b><br>This value shall be used to report the active diagnostic session in the server. The values are defined by the diagnosticSessionType subfunction parameter in the DiagnosticSessionControl service.                                                                                                                                                                                           | U  | ADSDID      |
| F187 | <b>VehicleManufacturerSparePartNumberDataIdentifier</b><br><b>Notation – Vehicle Part Number :</b><br><b>or - Vehicle Part Number(Location) :</b><br><b>*Location : e.g) FL,FR,RL,RR</b><br>This value shall be used to reference the vehicle manufacturer spare part number.. Record data format shall be ASCII and size shall be 10 ~ 15 bytes. Color code could be max. 15 bytes.<br>(e.g. 959102T250 → 0x 39 35 39 31 30 32 54 32 35 30) | M  | VMSPDID     |
| F188 | <b>VehicleManufacturerECUSoftwareNumberDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer ECU (server) software number. Record data content and format shall be server specific and defined by the vehicle manufacturer.                                                                                                                                                                                   | U  | VMECUSNDID  |
| F189 | <b>VehicleManufacturerECUSoftwareVersionNumberDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer ECU (server) software version number. Record data content and format shall be server specific and defined by the vehicle manufacturer.                                                                                                                                                                    | U  | VMECUSVNDID |
| F18A | <b>SystemSupplierIdentifierDataIdentifier</b><br>This value shall be used to reference the system supplier name and address information. Record data content and format shall be server specific and defined by the system supplier.                                                                                                                                                                                                         | U  | SSIDDID     |
| F18B | <b>ECUManufacturingDateDataIdentifier</b><br><b>Notation – Supplier H/W manufacturing Date :</b><br><b>Or - Supplier H/W manufacturing Date(Location) :</b><br><b>*Location : e.g.) FL, FR, RL,RR</b><br>This value shall be used to reference the ECU (server) manufacturing date. Record data format shall be HEX and size shall be 4 bytes. The content shall be ordered as Year, Month, Day.(e.g. 2018.03.06 → 0x20180306)               | M  | ECUMDDID    |
| F18C | <b>ECUSerialNumberDataIdentifier</b><br><b>Notation – Serial Number :</b><br><b>Or - Serial Number(Nocation) :</b><br><b>*Location : e.g.) FL,FR,RL,RR</b><br>This value shall be used to reference the ECU (server) serial number. Data shall be defined as ASCII code.                                                                                                                                                                     | M  | ECUSNDID    |
| F18D | <b>SupportedFunctionalUnitsDataIdentifier</b><br>This value shall be used to request the functional units implemented in a server.                                                                                                                                                                                                                                                                                                           | U  | SFUDID      |
| F18E | <b>VehicleManufacturerKitAssemblyPartNumberDataIdentifier</b><br>This value shall be used to reference the vehicle manufacturer order number for a kit (assembled parts bought as a whole for production e.g. cockpit), when the spare part number designates only the server (e.g. for aftersales). The record data content and format shall be server specific and defined by the vehicle manufacturer.                                    | U  | VMKAPNDID   |
| F18F | <b>RXSWINDataIdentifier</b><br><b>Notation – RXSWIN Data:</b><br>RXSWIN is information that combines local RXSWIN information into one RXSWIN. Refer to Annex J                                                                                                                                                                                                                                                                              | C2 |             |
| F190 | <b>VINDataIdentifier</b><br>This value shall be used to reference the VIN number. VIN data shall be defined as ASCII code                                                                                                                                                                                                                                                                                                                    | U  | VINDID      |



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| F191 | <b>VehicleManufacturerECUHardwareNumberDataIdentifier</b><br><b>Notation – Vehicle H/W Version :</b><br>This value is <b>HMCKIA</b> defined ECU hardware version number shown on the drwaing of each system. H/W version shall be defined as ASCII code.                                                                                                                  | M | VMECUHNDI<br>D   |
| F192 | <b>SystemSupplierECUHardwareNumberDataIdentifier</b><br>This value shall be used to reference the system supplier specific ECU (server) hardware number. This value shall be defined as ASCII code.                                                                                                                                                                       | U | SSECUHWN<br>DID  |
| F193 | <b>SystemSupplierECUHardwareVersionNumberDataIdentifier</b><br><b>Notation – Supplier H/W Version :</b><br><b>Or - Supplier H/W Version(Location) :</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the system supplier specific ECU (server) hardware version number. This version shall be defined as ASCII code.               | M | SSECUHWVN<br>DID |
| F194 | <b>SystemSupplierECUSoftwareNumberDataIdentifier</b><br>This value shall be used to reference the system supplier specific ECU (server) software number. This value shall be defined as ASCII code.                                                                                                                                                                       | U | SSECUSWND<br>ID  |
| F195 | <b>SystemSupplierECUSoftwareVersionNumberDataIdentifier</b><br><b>Notation – Supplier S/W Version :</b><br>This value shall be used to reference the system supplier specific ECU (server) software version number. This value shall be defined as ASCII code.                                                                                                            | U | SSECUSWVN<br>DID |
| F196 | <b>ExhaustRegulationOrTypeApprovalNumberDataIdentifier</b><br>This value shall be used to reference the exhaust regulation or type approval number (valid for those systems which require type approval). Record data content and format shall be server specific and defined by the vehicle manufacturer.                                                                | U | EROTANDID        |
| F197 | <b>SystemNameOrEngineTypeDataIdentifier</b><br><b>Notation – System Name :</b><br>This value shall be used to reference the system name or engine type. Record data content and format shall be server specific and defined by the vehicle manufacturer.<br>Data content format shall be ASCII<br>e.g.) RR_C_RDR -> 8byte ASCII (0x82 0x82 0x95 0x67 0x95 0x82 0x68 0x82) | M | SNOETDID         |
| F198 | <b>RepairShopCodeOrTesterSerialNumberDataIdentifier</b><br>This value shall be used to reference the repair shop code or tester (client) serial number (e.g., to indicate the most recent service client used re-program server memory). Record data content and format shall be server specific and defined by the vehicle manufacturer.                                 | U | RSCOTSNDI<br>D   |
| F199 | <b>ProgrammingDateDataIdentifier</b><br>This value shall be used to reference the date when the server was last programmed. Record data content and format shall be unsigned numeric, ASCII or BCD, and shall be ordered as Year, Month, Day.                                                                                                                             | U | PDDID            |
| F19A | <b>CalibrationRepairShopCodeOrCalibrationEquipmentSerialNumberDataIdentifier</b><br>This value shall be used to reference the repair shop code or client serial number (e.g., to indicate the most recent service client used re-calibrate the server). Record data content and format shall be server specific and defined by the vehicle manufacturer.                  | U | CRSCOCESN<br>DID |
| F19B | <b>CalibrationDateDataIdentifier</b><br>This value shall be used to reference the date when the server was last calibrated. Record data content and format shall be unsigned numeric, ASCII or BCD, and shall be ordered as Year, Month, Day.                                                                                                                             | U | CDDID            |
| F19C | <b>CalibrationEquipmentSoftwareNumberDataIdentifier</b><br>This value shall be used to reference software version within the client used to calibrate the server. Record data content and format shall be server specific and defined by the vehicle manufacturer.                                                                                                        | U | CESWNDID         |
| F19D | <b>ECUInstallationDateDataIdentifier</b><br>This value shall be used to reference the date when the ECU (server) was installed in the vehicle. Record data content and format shall be either unsigned numeric, ASCII or BCD, and shall be ordered as Year, Month, Day.                                                                                                   | U | EIDDID           |
| F19E | <b>ODXFileDataIdentifier</b><br>This value shall be used to reference the ODX (Open Diagnostic Data Exchange) file of the server to be used to interpret and scale the server data.                                                                                                                                                                                       | U | ODXFDID          |
| F19F | <b>EntityDataIdentifier</b><br>This value shall be used to reference the entity data identifier as defined in ISO 15764 for a secured data transmission.                                                                                                                                                                                                                  | U | EDID             |



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| F1A0      | <b>SoftwareVersionfor HMCKIA VehicleManufactureDataIdentifier</b><br><b>Notation – Vehicle S/W Version</b><br><b>Or - Vehicle S/W Version(Location) :</b><br><b>*Location : e.g.) FL,FR,RL,RR</b><br>This value is HMCKIA defined ECU software version number. This version is used to assist the unit version or to represent all SW units. S/W version shall be defined as ASCII code. Dataidentifier parameter.<br>※ In case of a ECU has sub ECUs, 4 * n bytes would be used.<br>('n' is the number of sub ECUs) In this case the order of ECU version shall be fixed.                                                                                           | U  | SVHVMDID   |
| F1A1      | <b>ECUSupplierCodeDataIdentifier</b><br><b>Notation – System Supplier Code :</b><br><b>Or - System Supplier Code(Location) :</b><br><b>*Location : e.g.) FL,FR,RL,RR</b><br>This value shall be used to reference the ECU (server) Supplier Code.<br>Record data format shall be ASCII and size shall be 4 bytes.<br>(e.g. DELPHI CORPORATION : AB9H → 0x41 42 39 48)                                                                                                                                                                                                                                                                                                | M  | ECUSCD     |
| F1A2      | <b>ECUApplicationVersionDataIdentifier</b><br>This value shall be used to reference the Application version. following format shall be used.<br>{App ID (ASCII 5bytes) + Length(Hex 1byte) + Version(ASCII 'n' bytes)}<br>*'m'<br>The number of iteration 'm' depends on the number of App installed in the ECU.<br>The length of the Version is depends on application specific.                                                                                                                                                                                                                                                                                    | U  | ECUAVD     |
| F1A3      | <b>ECUApplicationIVDHashDataIdentifier</b><br>This value shall be used to reference the Application IVD Hash. following format shall be used. {App ID (ASCII 5bytes) + Length(Hex 1byte) + IVD(32bytes)}                                                                                                                                                                                                                                                                                                                                                                                                                                                             | C3 | ECUAIVDHD  |
| F1A4-F1AF | <b>Reserved for HMCKIA Specific – Identification Option</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |    |            |
| F1B0      | <b>ECUSoftwareUNITnumberDataIdentifier</b><br><b>Nontation – The number of S/W Unit :</b><br>This Identifier shall be used to reference unique system specific SW version. This value shall be used to reference the number of system specific S/W Unit in the ECU(server). The number of S/W Unit must consist of within 1byte and data type is unit8. If there is only one S/W unit, it is expressed as 1(dec). (Increase the value by 1 according to the number of S/W units.), and It can be expressed up to 15(dec). DID starts from F1B1 and be increased by 1.<br>e.g) The number of SW unit is 3 then only three DIDs are allowed to use. (F1B1, F1B2, F1B3) | M  | ECUSUNDID  |
| F1B1      | <b>ECUSoftwareUNIT1VersionDataIdentifier</b><br><b>Notation –S/W Unit1 Version :</b><br><b>Or - S/W Unit1 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 1 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code.                                                                                                                                                                                                                                     | M  | ECUSU1VDID |
| F1B2      | <b>ECUSoftwareUNIT2VersionDataIdentifier</b><br><b>Notation –S/W Unit2 Version :</b><br><b>Or - S/W Unit2 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 2 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code.                                                                                                                                                                                                                                     | C1 | ECUSU2VDID |



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| F1B3 | <b>ECUSoftwareUNIT3VersionDataIdentifier</b><br><b>Notation –S/W Unit3 Version :</b><br><b>Or - S/W Unit3 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 3 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU3VDID |
| F1B4 | <b>ECUSoftwareUNIT4VersionDataIdentifier</b><br><b>Notation –S/W Unit4 Version :</b><br><b>Or - S/W Unit4 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 4 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU4VDID |
| F1B5 | <b>ECUSoftwareUNIT5VersionDataIdentifier</b><br><b>Notation –S/W Unit5 Version :</b><br><b>Or - S/W Unit5 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 5 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU5VDID |
| F1B6 | <b>ECUSoftwareUNIT6VersionDataIdentifier</b><br><b>Notation –S/W Unit6 Version :</b><br><b>Or - S/W Unit6 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 6 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU6VDID |
| F1B7 | <b>ECUSoftwareUNIT7VersionDataIdentifier</b><br><b>Notation –S/W Unit7 Version :</b><br><b>Or - S/W Unit7 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 7 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU7VDID |
| F1B8 | <b>ECUSoftwareUNIT8VersionDataIdentifier</b><br><b>Notation –S/W Unit8 Version :</b><br><b>Or - S/W Unit8 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 8 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU8VDID |
| F1B9 | <b>ECUSoftwareUNIT9VersionDataIdentifier</b><br><b>Notation –S/W Unit9 Version :</b><br><b>Or - S/W Unit9 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 9 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU9VDID |



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| F1BA | <b>ECUSoftwareUNIT10VersionDataIdentifier</b><br><b>Notation –S/W Unit10 Version :</b><br><b>Or - S/W Unit10 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 10 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU10VDI<br>D |
| F1BB | <b>ECUSoftwareUNIT11VersionDataIdentifier</b><br><b>Notation –S/W Unit11 Version :</b><br><b>Or - S/W Unit11 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 11 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU11VDI<br>D |
| F1BC | <b>ECUSoftwareUNIT12VersionDataIdentifier</b><br><b>Notation –S/W Unit12 Version :</b><br><b>Or - S/W Unit12 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 12 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU12VDI<br>D |
| F1BD | <b>ECUSoftwareUNIT13VersionDataIdentifier</b><br><b>Notation –S/W Unit13 Version :</b><br><b>Or - S/W Unit13 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 13 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU13VDI<br>D |
| F1BE | <b>ECUSoftwareUNIT14VersionDataIdentifier</b><br><b>Notation –S/W Unit14 Version :</b><br><b>Or - S/W Unit14 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 14 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU14VDI<br>D |
| F1BF | <b>ECUSoftwareUNIT15VersionDataIdentifier</b><br><b>Notation –S/W Unit15 Version :</b><br><b>Or - S/W Unit15 Version(Name or Location) :</b><br><b>*Name : e.g.) MCU, AP</b><br><b>*Location : e.g.) FL, FR, RL, RR</b><br>This value shall be used to reference the ECU S/W unit 15 as a system specific version that must be unique during entire ECU lifecycle. Record data content and format should be expressed by ASCII code. | C1 | ECUSU15VDI<br>D |



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|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------|----|------------|-----------|
| F1C0                   | <b>ECUSecurityInformationDataIdentifier</b><br>This value shall be used to reference the security status of the ECU, such as Secure boot, Secure Debug, and HSM configuration status.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                               |                                | M  | ECUSSDID   |           |
|                        | Byte #                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Description                                                                                                   | Conversion rule                |    |            | Data Size |
|                        | #0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>HSM Configuration Lock state</b><br>This value shall be used to check HSM configuration Lock state of ECU. | (bit 7..0)                     |    |            | 1byte     |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x00 = Disabled                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x01 = Enabled                 |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x02 ~ 0xFE = Reserved         |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0xFF = N/A                     |    |            |           |
|                        | #1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Secure Boot State</b><br>This value shall be used to check Secure Boot state of ECU.                       | (bit 7..0)                     |    |            | 1byte     |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x00 = Disabled                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x01 = Enabled                 |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x02 ~ 0xFE = Reserved         |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0xFF = N/A                     |    |            |           |
|                        | #2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Secure Debug State</b><br>This value shall be used to check Secure Debug state of ECU.                     | (bit 7..0)                     |    |            | 1byte     |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x00 = Disabled                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x01 = Enabled                 |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               | 0x02 = DebugProtectionTempStop |    |            |           |
| 0x03 = DebugDisable    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               |                                |    |            |           |
| 0x04 ~ 0xFE = Reserved |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               |                                |    |            |           |
| 0xFF = N/A             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                               |                                |    |            |           |
| #3                     | <b>External bootloader state</b><br>This value shall be used to check External bootloader state of ECU.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | (bit 7..0)                                                                                                    | 1byte                          |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0x00 = Unsupported                                                                                            |                                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0x01 = Disabled                                                                                               |                                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0x02 = Enabled                                                                                                |                                |    |            |           |
|                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 0xFF = N/A                                                                                                    |                                |    |            |           |
| F1C1                   | <b>ECUSoftwareUNIT1IVDDataIdentifier</b><br><b>Notation – S/W Unit1 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 1. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request.<br>If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> |                                                                                                               |                                | M  | ECUSU1HDID |           |
| F1C2                   | <b>ECUSoftwareUNIT2IVDDataIdentifier</b><br><b>Notation – S/W Unit2 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 2. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b>    |                                                                                                               |                                | C1 | ECUSU2HDID |           |



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|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------|
| F1C3 | <b>ECUSoftwareUNIT3IVDDataIdentifier</b><br><b>Notation – S/W Unit3 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 3. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU3HDID |
| F1C4 | <b>ECUSoftwareUNIT4IVDDataIdentifier</b><br><b>Notation – S/W Unit4 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 4. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU4HDID |
| F1C5 | <b>ECUSoftwareUNIT5IVDDataIdentifier</b><br><b>Notation – S/W Unit5 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 5. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU5HDID |
| F1C6 | <b>ECUSoftwareUNIT6IVDDataIdentifier</b><br><b>Notation – S/W Unit6 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 6. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU6HDID |
| F1C7 | <b>ECUSoftwareUNIT7IVDDataIdentifier</b><br><b>Notation – S/W Unit7 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 7. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU7HDID |



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|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|-----------------|
| F1C8 | <b>ECUSoftwareUNIT8IVDDataIdentifier</b><br><b>Notation – S/W Unit8 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 8. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b>    | C1 | ECUSU8HDID      |
| F1C9 | <b>ECUSoftwareUNIT9IVDDataIdentifier</b><br><b>Notation – S/W Unit9 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 9. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b>    | C1 | ECUSU9HDID      |
| F1CA | <b>ECUSoftwareUNIT10IVDDataIdentifier</b><br><b>Notation – S/W Unit10 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 10. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU10HDI<br>D |
| F1CB | <b>ECUSoftwareUNIT11IVDDataIdentifier</b><br><b>Notation – S/W Unit11 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 11. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU11HDI<br>D |
| F1CC | <b>ECUSoftwareUNIT12IVDDataIdentifier</b><br><b>Notation – S/W Unit12 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 12. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU12HDI<br>D |



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|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|------------------|
| F1CD      | <b>ECUSoftwareUNIT13IVDDataIdentifier</b><br><b>Notation – S/W Unit13 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 13. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU13HDI<br>D  |
| F1CE      | <b>ECUSoftwareUNIT14IVDDataIdentifier</b><br><b>Notation – S/W Unit14 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 14. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU14HDI<br>D  |
| F1CF      | <b>ECUSoftwareUNIT15IVDDataIdentifier</b><br><b>Notation – S/W Unit15 IVD Data :</b><br>This value shall be used to reference the IVD value of SW unit 15. Each unit shall have own IVD value to prove SW integrity. This value has to be same with IVD which is in the SUMS. If calculation of IVD affects hard realtime performance of ECU during the drive, ECU shall stop calculation and return NRC(22) for the DID request. If SW UNIT#n is not available(e.g. The value of the ECUSoftwareUNIT#n Version is "NULL" or SW UNIT#n is not supported), ECUSoftwareUNIT#nIVDDataIdentifier shall not be supported. That means server returns NRC31.<br>*IVD : Integrity Validation Data<br>Data type : HEX<br>Data length : <b>SHA256 (32bytes)</b> | C1 | ECUSU15HDI<br>D  |
| F1D0-F1ED | <b>Reserved for HMCKIA Specific – Identification Option</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | U  |                  |
| F1EE      | <b>RXSWINEOLResultIdentifier</b><br>This value shall be used to reference the result of Routine control 0x04F0<br>This value shall be maintained until new result has been stored.<br>Result shall be maintained even if driving cycle has been changed.<br>Data Type : Hex<br>Data Length : 1Byte<br>Volatile : Non-volatile data<br>R/W : Read only<br>Data definition(text table) :<br>0x00 : No result (Routine 0x04F0 has never been done before)<br>0x01 : RXSWIN Done (Routine 0x04F0 has been done)<br>0x02 : RXSWIN Aborted (Routine 0x04F0 has been aborted)                                                                                                                                                                                | C2 | RXSWINEOL<br>RES |
| F1EF      | <b>LocalRXSWINDataIdentifier</b><br><b>Notation – Local RXSWIN Data :</b><br>RXSWIN information which is in a ECU. Refer to Annex J                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | M  | LRXSWINDID       |
| F1F0-F1FF | <b>Reserved for System Specific – Identification Option</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | U  |                  |
| F200-F2FF | <b>Reserved – Periodic Data Identifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | U  |                  |
| F300-F3FF | <b>Reserved – Dynamically Defined Data Identifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | U  |                  |
| F400-F8FF | <b>Reserved for OBD – IMPORTANT : Do not use other than legislative ECU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | M  |                  |
| F900-F9FF | <b>TachographDataIdentifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | U  | TACHODID         |
| FA00-FA0F | <b>Airbag Deployment Data Identifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | U  | ADDID            |
| FA10-FA18 | <b>EDR Data Identifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | U  | EDRDID           |
| FA19-FAFF | <b>Safety System Data Identifier</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | U  | SSDID            |
| FB00-FCFF | <b>Reserved for Legislative Use – IMPORTANT : Do not use other than legislative ECU</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | M  |                  |

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|-----------|------------------------------|---|--|
| FD00-FEFF | Reserved for System Specific | M |  |
| FF02-FFFF | Reserved for ISOSAE          | M |  |

C1 : More than 1 SW unit is available on the system.  
C2 : Central Gateway ECU  
C3 : If Controller supports DID F1A2, This DID shall be supported.

Table C-4. Example#3 of DataIdentifier Parameter Table

Generic DID shall be shown as following order to the diagnostic tester.

| No | DID       | Notation of Diagnostic tool                             |
|----|-----------|---------------------------------------------------------|
| 1  | F187      | Vehicle Part Number                                     |
| 2  | F1B0      | The number of S/W Unit                                  |
| 3  | F1B1~F1BA | S/W Unit(*N) Version : (*N is the numbering of SW unit) |
| 4  | F197      | System Name :                                           |
| 5  | F193      | Supplier H/W Version                                    |
| 6  | F18B      | Supplier H/W Manufacturing Date :                       |
| 7  | F1A1      | System Supplier Code :                                  |
| 8  | C0DE      | EOL Coding Number :                                     |
| 9  | F18C      | Serial Number :                                         |

Table C-4.1. The order of DID in the diagnostic tool and Notation

| DID  | Usage                 |
|------|-----------------------|
| F1A0 | S/W version of HMCKIA |
| F187 | Vehicle Part Number   |
| F1B0 | The number of SWUnits |
| F1B1 | SW Unit1 version      |
| F1B2 | SW Unit2 version      |
| F1B3 | SW Unit3 version      |
| F1B4 | SW Unit4 version      |
| F1B5 | SW Unit5 version      |
| F1B6 | SW Unit6 version      |
| F1B7 | SW Unit7 version      |
| F1B8 | SW Unit8 version      |
| F1B9 | SW Unit9 version      |
| F1BA | SW Unit10 version     |

Table C-4.2. List of SUMS related DID



C.1.1 ControlEnableMaskRecord

| Byte# | Bit Evaluation |                |                                                                     |
|-------|----------------|----------------|---------------------------------------------------------------------|
|       | Bit            | PID #<br>(HEX) | Conversion                                                          |
| 1     | Bit 7          | 01             | 0 = doesn't affected by the request,<br>1 = affected by the request |
|       | Bit 6          | 02             |                                                                     |
|       | ...            | ...            |                                                                     |
|       | Bit 1          | 07             |                                                                     |
|       | Bit 0          | 08             |                                                                     |
| 2     | Bit 7          | 09             |                                                                     |
|       | Bit 6          | 0A             |                                                                     |
|       | ...            | ...            |                                                                     |
|       | Bit 1          | 0E             |                                                                     |
|       | Bit 0          | 0F             |                                                                     |
| ...   | ...            | ...            |                                                                     |
| N     | Bit 7          | (N-1)*8+1      |                                                                     |
|       | Bit 6          | (N-1)*8+2      |                                                                     |
|       | ...            | ...            |                                                                     |
|       | Bit 1          | (N-1)*8+7      |                                                                     |
|       | Bit 0          | (N-1)*8+8      |                                                                     |

Table C-5. Supported PID Check Byte Definitions



### C.1.2 Supported PID Check Byte Definitions

Not all PIDs are applicable or supported by all ECUs or systems. PID \$00 is a bit-encoded PID that indicates, for each ECU, which PIDs that ECU supports.

If there are more than 32 PIDs in each LID, 32\*n-th PID (0x00, 0x20, 0x40, 0x60 ...) will be a supported PID byte. The definition of Supported PID Byte is as below.

| Requested PID<br>(HEX) | Bit Evaluation  |                                            |                                    |
|------------------------|-----------------|--------------------------------------------|------------------------------------|
|                        | Bit             | PID #<br>(HEX)                             | Conversion                         |
| 00                     | 1st Byte, Bit 7 | 01                                         | 0 = not supported<br>1 = supported |
|                        | 1st Byte, Bit 6 | 02                                         |                                    |
|                        | ...             | ...                                        |                                    |
|                        | 4th Byte, Bit 1 | 1F                                         |                                    |
|                        | 4th Byte, Bit 0 | 20                                         |                                    |
| 20                     | 1st Byte, Bit 7 | 21                                         |                                    |
|                        | 1st Byte, Bit 6 | 22                                         |                                    |
|                        | ...             | ...                                        |                                    |
|                        | 4th Byte, Bit 1 | 3F                                         |                                    |
|                        | 4th Byte, Bit 0 | 40                                         |                                    |
| ...                    | ...             | ...                                        |                                    |
| C0                     | 1st Byte, Bit 7 | C1                                         |                                    |
|                        | 1st Byte, Bit 6 | C2                                         |                                    |
|                        | ...             | ...                                        |                                    |
|                        | 4th Byte, Bit 1 | DF                                         |                                    |
|                        | 4th Byte, Bit 0 | E0                                         |                                    |
| E0                     | 1st Byte, Bit 7 | E1                                         |                                    |
|                        | 1st Byte, Bit 6 | E2                                         |                                    |
|                        | ...             | ...                                        |                                    |
|                        | 4th Byte, Bit 1 | FF                                         |                                    |
|                        | 4th Byte, Bit 0 | Reserved<br>Always zero(0) = not supported |                                    |

Table C-6. Supported PID Check Byte Definitions

## C.2 ScalingByte Parameter Definitions

The parameter scalingByte (SBYT) consists of one byte (high and low nibble). The scalingByte high nibble defines the data type, which is used to represent the dataIdentifier (DID). The scalingByte low nibble defines the number of bytes used to represent the parameter in a datastream.

| Encoding of High Nibble (HEX) | Description of Data Type                                                                                                                                                                                                                                                                                                                                                                                                     | Cvt | Mnemonic |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 0                             | <b>unSignedNumeric</b> (1 to 4 bytes)<br>This encoding uses a common binary weighting scheme to represent a value by mean of discrete incremental steps. One byte affords 256 steps; two bytes yields 65536 steps, etc.                                                                                                                                                                                                      | U   | USN      |
| 1                             | <b>signedNumeric</b> (1 to 4 bytes)<br>This encoding uses a two's complement binary weighting scheme to represent a value by mean of discrete incremental steps. One byte affords 256 steps; two bytes yields 65536 steps, etc.                                                                                                                                                                                              | U   | SN       |
| 2                             | <b>bitMappedReportedWithOutMask</b><br>Bit mapped encoding uses individual bits or small groups of bits to represent status. For every bit which represents status, a corresponding mask bit is required as part of the parameter definition. The mask indicates the validity of the bit for particular applications. This type of bit mapped parameter does not contain additional bytes to report the validity mask.       | U   | BMRWOM   |
| 3                             | <b>bitMappedReportedWithMask</b><br>Bit mapped encoding uses individual bits or small groups of bits to represent status. For every bit which represents status, a corresponding mask bit is required as part of the parameter definition. The mask indicates the validity of the bit for particular applications. This type of bit mapped parameter contains one validity mask byte for each status byte representing data. | U   | BMRWM    |
| 4                             | <b>BinaryCodedDecimal</b><br>Conventional Binary Coded Decimal encoding is used to represent two numeric digits per byte. The upper nibble is used to represent the most significant digit (0 - 9), and the lower nibble the least significant digit (0 -9).                                                                                                                                                                 | U   | BCD      |
| 5                             | <b>stateEncodedVariable</b> (1 byte)<br>This encoding uses a binary weighting scheme to represent up to 256 distinct states.<br>An example is a parameter, which represents the status of the Ignition Switch. Codes "00", "01", "02" and "03" may indicate ignition off, locked, run, and start, respectively. The representation is always limited to one (1) byte.                                                        | U   | SEV      |
| 6                             | <b>ASCII</b> (1 to 15 bytes for each scalingByte)<br>Conventional ASCII encoding is used to represent up to 128 standard characters with the MSB = logic 0. An additional 128 custom characters may be represented with the MSB = logic 1.                                                                                                                                                                                   | U   | ASCII    |
| 7                             | <b>signedFloatingPoint</b><br>Floating point encoding is used for data that needs to be represented in floating point or scientific notation. Standard IEEE formats shall be used according to ANSI/IEEE Std 754 - 1985.                                                                                                                                                                                                     | U   | SFP      |
| 8                             | <b>packet</b><br>Packets contain multiple data values, usually related, each with unique scaling. Scaling information is not included for the individual values. Refer to section C.3.1 scalingByteExtension for scalingByte high nibble of bitMappedReportedWithOutMask.                                                                                                                                                    | U   | P        |



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|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |      |
|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------|
| 9     | <b>formula</b><br>A formula is used to calculate a value from the raw data. Formula Identifiers are specified in the table defining the formulaIdentifier encoding. Refer to section C.3.2 scalingByteExtension for scalingByte high nibble of formula.                                                                                                                                                                                                                                                 | U | F    |
| A     | <b>unit/format</b><br>The units and formats are used to present the data in a more user-friendly format. Unit and Format Identifiers are specified in the Table defining the formulaIdentifier encoding.<br><br>NOTE) If combined units and/or formats are used, e.g. mV, then one scalingByte (and scalingData) for each unit/format shall be included in the readScalingDataByIdentifier positive response. Refer to section C.3.3 scalingByteExtension for scalingByte high nibble of unit / format. | U | U    |
| B     | <b>stateAndConnectionType (1 byte)</b><br>This encoding is used especially for input and output signals. The information encoded in the data byte specifies the high level physical layout, electrical levels and functional state. It is recommended to use this option for digital input and output parameters. Refer to section C.3.4 scalingByteExtension for scalingByte high nibble of stateAndConnectionType.                                                                                    | U | SACT |
| C - F | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |   |      |

Table C-7. ScalingByte (High Nibble) Parameter Definitions

| Encoding of High Nibble (HEX) | Description of Data Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Cvt | Mnemonic |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 0 - F                         | <b>numberOfBytesOfParameter</b><br>This range of values specifies the number of data bytes in a data stream referenced by a parameter identifier. The length of a parameter is defined by the scaling byte(s), which is always preceded by a parameter identifier (one or multiple bytes). If multiple scaling bytes follow a parameter identifier the length of the data referenced by the parameter identifier is the summation of the content of the low nibbles in the scaling bytes.<br><br>e.g. VIN is identified by a single byte parameter identifier and followed by two scaling bytes.<br>The length is calculated up to 17 data bytes. The content of the two low nibbles may have any combination of values that add up to 17 data bytes.<br><br>Note) For the scalingByte with high nibble encoded as formula or unit/format this value is \$0. | U   | NROBOP   |

Table C-8. ScalingByte (Low Nibble) Parameter Definitions

### C.3 ScalingByteExtension Parameter Definitions

#### C.3.1 ScalingByteExtension for ScalingByte high nibble of BitMappedReportedWithoutMask

The parameter scalingByteExtension (SBYE) is only supported for scalingByte parameters with the high nibble encoded as formula, unit/format, or bitMappedReportedWithoutMask.

A scalingByte with high nibble encoded as bitMappedReportedWithoutMask shall be followed by scalingByteExtension bytes representing the validity mask for the bit mapped dataIdentifier. Each byte shall indicate which bits of the corresponding dataIdentifier byte are supported for the current application.

| ScalingByteExtension Byte | Description                               | Cvt |
|---------------------------|-------------------------------------------|-----|
| #1                        | dataIdentifier dataRecord#1 validity mask | M   |
| ...                       | ...                                       | C1  |
| #p                        | dataIdentifier dataRecord#p validity mask | C1  |

C1 : The presence of this parameter depends on the size of the dataIdentifier the information is being requested for. The validity mask shall have as many bytes as the dataIdentifier has dataRecords.

Table C-9. ScalingByteExtension Bytes for BitMappedReportedWithoutMask

#### C.3.2 ScalingByteExtension for ScalingByte high nibble of Formula

The parameter scalingByteExtension (SBYE) is only supported for scalingByte parameters with the high nibble encoded as formula, unit/format, or bitMappedReportedWithoutMask.

A scalingByte with high nibble encoded as formula shall be followed by scalingByteExtension bytes defining the formula. The scalingByteExtension consists a of one byte formulaIdentifier and constants as described in Table below.

| ScalingByteExtension Byte | Description                                                                            | Cvt |
|---------------------------|----------------------------------------------------------------------------------------|-----|
| #1                        | formulaIdentifier (refer to Table defining the formulaIdentifier encoding for details) | M   |
| #2                        | C0 high byte                                                                           | M   |
| #3                        | C0 low byte                                                                            | M   |
| #4                        | C1 high byte                                                                           | U   |
| #5                        | C1 low byte                                                                            | U   |
| ...                       | ...                                                                                    | U   |
| #2n+2                     | Cn high byte                                                                           | U   |
| #2n+3                     | Cn low byte                                                                            | U   |

Table C-10. ScalingByteExtension Bytes for Formula

| FormulaIdentifier (HEX) | Description              | Cvt |
|-------------------------|--------------------------|-----|
| 00                      | $y = C0 * x + C1$        | U   |
| 01                      | $y = C0 * (x + C1)$      | U   |
| 02                      | $y = C0 / (x + C1) + C2$ | U   |
| 03                      | $y = x / C0 + C1$        | U   |

|         |                                                |   |
|---------|------------------------------------------------|---|
| 04      | $y = (x + C0) / C1$                            | U |
| 05      | $y = (x + C0) / C1 + C2$                       | U |
| 06      | $y = C0 * x$                                   | U |
| 07      | $y = x / C0$                                   | U |
| 08      | $y = x + C0$                                   | U |
| 09      | $y = x * C0 / C1$                              | U |
| 0A – 7F | Reserved                                       | M |
| 80 – FF | Reserved (vehicle manufacturer specific range) | U |

Table C-11. FormulaIdentifier Encoding

Formulas are defined using variables (y, x, etc.) and constants (C0, C1, C2, etc.). The variable y is the calculated value. The other variables, in consecutive order, are part of the data stream referenced by a dataIdentifier. Each constant is expressed as a two byte real number defined in Table C.8 — Two byte real number format. The two byte real numbers ( $C = M * 10^E$ ) contain a 12 bit signed (2's complement) mantissa (M) and a 4 bit signed (2's complement) exponent (E). The mantissa can hold values within the range -2048 to +2047, and the exponent can scale the number by  $10^{-8}$  to  $10^7$ . The exponent is encoded in the high nibble of the high byte of the two byte real number. The mantissa is encoded in the low nibble of the high byte and the complete low byte of the two byte real number.

| High Byte   |    |    |    |            |    |   |   | Low Byte    |   |   |   |            |   |   |   |
|-------------|----|----|----|------------|----|---|---|-------------|---|---|---|------------|---|---|---|
| High Nibble |    |    |    | Low Nibble |    |   |   | High Nibble |   |   |   | Low Nibble |   |   |   |
| 15          | 14 | 13 | 12 | 11         | 10 | 9 | 8 | 7           | 6 | 5 | 4 | 3          | 2 | 1 | 0 |
| Exponent    |    |    |    | Mantissa   |    |   |   |             |   |   |   |            |   |   |   |

Table C-12. Two Byte Real Number Format

### C.3.3 ScalingByteExtension for ScalingByte high nibble of Unit / Format

The parameter scalingByteExtension (SBYE) is only supported for scalingByte parameters with the high nibble encoded as formula, unit/format, or bitMappedReportedWithoutMask.

A scalingByte with high nibble encoded as unit / format shall be followed by a single scalingByteExtension byte defining the unit / format. The one byte scalingByteExtension is defined in Table C.9 — Unit / format scalingByteExtension encoding. If combined units and/or formats are used, e.g. mV, then one scalingByte (and scalingByteExtension ) shall be included for each unit / format.

| ScalingByteExtension Byte #1 | Name               | Symbol | Description | Cvt |
|------------------------------|--------------------|--------|-------------|-----|
| 00                           | No unit, no prefix |        |             | U   |
| 01                           | Meter              | m      | Length      | U   |
| 02                           | Foot               | ft     | Length      | U   |
| 03                           | Inch               | in     | Length      | U   |
| 04                           | Yard               | yd     | Length      | U   |
| 05                           | mile (English)     | mi     | length      | U   |
| 06                           | Gram               | g      | mass        | U   |
| 07                           | ton (metric)       | t      | mass        | U   |
| 08                           | Second             | s      | time        | U   |



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|    |                         |       |                           |   |
|----|-------------------------|-------|---------------------------|---|
| 09 | Minute                  | min   | time                      | U |
| 0A | Hour                    | h     | time                      | U |
| 0B | Day                     | d     | time                      | U |
| 0C | year                    | y     | time                      | U |
| 0D | ampere                  | A     | current                   | U |
| 0E | volt                    | V     | voltage                   | U |
| 0F | coulomb                 | C     | electric charge           | U |
| 10 | ohm                     | W     | resistance                | U |
| 11 | farad                   | F     | capacitance               | U |
| 12 | henry                   | H     | inductance                | U |
| 13 | siemens                 | S     | electric conductance      | U |
| 14 | weber                   | Wb    | magnetic flux             | U |
| 15 | tesla                   | T     | magnetic flux density     | U |
| 16 | kelvin                  | K     | thermodynamic temperature | U |
| 17 | Celsius                 | °C    | thermodynamic temperature | U |
| 18 | Fahrenheit              | °F    | thermodynamic temperature | U |
| 19 | candela                 | cd    | luminous intensity        | U |
| 1A | radian                  | rad   | plane angle               | U |
| 1B | degree                  | °     | plane angle               | U |
| 1C | hertz                   | Hz    | frequency                 | U |
| 1D | joule                   | J     | energy                    | U |
| 1E | Newton                  | N     | force                     | U |
| 1F | kilopond                | kp    | force                     | U |
| 20 | pound force             | lbf   | force                     | U |
| 21 | watt                    | W     | power                     | U |
| 22 | horse power (metric)    | hk    | power                     | U |
| 23 | horse power (UK and US) | hp    | power                     | U |
| 24 | Pascal                  | Pa    | pressure                  | U |
| 25 | bar                     | bar   | pressure                  | U |
| 26 | atmosphere              | atm   | pressure                  | U |
| 27 | pound force per square  | psi   | pressure                  | U |
| 28 | becquerel               | Bq    | radioactivity             | U |
| 29 | lumen                   | lm    | light flux                | U |
| 2A | lux                     | lx    | illuminance               | U |
| 2B | liter                   | l     | volume                    | U |
| 2C | gallon (British)        | -     | volume                    | U |
| 2D | gallon (US liq)         | -     | volume                    | U |
| 2E | cubic inch              | cu in | volume                    | U |

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|    |                         |                    |                              |   |
|----|-------------------------|--------------------|------------------------------|---|
| 2F | meter per second        | m/s                | speed                        | U |
| 30 | kilometer per hour      | km/h               | speed                        | U |
| 31 | mile per hour           | mph                | speed                        | U |
| 32 | revolutions per second  | rps                | angular velocity             | U |
| 33 | revolutions per minute  | rpm                | angular velocity             | U |
| 34 | counts                  | -                  | -                            | U |
| 35 | percent                 | %                  | -                            | U |
| 36 | milligram per stroke    | mg/stroke          | mass per engine stroke       | U |
| 37 | meter per square second | m/s <sup>2</sup>   | acceleration                 | U |
| 38 | Newton meter            | Nm                 | moment (e.g. torsion moment) | U |
| 39 | liter per minute        | l/min              | flow                         | U |
| 3A | Watt per square meter   | W/m <sup>2</sup>   | Intensity                    | U |
| 3B | Bar per second          | bar/s              | Pressure change              | U |
| 3C | Radians per second      | rad/s              | Angular velocity             | U |
| 3D | Radians per square      | rad/s <sup>2</sup> | Angular acceleration         | U |
| 3E | Kilogram per square     | kg/m <sup>2</sup>  | -                            | U |
| 3F | -                       | -                  | Reserved                     | M |
| 40 | exa (prefix)            | E                  | 10 <sup>18</sup>             | U |
| 41 | peta (prefix)           | P                  | 10 <sup>15</sup>             | U |
| 42 | tera (prefix)           | T                  | 10 <sup>12</sup>             | U |
| 43 | giga (prefix)           | G                  | 10 <sup>9</sup>              | U |
| 44 | mega (prefix)           | M                  | 10 <sup>6</sup>              | U |
| 45 | kilo (prefix)           | k                  | 10 <sup>3</sup>              | U |
| 46 | hecto (prefix)          | h                  | 10 <sup>2</sup>              | U |
| 47 | deca (prefix)           | da                 | 10                           | U |
| 48 | deci (prefix)           | d                  | 10 <sup>-1</sup>             | U |
| 49 | centi (prefix)          | c                  | 10 <sup>-2</sup>             | U |
| 4A | milli (prefix)          | m                  | 10 <sup>-3</sup>             | U |
| 4B | micro (prefix)          | μ                  | 10 <sup>-6</sup>             | U |
| 4C | nano (prefix)           | n                  | 10 <sup>-9</sup>             | U |
| 4D | pico (prefix)           | p                  | 10 <sup>-12</sup>            | U |
| 4E | femto (prefix)          | f                  | 10 <sup>-15</sup>            | U |
| 4F | atto (prefix)           | a                  | 10 <sup>-18</sup>            | U |
| 50 | Date1                   | -                  | Year-Month-Day               | U |
| 51 | Date2                   | -                  | Day/Month/Year               | U |
| 52 | Date3                   | -                  | Month/Day/Year               | U |

|       |                   |       |                                                                                |   |
|-------|-------------------|-------|--------------------------------------------------------------------------------|---|
| 53    | week              | W     | calendar week                                                                  | U |
| 54    | Time1             | -     | UTC<br>Hour/Minute/Second                                                      | U |
| 55    | Time2             | -     | Hour/Minute/Second                                                             | U |
| 56    | DateAndTime1      | -     | Second/Minute/Hour/Day<br>/Month/Year                                          | U |
| 57    | DateAndTime2      | -     | Second/Minute/Hour/Day<br>/Month/Year/Local minute<br>offset/Local hour offset | U |
| 58    | DateAndTime3      | -     | Second/Minute/Hour/Mon<br>th/Day/Year                                          | U |
| 59    | DateAndTime4      | -     | Second/Minute/Hour/Mon<br>th/Day/Year/Local minute<br>offset/Local hour offset | U |
| B0    | degree per second | deg/s | degree/second                                                                  | U |
| 5A-FF | -                 | -     | Reserved                                                                       | M |

Table C-13. Unit / Format ScalingByteExtension Encoding

### C.3.4 ScalingByteExtension for ScalingByte high nibble of StateAndConnectionType

A scalingByte with high nibble encoded as stateAndConnectionType shall be followed by a single scalingByteExtension byte defining the stateAndConnectionType. The one byte scalingByteExtension is defined in Table C.9 — Encoding of scalingByte High Nibble of stateAndConnectionType. The stateAndConnectionType encoding is used specially for input and output signals. Encoded in the scalingByteExtension data byte is information about the physical layout, electrical levels and functional state.

| Encoding of Bit | Value | Used with Input Signal                                                   | Used with Output Signal                                                  |
|-----------------|-------|--------------------------------------------------------------------------|--------------------------------------------------------------------------|
| 0 – 2           | 0     | State : Not Active                                                       | State : Not Activated                                                    |
|                 | 1     | State : Active, Function 1                                               | State : Active, Function 1                                               |
|                 | 2     | State : Error Detected                                                   | State : Plausibility Error Detected                                      |
|                 | 3     | State : Not Available                                                    | State : Not Available                                                    |
|                 | 4     | State : Active, Function 2<br>(only in combination with 3 states)        | State : Active, Function 2<br>(only in combination with 3 states)        |
|                 | 5 – 7 | Reserved                                                                 | Reserved                                                                 |
| 3 – 4           | 0     | Signal at Low Level (Ground)                                             | Signal at Low Level (Ground)                                             |
|                 | 1     | Signal at Middle Level<br>(Between Ground Level and +)                   | Signal at Middle Level<br>(Between Ground Level and +)                   |
|                 | 2     | Signal at High Level (+)                                                 | Signal at High Level (+)                                                 |
|                 | 3     | Reserved                                                                 | Reserved                                                                 |
| 5               | 0     | Input Signal                                                             | Not Defined                                                              |
|                 | 1     | Not Defined                                                              | Output Signal                                                            |
| 6 – 7           | 0     | Internal Signal or via CAN not exclusively<br>available in ECU connector | Internal Signal or via CAN not exclusively<br>available in ECU connector |
|                 | 1     | Pull-down Resistor input type (2 states)                                 | Low side switch (2 states)                                               |
|                 | 2     | Pull-up Resistor input type (2 states)                                   | High side switch (2 states)                                              |
|                 | 3     | Pull-up and Pull-down resistor input type (3<br>state)                   | Low side and High side switch (3 state)                                  |

Table C-14. Encoding of scalingByte High Nibble of stateAndConnectionType



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| HEX     | Description                                                                                                                                                                                                                                                                                                                                                                                 | Cvt | Mnemonic |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00      | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                             |     |          |
| 01      | <b>sendAtSlowRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a slow rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend). The repetition rate specified by the transmissionMode parameter slow is vehicle manufacturer specific, and pre-defined in the server.       | U   | SASR     |
| 02      | <b>sendAtMediumRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a medium rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend). The repetition rate specified by the transmissionMode parameter medium is vehicle manufacturer specific, and pre-defined in the server. | U   | SAMR     |
| 03      | <b>sendAtFastRate</b><br>This parameter specifies that the server shall transmit the requested dataRecord information at a fast rate in response to the request message (where the # of responses to be sent = maximumNumberOfResponsesToSend). The repetition rate specified by the transmissionMode parameter fast is vehicle manufacturer specific, and pre-defined in the server.       | U   | SAFR     |
| 04      | <b>stopSending</b><br>The server stops transmitting positive response messages send periodically/repeatedly. Note that maximumNumberOfResponsesToSend parameter should be set to 01 hex if transmissionMode = stopSending (otherwise, server operation could be undefined).                                                                                                                 | C   | SS       |
| 05 - FF | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                             |     |          |

C : stopSending shall be supported if sendAtSlowRate, sendAtMediumRate, and/or sendAtFastRate are supported.

Table C-15. Encoding of scalingByte High Nibble of stateAndConnectionType



## Annex. D Stored Data Transmission Functional Unit Data Parameter Definitions

### D.1 GroupOfDTC Parameter Definition

Table D-1. Definition of GroupOfDTC and range of DTC numbers provides group of DTC definitions.

| HEX             | Description                                  | Cvt | Mnemonic |
|-----------------|----------------------------------------------|-----|----------|
| 000000 - 0000FF | Reserved for future legislative requirements | M   | RFLU     |
| 0001FF          | Powertrain Group                             | U   | PG       |
| 0001XX - 3FFFXX | Powertrain DTC's (except 0001FF)             | U   | PDTC_    |
| 4000FF          | Chassis Group                                | U   | CG       |
| 4001XX - 7FFFXX | Chassis DTC's                                | U   | CDTC_    |
| 8000FF          | Body Group                                   | U   | BG       |
| 8001XX - BFFFXX | Body DTC's                                   | U   | BDTC_    |
| C000FF          | Network Communication Group                  | U   | NCG      |
| C001XX - FFFEXX | Network Communication DTC's                  | U   | NCDTC_   |
| FFFF33          | Emissions-system group                       | C   | ESG      |
| FFFFD0          | Safety-system group                          | U   | SSG      |
| FFFFFF          | All Groups (All DTC's)                       | M   | AG       |

C : This parameter selects the emission-related systems.

Table D-1. Definition of GroupOfDTC and range of DTC numbers

## D.2 DTCStatusMask and StatusOfDTC Bit Definitions

### D.2.1 Convention

This section defines the mapping of the DTCStatusMask / statusOfDTC parameters used with the ReadDTCInformation service. Every server shall adhere to the convention for storing bit-packed DTC status information as defined in table below. Actual usage of the bit-fields shall be defined in the implementation standards.

The following is a list of definitions used for the description of the DTC status bit definitions.

- **Test** : A test is an on-board diagnostic software algorithm that determines the malfunction status of a component or system. Some tests run only once during an operation cycle. Other tests can run every program loop, as often as every few milliseconds. The result of a test represents a completely matured / qualified failure condition. That means a test which needs a failing condition over a specific time or evaluation of additional plausibility checks before a component is considered to be failing will return a "Failed" condition after all maturation criteria have been fulfilled. Each test is associated with a unique DTC representing the root failure and detectable fault symptom.
- **Complete** : Complete is an indication that a test was able to determine whether a malfunction exists or does not exist for the current operation cycle (complete does not imply failed).
- **Test results** : While a test runs or after it has completed it may indicate one the following results to the internal failure handler :
  - **PreFailed** : This status may be used by tests in ECUs to indicate that the test is currently maturing a failure condition. One use case for this information is in manufacturing to speed up failure detection for optimised workflow while maintaining fault tolerance in the field.
  - **Failed** : This status is available after a monitor has run to its completion and indicates a completely matured failing condition.
  - **Passed** : This status is available after a monitor has run to its completion and indicates that the system or component is not failing.
- **Failure** : A failure is the inability of a component or system to meet its intended function. A failure has occurred when fault conditions have been detected for a sufficient period of time to warrant storage of a pending (for emission related components only) or confirmed DTC implying that a test returned a "Failed" result. The terms "failure" and "malfunction" are interchangeable.
- **Monitor** : A monitor consists of one or more tests used to determine the proper functioning of a component or system.
- **Monitoring cycle** : A monitoring cycle is the time in which a monitor runs to its completeness. This is a manufacturer defined set of conditions during which the tests of a monitor can run. A monitoring cycle may be executed several times during an operation cycle.
- **Operation Cycle** : An operation cycle defines the start and end conditions for monitors to run. During an operation cycle several monitoring cycles may have passed (regardless of their test results). An ECU may support several operation cycles. For body and chassis ECU's it is up to the manufacturer to define an operation cycle (e.g. time between powering up and powering down the ECU or between ignition on and ignition off). For powertrain ECU's, there are additional criteria defining an operation cycle. Emissionsrelated powertrain ECU's use an engine-running or engine-off time period to define an operation cycle which is referred to as driving cycle.

NOTE) For emissions-related monitors the criteria for the beginning and the end of an operation cycle are defined by



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legislation.

- **Pending** : The pending status of a failure is defined as a Test Result having reported a “Failed” for this failure during the current and/or the last completed operation cycle. Once the test has reported a “Passed” condition for a complete operation cycle of this failure the pending status is reset.
- **Driving cycle** : A specific type of operation cycle used for emissions-related ECU's. Refer to “OperationCycle” for further details. In emissions-related ECU's only one operation cycle shall be supported, which is identical to the driving cycle as defined by legislation.

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### D.2.2 StatusOfDTC Bit Definition

| Bit | Description                                                                                                                                                                                                                                                                                                                                                                | Default Value | Cvt | Mnemonic |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-----|----------|
| 0   | <b>TestFailed</b>                                                                                                                                                                                                                                                                                                                                                          | 0             | M   | TF       |
|     | '0' = most recent result from DTC test indicated no failure detected<br>'1' = most recent result from DTC test indicated a matured failing result.                                                                                                                                                                                                                         |               |     |          |
| 1   | <b>TestFailedThisOperationCycle</b>                                                                                                                                                                                                                                                                                                                                        | 0             | C1  | TFTOC    |
|     | '0' = testFailed: result has not been reported during the current operation cycle or after a call was made to ClearDiagnosticInformation during the current operation cycle.<br>'1' = testFailed: result was reported at least once during the current operation cycle.                                                                                                    |               |     |          |
| 2   | <b>PendingDTC</b>                                                                                                                                                                                                                                                                                                                                                          | 0             | U   | PDTC     |
|     | '0' = This bit shall be set to 0 after completing an operation cycle during which the test completed and a malfunction was not detected or upon a call to the ClearDiagnosticInformation service.<br>'1' = This bit shall be set to 1 and latched if a malfunction is detected during the current operation cycle.                                                         |               |     |          |
| 3   | <b>ConfirmedDTC</b>                                                                                                                                                                                                                                                                                                                                                        | 0             | M   | CDTC     |
|     | '0' = DTC has never been confirmed since the last call to ClearDiagnosticInformation or after the aging criteria have been satisfied for the DTC (or DTC has been erased due to fault memory overflow).<br>'1' = DTC confirmed at least once since the last call to ClearDiagnosticInformation and aging criteria have not yet been satisfied.                             |               |     |          |
| 4   | <b>TestNotCompletedSinceLastClear</b>                                                                                                                                                                                                                                                                                                                                      | 1             | C2  | TNCSLC   |
|     | '0' = DTC test has returned either a passed or failed test result at least one time since the last time diagnostic information was cleared.<br>'1' = DTC test has not run to completion since the last time diagnostic information was cleared.                                                                                                                            |               |     |          |
| 5   | <b>TestFailedSinceLastClear</b>                                                                                                                                                                                                                                                                                                                                            | 0             | C2  | TFSLC    |
|     | '0' = DTC test has not indicated a failed result since the last time diagnostic information was cleared.<br>'1' = DTC test returned a failed result at least once since the last time diagnostic information was cleared.                                                                                                                                                  |               |     |          |
| 6   | <b>TestNotCompleteThisOperationCycle</b>                                                                                                                                                                                                                                                                                                                                   | 1             | U   | TNCTOC   |
|     | '0' = DTC test has returned either a passed or testFailedThisOperationCycle = '1' result during the current drive cycle (or since the last time diagnostic information was cleared during the current operation cycle).<br>'1' = DTC test has not run to completion this operation cycle (or since the last time diagnostic information was cleared this operation cycle). |               |     |          |
| 7   | <b>WarningIndicatorRequested</b>                                                                                                                                                                                                                                                                                                                                           | 0             | M   | WIR      |
|     | '0' = Server is not requesting warningIndicator to be active<br>'1' = Server is requesting warningIndicator to be active                                                                                                                                                                                                                                                   |               |     |          |

C1 : Bit 1 (TestFailedThisOperationCycle) is Mandatory if Bit 2 (PendingDTC) is supported. Bit 1 is User Optional if Bit 2 is not supported.  
C2 : Bit 4 (TestNotCompletedSinceLastClear) and Bit 5 (TestFailedSinceLastClear) shall always be supported together.

\*. DTCStatusMask will be determined by setting each bit of StatusOfDTC '0 = not supported' or '1 = supported'.

Table D-2. DTC Status Bit Definition Overview



기술 표준  
ENGINEERING STANDARD

규격번호 (SPEC No)  
ES95486-02

페이지  
(SHT/SHTS) 217/244

| DTC Case | Status Of Bit<br>(Lower 4Bit) |         |                              |            | DTC Status Mask (request)<br>(※ If the result is greater than or equal to 1, responds corresponding DTC) |                              | Display of tester | Reason for not display    |
|----------|-------------------------------|---------|------------------------------|------------|----------------------------------------------------------------------------------------------------------|------------------------------|-------------------|---------------------------|
|          | Bit3                          | Bit2    | Bit1                         | Bit0       | 09<br>(EOL HADS, A/S GDS)                                                                                | 0D<br>(EOL U-Robot, A/S GDS) |                   |                           |
|          | Confirmed                     | Pending | TestFailedThisOperationCycle | TestFailed |                                                                                                          |                              |                   |                           |
| #1       | 0                             | 0       | X                            | 0          | 0 (no response)                                                                                          | 0 (no response)              | -                 | Not DTC occur and masking |
| #2       | 0                             | 0       | X                            | 1          | 1                                                                                                        | 1                            | -                 | Not occur <sup>1)</sup>   |
| #3       | 0                             | 1       | X                            | 0          | 0 (no response)                                                                                          | 1                            | Pending DTC       |                           |
| #4       | 0                             | 1       | X                            | 1          | 1                                                                                                        | 2                            | Pending DTC       |                           |
| #5       | 1                             | 0       | X                            | 0          | 1                                                                                                        | 1                            | Historic DTC      |                           |
| #6       | 1                             | 0       | X                            | 1          | 2                                                                                                        | 2                            | -                 | Not occur <sup>1)</sup>   |
| #7       | 1                             | 1       | X                            | 0          | 1                                                                                                        | 2                            | Historic DTC      |                           |
| #8       | 1                             | 1       | X                            | 1          | 2                                                                                                        | 3                            | Active DTC        |                           |

<sup>1)</sup> In case of supporting Pending DTC, if Test Failed Bit is set, Pending DTC is also set.

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표 D-3. DTC Status Definition and Display of tester (ECU supported Pending DTC)

| DTC Case | Status Of Bit |            | DTC Status Mask (request)<br>(※ If the result is greater than or equal to 1, responds corresponding DTC) | Display of tester | Reason for not display              |
|----------|---------------|------------|----------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------|
|          | Bit3          | Bit0       |                                                                                                          |                   |                                     |
|          | Confirmed     | TestFailed | 08<br>(EOL, A/S GDS)                                                                                     |                   |                                     |
| #1       | 0             | 0          | 0 (no response)                                                                                          | -                 | Not DTC occur and masking           |
| #2       | 0             | 1          | 0 (no response)                                                                                          | -                 | Not Occur <sup>2)</sup> and masking |
| #3       | 1             | 0          | 1                                                                                                        | Historic DTC      |                                     |
| #4       | 1             | 1          | 1                                                                                                        | Active DTC        |                                     |

<sup>2)</sup> In case of not supporting Pending DTC, if Test Failed Bit is set, Confirmed DTC is also set.  
(Confirmation threshold : 1)

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표 D-4. DTC Status Definition and Display of tester (ECU not supported Pending DTC)

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                            | Description       | Cvt : Emission                            | Cvt : Non-Emission | Mnemonic |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------|--------------------|----------|
| 0                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>TestFailed</b> | U                                         | U                  | TF       |
| This bit shall indicate the result of the most recently performed test. A logical '1' shall indicate that the last test failed meaning that the failure is completely matured. Reset to logical '0' if the result of the most recently performed test returns a “pass” result meaning that all de-mature criteria have been fulfilled. Additional reset conditions may be defined by the vehicle manufacturer / implementation |                   |                                           |                    |          |
| Bit state after a successful ClearDiagnosticInformation service                                                                                                                                                                                                                                                                                                                                                                |                   |                                           | Logical '0'        |          |
| Reset to logical '0' if a call has been made to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                                    |                   |                                           |                    |          |
| Bit state definition                                                                                                                                                                                                                                                                                                                                                                                                           |                   | Test Equipment Display Text               |                    |          |
| '0' = most recent result from DTC test indicated no failure detected (or test has not completed this operation cycle).                                                                                                                                                                                                                                                                                                         |                   | DTC test is not failed at time of request |                    |          |
| '1' = most recent result from DTC test indicated a matured failing result.                                                                                                                                                                                                                                                                                                                                                     |                   | DTC test failed at time of request        |                    |          |

Table D-4. DTC Status Bit 0 TestFailed Definitions

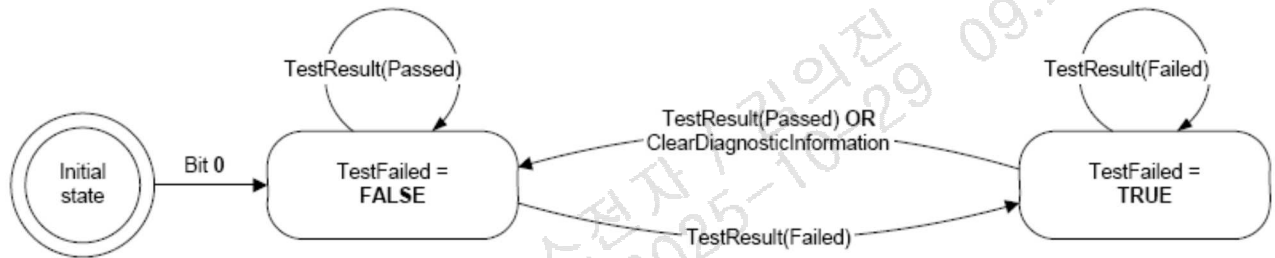


Figure D-1. DTC Status Bit 0 TestFailed Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                            | Description                         | Cvt : Emission                               | Cvt : Non-Emission | Mnemonic |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|----------------------------------------------|--------------------|----------|
| 1                                                                                                                                                                                                                                                                                                                                                                                                              | <b>TestFailedThisOperationCycle</b> | M                                            | C1                 | TFTOC    |
| This bit shall indicate whether or not a diagnostic test has reported a testFailed result at any time during the current operation cycle (or that a testFailed result has been reported during the current operation cycle and after the last time a call was made to ClearDiagnosticInformation). Reset to logical '0' when a new operation cycle is initiated or after a call to ClearDiagnosticInformation. |                                     |                                              |                    |          |
| If this bit is set to logical '1', it shall remain a '1' until a new operation cycle is started.                                                                                                                                                                                                                                                                                                               |                                     |                                              |                    |          |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                                                                         |                                     |                                              | Logical '0'        |          |
| Reset to logical '0' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                               |                                     |                                              |                    |          |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                                                                    |                                     | <b>Test Equipment Display Text</b>           |                    |          |
| '0' = testFailed: result has not been reported during the current operation cycle or after a call was made to ClearDiagnosticInformation during the current operation cycle.                                                                                                                                                                                                                                   |                                     | DTC test has not failed this operation cycle |                    |          |
| '1' = testFailed: result was reported at least once during the current operation cycle.                                                                                                                                                                                                                                                                                                                        |                                     | DTC test failed this operation cycle         |                    |          |
| C1: Bit 1 (testFailedThisOperationCycle) is Mandatory if bit 2 (pendingDTC) is supported. Bit 1 (testFailedThisOperationCycle) is User Optional if bit 2 (pendingDTC) is not supported.                                                                                                                                                                                                                        |                                     |                                              |                    |          |

Table D-5. DTC Status Bit 1 TestFailedThisOperationCycle Definitions

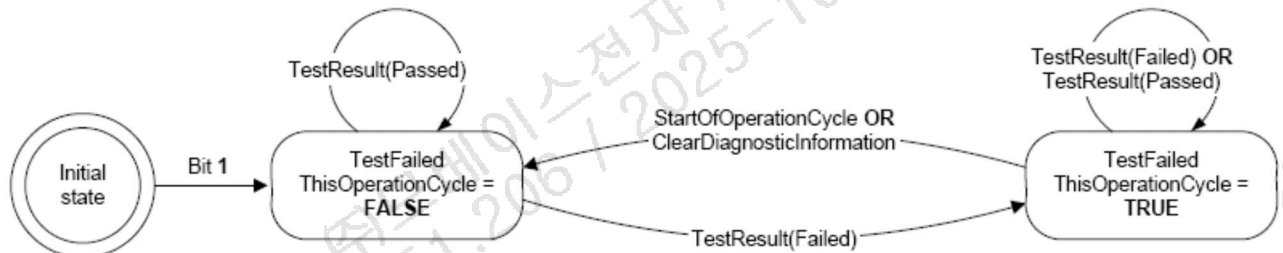


Figure D-2. DTC Status Bit 1 TestFailedThisOperationCycle Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Description | Cvt : Emission                                                               | Cvt : Non-Emission | Mnemonic |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|------------------------------------------------------------------------------|--------------------|----------|
| 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | PendingDTC  | M                                                                            | U                  | PDTC     |
| <p>This bit shall indicate whether or not a diagnostic test has reported a testFailed result at any time during the current or last completed operation cycle. The status shall only be updated if the test runs and completes. The criteria to set the pendingDTC bit and the TestFailedThisOperationCycle bit are the same. The difference is that the testFailedThisOperationCycle is cleared at the end of the current operation cycle and the pendingDTC bit is not cleared until an operation cycle has completed where the test has passed at least once and never failed.</p> <p>If the test did not complete during the current operation cycle, the status bit shall not be changed. For example, if a monitor stops running after a confirmed DTC is set, the pendingDTC must remain set = '1'. For an OBD DTC, a pending DTC is required to be stored after a malfunction is detected during the first driving cycle.</p> |             |                                                                              |                    |          |
| Bit state after a successful ClearDiagnosticInformation service                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |             |                                                                              | Logical '0'        |          |
| Reset to logical '0' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |                                                                              |                    |          |
| Bit state definition                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |             | Test Equipment Display Text                                                  |                    |          |
| '0' = This bit shall be set to 0 after completing an operation cycle during which the test completed and a malfunction was not detected or upon a call to the ClearDiagnosticInformation service.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |             | DTC test has not failed during the current or last completed operation cycle |                    |          |
| '1' = This bit shall be set to 1 and latched if a malfunction is detected during the current operation cycle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |             | DTC test failed during the current or last completed operation cycle         |                    |          |

Table D-6. DTC Status Bit 2 PendingDTC Definitions

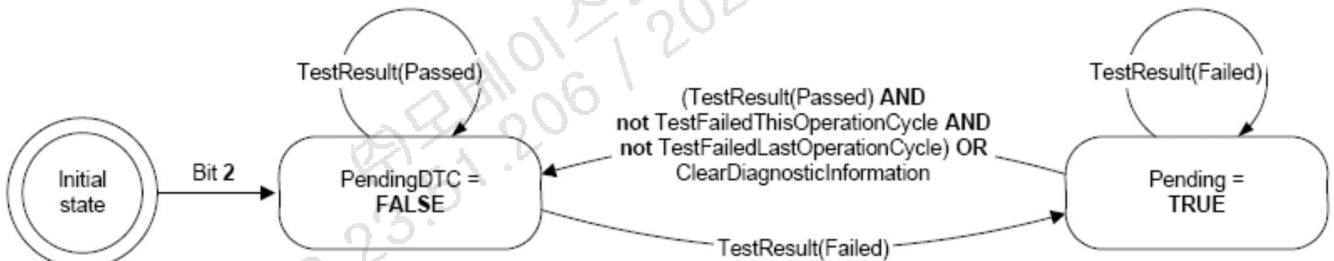


Figure D-3. DTC Status Bit 2 PendingDTC Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description         | Cvt : Emission                            | Cvt : Non-Emission | Mnemonic |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------------------------------------|--------------------|----------|
| 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>ConfirmedDTC</b> | M                                         | M                  | CDTC     |
| <p>This bit shall indicate whether a malfunction was detected enough times to warrant that the DTC is stored in long-term memory (e.g. pendingDTC has been set = '1' one or more times, depending on the DTC confirmation criteria).</p> <p>A confirmedDTC does not always indicate that the malfunction is present at the time of the request. (testFailed can be used to determine if a malfunction is present at the time of the request).</p> <p>Reset to logical '0' after a call to ClearDiagnosticInformation or after aging criteria has been satisfied (e.g., 40 engine warm-ups without another detected malfunction). Furthermore this bit is reset when the fault record associated with this DTC is overwritten by a newer DTC based upon vehicle manufacturer specific fault memory overflow requirements.</p> <p>DTC confirmation and aging criteria are defined by the vehicle manufacturer or mandated by On Board Diagnostic regulations.</p> |                     |                                           |                    |          |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                     |                                           | Logical '0'        |          |
| Reset to logical '0' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                     |                                           |                    |          |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                     | <b>Test Equipment Display Text</b>        |                    |          |
| '0' = DTC has never been confirmed since the last call to ClearDiagnosticInformation or after the aging criteria have been satisfied for the DTC (or DTC has been erased due to fault memory overflow).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                     | A confirmed DTC is not stored in the ECU. |                    |          |
| '1' = DTC confirmed at least once since the last call to ClearDiagnosticInformation and aging criteria have not yet been satisfied.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                     | A confirmed DTC is stored in the ECU.     |                    |          |

Table D-7. DTC Status Bit 3 ConfirmedDTC Definitions

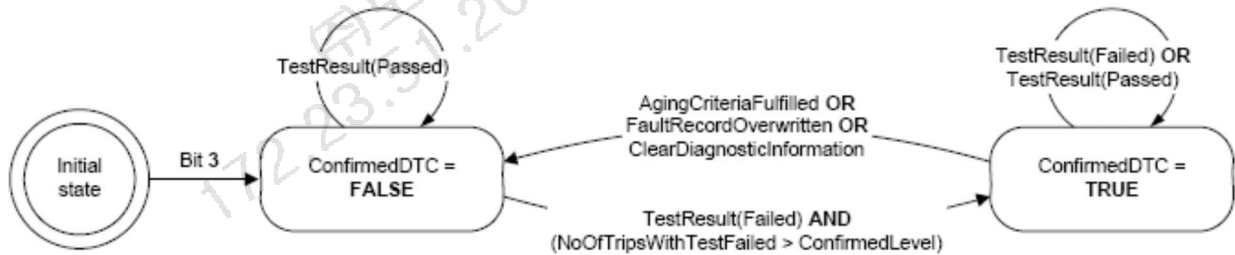


Figure D-4. DTC Status Bit 3 ConfirmedDTC Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                  | Description                          | Cvt : Emission                                   | Cvt : Non-Emission | Mnemonic |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------------------------|--------------------|----------|
| 4                                                                                                                                                                                                                                                                                                                                                                    | <b>TestNotCompleteSinceLastClear</b> | C2                                               | C2                 | TNCSLC   |
| This bit shall indicate whether a DTC test has ever run and completed since the last time a call was made to ClearDiagnosticInformation. One ('1') shall indicate that the DTC test has not run to completion. If the test runs and passes or if the test runs and fails (e.g. testFailedThisOperationCycle = '1') then the bit shall be set to a '0' (and latched). |                                      |                                                  |                    |          |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                               |                                      |                                                  | Logical '1'        |          |
| Reset to logical '1' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                     |                                      |                                                  |                    |          |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                          |                                      | <b>Test Equipment Display Text</b>               |                    |          |
| '0' = DTC test has returned either a passed or failed test result at least one time since the last time diagnostic information was cleared.                                                                                                                                                                                                                          |                                      | DTC test completed since the last code clear     |                    |          |
| '1' = DTC test has not run to completion since the last time diagnostic information was cleared.                                                                                                                                                                                                                                                                     |                                      | DTC test not completed since the last code clear |                    |          |
| C2: Bit 4 (testNotCompletedSinceLastClear) and bit 5 (testFailedSinceLastClear) shall always be supported together.                                                                                                                                                                                                                                                  |                                      |                                                  |                    |          |

Table D-8. DTC Status Bit 4 TestNotCompleteSinceLastClear Definitions

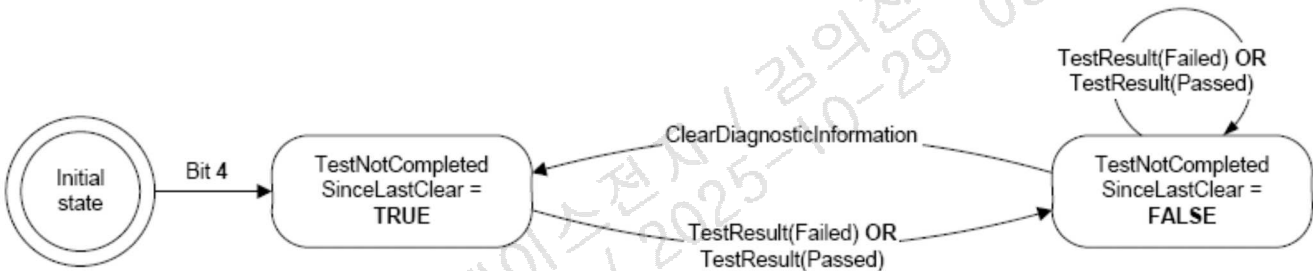


Figure D-5. DTC Status Bit 4 TestNotCompleteSinceLastClear Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Description                     | Cvt : Emission                                      | Cvt : Non-Emission | Mnemonic |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------|--------------------|----------|
| 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>TestFailedSinceLastClear</b> | C2                                                  | C2                 | TFSLC    |
| <p>This bit shall indicate whether a DTC test has ever completed with a failed result since the last time a call was made to ClearDiagnosticInformation (i.e., this is a latched testFailedThisOperationCycle = '1').</p> <p>Zero ('0') shall indicate that the test has not run or that the DTC test ran and passed (but never failed). If the test runs and fails then the bit shall remain latched at a '1'. Unlike confirmedDTC, this bit is not reset by aging-criteria or due to an overflow of the fault memory.</p> |                                 |                                                     |                    |          |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                 |                                                     | Logical '0'        |          |
| Reset to logical '0' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                 |                                                     |                    |          |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                 | <b>Test Equipment Display Text</b>                  |                    |          |
| '0' = DTC test has not indicated a failed result since the last time diagnostic information was cleared.                                                                                                                                                                                                                                                                                                                                                                                                                    |                                 | DTC test never failed since last code clear         |                    |          |
| '1' = DTC test returned a failed result at least once since the last time diagnostic information was cleared.                                                                                                                                                                                                                                                                                                                                                                                                               |                                 | DTC test failed at least once since last code clear |                    |          |
| C2: Bit 4 (testNotCompletedSinceLastClear) and bit 5 (testFailedSinceLastClear) shall always be supported together.                                                                                                                                                                                                                                                                                                                                                                                                         |                                 |                                                     |                    |          |

Table D-9. DTC Status Bit 5 TestFailedSinceLastClear Definitions

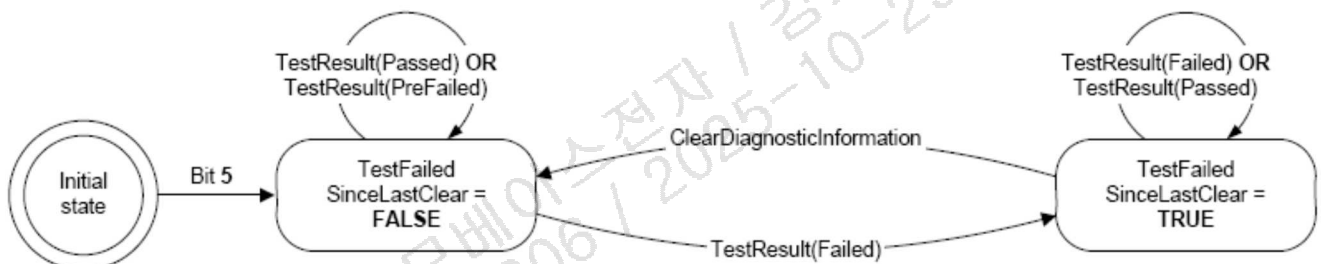


Figure D-6. DTC Status Bit 5 TestFailedSinceLastClear Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Description                              | Cvt : Emission                              | Cvt : Non-Emission | Mnemonic |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|---------------------------------------------|--------------------|----------|
| 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>TestNotCompleteThisOperationCycle</b> | M                                           | U                  | TNCTOC   |
| <p>This bit shall indicate whether a DTC test has ever run and completed during the current operation cycle (or completed during the current operation cycle after the last time a call was made to ClearDiagnosticInformation).</p> <p>A logical '1' shall indicate that the DTC test has not run to completion during the current operation cycle. If the test runs and passes or fails then the bit shall be set (and latched) to '0' until a new operation cycle is started.</p> |                                          |                                             |                    |          |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                                                                                                                                               |                                          |                                             | Logical '1'        |          |
| Reset to logical '1' after a call to ClearDiagnosticInformation.                                                                                                                                                                                                                                                                                                                                                                                                                     |                                          |                                             |                    |          |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                          | <b>Test Equipment Display Text</b>          |                    |          |
| '0' = DTC test has returned either a passed or testFailedThisOperationCycle = '1' result during the current drive cycle (or since the last time diagnostic information was cleared during the current operation cycle).                                                                                                                                                                                                                                                              |                                          | DTC test completed this operation cycle     |                    |          |
| '1' = DTC test has not run to completion this operation cycle (or since the last time diagnostic information was cleared this operation cycle).                                                                                                                                                                                                                                                                                                                                      |                                          | DTC test not completed this operation cycle |                    |          |

Table D-10. DTC Status Bit 6 TestNotCompleteThisOperationCycle Definitions

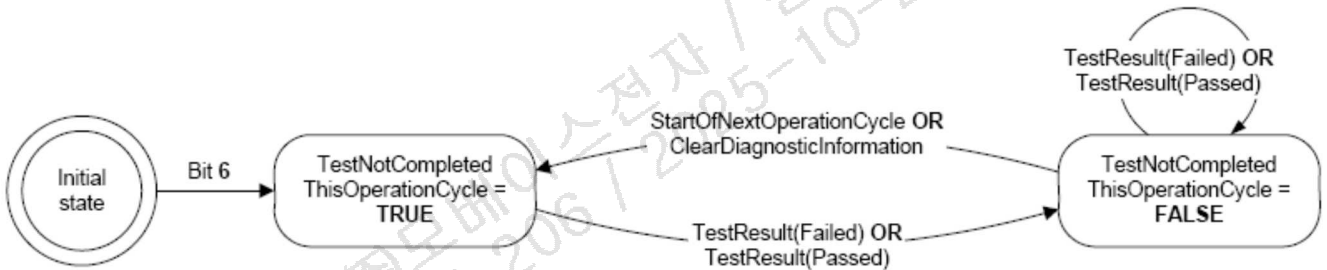
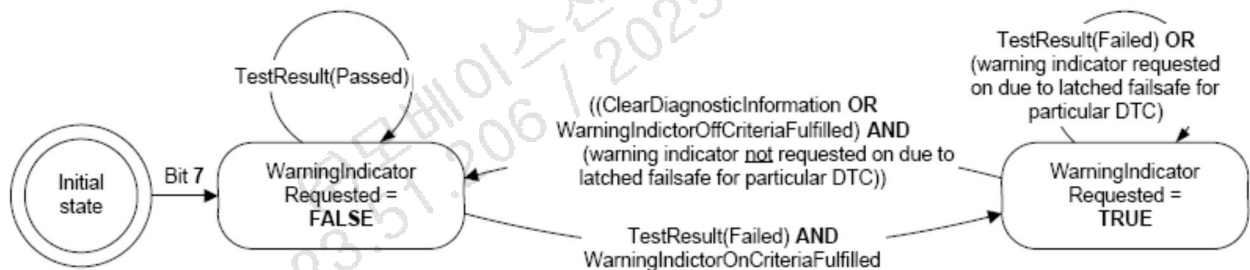


Figure D-7. DTC Status Bit 6 TestNotCompleteThisOperationCycle Logic

| Bit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Description                      | Cvt : Emission                                     | Cvt : Non-Emission | Mnemonic    |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|----------------------------------------------------|--------------------|-------------|
| 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <b>WarningIndicatorRequested</b> | M                                                  | M                  | WIR         |
| <p>This bit shall report the status of any warning indicators associated with a particular DTC. Warning outputs may consist of indicator lamp(s), displayed text information, etc. If no warning indicators exist for a given system or particular DTC, this status shall default to a logic '0' state.</p> <p>Conditions for activating the warning indicator shall be defined by the vehicle manufacturer / implementation, but if the warning indicator is on for a given DTC, then confirmedDTC shall also be set to '1' (with the exception of a confirmed fault with a latched failsoft strategy as described below).</p> |                                  |                                                    |                    |             |
| <b>Bit state after a successful ClearDiagnosticInformation service</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                  |                                                    |                    | Logical '0' |
| <p>Reset to a logical '0' after a call to ClearDiagnosticInformation. Some ECUs may latch the failsoft strategy associated with a particular confirmed fault for the current operation cycle. If the warning indicator is still requested due to this latched failsoft following a call to ClearDiagnosticInformation, this bit shall not be cleared to a logical '0'. Rather, this bit shall remain set to logical '1' until the failsoft strategy is no longer active (e.g., test completes and passes). Additional reset conditions shall be defined by the vehicle manufacturer / implementation.</p>                       |                                  |                                                    |                    |             |
| <b>Bit state definition</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                  | <b>Test Equipment Display Text</b>                 |                    |             |
| '0' = Server is not requesting warningIndicator to be active                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                  | DTC is currently not requesting warning indication |                    |             |
| '1' = Server is requesting warningIndicator to be active                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                  | DTC is currently requesting warning indication     |                    |             |

Table D-11. DTC Status Bit 7 WarningIndicatorRequested Definitions



WarningIndicatorOnCriteriaFulfilled = warning indicator exists for given system and particular DTC AND (confirmedDTC = 1 OR vehicle manufacturer or implementation-specific warning indicator enable criteria are satisfied)

WarningIndicatorOffCriteriaFulfilled = TestResult(Passed) OR vehicle manufacturer or implementation-specific warning indicator disable criteria are satisfied

Figure D-8. DTC Status Bit 7 WarningIndicatorRequested Logic

### D.3 DTCSeverityMask and DTCSeverity Bit Definitions

This section defines the mapping of the DTCSeverityMask / DTCSeverity parameters used with the ReadDTCInformation service. Every server shall adhere to the convention for storing bit-packed DTC severity information as defined in Table D-11. DTCSeverity byte definition and Table D-12. DTC severity bit definitions (bit 7-5).

The severity information is reported in a 1-byte value. Only the upper 3 bits (bit 7-5) of the 1-byte value are used to represent the DTC severity information. The remaining bits (bit 4-0) have to be set to zero (0). Based on this the following bit coding applies to the 3 bit DTC severity information contained in the 1-byte DTCSeverity parameter.

| DTC Severity Byte          |       |       |                              |       |       |       |       |
|----------------------------|-------|-------|------------------------------|-------|-------|-------|-------|
| Bit 7                      | Bit 6 | Bit 5 | Bit 4                        | Bit 3 | Bit 2 | Bit 1 | Bit 0 |
| 3 Bit Severity Information |       |       | Reserved (To be set to zero) |       |       |       |       |

Table D-12. DTCSeverity byte definition

| Bit 7-5  | Description                                                                                                     | Cvt | Mnemonic |
|----------|-----------------------------------------------------------------------------------------------------------------|-----|----------|
| 000b     | <b>NoSeverityAvailable</b><br>There is no severity information available.                                       | M   | NSA      |
| 001b     | <b>MaintenanceOnly</b><br>This value indicates that the failure requests maintenance only.                      | M   | MO       |
| 010b     | <b>CheckAtNextHalt</b><br>This value indicates to the failure requires a check of the vehicle at the next halt. | M   | CHKANH   |
| 011b     | <b>Reserved</b>                                                                                                 | M   |          |
| 100b     | <b>CheckImmediately</b><br>This value indicates to the failure requires an immediate check of the vehicle.      | M   | CHKI     |
| 101~111b | <b>Reserved</b>                                                                                                 | M   |          |

Table D-13. DTC severity bit definitions (bit 7-5)



#### D.4 DTC Format Structure

| DTC High Byte                                                                |                                                                                                               |                                                      | DTC Middle Byte                                      |                                                      | DTC Low Byte                                            |                                                         |
|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------------|
| Bit 7-6                                                                      | Bit 5-4                                                                                                       | Bit 3-0                                              | Bit 7-4                                              | Bit 3-0                                              | Bit 7-4                                                 | Bit 3-0                                                 |
| 00 = powertrain (P)<br>01 = chassis (C)<br>10 = body (B)<br>11 = network (U) | 00 = ISO/SAE controlled<br>01 = manufacturer controlled<br>10 = ISO/SAE controlled<br>11 = ISO/SAE controlled | 1 <sup>st</sup><br>Character<br>Of Code<br>(Hex 0-F) | 2 <sup>nd</sup><br>Character<br>of Code<br>(Hex 0-F) | 3 <sup>rd</sup><br>Character<br>of Code<br>(Hex 0-F) | DTC Failure<br>Category<br><br>(refer to<br>ISO15031-6) | DTC Failure<br>Sub Type<br><br>(refer to<br>ISO15031-6) |

Table D-14. Structure of Diagnostic Trouble Code

##### D.4.1 DTC Failure Category and Sub Type definition

Please refer to the latest SAE J2012-DA

#### D 5 DTCStatusAvailabilityMask

A byte whose bits are defined the same as statusofDTC and represents the status bits that are supported by the server.

Bits that are not support by the server shall be set to '0'. Each supported bit (indicated by a value of '1') shall be implemented for every DTC supported by the server



## Annex. E Input Output Control Functional Unit Data Parameter Definitions

### E.1 InputOutputControlParameter Definitions

| HEX   | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Cvt | Mnemonic |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----------|
| 00    | <b>returnControlToECU</b><br><br>This value shall indicate to the server that the client does no longer have control about the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br><br>- Number of controlState bytes in request: 0<br>- Number of controlState bytes in pos. response: depends on the dataIdentifier                                                                                                                                                                                                   | U   | RCTECU   |
| 01    | <b>resetToDefault</b><br><br>This value shall indicate to the server that it is requested to reset the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier to its default state.<br><br>- Number of controlState bytes in request: 0<br>- Number of controlState bytes in pos. response: depends on the dataIdentifier                                                                                                                                                                                                      | U   | RTD      |
| 02    | <b>freezeCurrentState</b><br><br>This value shall indicate to the server that it is requested to freeze the current state of the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier.<br><br>- Number of controlState bytes in request: 0<br>- Number of controlState bytes in pos. response: depends on the dataIdentifier                                                                                                                                                                                                 | U   | FCS      |
| 03    | <b>shortTermAdjustment</b><br><br>This value shall indicate to the server that it is requested to adjust the input signal, internal parameter or output signal referenced by the inputOutputLocalIdentifier in RAM to the value(s) included in the controlOption parameter(s). (e.g. set Idle Air Control Valve to a specific step number, set pulse width of valve to a specific value/duty cycle).<br><br>- Number of controlState bytes in request: depends on the dataIdentifier<br>- Number of controlState bytes in pos. response: depends on the dataIdentifier | U   | STA      |
| 04~FF | <b>Reserved</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | M   |          |

Table E-1. InputOutputControlParameter Definition



## Annex. F Remote Activation of Routine Functional Unit Data Parameter Definitions

Refer to Table 8-3. Request message routineIdentifier definition

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**Annex. G Address And Length Format Identifier Parameter Values**

**G.1 AddressAndLengthFormatIdentifier Definitions**

This parameter is a one byte value with each nibble encoded separately. See below table.

| AddressAndLengthFormatIdentifier    |   |   |   |                                        |   |   |   |
|-------------------------------------|---|---|---|----------------------------------------|---|---|---|
| High Nibble                         |   |   |   | Low Nibble                             |   |   |   |
| 7                                   | 6 | 5 | 4 | 3                                      | 2 | 1 | 0 |
| Bytes used for MemorySize Parameter |   |   |   | Bytes used for MemroyAddress Parameter |   |   |   |

Table G-1. AddressAndLengthFormatIdentifier Definition

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## Annex. H Diagnostic CAN Identifiers

### H.1 Definition of CAN Identifier for Enhanced Diagnostic Communication

※ The latest Diagnostic CAN Identifiers refer to the iDIMS and NIDS

| CAN ID (HEX)                 | Description                                                                                                                                          |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Functional Addressing</b> |                                                                                                                                                      |
| <b>07DF</b>                  | Functional request CAN ID sent by the external test equipment<br>(All controllers must support this CAN Identifier)                                  |
| <b>Physical Addressing</b>   |                                                                                                                                                      |
| <b>0580~05FF</b>             | HMC specific. See also iDIMS and NIDS ※Only for Third Generation of CAN (CAN-FD)                                                                     |
| <b>0700~07DE</b>             | HMC specific. See also iDIMS and NIDS                                                                                                                |
| <b>07E0</b>                  | Physical request CAN ID to EMS (Engine Management System), FCU (Fuelcell Control Unit)                                                               |
| <b>07E8</b>                  | Physical response CAN ID from EMS (Engine Management System), FCU (Fuelcell Control Unit)                                                            |
| <b>07E1</b>                  | Physical request CAN ID to TCU (Transmission Control Unit)                                                                                           |
| <b>07E9</b>                  | Physical response CAN ID from TCU (Transmission Control Unit)                                                                                        |
| <b>07E2</b>                  | Physical request CAN ID to HCU (Hybrid Control Unit) / VCU (Vehicle Control Unit) / HCULDC (HCU & Low DC-DC Converter)                               |
| <b>07EA</b>                  | Physical response CAN ID from HCU (Hybrid Control Unit) / VCU (Vehicle Control Unit) / HCULDC (HCU & Low DC-DC Converter)                            |
| <b>07E3</b>                  | Physical request CAN ID to MCU (Motor Control Unit) / MHSG (Mild Hybrid Starter Generator) / F_MCU (Front MCU) / R_MCU (Rear MCU)                    |
| <b>07EB</b>                  | Physical response CAN ID from MCU (Motor Control Unit) / MHSG (Mild Hybrid Starter Generator) / F_MCU (Front MCU) / R_MCU (Rear MCU)                 |
| <b>07E4</b>                  | Physical request CAN ID to BMS (Battery Management System)                                                                                           |
| <b>07EC</b>                  | Physical response CAN ID from BMS (Battery Management System)                                                                                        |
| <b>07E5</b>                  | Physical request CAN ID to OBC (On Board Charger) : From LF/JF PHEV, AE/DE EV/PHEV<br>ICCU (Integrated Charge Control Unit) : From Osc EV LP1        |
| <b>07ED</b>                  | Physical response CAN ID from OBC (On Board Charger) : From LF/JF PHEV, AE/DE EV/PHEV<br>ICCU (Integrated Charge Control Unit) : From Osc EV LP1     |
| <b>07E6</b>                  | Physical request CAN ID to SCU_FF (SBW Control Unit) : From LX2<br>EVSCU_FF (EV SBW Control Unit) : EV SCU                                           |
| <b>07EE</b>                  | Physical request CAN ID from SCU_FF (SBW Control Unit) : From LX2<br>EVSCU_FF (EV SBW Control Unit) : EV SCU                                         |
| <b>07E7</b>                  | Physical request CAN ID to DCU (Doesing Control Unit) : Diesel Engine Only<br>ESC(IEB) (Electronic Stability Control Integrated Electronic Brake)    |
| <b>07EF</b>                  | Physical response CAN ID from DCU (Doesing Control Unit) : Diesel Engine Only<br>ESC(IEB) (Electronic Stability Control Integrated Electronic Brake) |
| <b>07F0~07F3</b>             | HMC specific. See also iDIMS and NIDS                                                                                                                |
| <b>07F4~07F6</b>             | ISO15765-2 Reserved                                                                                                                                  |
| <b>07F7~07F9</b>             | HMC specific. See also iDIMS and NIDS                                                                                                                |
| <b>07FA~07FC</b>             | ISO15765-2 Reserved                                                                                                                                  |
| <b>07FD~07FF</b>             | HMC specific. See also iDIMS and NIDS                                                                                                                |
| <b>17FF0001~17FFFFFFE</b>    | HMC specific. See also iDIMS and NIDS                                                                                                                |

Table H-1. Definition of CAN Identifier for Enhanced Diagnostic Communication

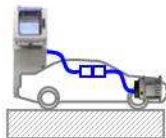
## Annex. I EOL (End Of Line) Diagnostic Mode

### I.1 Dual Expression of DTC in EOL Process (HADS, U-Robot equipment)

차량제작



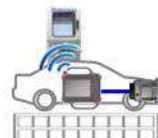
검사장비(HADS)



주행검사



U-로봇



- [HADS] Excluding pending DTC expression (DTC Case #1)
- [U-Robot] Check all DTC
- Standardization of DTC expression standard of PT Gasoline / Diesel system (Including of Bit 2 Pending Concept)
- Applying pending DTC concepts of UDS on CAN, ECUs are compliant with the following table.

| DTC Case    | Status Of Bit |         |            | DTC Status Mask<br>(※ responds to corresponding DTC in case the result is more than "1") |         | Display of External diagnostic equipment |
|-------------|---------------|---------|------------|------------------------------------------------------------------------------------------|---------|------------------------------------------|
|             | Bit3          | Bit2    | Bit0       | HADS                                                                                     | U-Robot |                                          |
|             | Confirmed     | Pending | TestFailed | 09                                                                                       | 0D      |                                          |
| DTC Case #1 | 0             | 1       | 0          | 0 (No response)                                                                          | 1       | Pending DTC                              |
| DTC Case #2 | 0             | 1       | 1          | 1                                                                                        | 2       | Pending DTC                              |
| DTC Case #3 | 1             | 0       | 0          | 1                                                                                        | 1       | Historic DTC                             |
| DTC Case #4 | 1             | 1       | 0          | 1                                                                                        | 2       | Historic DTC                             |
| DTC Case #5 | 1             | 1       | 1          | 2                                                                                        | 3       | Active DTC                               |

Table I-1. Meaning of DTC according to Status Of Bit and DTC Status Mask

## I.2 EOL Batch Clear DTC (Error Eraser equipment)

When conducting Batch Clear DTC of all ECUs that comply with ES95486-00, Communication processes between ECU and external test equipment shall be compliant with following details.

### I.2.1 CAN ID

When conducting Clear DTC of each ECU, it communicates by using each CAN ID specified in specification. When conducting Batch Clear DTC, it communicates by using 0x7DF (Functional Request CAN ID)

### I.2.2 EOL Standard Process (Error Eraser ↔ ECU)

| No. | Process                                       | Communication                  | Note                                                                                   |
|-----|-----------------------------------------------|--------------------------------|----------------------------------------------------------------------------------------|
| 1   | Vehicle Energize                              | -                              | Implements IG ON                                                                       |
| 2   | Start of Diagnostic Communication of each ECU | Tx : 02 10 03<br>Rx : 02 50 03 |                                                                                        |
| 3   | ...                                           |                                | Implements additional functions of each ECU (Coding, Calibration, Initialization etc.) |

|   |                                              |                                   |                                                                                   |
|---|----------------------------------------------|-----------------------------------|-----------------------------------------------------------------------------------|
| 4 | End of Diagnostic Communication of each ECU  | Tx : 10 01<br>Rx : 50 01          |                                                                                   |
| 5 | Start of communication of UDS on CAN ECU     | Tx : 02 10 03<br>Rx : 02 50 03    | Tx CAN ID : 0x7DF<br>Rx CAN ID : different depending on each ECU                  |
| 6 | Conducting Batch Clear DTC of UDS on CAN ECU | Tx : 04 14 FF FF FF<br>Rx : 01 54 | ↑                                                                                 |
| 7 | Completion of Error Erasing                  |                                   | Shows OK/NG results on the screen of equipment (Different depending on factories) |

Table I-2 EOL Standard Process of Batch Clear DTC

- Conducting of Individual Clear DTC of each ECU and additional functions is the same as existing methods. (1~4)
- Error Eraser equipment starts communication as 0x7DF CAN ID and sends ClearDTC request.

## Annex. J RXSWIN requirement

### J.1 RXSWIN Read

The EDT requests RXSWIN (RDBI 22 F18F) and the Central gateway requests a local RXSWIN of each ECU (RDBI 22 F1EF), stores the result, filters (remove redundancy) and transmits the result.

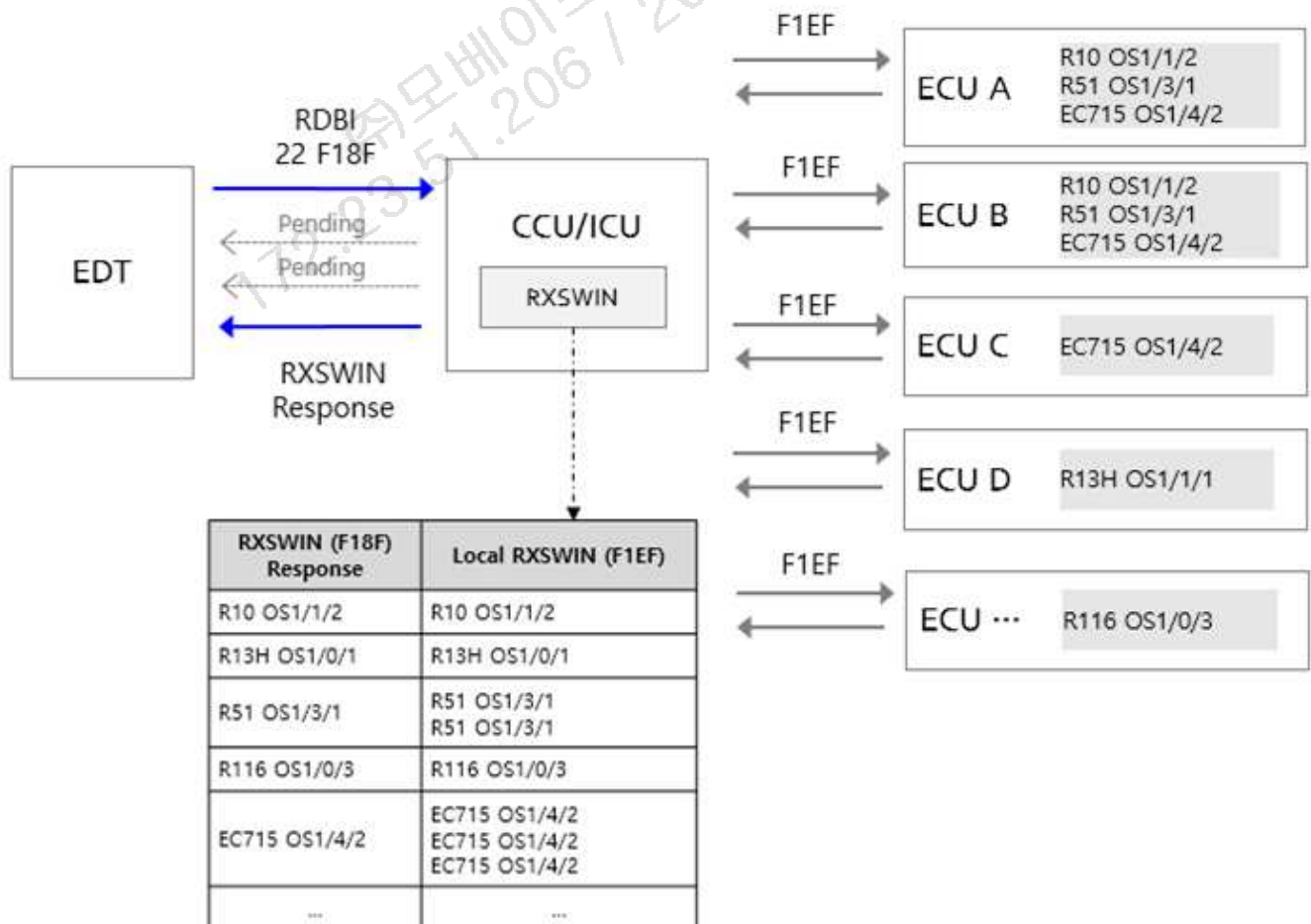


Figure J-7. RXSWIN Overview

| Hex  | Description                                                                                    | Cvt | Mnemonic |
|------|------------------------------------------------------------------------------------------------|-----|----------|
| F18F | <b>RXSWIN</b><br>RXSWIN is information that combines local RXSWIN information into one RXSWIN. | C1  |          |
| F1EF | <b>Local RXSWIN</b><br>RXSWIN information which is in a ECU.                                   | C2  |          |

C1: Only the central gateway shall support this DID.

C2: All ECUs related to RXSWIN shall support this Identifier.

Table J-1. RXSWIN Identification

## J.2 RXSWIN Request and Response sequence

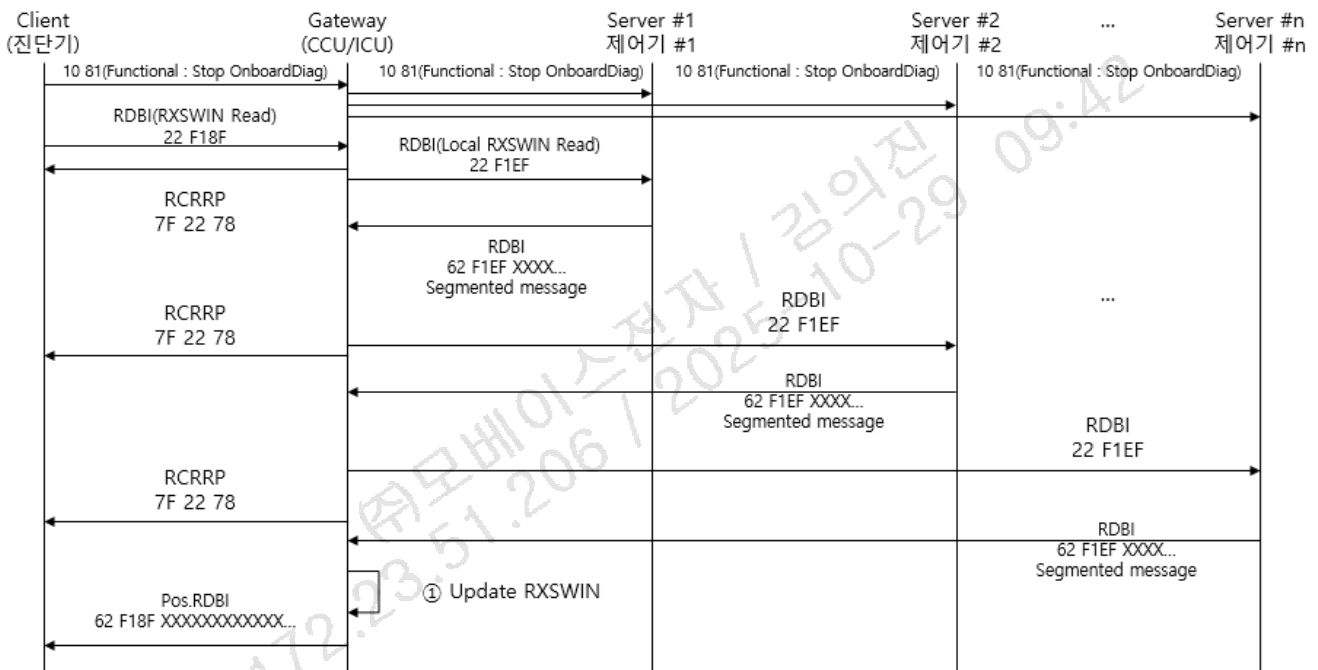


Figure J-8. RXSWIN gathering and read sequence caused by RDBI(F18F)

Exception : 10 81(Functional:Stop OnboardDiag) Although Functional addressing 10 81 causes Stop onboard diagnostics, Gateway can gather Local RXSWIN

\*OnBoardDiag : Diagnostics from ECU to ECU (Client : ECU, Server : ECU)

When the diagnosis device requests RXSWIN in the IGN off state, Gateway does not request LocalRXSWIN but responds by referring to the RXSWIN table updated previously.

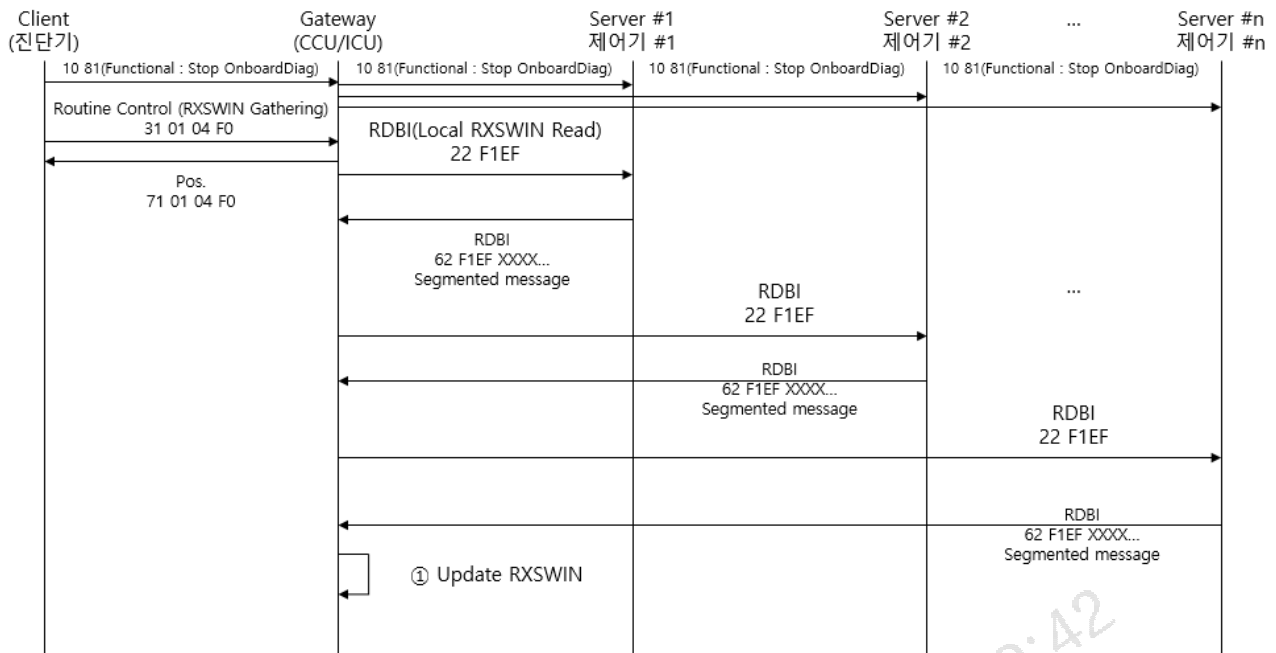


Figure J-9. Start RXSWIN gathering – routine control

The gateway can perform the gathering and reading separately. The gathering can be started by routine control(\$31). After the gateway finish gathering RXSWIN, RXSWIN shall be updated in NVM.

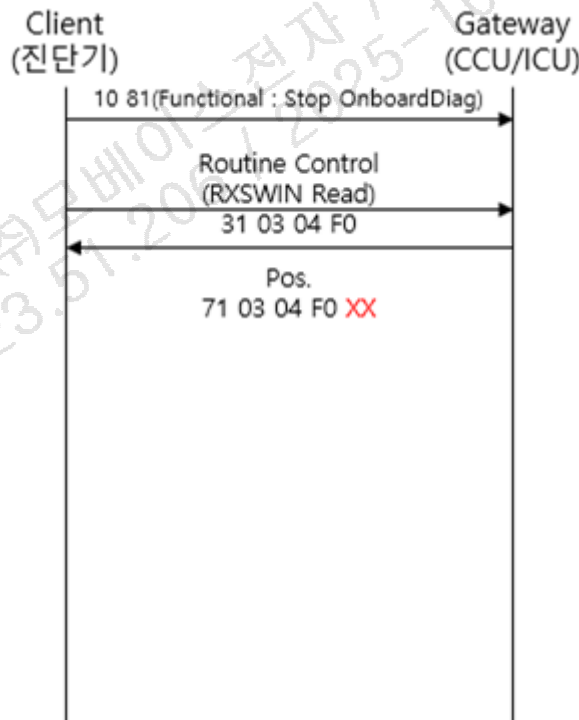


Figure J-10. Request routine result - Routine Control

| Value | Routine Result | Description                                                          |
|-------|----------------|----------------------------------------------------------------------|
| 0     | Done           | Routine has been finished correctly – RXSWIN gathering has been done |

|   |            |                                                                   |
|---|------------|-------------------------------------------------------------------|
| 1 | Collecting | Routine is running – Central Gateway is gathering RXSWIN          |
| 2 | Aborted    | Because of IGN off or other problems, Gathering has been aborted. |

Table J-6. RXSWIN Routine result

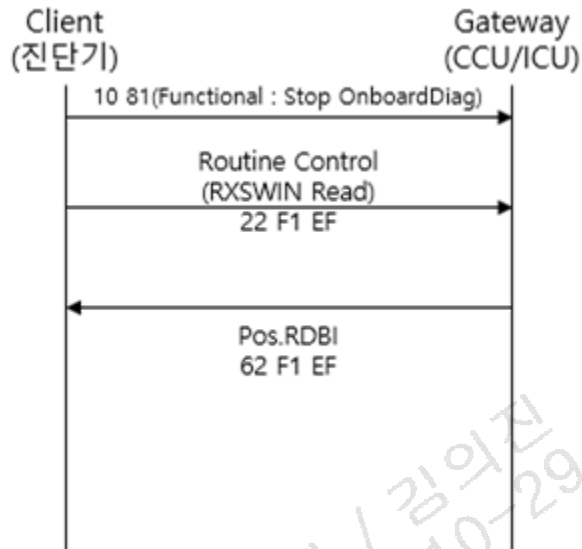


Figure J-11. Read local RXSWIN from the central gateway

If diagnostic tester requests RDBI with F1EF(local RXSWIN) to the central gateway, the central gateway doesn't gather RXSWIN and respond positive with already gathered RXSWIN which is stored in NVM.

If RXSWIN never has been gathered, the central gateway respond positive with only gateway own RXSWIN.

If the central gateway received RDBI with F1EF or F18F during the gathering RXSWIN by routine control, the central gateway shall respond positive with already gathered RXSWIN which is stored in NVM. And it doesn't affect gathering RXSWIN.

### J.3 Local RXSWIN collection example for master - slave ECU





# 기술 표준

## ENGINEERING STANDARD

규격번호 (SPEC No)  
ES95486-02

페이지  
(SHT/SHTS) 238/244

Length value shall be 0x0C (Decimal 12)

### Length Example2)

R83 DN8/1/1

The length of RXSWIN string is 11bytes

Length field shall be 0x0B (Decimal 11)

| RXSWIN |                  | Length | Reg. Authority | Reg. No.  | Space | Vehicle Project | /Version | /Extension | RXSWIN Result (Length)String |
|--------|------------------|--------|----------------|-----------|-------|-----------------|----------|------------|------------------------------|
| Ex1)   | RXSWIN Initial   | 0x13   | EC             | 1008/2010 | “ ”   | CN7             | /1       | /0         | (0x13)EC1008/2010 CN7/1/0    |
|        | RXSWIN Extension | 0x13   | EC             | 1008/2010 | “ ”   | CN7             | /1       | /1         | (0x13)EC1008/2010 CN7/1/1    |
| Ex2)   | Type Gasoline    | 0x0B   | R              | 83        | “ ”   | DN8             | /1       | /0         | (0x0B)R83 DN8/1/0            |
|        | Type Diesel      | 0x0B   | R              | 83        | “ ”   | DN8             | /2       | /0         | (0x0B)R83 DN8/2/0            |

Table J-3. RXSWIN structure

### J.5 RXSWIN Update Example

As is (RXSWWIN table – Central Gateway)

| List No. | RXSWIN STRING |   |   |   |  |   |   |   |   |   |   |   |  |  |  |
|----------|---------------|---|---|---|--|---|---|---|---|---|---|---|--|--|--|
| 0        | R             | 1 | 5 | 6 |  | D | N | 8 | / | 1 | / | 1 |  |  |  |
| 1        | R             | 8 | 3 |   |  | D | N | 8 | / | 1 | / | 0 |  |  |  |
| 2        |               |   |   |   |  |   |   |   |   |   |   |   |  |  |  |
| ...      |               |   |   |   |  |   |   |   |   |   |   |   |  |  |  |

Example) Local RXSWIN Response 1

|               |             |     |     |            |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |  |  |  |
|---------------|-------------|-----|-----|------------|--------|---------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|---------------------|----|----|----|----|----|----|----|----|----|--|--|--|
| Gate way Req. | HEX         | 22  | F1  | EF         |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |  |  |  |
|               | Description | SID | DID |            |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |  |  |  |
| ECU Resp.     | HEX         | 62  | F1  | EF         | 07     | 00                  | 0C | 52 | 31 | 35 | 36 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 31 | 0B | 52     | 34                  | 38 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 30 |  |  |  |
|               | Description | SID | DID | REQ CAN ID | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |    |    |    |    |    | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |  |  |  |
|               |             |     |     |            |        | R                   | 1  | 5  | 6  |    | D  | N  | 8  | /  | 1  | /  | 1  |    | R  | 8  |        | 3                   |    | D  | N  | 8  | /  | 1  | /  | 0  |    |  |  |  |

Example) Local RXSWIN Response 2

|               |             |     |     |            |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |   |  |
|---------------|-------------|-----|-----|------------|--------|---------------------|----|----|----|----|----|----|----|----|----|----|----|----|----|----|--------|---------------------|----|----|----|----|----|----|----|----|----|---|--|
| Gate way Req. | HEX         | 22  | F1  | EF         |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |   |  |
|               | Description | SID | DID |            |        |                     |    |    |    |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |   |  |
| ECU Resp.     | HEX         | 62  | F1  | EF         | 07     | 00                  | 0C | 52 | 31 | 35 | 36 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 32 | 0B | 52     | 34                  | 38 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 30 |   |  |
|               | Description | SID | DID | REQ CAN ID | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |    |    |    |    |    | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |   |  |
|               |             |     |     |            |        |                     | R  | 1  | 5  | 6  |    | D  | N  | 8  | /  | 1  | /  | 2  |    |    | R      | 8                   | 3  |    |    | D  | N  | 8  | /  | 1  | /  | 0 |  |

To be (RXSWWIN table – Central Gateway)

| List No. | RXSWIN STRING |   |   |   |  |   |   |   |   |   |   |   |  |  |  |
|----------|---------------|---|---|---|--|---|---|---|---|---|---|---|--|--|--|
| 0        | R             | 1 | 5 | 6 |  | D | N | 8 | / | 1 | / | 2 |  |  |  |

|     |   |   |   |  |   |   |   |   |   |   |   |  |  |  |  |
|-----|---|---|---|--|---|---|---|---|---|---|---|--|--|--|--|
| 1   | R | 8 | 3 |  | D | N | 8 | / | 1 | / | 0 |  |  |  |  |
| 2   |   |   |   |  |   |   |   |   |   |   |   |  |  |  |  |
| ... |   |   |   |  |   |   |   |   |   |   |   |  |  |  |  |

Example) RXSWIN Response (EDT-Central Gateway)

|             |             |     |     |        |                     |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |    |    |  |  |  |
|-------------|-------------|-----|-----|--------|---------------------|----|----|----|----|----|----|----|----|----|----|----|--------|---------------------|----|----|----|----|----|----|----|----|----|----|----|--|--|--|
| EDT Request | HEX         | 22  | F1  | 8F     |                     |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |    |    |  |  |  |
|             | Description | SID | DID |        |                     |    |    |    |    |    |    |    |    |    |    |    |        |                     |    |    |    |    |    |    |    |    |    |    |    |  |  |  |
| ECU Resp.   | HEX         | 62  | F1  | 8F     | 0C                  | 52 | 31 | 35 | 36 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 32     | 0B                  | 52 | 34 | 38 | 20 | 44 | 4E | 38 | 2F | 31 | 2F | 30 |  |  |  |
|             | Description | SID | DID | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |    |    | Length | RXSWIN ASCII STRING |    |    |    |    |    |    |    |    |    |    |    |  |  |  |
|             |             |     |     |        | R                   | 1  | 5  | 6  |    | D  | N  | 8  | /  | 1  | /  | 2  |        | R                   | 8  | 3  |    | D  | N  | 8  | /  | 1  | /  | 0  |    |  |  |  |

The gateway shall store the RXSWIN table in memory. The Local RXSWIN received from the ECU shall be updated to the latest value after removing redundancy.

### Case 1. Update RXSWIN

| No | RXSWIN       |
|----|--------------|
| 1  | R156 DN8/1/0 |
| 2  | R155 DN8/1/0 |
| 3  | R48 DN8/1/1  |
| 4  | R83 DN8/1/1  |
| 5  | ...          |

RAM

Server #n Response

R48 DN8/1/1  
R83 DN8/1/2  
R156 DN8/1/2

| No | RXSWIN       |
|----|--------------|
| 1  | R156 DN8/1/2 |
| 2  | R155 DN8/1/0 |
| 3  | R48 DN8/1/1  |
| 4  | R83 DN8/1/2  |
| 5  | ...          |

RAM  
(Gateway Response : F18F)

### Case 2. Add RXSWIN

| No | RXSWIN       |
|----|--------------|
| 1  | R156 DN8/1/2 |
| 2  | R155 DN8/1/0 |
| 3  | R48 DN8/1/1  |
| 4  | R83 DN8/1/1  |
| 5  |              |
| 6  |              |
| 7  |              |

RAM

Server #n-1 Response

XXX DN8/1/1  
R83 DN8/1/1  
R156 DN8/2/1

Server #n Response

| No | RXSWIN       |
|----|--------------|
| 1  | R156 DN8/1/2 |
| 2  | R155 DN8/1/0 |
| 3  | R48 DN8/1/1  |
| 4  | R83 DN8/1/1  |
| 5  | XXX DN8/1/1  |
| 6  | R156 DN8/2/1 |
| 7  |              |

RAM  
(Gateway Response : F18F)

Figure J-13. RXSWIN memory handling in the Central gateway-1

NOTE : Each case is not sequential.

This is a example to show how to gathering RXSWIN on the central gateway side.

The meaning of Gateway Response in the picture is that Response of RXSWIN for DID F18F is made from the RAM which is result of gathering local RXSWIN.

If one or more fields (Reg. Authority / Reg.No / Project / Version), which are the local RXSWIN of server #n, are not the same as the gathered RXSWIN fields, RXSWIN shall be added. If each field between the gathered RXSWIN and the local RXSWIN

response is the same except the extension field and the local RXSWIN extension value is greater than the gathered RXSWIN, the extension shall be updated.

Under the following conditions, the central gateway responds to the RXSWIN stored on the NVM. Under the following conditions, The central gateway respond RXSWIN stored in the NVM :

- 1) IGN off
- 2) DID is F1EF

### Case 3. Finalize RXSWIN gathering

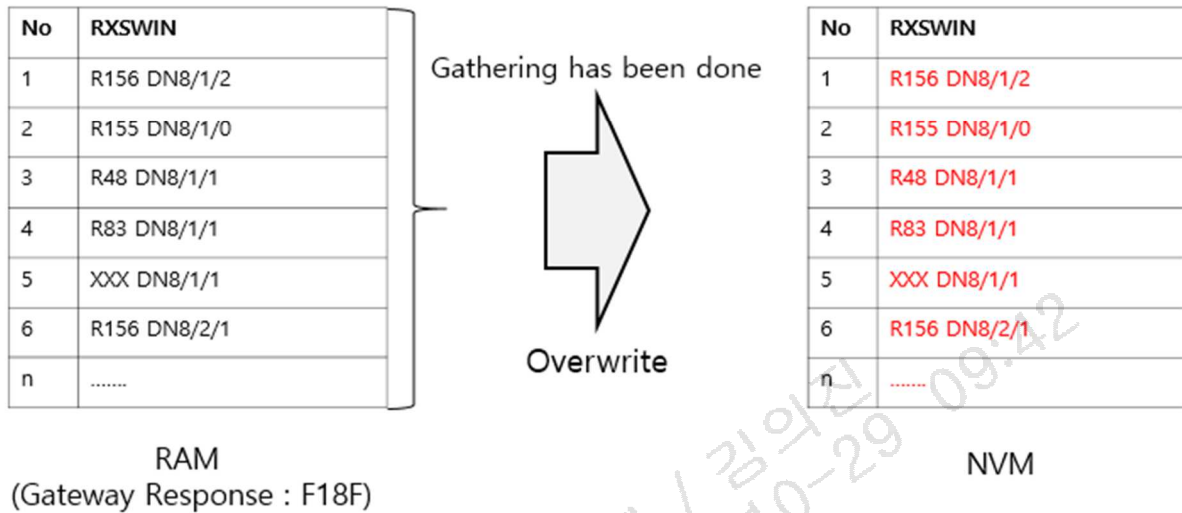


Figure J-14. RXSWIN memory handling in the Central gateway-2

After finish gathering RXSWIN to all servers, The gateway stores RXSWIN(RAM) in the NVM(overwrite).

### J.6 RXSWIN Session Timer

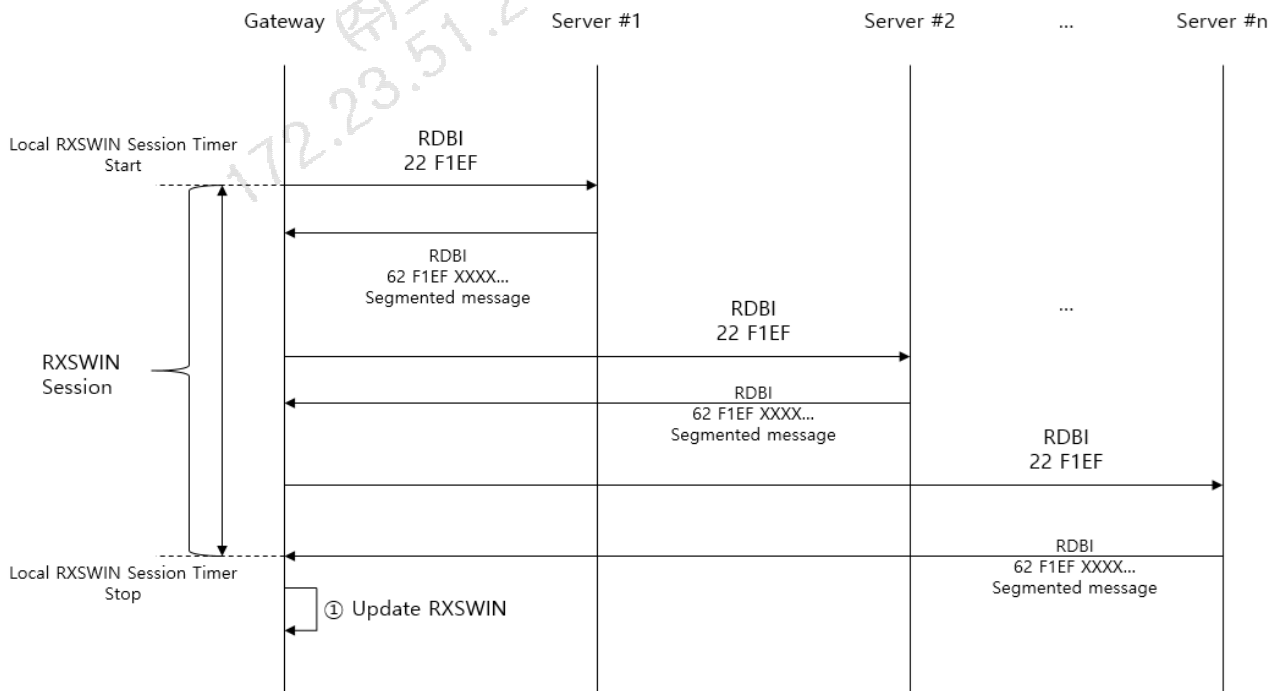


Figure J-15. RXSWIN Session timer sequence diagram

| Timing parameter     | Action | Condition                                                                                                                                                                            |
|----------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Local RXSWIN Session | Start  | Gateway starts request of LocalRXSWIN                                                                                                                                                |
| Local RXSWIN Session | Stop   | The end of LocalRXSWIN response from All ECU which has to respond to the gateway.                                                                                                    |
|                      |        | 60seconds(+3sec) elapsed from Timer start.<br>(If the 60-second timer has been terminated but Gateway has not completed the RXSWIN gathering, Gateway shall stop gathering RXSWIN. ) |

Table J-4 RXSWIN Session

### J.7 Timing Parameter and Retry logic

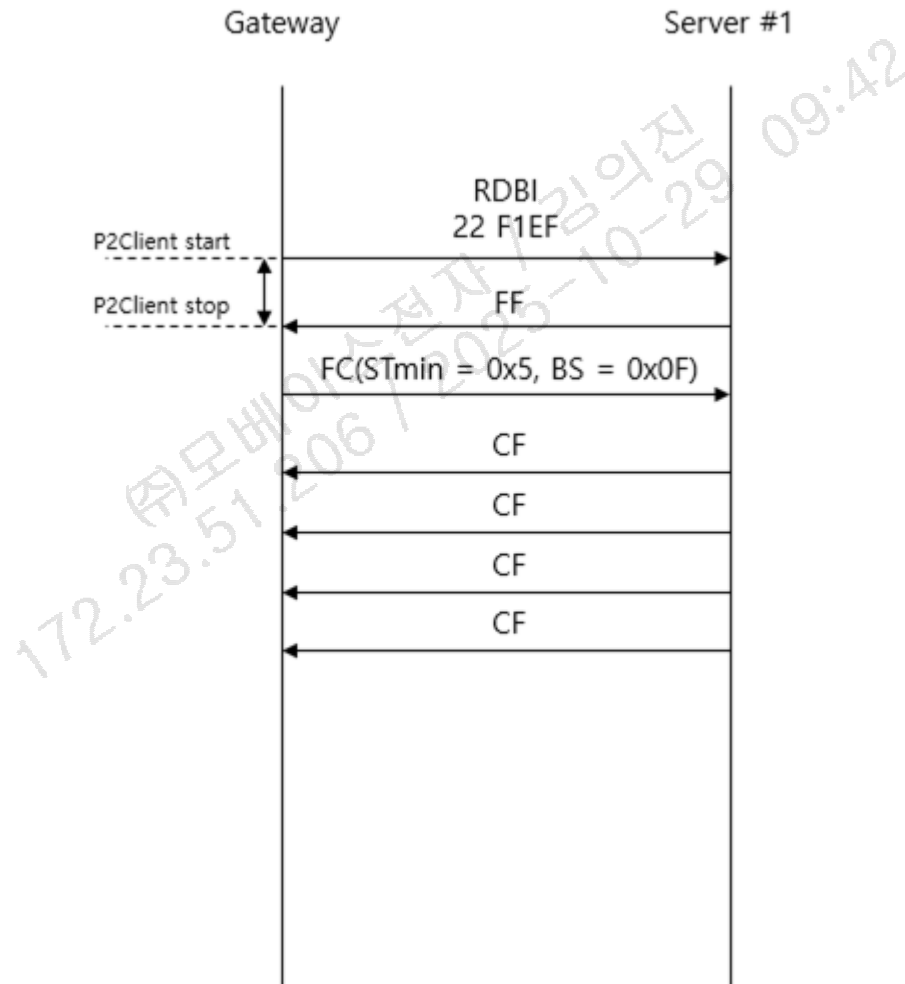


Figure J-16. RXSWIN Timing parameter diagram

| Parameter | Time  | Description                                                                                                                                                                |
|-----------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| P2Server  | 50ms  | ECU(Server) shall respond within 50ms                                                                                                                                      |
| P2Client  | 200ms | Gateway shall await response from ECU(Server).<br>If there is no response within P2Client, Gateway shall abort awaiting and send request to the next server(e.g. Server#2) |

Table J-5 RXSWIN Timing parameter

Server shall not send RCRRP(requestCorrectlyReceivedResponsePending 0x78) for the RDBI request.

If there was P2 Client timeout or error, Gateway shall retry gathering RXSWIN after all request to the ECU has been completed.

e.g.) If there is no response from Server #3, Gateway shall proceed with RXSWIN request to the next server (Server #4). After Gateway complete gathering RXSWIN from the last server (Server#n), Gateway can retry to gather RXSWIN for failed Server#3

Gateway shall not retry gathering RXSWIN if gateway receives NRC(Except Pending) from the Server(ECU).

When the gateway retries gathering RXSWIN, P2Client is constantly 200ms.

### J.8 Error handling

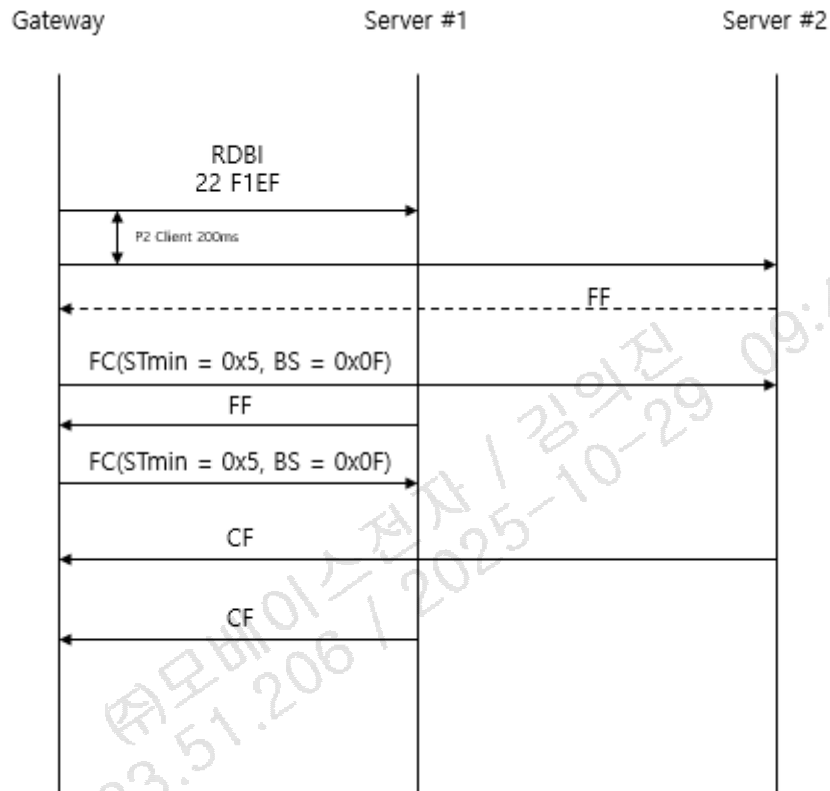


Figure J-17. RXSWIN error handling-1

In case Server #1 which is timedout responds to RDBI while Server #2 is sending RXSWIN, Gateway shall discard each response sent from Server#1 Server#2. And a retry is also required according to the J.7 Timing parameter and Retry logic.

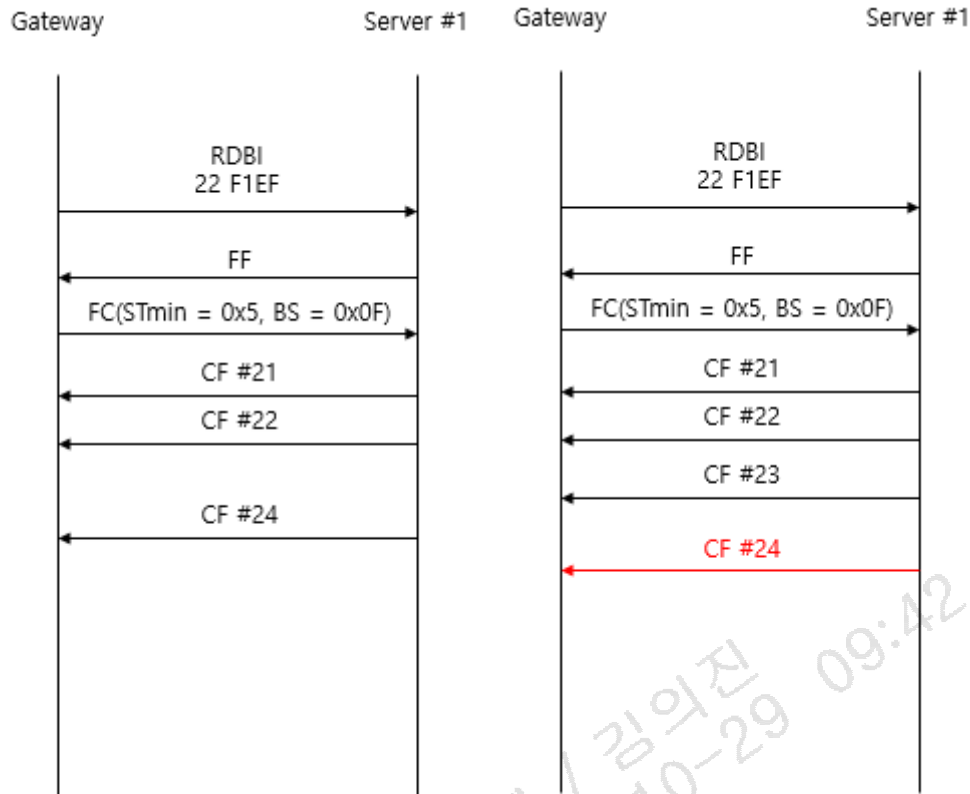


Figure J-18. RXSWIN error handling-2

In case a CF omission occurs, Gateway discard all responses. And a retry is also required according to the J.7 Timing parameter and Retry logic.

N\_CR timeout is 200ms.

In case expected CF is not sent from Server(ECU) within 200ms(N\_Cr), Gateway discard all responses. And a retry is also required according to the J.7 Timing parameter and Retry logic.

### J.9 Addressing and DID requirement

Only a central gateway can have DID F18F.

According to the response behavior, ECU except central gateway shall respond with NRC31 in case client requests DID F18F with physical addressing.

According to the response behavior, No response is required in case client requests DID F18F with functional addressing to the ECU except central gateway.

## Annex. K IVD Requirements

### K.1 IVD Requirements

Controllers that use aSIMS (or FST) to create Secure Flash firmware use the hash provided when firmware is created as an IVD.

The area used to calculate the IVD shall be the entire area excluding the signature.

For controllers that do not utilize aSIMS (or FST) or have own IVD calculation algorithm, the following shall be satisfied:

- Algorithm shall be SHA256
- SHA384, SHA512, SHA3 series could be allowed but need to be announced to the HMC
- The IVD calculation area shall be the entire area. However, if it is not possible to calculate the IVD value as a whole area, it may be excluded only in areas not affected by type approval.

## K.2 Hash value provision and verification method

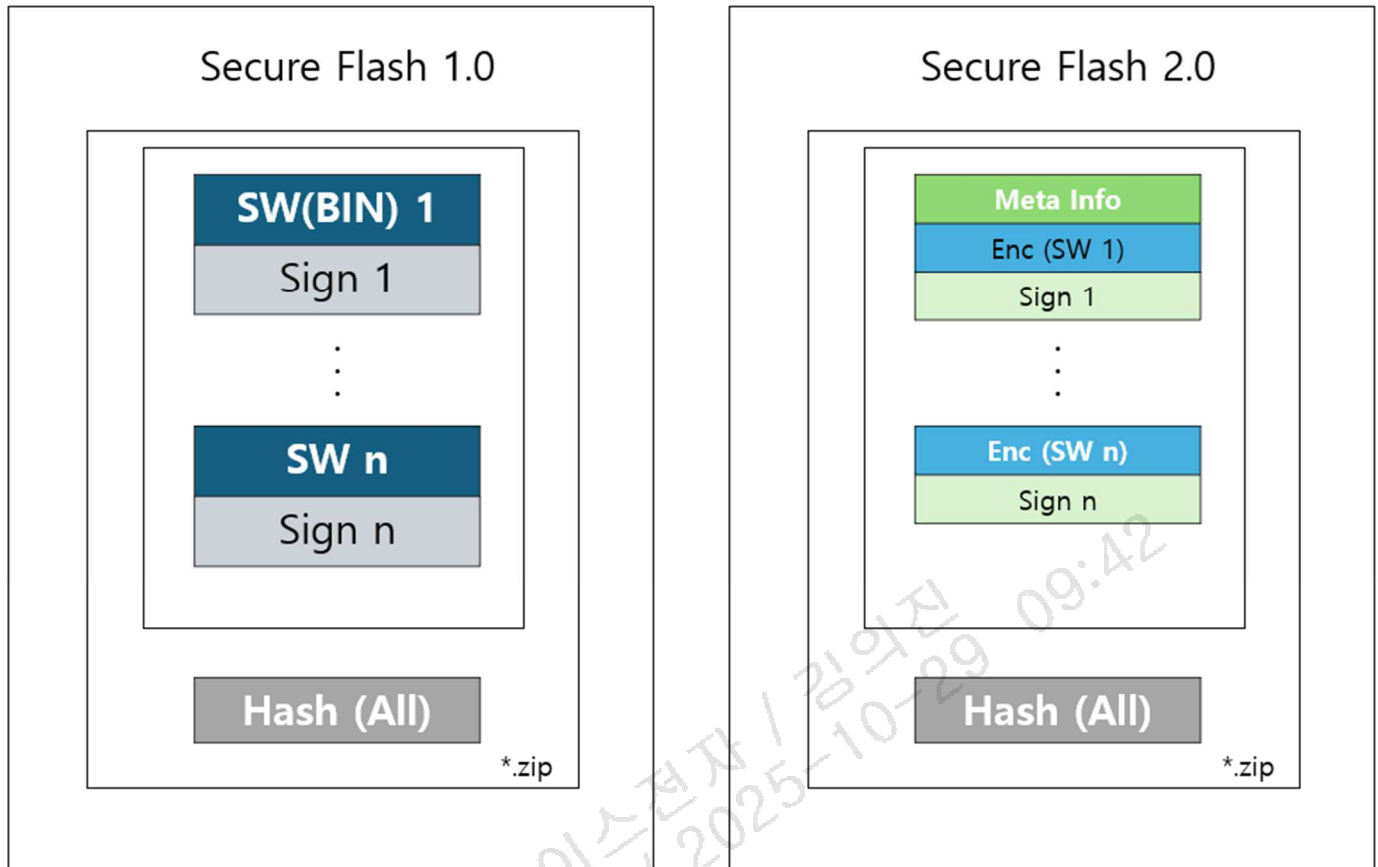


Figure K-1. Secure flash hash provision

- ※ Max n =8 (aSIMS) or 10(FST)
- ※ The signature of Secure Flash 2.0 is generated using plaintext firmware.
- ※ A hash is provided for use in the SUMS.
- ※ Except hash, Entire area shall be transmitted and reprogrammed.

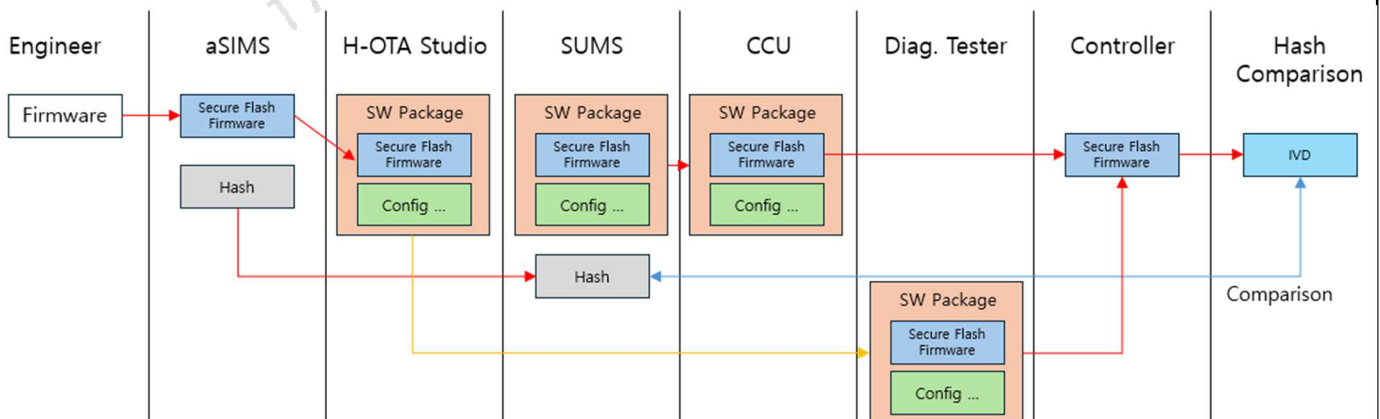


Figure K-2. Hash-IVD verification method